# NAZORATCHI — Infratuzilma monitoringi platformasi

## Texnik topshiriq (TZ)

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|---|---|
| **Hujjat turi** | Texnik topshiriq (Technical Specification) |
| **Loyiha ishchi nomi** | Nazoratchi (Nazoratchi Monitor) |
| **Versiya** | 1.17 (loyiha) |
| **Sana** | 2026-10-26 |
| **Holat** | Tasdiqlash uchun taqdim etilgan |
| **Buyurtmachi / mahsulot egasi** | Jasurbek (Avlo loyihalari egasi) |
| **Hujjat tili** | O'zbek (lotin), texnik atamalar va identifikatorlar asl holida |
| **Hujjat formati** | Markdown manba + Word (.docx) |

## 0.1. Hujjat tarixi

| Versiya | Sana | Muallif | O'zgarish |
|---|---|---|---|
| 0.1 | 2026-10-22 | Claude (loyiha yordamchisi) | Birinchi qoralama: tuzilma va inventar |
| 0.5 | 2026-10-22 | Claude | Funksional va nofunksional talablar, arxitektura |
| 1.0 | 2026-10-22 | Claude | To'liq nashr: ma'lumotlar modeli, API, UI, test, reja, ilovalar |
| 1.1 | 2026-10-26 | Claude | Loyiha nomi "Qarovul" dan "Nazoratchi" ga o'zgartirildi (QARORLAR.md, Q-06). Core va offsite S-00 da; S-03 ishlatilmaydi (QARORLAR.md) |
| 1.2 | 2026-10-26 | Claude | Q-12…Q-14 qarorlari: GCP IP yopilgan (`retired_ips`), Nazoratchi o'z zaxirasi `age` bilan shifrlanib Owner'ning shaxsiy Telegram chatiga yuboriladi, watchdog S-01 da (S-00 va S-01 bir-birini kuzatadi). `imraninc.com` loyihadan chiqarildi. NFR-SEC-010 M/F1 ga ko'tarildi |
| 1.3 | 2026-10-26 | Claude | P-02 natijalari: soat siljishi `sent_at` bilan aniqlanadi (6.10, 7.5.2, T-U-ING-04, -12…-14); JOIN li `rollup` agregati tasdiqlandi (8.4.2); siqilgan bo'lakka yozuv cheklovi (8.4.1); o'lchangan hajm 6.9 bayt/namuna (5.1.3); log maskalash regexi tuzatildi (14.9); R-102 hal bo'ldi |
| 1.4 | 2026-10-26 | Claude | E-01: `appctl` `state.json`/`_run.json` amalga oshirildi va sinaldi (10.3.1, 10.3.3, FR-INT-003 aniqlashtirildi: offsite `_run.json` da; `daily` `null` bo'lishi mumkin; haftalik `tar` xatosi endi zaxirani fail qiladi) |
| 1.5 | 2026-10-26 | Claude | E-02: `appctl list --json` / `status --json` amalga oshirildi va sinaldi (18.1 aniqlashtirildi: `state` ning `stopped`/`unknown` ma'nosi, `null` = olib bo'lmadi, xato JSON'i) |
| 1.6 | 2026-10-26 | Claude | E-03: heartbeat `appctl` ichiga qo'yildi (cron.d ko'p qatorli namuna ishlamaydi; token `appctl.secret` 0600 da), `setup.sh --heartbeat-url`; 9.14 va 10.3.5 yangilandi. Buzuq `.appctl` endi loyihani `fail` qiladi |
| 1.7 | 2026-10-27 | Claude | F1 S1 topilmalari: sessiya jadvali (`user_sessions`) va `public_id` unique emasligi (8.8); `__Host-` cookie faqat prod'da (14.3.4); `/healthz` ga loopback host orqali kirish (15.4, 15.5); prototip agent `enroll` (7.5.1) |
| 1.8 | 2026-10-27 | Claude | F1 S2: Go agent (system/disk/net) va Telegram kanali amalga oshirildi. Aniqlashtirishlar: core xatosi `{"error":{...}}` (7.5.1); core buyruqlarini agent hali bajarmaydi (7.5.3); `notifications.payload`/`next_attempt_at` (8.8); kanal API: `config` faqat yoziladi, `config_masked`, test `502`, tarix sahifalash (9.9); qayta urinish jadvali (12.9.0); `notifier` servisi va `TELEGRAM_API_BASE` (15.4, 15.5) |
| 1.9 | 2026-10-27 | Claude | F1 S3: alert dvigateli amalga oshirildi (16 standart qoida). Aniqlashtirishlar: `state` manbalari va "ma'lumot yo'q" (12.3.5), `of` kaliti (12.3.6), ahamiyat bo'yicha bostirish va S3 da bo'lmagan qismlar (12.5), incident boshlanishi = shart rost bo'la boshlagan vaqt (6.7.2), `pending` uzluksizligi (6.7.1), qo'shimcha ustunlar (8.7), `evaluator`/`notifier` komandalari va healthcheck (15.4) |
| 1.10 | 2026-10-27 | Claude | F1 S4: reyestr CRUD (serverlar, loyihalar, domenlar), seed, prober (HTTP/TCP), Telegram tugmalari. Aniqlashtirishlar: pauza `until`/sabab (8.3, 9.3), port to'qnashuvi va keyingi bo'sh port qoidasi (9.4), probe `DELETE` = arxiv va `allow_private` faqat Owner (9.6), webhook sirlari va `callback_data` formati (9.12), prober amalga oshirilishi: IP-pinning, qayta tekshiruv `next_run_at` orqali (6.8), `primary` kaliti (18.4), AL-049/049H/085 va `probe`/`prober` manbalari (12.3.5, 12.7), `prober`, `TELEGRAM_WEBHOOK_SECRET`, seed (15.4, 15.5) |
| 1.11 | 2026-10-27 | Claude | F1 S5: TLS va DNS tekshiruvlari, sertifikat/DNS/konteyner qoidalari (AL-035, 036, 057-060, 062, 063), Go agent `docker` kollektori, metrikalar so'rovi API (B4). Aniqlashtirishlar: kutilgan konteyner agent xotirasida (7.4.4, 11.5.1), TLS/DNS amalga oshirilishi (6.8), `cert_status`/`dns_results`/`dns_changes`/`settings` (8.5), so'rov API qoidalari va `null` bo'shliqlar (9.5), `/certs/`, `/domains/{id}/dns`, konteyner va fayl tizimi endpointlari (9.3, 9.4, 9.6), `when` sharti va `cert`/`dns` manbalari (12.3), WHOIS/RDAP F2 ga qoldi |

| Versiya | Sana | Muallif | O'zgarish |
|---|---|---|---|
| 1.12 | 2026-10-27 | Claude | F1 S6: zaxira monitoringi (B10) va agent `systemd/backup` kollektorlari (B5c). Aniqlashtirishlar: `systemd/backup/beat` amalga oshirilishi (7.4.5, 7.4.6), `projects.backup` siyosati va `backup_runs` agent metrikalaridan hosil bo'lishi (8.3, 8.6), `/backups/` API shakli (9.7), `backup` state manbasi va AL-068…072, 075 (12.3.5, 12.7.10) |
| 1.13 | 2026-10-28 | Claude | F1 S7/S8: o'zini kuzatish (B11), mustaqil watchdog (S-01), o'z zaxirasi, agent `install.sh/uninstall`, frontend asosi (S8). Aniqlashtirishlar: `/healthz` (liveness), `/readyz` (tayyorlik), `/metrics` (ichki) va `WATCHDOG_URL` ping oqimi (9.15), AL-084 watchdog orqali amalga oshirilishi, AL-090/091 o'z zaxirasi qoidalari (12.7), `NAZORATCHI_BACKUP_AGE_PUBKEY`, `METRICS_TOKEN`, `MINISIGN_PUBKEY`, `AGENT_DIST_DIR` (15.5), `install.sh` imzo fayli `SHA256SUMS.minisig` (7.7.1), `GET /agent/v1/<fayl>` (9.14), `warning` rangi `#c28400` (13.x, WCAG 3:1) |
| 1.14 | 2026-10-28 | Claude | F1 S9: frontend sahifalari (zaxira, domen, incident, texnik ish/silence, sozlamalar) va B12 deploy. Aniqlashtirishlar: prod compose `evaluator/prober/notifier/self-backup` servislari, Celery yo'q (15.4); backend image build context repo ildizi, `install.sh` va `seed` image ichida (15.3, 15.4); frontend image `nginx-unprivileged` 8080 (15.4); `/metrics` va `/readyz` nginx'da tashqariga berilmaydi, SSE `stream` yo'q (15.6); `deploy/bootstrap.sh` (15.7); `release.yml` va CI joblari (15.10); RB-044 haqiqiy `restore_self_backup` buyruqlari bilan, runbook'lar `docs/runbooks/` da (15.9, 18.2); mobil pastki panel 5 ustun (13.9) |
| 1.15 | 2026-10-28 | Claude | F1 S10: sinov va mustahkamlash (B13). Aniqlashtirishlar: `evaluator` va `prober` leader qulfi `pg_try_advisory_lock` bilan amalga oshirildi (15.4); ulanish `connect_timeout=5`, baza o'chiq bo'lsa `/healthz` 5 s da `503` (16.6, T-C-01); noma'lum `/api/`, `/ingest/`, `/agent/` yo'llari JSON `404`, `500` tafsilotsiz JSON (9.1.1); 14.13 va 16.5–16.7 bandlari holati `docs/sinov/B13-natijalar.md` da; `tools/simulator` yuklama simulyatori (16.9) |
| 1.16 | 2026-10-28 | Claude | F1 S10 davomi: restore drill (UC-11, FR-BKP-016) F2 dan F1 ga olindi. Aniqlashtirishlar: `restore_drills` jadvali va `/backups/drills` API (9.7), zaxira jadvali/kartasida `drill` bloki — "hech qachon" va muddati o'tgani ko'rinadi (R-117); frontend: karta sahifasida forma va tarix, ro'yxatda drill belgisi (13, SCR-07). AL-081…083 F2 da qoladi |
| 1.17 | 2026-10-28 | Claude | F1 S10 davomi: AL-081, AL-082, AL-083 (restore drill muddati/natijasi) F2 dan F1 ga olindi, F1 alertlar soni 32 → 35 (12.7.10, 12.9). Aniqlashtirishlar: yangi `drill` state manbasi (`due_soon`, `overdue`, `failed`), faqat faol, ko'chirilgan `expect_daily` loyihalar; AL-081 muddat o'tmaguncha yonadi, o'tgach AL-082 (takroriy shovqin yo'q); hech qachon drill o'tkazilmagan loyiha `overdue`; muvaffaqiyatsiz drill muddat taymerini qayta boshlamaydi; RB-034 to'ldirildi |

Hujjatga o'zgartirish kiritish tartibi: har bir o'zgarish yangi qator sifatida shu jadvalga yoziladi; talab identifikatorlari (FR-..., NFR-...) hech qachon qayta ishlatilmaydi, o'chirilgan talab "BEKOR QILINGAN" deb belgilanadi, raqami saqlanadi.

## 0.2. Mundarija


| № | Bo'lim | Fayl |
|---|---|---|
| 0 | Titul, atamalar, hujjatdan foydalanish | 00-titul.md |
| 1 | Kirish: maqsad, muammo, doira, KPI | 01-kirish.md |
| 2 | Hozirgi infratuzilma inventari | 02-inventar.md |
| 3 | Foydalanuvchilar, rollar va foydalanish stsenariylari | 03-foydalanuvchilar.md |
| 4 | Funksional talablar | 04a-funksional-talablar.md, 04b-funksional-talablar.md |
| 5 | Nofunksional talablar | 05-nofunksional-talablar.md |
| 6 | Tizim arxitekturasi | 06-arxitektura.md |
| 7 | Agent spetsifikatsiyasi va metrikalar katalogi | 07-agent-va-metrikalar.md |
| 8 | Ma'lumotlar modeli | 08-malumotlar-modeli.md |
| 9 | API spetsifikatsiyasi | 09-api.md |
| 10 | Zaxira nusxalar (backup) monitoringi | 10-backup-monitoring.md |
| 11 | Loyiha, servis, domen va sertifikat monitoringi | 11-servis-monitoring.md |
| 12 | Alert (ogohlantirish) tizimi | 12-alert-tizimi.md |
| 13 | Foydalanuvchi interfeysi | 13-interfeys.md |
| 14 | Xavfsizlik | 14-xavfsizlik.md |
| 15 | O'rnatish, deploy va ekspluatatsiya | 15-deploy.md |
| 16 | Test rejasi va qabul mezonlari | 16-test-rejasi.md |
| 17 | Loyiha rejasi, resurslar va xavflar | 17-reja-va-xavflar.md |
| 18 | Ilovalar (JSON sxemalar, runbook'lar, tekshiruv sxemalari, seed reyestr, sanoq) | 18a-ilovalar.md, 18b-ilovalar.md |


## 0.3. Hujjatdan foydalanish qoidalari

1. **Talablar identifikatori.** Har bir talab noyob kodga ega: `FR-<MODUL>-<NNN>` (funksional), `NFR-<TOIFA>-<NNN>` (nofunksional), `US-<NNN>` (foydalanuvchi hikoyasi), `AL-<NNN>` (alert qoidasi), `RB-<NNN>` (runbook), `R-<NNN>` (xavf), `T-<NNN>` (test).

2. **Ustuvorlik (MoSCoW).** Har talabda ustuvorlik belgisi bor:
   - **M** (Must) — usiz tizim qabul qilinmaydi;
   - **S** (Should) — muhim, lekin birinchi reliz keyinga qoldirilishi mumkin;
   - **C** (Could) — foydali, vaqt bo'lsa;
   - **W** (Won't) — bu versiyada qilinmaydi, kelajak uchun qayd etilgan.
3. **Bosqich (Faza).** Har talab qaysi bosqichda amalga oshirilishi ko'rsatiladi: F1 (MVP), F2 (kengaytirilgan), F3 (yetuklik). Bosqichlar 17-bo'limda tushuntirilgan.
4. **Kalit so'zlar.** "kerak" (shall) — majburiy; "tavsiya etiladi" (should) — tavsiya; "mumkin" (may) — ixtiyoriy. Raqamli chegaralar (masalan, 60 soniya) o'zgartirilishi mumkin bo'lgan sozlamalar bo'lsa, "standart qiymat" deb yoziladi.
5. **Misollar.** Hujjatdagi IP-manzillar, domenlar va portlar 2026-10-22 holatiga ko'ra haqiqiy infratuzilmadan olingan. Parollar, tokenlar va kalitlar hujjatga **hech qachon** kiritilmaydi; ular o'rnida `***` yoki `<TOKEN>` yoziladi.

## 0.4. Atamalar va qisqartmalar

| Atama | Ma'nosi |
|---|---|
| **Agent** | Kuzatiladigan serverda ishlaydigan kichik dastur; metrikalarni yig'ib markazga yuboradi |
| **Markaziy server (Core)** | Metrikalarni qabul qiladigan, saqlaydigan, alert hisoblaydigan va UI beradigan server |
| **Host** | Kuzatiladigan jismoniy yoki virtual server (VPS) |
| **Loyiha (Project)** | Bitta mahsulot: bir yoki bir nechta servis, domen va bazadan iborat (masalan, Nazar, BiznesRadar) |
| **Servis (Service)** | Loyihaning ishlaydigan qismi: Docker konteyner, systemd unit yoki nginx sayti |
| **Metrika (Metric)** | Vaqt bo'yicha o'zgaruvchi son (CPU foizi, bo'sh xotira va h.k.) |
| **Namuna (Sample)** | Metrikaning bitta vaqtdagi qiymati |
| **Tekshiruv (Check)** | Faol (active) tekshirish: HTTP so'rov, TCP ulanish, DNS so'rov, sertifikat muddati |
| **Alert** | Shartga asoslangan ogohlantirish (masalan, "disk 85% dan oshdi") |
| **Incident** | Bir yoki bir nechta alertdan tashkil topgan voqea, yechilguncha ochiq turadi |
| **Heartbeat** | Agentning "tirikman" signali |
| **Retention** | Ma'lumotni saqlash muddati |
| **Downsampling** | Eski ma'lumotni siyraklashtirib (o'rtachalab) saqlash |
| **RPO** | Recovery Point Objective — ma'lumotning yo'qolishi mumkin bo'lgan maksimal vaqt |
| **RTO** | Recovery Time Objective — tiklash uchun maksimal vaqt |
| **SLO / SLA** | Xizmat darajasi maqsadi / kafolati |
| **MTTR / MTTD** | Tiklashning / aniqlashning o'rtacha vaqti |
| **appctl** | `/usr/local/bin/appctl` — serverdagi loyihalarni boshqarish skripti (mavjud) |
| **RBAC** | Rolga asoslangan kirish nazorati |
| **TSDB** | Vaqt qatorlari bazasi (time-series database) |
| **SSE** | Server-Sent Events — serverdan brauzerga bir yo'nalishli oqim |
| **ACME / HTTP-01** | Let's Encrypt sertifikat olish protokoli / tekshiruv usuli |
| **PROXY protocol** | TCP proksi orqali haqiqiy mijoz IP sini uzatish protokoli |
| **Runbook** | Muayyan alert chiqqanda bajariladigan qadamma-qadam yo'riqnoma |
| **Maintenance window** | Rejalashtirilgan ishlar vaqti; alertlar jim qilinadi |

# 1. Kirish

## 1.1. Hujjatning maqsadi

Ushbu texnik topshiriq **Nazoratchi** nomli infratuzilma monitoringi platformasini yaratish uchun barcha talablarni belgilaydi. Hujjat quyidagi savollarga aniq javob berishi kerak:

- Tizim nimani kuzatadi va nima uchun?
- Qaysi ma'lumot qanday yig'iladi, qayerda va qancha vaqt saqlanadi?
- Qachon va kimga ogohlantirish yuboriladi?
- Dasturchi qaysi texnologiya bilan, qaysi tartibda va qanday sifat mezoni bilan ishlaydi?
- Tizim tayyor ekani qanday tekshiriladi?

Hujjat bir vaqtning o'zida (a) dasturchi uchun ishlab chiqish qo'llanmasi, (b) buyurtmachi uchun qabul qilish shartnomasi va (c) keyinchalik tizimni ekspluatatsiya qiladigan odam uchun ma'lumotnoma bo'lib xizmat qiladi.

## 1.2. Loyihaning qisqacha tavsifi

**Nazoratchi** — buyurtmachining barcha serverlari, ularda ishlayotgan loyihalar, zaxira nusxalar, sertifikatlar va domenlarning holatini bitta oynada ko'rsatuvchi hamda muammo paydo bo'lishidan **oldin** yoki paydo bo'lgan zahoti Telegram orqali xabar beruvchi platforma.

Platforma uch qismdan iborat:

1. **Agent** — har bir serverga o'rnatiladi, CPU, RAM, disk (SSD), tarmoq, Docker konteynerlar, systemd servislar va zaxira papkalari haqida ma'lumot yig'adi.
2. **Markaziy server** — ma'lumotni qabul qiladi, saqlaydi, qoidalar bo'yicha tahlil qiladi, faol tekshiruvlarni (HTTP, TLS, DNS) o'zi bajaradi, alert yuboradi.
3. **Veb-panel** — barcha ma'lumotni ko'rsatadi (jonli grafiklar, loyihalar ro'yxati, zaxiralar holati, incidentlar tarixi).

## 1.3. Muammo bayoni va loyiha asosi

### 1.3.1. 2026-10 hodisasi (post-mortem qisqacha)

2026-10-18 kuni 15:25 da toza `poweroff` bilan to'xtagan "Avlo main server" (`185.214.135.9`, Contabo vmi3628663) qayta ishga tushmadi: VNC'da faqat `Booting from Hard Disk...` ko'rinib, foydalanuvchi darajasiga hech qachon yetib bormadi. Bir necha kunlik diagnostika (VNC, Rescue System, `fsck`, GRUB sozlamalari, boshqa kernel tanlash) natija bermadi. Barcha ma'lumot rescue tizimidan `rsync` orqali yangi serverga (`164.68.105.194`) ko'chirildi va loyihalar bir-bir qayta ko'tarildi. (2026-10-26 holati: eski server tozalanib, Nazoratchi core serveri sifatida qayta o'rnatilmoqda, QARORLAR.md.)

Hodisadan olingan saboqlar, ular shu loyiha talablarining asosi bo'ldi:

| № | Saboq | Qaysi talabga olib keldi |
|---|---|---|
| L-1 | Server ishdan chiqqanini foydalanuvchilar yoki egasi o'zi bilib qoldi, avtomatik xabar bo'lmadi | Heartbeat va "server o'chdi" alerti (FR-ALR-*) |
| L-2 | Zaxira nusxalar faqat shu serverning o'zida edi; serverga kirib bo'lmagach, ularga ham kirib bo'lmadi | Offsite zaxira monitoringi (FR-BKP-*) |
| L-3 | Server GCP'dan ko'chirilgan, GCP'ga xos kernel va agentlar qolgan edi; bu xavf oldindan ko'rinmagan | Inventar va "kutilayotgan kernel yangilanishi" tekshiruvi (FR-HST-*) |
| L-4 | Kernel yangilanishi/reboot dan oldin snapshot olinmagan | Reboot oldidan nazorat ro'yxati va "reboot kerak" belgisi |
| L-5 | Loyihalar qaysi serverda, qaysi portda, qaysi domen bilan ishlashi bitta joyda yozilmagan edi | Reyestr (registry) moduli (FR-REG-*) |
| L-6 | DNS yozuvlari turli provayderlarda (ahost.uz, Namecheap, Google Cloud DNS, Beget); qaysi domen qayerga qarashi noaniq | DNS monitoring (FR-DNS-*) |
| L-7 | Sertifikat muddati qo'lda kuzatilgan | TLS monitoring (FR-TLS-*) |
| L-8 | Disk va log hajmi nazoratsiz | Disk bashorati va Docker log hajmi tekshiruvi |

### 1.3.2. Hozirgi kuzatuv holati

Hozir (2026-10-22) monitoring **yo'q**: faqat `appctl status` buyrug'i qo'lda ishga tushiriladi va diskning, xotiraning, konteynerlarning, sertifikat muddatlarining hamda oxirgi zaxiralarning bir lahzalik holatini ko'rsatadi. Tarix saqlanmaydi, ogohlantirish yo'q.

### 1.3.3. Ishlatilmagan imkoniyatlar

- Zaxira kunlik 03:15 da avtomatik olinadi (`/etc/cron.d/appctl-backup`), lekin natijasi faqat `/var/log/appctl-backup.log` ga yoziladi. Hech kim uni o'qimaydi; muvaffaqiyatsiz zaxira bir hafta jim qolishi mumkin.
- Sertifikatlar `certbot` bilan yangilanadi, lekin yangilanish xatosi haqida xabar yo'q.
- Docker konteynerlar `restart: unless-stopped` bilan qayta ishga tushadi, lekin cheksiz qayta ishga tushish (crash loop) bildirilmaydi.

## 1.4. Loyiha maqsadlari

| ID | Maqsad | O'lchov |
|---|---|---|
| G-1 | Har bir server holatini (CPU, RAM, disk, tarmoq) real vaqtga yaqin ko'rish | Yangilanish kechikishi ≤ 30 soniya |
| G-2 | Server yoki servis ishdan chiqqanini tez bilish | MTTD ≤ 2 daqiqa (server), ≤ 90 soniya (HTTP servis) |
| G-3 | Disk to'lib qolishidan oldin ogohlantirish | Disk to'lishiga ≥ 7 kun qolganda xabar |
| G-4 | Zaxira nusxalar mavjudligi, yangiligi va to'liqligiga ishonch | Har loyiha uchun "oxirgi muvaffaqiyatli zaxira yoshi" ko'rinadi; > 26 soat bo'lsa alert |
| G-5 | Sertifikat va domen muddati o'tib ketmasligi | 21, 7, 3 kun oldin ogohlantirish |
| G-6 | Barcha loyiha/domen/port/baza ma'lumoti bitta reyestrda | Reyestr 100% inventarga mos |
| G-7 | Ogohlantirishlar "shovqin" bo'lmasligi | Soxta alert ulushi ≤ 5% (oyiga) |
| G-8 | Tizimning o'zi ishonchli bo'lishi | Nazoratchi'ning o'zi o'chsa, boshqa serverdagi (S-01) "watchdog" xabar beradi |
| G-9 | Kam resurs sarfi | Agent ≤ 1% CPU, ≤ 60 MB RAM; markaziy server 1 vCPU/2 GB da ishlaydi |
| G-10 | Sodda boshqaruv | Yangi serverni 5 daqiqada qo'shish (bitta buyruq) |

## 1.5. Loyiha doirasi

### 1.5.1. Doira ichida

1. Linux serverlarni (Ubuntu 22.04/24.04) agent orqali kuzatish.
2. Tizim metrikalari: CPU, yuklama (load), RAM, swap, disk (hajm, inode, I/O), tarmoq, jarayonlar soni, uptime, kernel versiyasi, kutilayotgan reboot.
3. Docker: konteynerlar holati, qayta ishga tushishlar, resurs sarfi, log hajmi, image/volume hajmi.
4. systemd: berilgan servislar holati (masalan, `avlo-contact`, `redis`, `nginx`).
5. Faol tekshiruvlar: HTTP(S), TCP port, DNS, TLS sertifikat muddati, domen (WHOIS) muddati.
6. Zaxira nusxalar: `/opt/applications/backups/<loyiha>/{daily,weekly}` ni kuzatish, offsite sinxronlashni tekshirish, tiklash testini (restore drill) rejalashtirish va qayd etish.
7. Reyestr: serverlar, loyihalar, domenlar, portlar, bazalar, egalar.
8. Alert tizimi: qoidalar, jim qilish (silence), texnik ish oynalari, eskalatsiya.
9. Kanallar: Telegram (asosiy), elektron pochta (ikkilamchi), webhook (kengaytma).
10. Veb-panel (uz/ru/en), mobil brauzerda qulay.
11. Rollar: egasi (owner), operator, kuzatuvchi (read-only).
12. Audit jurnali: kim nimani o'zgartirdi.
13. O'zini-o'zi kuzatish va alohida serverdagi watchdog (S-01).

### 1.5.2. Doira tashqarisida (bu versiyada)

| ID | Nima | Sababi |
|---|---|---|
| X-1 | Ilova ichidagi APM / distributed tracing (OpenTelemetry) | Alohida katta yo'nalish; F3 dan keyin |
| X-2 | Log yig'ish va qidirish (Loki/ELK analogi) | Faqat log hajmi nazorati kiradi, mazmuni emas |

| ID | Nima | Sababi |
|---|---|---|
| X-3 | Avtomatik tuzatish (auto-remediation), masalan, konteynerni o'zi qayta ishga tushirish | Xavfli; faqat F3 da, cheklangan "restart" amali |
| X-4 | Windows/macOS serverlar | Infratuzilmada yo'q |
| X-5 | Kubernetes | Infratuzilmada yo'q |
| X-6 | Ko'p ijarachili (multi-tenant) SaaS rejimi | Bitta egasi uchun |
| X-7 | Zaxiralarni o'zi olish | Zaxirani `appctl` oladi; Nazoratchi faqat kuzatadi (restore drill bundan mustasno, 10-bo'lim) |
| X-8 | Serverni boshqarish (SSH terminal, fayl menejer) | Xavfsizlik xavfi; kuzatuv ≠ boshqaruv |

## 1.6. Manfaatdor tomonlar

| Tomon | Qiziqishi |
|---|---|
| Egasi (Jasurbek) | Barcha loyiha ishlashini bilish, mijozlar oldida javobgarlik, pul yo'qotishdan saqlanish |
| Dasturchi/DevOps (egasi yoki yollangan) | Muammoni tez topish, aniq ko'rsatma, kam shovqin |
| Mijozlar (masalan, wmg.uz egasi, Nazar foydalanuvchilari) | Xizmat uzluksizligi; to'g'ridan-to'g'ri Nazoratchi'dan foydalanmaydi |
| Hamkor hosting provayderlari (Contabo va boshqalar) | Faqat tashqi omil: server holati ularga bog'liq |

## 1.7. Muvaffaqiyat mezonlari (KPI)

Tizim ishga tushirilgandan keyin 90 kun ichida quyidagilar bajarilishi kerak:

| KPI | Maqsad |
|---|---|
| Barcha serverlar va loyihalar reyestrda | 100% |
| Server o'chganda xabar kelish vaqti | ≤ 2 daqiqa |
| HTTP servis ishdan chiqqanda xabar kelish vaqti | ≤ 90 soniya |
| Muvaffaqiyatsiz zaxira aniqlanishi | Keyingi tekshiruvda (≤ 15 daqiqa); eng ko'pi bilan 26 soatda |
| Sertifikat muddati o'tib ketgan holatlar | 0 |
| Disk to'lib ketgan holatlar | 0 |
| Soxta alertlar | ≤ 5% |
| Haqiqiy incidentlarning aniqlanish ulushi | ≥ 95% (qo'lda baholash) |
| Nazoratchi'ning o'zi uptime'i | ≥ 99.5% |
| Restore drill | Har loyiha uchun kamida chorakda 1 marta muvaffaqiyatli |

## 1.8. Taxminlar va cheklovlar

### 1.8.1. Taxminlar

1. Barcha kuzatiladigan serverlar Linux (Ubuntu), `systemd` va Docker (ba'zilarida) bilan.
2. Markaziy server **kuzatiladigan serverlardan alohida** joyda turadi (kamida bitta server boshqasini kuzatishi uchun). Tavsiya: alohida arzon VPS (1 vCPU / 2 GB / 20 GB).
3. Agentlar markazga HTTPS orqali chiqish (outbound) ulanishi bilan ma'lumot yuboradi; kuzatiladigan serverda yangi kiruvchi (inbound) port ochish talab qilinmaydi.
4. Telegram bot tokeni mavjud va kanal (chat) ochilgan bo'ladi.
5. Egasi bitta shaxs yoki kichik jamoa; 10–30 server va 20–60 loyiha miqyosi.

### 1.8.2. Cheklovlar

| ID | Cheklov |
|---|---|
| C-1 | Byudjet cheklangan: ochiq manbali komponentlar ishlatiladi, pullik SaaS yo'q |
| C-2 | Asosiy texnologiyalar buyurtmachi tanish stekka mos: Python/Django, PostgreSQL, Docker Compose, nginx |
| C-3 | Markaziy server resursi kam (1–2 vCPU, 2–4 GB RAM) |
| C-4 | Maxfiy ma'lumot (parol, token) metrikalarga yoki loglarga tushmasligi kerak |
| C-5 | Barcha interfeys matnlari o'zbek tilida (lotin); qo'shimcha ru, en |

| ID | Cheklov |
|---|---|
| C-6 | Agent o'rnatilishi mavjud ishlayotgan loyihalarni to'xtatmasligi kerak |

## 1.9. Mavjud yechimlar bilan taqqoslash va tanlov asosi

Bu bo'lim halollik uchun kiritilgan: tayyor yechimlar bor, ularning ba'zilari ushbu TZ ning katta qismini allaqachon bajaradi. Qaror egasiga tegishli.

| Yechim | Kuchli tomoni | Zaif tomoni (bizning ehtiyoj uchun) |
|---|---|---|
| **Uptime Kuma** | HTTP/TCP/DNS/sertifikat tekshiruvi, chiroyli panel, Telegram — 10 daqiqada ishga tushadi | CPU/RAM/disk yo'q; zaxira fayllarini kuzatmaydi |
| **Netdata** | Server metrikalari va Docker, juda boy, tayyor alertlar | Zaxira, reyestr, o'zbekcha UI, bizning `appctl` bilan integratsiya yo'q; markaziy panel (Netdata Cloud) tashqi xizmat |
| **Prometheus + Grafana + Alertmanager** | Sanoat standarti, cheksiz moslashuvchan | Sozlash va ekspluatatsiya murakkab; zaxira/reyestr uchun baribir o'z kod kerak |
| **Beszel / Glances** | Yengil agent + panel | Funksiyalar cheklangan |
| **Zabbix** | To'liq korxona yechimi | Og'ir, ko'p resurs, o'rganish qiyin |
| **Healthchecks.io** (o'zi-hosting) | Cron/zaxira "tirikman" signallari uchun juda yaxshi | Faqat heartbeat |

**Tanlov asosi.** Nazoratchi yaratilishining sababi bitta: ehtiyojlarning aynan **birlashmasi** — (server metrikasi) + (loyiha reyestri) + (appctl bilan zaxira nazorati va restore drill) + (DNS/domen/sertifikat) + (o'zbekcha, sodda UI) — tayyor yechimda yo'q. Shu bilan birga:

> **Tavsiya (oraliq yechim).** Nazoratchi'ning birinchi ishlaydigan versiyasi (F1) bir dasturchi bilan taxminan 4–5 oy, ikki dasturchi bilan 2–2.5 oy talab qiladi (17.3.5). Shu vaqt davomida serverlar kuzatuvsiz qolmasligi uchun **Uptime Kuma** ni darhol o'rnatish tavsiya etiladi (1–2 soat ish). Nazoratchi F1 tugagach, Uptime Kuma o'chiriladi yoki ikkilamchi tekshiruv sifatida qoladi.

**Yaratish yoki tayyorini olish (Build vs Buy) bo'yicha xulosa:** Agar egasi faqat "server o'chdi / sayt ishlamayapti" xabarini xohlasa — tayyor yechim yetarli. Agar zaxira va reyestr nazorati ham muhim bo'lsa (hodisa saboqlari L-2, L-5), Nazoratchi asoslangan.

## 1.10. Talablar ustuvorligi

Talablar umumiy soni (4–5-bo'limlar): **243 ta** (178 funksional + 65 nofunksional). Taqsimot:

| Ustuvorlik | Funksional | Nofunksional | Jami | Izoh |
|---|---|---|---|---|
| M (Must) | 95 | 47 | 142 | F1 va F2 ning asosi |
| S (Should) | 59 | 15 | 74 | Asosan F2 |
| C (Could) | 24 | 3 | 27 | Asosan F3 |
| W (Won't) | 0 | 0 | 0 | Bu versiyada qilinmaydiganlar 1.5.2 da (X-1..X-8) alohida ro'yxatlangan |

Aniq ro'yxat va bosqichlar 4–5 hamda 17-bo'limlarda; bosqichlar bo'yicha taqsimot — 18.7.

# 2. Hozirgi infratuzilma inventari

Bu bo'lim 2026-10-22 holatiga ko'ra mavjud tizimni tasvirlaydi. Nazoratchi reyestri (FR-REG-*) birinchi marta shu ma'lumot bilan to'ldiriladi. Aniq noma'lum qiymatlar "aniqlanadi" deb belgilangan; ularni yig'ish buyruqlari 2.9-bo'limda.

## 2.1. Serverlar

| ID | Nomi | Provayder | IP | Rol | Holat |
|---|---|---|---|---|---|
| S-01 | vmi3679093 ("main-2") | Contabo VPS | 164.68.105.194 | Asosiy ishchi server: Nazar, BiznesRadar, lutfan_ai, wmg, avlo.ai, landinglar; keyinchalik Nazoratchi watchdog | Ishlayapti |
| S-02 | vmd208589 | (aniqlanadi) | 173.249.37.152 | Aaclub staging + production stek, asadbekashurov.uz landingi | Ishlayapti |
| S-03 | (nomi aniqlanadi) | (aniqlanadi) | 38.242.143.248 | Egasiniki; zaxira uchun **ishlatilmaydi** (Owner qarori, 2026-10-26) | Zaxiralanmaydi |
| S-00 | vmi3628663 ("Avlo main server") | Contabo VPS, 200 GB SSD | 185.214.135.9 | **Nazoratchi core + S-01 uchun offsite zaxira** (Owner qarori). Oldin yuklanmagan, tozalanib qayta o'rnatilmoqda | Qayta o'rnatilmoqda |
| S-GCP | (Google Cloud) | GCP | 34.173.220.247 | Eski bulut, **endi mavjud emas** (Owner, 2026-10-26); lutfanai.uz, wmg.uz subdomenlari hali shu IP ga qaraydi | O'CHIRILGAN, DNS yangilanishi kerak |

### 2.1.1. S-01 batafsil

| Parametr | Qiymat |
|---|---|
| OS | Ubuntu 24.04 LTS |
| Kernel (ishlayotgan) | 6.8.0-139-generic |
| Kernel (o'rnatilgan, kutilayotgan) | 6.8.0-147-generic (reboot kerak) |
| Docker | Docker Engine + Compose v5; `daemon.json`: `json-file` 10m×3, `live-restore: true` |
| Veb-server | nginx (host), konfiglar `/opt/applications/nginx/*.conf` |
| Sertifikat | certbot (nginx authenticator); `/etc/letsencrypt` |
| Node.js | 22.23.3 (NodeSource), `avlo-contact` uchun |
| Tarmoq | Tashqariga faqat 22, 80, 443; qolgan portlar `127.0.0.1` da |
| Boshqaruv | `appctl` (`/usr/local/bin/appctl`), `/opt/applications/{projects,nginx,scripts,backups,sites,services}` |
| Zaxira | Kunlik 03:15 `appctl backup all`; log `/var/log/appctl-backup.log` |
| SSH | Parol bilan (kalit keyinroq rejalashtirilgan) |
| CPU / RAM / disk | aniqlanadi (2.9-bo'lim) |

### 2.1.2. S-02 batafsil

| Parametr | Qiymat |
|---|---|
| Disk | 193 GB, ~19 GB band (10%) |
| nginx | 1.24.0, faqat front-proksi (`stream` + `http` map), konfig yagona fayl `/etc/nginx/nginx.conf` |
| Yo'naltirish | `*.staging.*` → 127.0.0.1:28080/28443; qolgan hamma → 127.0.0.1:18080/18443 (production, hozir ishlamayapti); `asadbekashurov.uz` → 127.0.0.1:38080/38443 |
| Docker | `aaclub-staging-*` (≈ 25 konteyner: api, worker-*, beat, postgres, redis, minio, qdrant, prometheus, loki, grafana, alertmanager, node-exporter, cadvisor va h.k.), `asadbekashurov-web-1` |
| Himoya | ufw (22, 80, 443), fail2ban (sshd) |
| Eslatma | Serverda **allaqachon** Prometheus+Grafana+node-exporter (staging steki ichida) bor; Nazoratchi ularni almashtirmaydi, lekin ularning mavjudligini hisobga oladi |
| Maxsus | nginx.conf "TO'LIQ fayl" deb belgilangan, aaclub repo'sidan boshqariladi; bizning qo'shgan qoidalarimiz qayta yozilishi xavfi bor (R-009) |

## 2.2. Loyihalar (S-01)

Barcha loyihalar `/opt/applications/projects/<Nomi>/` da. Portlar `127.0.0.1` ga bog'langan.

| Loyiha | Domen(lar) | Backend port | Frontend port | Turi | Baza | Zaxira |
|---|---|---|---|---|---|---|
| Nazar | nazar.avlo.ai, app.beyondlinks.uz | 8020 | 3020 | Docker Compose (`docker-compose.prod.yml`): db, redis, web, worker, worker-yandex, beat, frontend | Postgre SQL 16 | pg_dump kunlik |

| Loyiha | Domen(lar) | Backend port | Frontend port | Turi | Baza | Zaxira |
|---|---|---|---|---|---|---|
| BiznesRadar | biznesradar.uz, api./admin./cabinet. | 8030 | 4321 | Docker Compose | Postgre SQL 17 | pg_dump kunlik + media haftalik |
| beyondlinks_landing | beyondlinks.uz | — | statik | nginx | — | — |
| freighthubllc | freighthubllc.co / .com | — | statik | nginx | — | — |
| lutfan_ai | lutfanai.uz, lutfan.israilov.uz | — | 8080 (+MinIO 9000) | Docker Compose + `docker-compose.server.yml` | Postgre SQL 15 | pg_dump + media/minio haftalik |
| wmg | moliyaviyerkinlik.wmg.uz, investitsiya.wmg.uz | — | 8011, 8012 | Docker Compose (2 servis, gunicorn) | SQLite | sqlite+media haftalik |
| avlo.ai sayt | avlo.ai, www.avlo.ai | 3017 (`avlo-contact`) | statik `/opt/applications/sites/avlo` | nginx + systemd | — | — |

### 2.2.1. Servis nomlari va `appctl` konfiguratsiyasi

Har loyiha papkasida `.appctl` fayli (bash sintaksisi) bor:

| O'zgaruvchi | Ma'nosi |
|---|---|
| `DB_SERVICE`, `DB_USER`, `DB_NAME` | pg_dump uchun |
| `PROJECT_NAME` | Compose loyiha nomi |
| `COMPOSE_EXTRA` | Qo'shimcha compose fayl (serverga xos portlar) |
| `HEALTH_URL`, `HEALTH_CURL_OPTS` | Sog'liq tekshiruvi (masalan, `-H Host:nazar.avlo.ai -H X-Forwarded-Proto:https`) |
| `BACKUP_PATHS` | Haftalik fayl zaxirasi yo'llari |

Nazoratchi bu faylni **o'qimaydi**; ma'lumotni `appctl list --json` / `appctl status --json` orqali oladi (talab FR-INT-001..003, 4-bo'lim).

## 2.3. Loyihalar (S-02)

| Loyiha | Domen | Port | Izoh |
|---|---|---|---|
| Aaclub staging | *.staging.<domen> | 28080/28443 | Docker, to'liq stek |
| Aaclub production | (kutilmoqda) | 18080/18443 | `/opt/aaclub`; hozir ishlamayapti |
| asadbekashurov.uz | asadbekashurov.uz, www | 38080/38443 | Statik landing, `nginx:1.27-alpine`, `/opt/asadbekashurov`; sertifikat kunlik cron bilan yangilanadi |

## 2.4. Hali ko'chirilmagan loyihalar (eski serverdagi nusxa `/root/old-server`)

| Loyiha | Domen | Ishlash usuli (eski serverda) | Izoh |
|---|---|---|---|
| avlocorp.com (Quran) | avlocorp.com | `projects/Quran`, backend :8200 | Keyinga qoldirilgan |
| Matnchi | matnchi.avlo.ai | Django + PostgreSQL | Tavsiya etilgan keyingi loyiha |
| CallAI | call.avlo.ai | Django (`CallAI_back`) + front | |
| Transcription / TranscriptionBot | transcription.avlo.ai | systemd (`transcriptionbot-django`, `audio-analysis`) | |
| bot.avlo.ai, najotnur.avlo.ai, asror-iskandarov.avlo.ai | | aniqlanmagan | |
| Bitrix* | | systemd (`bitrix`, `bitrix-celery`, `celery-beat`) | |
| receipt_bot, xatmbot, AvloAIBot-IG, consultation | | systemd/boshqa | |
| avloai.uz | avloai.uz | statik (avlo.ai ning nusxasi) | Egasi ko'chirmaslikka qaror qildi |
| ai-journalist | | `/opt/ai-journalist` | |

Nazoratchi reyestri bu loyihalarni "ko'chirilmagan" holati bilan ham saqlashi kerak (FR-REG-007).

## 2.5. Domenlar va DNS

| Domen | NS / DNS provayder | Hozirgi A yozuvi | Egasi / nazorati |
|---|---|---|---|
| avlo.ai | Namecheap (registrar-servers.com) | 164.68.105.194 | Buyurtmachi |
| avloai.uz, asadbekashurov.uz | ahost.uz (rdns1/2) | avloai.uz → 185.214.135.9 (endi core serveri IP si, ko'chirilishi kerak); asadbekashurov.uz → 173.249.37.152 | Buyurtmachi |
| beyondlinks.uz, biznesradar.uz, freighthubllc.co/.com | (aniqlanadi) | 164.68.105.194 | Buyurtmachi/mijoz |
| lutfanai.uz, lutfan.israilov.uz | (aniqlanadi) | 34.173.220.247 (GCP, mavjud emas) | DNS yangilanishi kerak |
| *.wmg.uz | Beget | 34.173.220.247 (GCP, mavjud emas) | **Mijoz** nazorati; DNS yangilanishi kerak |

## 2.6. Zaxira tuzilmasi (S-01)

```
/opt/applications/backups/
  <loyiha>/
    daily/    <loyiha>-YYYYMMDD-HHMMSS.sql.gz        (7 kun)
    weekly/   <loyiha>-YYYYMMDD-HHMMSS.sql.gz        (yakshanba nusxasi, 5 hafta)
              <loyiha>-files-YYYYMMDD-HHMMSS.tar.gz  (BACKUP_PATHS, haftalik)
```

Qoidalar (appctl):

- `pg_dump | gzip`, fayl `.tmp` ga yoziladi, hajmi > 1024 bayt bo'lsa qayta nomlanadi.
- `KEEP_DAILY=7`, `KEEP_WEEKLY=5` (kun hisobida `*7`).
- `OFFSITE` (`/opt/applications/appctl.conf`): bo'sh bo'lsa offsite o'chiq; to'ldirilsa `rsync -a -e "ssh -o BatchMode=yes"` bilan butun papka yuboriladi (`--delete` yo'q).
- Cron: `15 3 * * * root /usr/local/bin/appctl backup all >> /var/log/appctl-backup.log 2>&1`.

Hozirgi holat: `OFFSITE` hali **sozlanmagan** (egasi "keyinroq" degan). Bu eng katta ochiq xavf (R-001) va Nazoratchi'ning birinchi muhim tekshiruvlaridan biri bo'ladi.

## 2.7. systemd servislar (S-01)

| Servis | Vazifa | Foydalanuvchi | Holat |
|---|---|---|---|
| `avlo-contact` | avlo.ai kontakt formasi → Telegram (Node, 127.0.0.1:3017) | `avlo` | Ishlayapti |
| `nginx`, `docker`, `cron`, `ssh`, `fail2ban` (rejada) | Tizim servislari | root | Ishlayapti |

Eski serverdan ko'chirilishi kutilayotganlar: 2.4-bo'limga qarang.

## 2.8. Ma'lum xavflar (joriy)

| ID | Xavf | Ta'sir | Nazoratchi qanday yordam beradi |
|---|---|---|---|
| R-001 | Offsite zaxira yo'q | Server o'lsa, barcha ma'lumot yo'qoladi | FR-BKP-010..014 |
| R-002 | Kernel yangilanishi kutilmoqda; reboot o'tgan safar muammo keltirgan | Reboot'dan keyin server ko'tarilmasligi mumkin | FR-HST-009 "reboot kerak" + nazorat ro'yxati |
| R-003 | SSH faqat parol bilan, bruteforce ehtimoli | Server buzilishi | FR-HST-014 (SSH urinishlari soni), tavsiya: kalitga o'tish |
| R-004 | DNS bir necha provayderda, bir nechtasi mijoz qo'lida | Domen noto'g'ri IP ga qarab qoladi | FR-DNS-* |
| R-005 | Sertifikat yangilanishi (S-01 certbot, S-02 konteyner cron) xato bersa jim qoladi | Sayt "xavfsiz emas" | FR-TLS-* |
| R-006 | Bitta server — bitta nosozlik nuqtasi (SPOF) | Hamma loyiha to'xtaydi | Heartbeat + offsite |
| R-007 | Ko'chirilmagan loyihalar eski serverga bog'liq | Ular hozir ishlamayapti | Reyestrda ko'rinadi |
| R-008 | Docker log hajmi (limit endi qo'yilgan, lekin eski konteynerlar qayta yaratilmagan bo'lishi mumkin) | Disk to'lishi | FR-DKR-006 |
| R-009 | S-02 nginx.conf qayta yozilishi mumkin | asadbekashurov.uz ishlamay qoladi | FR-TLS/HTTP tekshiruvlari tezda ko'rsatadi |

## 2.9. Ma'lumot yig'ish buyruqlari (inventarni to'ldirish uchun)

Quyidagilarni har serverda bir marta bajarib, natijani reyestrga kiritish kerak:

```
hostnamectl --static
lscpu | grep -E 'Model name|^CPU\(s\)'
free -h | sed -n 2p
df -h / | tail -n 1
lsblk -o NAME,SIZE,TYPE,MOUNTPOINT
uname -r; cat /etc/os-release | grep PRETTY_NAME
docker --version; docker compose version
systemctl list-units --type=service --state=running --no-pager | head -n 40
crontab -l; ls /etc/cron.d
nginx -T 2>/dev/null | grep -E 'server_name|proxy_pass|root ' | sort -u
ls /etc/letsencrypt/live/
```

# 3. Foydalanuvchilar, rollar va foydalanish stsenariylari

## 3.1. Foydalanuvchi turlari (personalar)

### P-1. Egasi (Owner) — "Jasurbek"

- Barcha loyihalarning egasi, ayni paytda DevOps ham o'zi.
- Kuniga bir necha marta telefondan (Telegram) va kompyuterdan panelga kiradi.
- Ehtiyoji: "Hammasi joyidami?" degan savolga 5 soniyada javob olish; muammo bo'lsa, nima qilishni darrov bilish; kechasi faqat haqiqiy muhim narsa uchun uyg'otilish.
- Texnik savodxonligi yuqori (Linux, Docker, nginx, Django), lekin vaqti kam.

### P-2. Operator

- Egasi ishonib topshirgan dasturchi yoki yordamchi.
- Incident'larni qabul qiladi (acknowledge), alertlarni jim qiladi, texnik ish oynalarini rejalashtiradi, reyestrga yangi loyiha qo'shadi.
- Serverga SSH orqali kira oladi, lekin Nazoratchi sozlamalarini (foydalanuvchilar, tokenlar) o'zgartira olmaydi.

### P-3. Kuzatuvchi (Viewer)

- Faqat o'qiydi: mijoz vakili yoki jamoa a'zosi. Masalan, wmg.uz mijozi o'z loyihasining holatini ko'radi.
- Hech narsani o'zgartira olmaydi; boshqa loyihalarni ko'rmaydi (loyiha bo'yicha cheklash, F2).

### P-4. Agent (mashina foydalanuvchi)

- Server tomonidagi dastur; faqat o'z hosti uchun ma'lumot yuboradi. Token bilan autentifikatsiya.

### P-5. Tashqi tekshiruvchi (Watchdog)

- Nazoratchi'ning o'zi ishlayotganini tashqaridan tekshiruvchi mustaqil xizmat. Nazoratchi tomonidan har 1 daqiqada heartbeat oladi; kelmasa Telegram'ga alohida xabar yuboradi.

## 3.2. Rollar va ruxsatlar matritsasi

Belgilar: ✓ — ruxsat; ✕ — taqiqlangan; (L) — faqat o'ziga biriktirilgan loyihalar.

| Amal | Owner | Operator | Viewer |
|---|---|---|---|
| Dashboard, grafiklar, ro'yxatlarni ko'rish | ✓ | ✓ | ✓ (L) |
| Incident'ni ko'rish | ✓ | ✓ | ✓ (L) |
| Incident'ni qabul qilish (ack) / izoh yozish | ✓ | ✓ | ✕ |
| Alertni jim qilish (silence) | ✓ | ✓ | ✕ |
| Maintenance window yaratish | ✓ | ✓ | ✕ |
| Host qo'shish / o'chirish | ✓ | ✓ (qo'shish) | ✕ |
| Agent tokenini yaratish / bekor qilish | ✓ | ✕ | ✕ |
| Loyiha / domen / servis qo'shish va tahrirlash | ✓ | ✓ | ✕ |
| Faol tekshiruv (HTTP/TCP/DNS) yaratish | ✓ | ✓ | ✕ |
| Alert qoidalarini yaratish / o'zgartirish | ✓ | ✓ (mavjudini sozlash) | ✕ |
| Bildirishnoma kanallarini sozlash | ✓ | ✕ | ✕ |
| Restore drill natijasini kiritish | ✓ | ✓ | ✕ |
| Foydalanuvchilarni boshqarish | ✓ | ✕ | ✕ |
| Audit jurnalini ko'rish | ✓ | ✕ | ✕ |
| Tizim sozlamalari, retention | ✓ | ✕ | ✕ |
| Hisobotlarni yuklab olish (CSV/PDF) | ✓ | ✓ | ✓ (L) |

## 3.3. Foydalanuvchi hikoyalari (User Stories)

Format: "Rol sifatida men ... xohlayman, shunda ...". Har hikoya talablarga bog'langan.

### 3.3.1. Umumiy ko'rinish

| ID | Hikoya | Ustuvorlik | Talablar |
|---|---|---|---|
| US-001 | Egasi sifatida men bosh sahifada barcha serverlar va loyihalarning umumiy holatini (yashil/sariq/qizil) bir qarashda ko'rishni xohlayman, shunda bir necha soniyada hammasi joyida ekanini bilaman | M | FR-UI-001, FR-UI-002 |
| US-002 | Egasi sifatida men har serverning CPU, RAM, disk va tarmoq grafiklarini oxirgi 1 soat / 24 soat / 7 kun / 30 kun uchun ko'rishni xohlayman | M | FR-HST-001..008, FR-UI-005 |
| US-003 | Egasi sifatida men barcha loyihalar ro'yxatini ko'rishni xohlayman: domen, server, port, holat, oxirgi zaxira, sertifikat muddati | M | FR-REG-001..004, FR-UI-003 |
| US-004 | Egasi sifatida men tizimni telefonda ham qulay ishlatishni xohlayman | S | NFR-USA-002 |
| US-005 | Egasi sifatida men interfeysni o'zbek tilida ko'rishni xohlayman | M | NFR-USA-001 |

### 3.3.2. Serverlar

| ID | Hikoya | Ustuvorlik | Talablar |
|---|---|---|---|
| US-006 | Operator sifatida men yangi serverni bitta buyruq bilan Nazoratchi'ga qo'shishni xohlayman | M | FR-AGT-001..003 |
| US-007 | Egasi sifatida men server o'chib qolsa 2 daqiqa ichida Telegram'da xabar olishni xohlayman | M | FR-ALR-005, FR-AGT-006 |
| US-008 | Egasi sifatida men disk 80% to'lganda va kunlar bo'yicha qachon to'lishini bashorat qilib xabar olishni xohlayman | M | FR-HST-005, FR-ALR-008 |
| US-009 | Operator sifatida men serverda kutilayotgan reboot (yangi kernel) borligini ko'rishni xohlayman | S | FR-HST-009 |
| US-010 | Egasi sifatida men RAM yoki CPU uzoq vaqt yuqori turgan serverni ko'rsatuvchi ro'yxatni xohlayman | S | FR-UI-007 |
| US-011 | Operator sifatida men SSH ga muvaffaqiyatsiz urinishlar sonini ko'rishni xohlayman, shunda bruteforce ni sezaman | C | FR-HST-014 |
| US-012 | Egasi sifatida men serverni vaqtincha kuzatuvdan to'xtatishni (pauza) xohlayman, masalan, ko'chirish paytida | M | FR-HST-016, FR-ALR-012 |

### 3.3.3. Loyihalar va servislar

| ID | Hikoya | Ustuvorlik | Talablar |
|---|---|---|---|
| US-013 | Egasi sifatida men har loyiha uchun saytning ishlayotganini (HTTP 200) tekshirishni xohlayman | M | FR-HTTP-001 |
| US-014 | Egasi sifatida men sayt javob vaqti sekinlashganini bilishni xohlayman | S | FR-HTTP-006, FR-ALR-009 |
| US-015 | Operator sifatida men qaysi Docker konteyner qayta-qayta ishga tushayotganini (crash loop) bilishni xohlayman | M | FR-DKR-004, FR-ALR-010 |
| US-016 | Operator sifatida men systemd servis (masalan, `avlo-contact`) to'xtasa xabar olishni xohlayman | M | FR-SVC-002 |
| US-017 | Egasi sifatida men yangi loyiha qo'shganda uni reyestrga bir formada kiritishni xohlayman | M | FR-REG-005 |
| US-018 | Egasi sifatida men eski serverdan hali ko'chirilmagan loyihalarni ham ro'yxatda ko'rishni xohlayman | S | FR-REG-007 |
| US-019 | Operator sifatida men kontainerning resurs sarfini (CPU/RAM) ko'rishni xohlayman | S | FR-DKR-003 |

### 3.3.4. Zaxira nusxalar

| ID | Hikoya | Ustuvorlik | Talablar |
|---|---|---|---|
| US-020 | Egasi sifatida men har loyihaning oxirgi muvaffaqiyatli zaxirasi qachon olinganini ko'rishni xohlayman | M | FR-BKP-001..003 |
| US-021 | Egasi sifatida men kunlik zaxira olinmasa yoki hajmi keskin kichraysa xabar olishni xohlayman | M | FR-BKP-004..006 |
| US-022 | Egasi sifatida men zaxiralar offsite (boshqa serverga) yetib borganini tasdiqlashni xohlayman | M | FR-BKP-010..013 |
| US-023 | Operator sifatida men zaxira tiklash testini (restore drill) rejalashtirib, natijasini qayd etishni xohlayman | S | FR-BKP-015..017 |
| US-024 | Egasi sifatida men zaxira hajmi o'sish tendensiyasini ko'rishni xohlayman | C | FR-BKP-008 |
| US-025 | Egasi sifatida men offsite serverdagi bo'sh joyni kuzatishni xohlayman | S | FR-BKP-014 |

### 3.3.5. Domen, DNS va sertifikat

| ID | Hikoya | Ustuvorlik | Talablar |
|---|---|---|---|
| US-026 | Egasi sifatida men har domen sertifikati tugashiga 21/7/3 kun qolganda xabar olishni xohlayman | M | FR-TLS-002, FR-TLS-003 |
| US-027 | Egasi sifatida men domen A yozuvi kutilgan IP dan boshqasiga qarab qolsa xabar olishni xohlayman | M | FR-DNS-001..003 |
| US-028 | Egasi sifatida men domen ro'yxatdan o'tgan muddati (WHOIS) tugashini kuzatishni xohlayman | S | FR-DNS-006 |
| US-029 | Operator sifatida men DNS o'zgartirilgandan keyin o'zgarish tarqalganini (propagation) bir nechta DNS serverdan tekshirishni xohlayman | C | FR-DNS-005 |

| ID | Hikoya | Ustuvorlik | Talablar |
|---|---|---|---|
| US-030 | Egasi sifatida men certbot yangilanishi muvaffaqiyatsiz bo'lsa xabar olishni xohlayman | S | FR-TLS-006 |

### 3.3.6. Alertlar va incidentlar

| ID | Hikoya | Ustuvorlik | Talablar |
|---|---|---|---|
| US-031 | Egasi sifatida men alertni Telegram'da olishni va xabardan to'g'ridan-to'g'ri paneldagi incident sahifasiga o'tishni xohlayman | M | FR-NTF-001..003 |
| US-032 | Operator sifatida men alertni "qabul qildim" deb belgilashni xohlayman, shunda takroriy eslatma to'xtaydi | M | FR-ALR-013 |
| US-033 | Operator sifatida men ma'lum muammo uchun alertni belgilangan muddatga jim qilishni xohlayman | M | FR-ALR-012 |
| US-034 | Egasi sifatida men kechasi faqat "kritik" alertlar kelishini xohlayman (sokin soatlar) | S | FR-NTF-006 |
| US-035 | Egasi sifatida men muammo hal bo'lganda "tiklandi" xabarini olishni xohlayman | M | FR-ALR-014 |
| US-036 | Operator sifatida men har alertda "nima qilish kerak" (runbook) havolasini ko'rishni xohlayman | S | FR-ALR-017, 18-bo'lim |
| US-037 | Egasi sifatida men o'tgan oydagi incidentlar tarixini va ularning davomiyligini ko'rishni xohlayman | S | FR-ALR-016, FR-UI-012 |
| US-038 | Egasi sifatida men kritik alert 15 daqiqa qabul qilinmasa, ikkinchi shaxsga yuborilishini xohlayman | C | FR-ALR-015 |

### 3.3.7. Xavfsizlik va boshqaruv

| ID | Hikoya | Ustuvorlik | Talablar |
|---|---|---|---|
| US-039 | Egasi sifatida men tizimga faqat parol + ikki bosqichli tasdiqlash bilan kirishni xohlayman | S | FR-AUTH-003 |
| US-040 | Egasi sifatida men har bir agent uchun alohida token berishni va kerak bo'lsa bekor qilishni xohlayman | M | FR-AGT-002, FR-AUTH-006 |
| US-041 | Egasi sifatida men kim nimani o'zgartirganini audit jurnalidan ko'rishni xohlayman | S | FR-AUD-001..003 |
| US-042 | Egasi sifatida men mijozga faqat o'z loyihasini ko'rsatuvchi hisob berishni xohlayman | C | FR-AUTH-005 |
| US-043 | Egasi sifatida men Nazoratchi'ning o'zi o'chib qolsa xabar olishni xohlayman | M | FR-SYS-002 |

## 3.4. Foydalanish stsenariylari (Use Cases)

### UC-01. Yangi serverni qo'shish

- **Aktyor:** Operator / Owner.
- **Old shart:** Panelga kirilgan; yangi serverda root huquqi bor.
- **Asosiy oqim:**
  1. Panelda "Serverlar → Server qo'shish" bosiladi.
  2. Nom (masalan, `main-2`), provayder, rol va teglar kiritiladi.
  3. Tizim bir martalik **o'rnatish buyrug'ini** ko'rsatadi: `curl -fsSL https://<core>/install.sh | sudo bash -s -- --token <ENROLL_TOKEN>`.
  4. Operator buyruqni yangi serverda ishga tushiradi.
  5. Skript agent faylini yuklaydi, yagona imzoni tekshiradi, systemd unit yaratadi, bir martalik enroll tokenini doimiy agent tokeniga almashtiradi.
  6. Agent birinchi ma'lumotni yuboradi; panelda server "Yangi" holatidan "Onlayn" ga o'tadi.
  7. Tizim hostning OS/CPU/RAM/disk ma'lumotini avtomatik to'ldiradi.
- **Muqobil oqim:** Enroll tokeni 1 soatdan keyin eskiradi → panelda qayta yaratish.
- **Natija:** Server reyestrda, grafiklar chiza boshlaydi, standart alert qoidalari qo'llanadi.

### UC-02. Server o'chganda aniqlash va xabar berish

- **Aktyor:** Tizim → Owner.
- **Asosiy oqim:**
  1. Agent heartbeat 15 soniyada bir marta yuboradi.
  2. Markaziy server 120 soniya heartbeat olmasa, host holatini "Aloqa yo'q" qiladi.
  3. Tizim parallel ravishda hostning ommaviy IP siga ICMP/TCP:22 tekshiruvi bajaradi (agent muammosini server muammosidan ajratish uchun).
  4. Ikkala tekshiruv ham muvaffaqiyatsiz bo'lsa — "Server o'chdi" (kritik) alerti yaratiladi. Faqat agent jim bo'lsa — "Agent javob bermayapti" (ogohlantirish).

5. Telegram'ga xabar: server nomi, oxirgi ko'rilgan vaqt, ta'sirlangan loyihalar ro'yxati, runbook RB-001 havolasi.
6. Server qaytganda "Tiklandi" xabari va to'xtash davomiyligi yuboriladi.

### UC-03. Disk to'lishi haqida ogohlantirish

- **Aktyor:** Tizim → Owner/Operator.
- **Asosiy oqim:**
  1. Agent har 30 soniyada `/` bo'limi bandligini yuboradi.
  2. Markaziy server oxirgi 6 soatlik chiziqli regressiya bilan to'lish vaqtini bashorat qiladi.
  3. Bashorat 7 kundan kam bo'lsa — ogohlantirish; 24 soatdan kam yoki bandlik ≥ 90% bo'lsa — kritik.
  4. Xabarda: bandlik, o'sish tezligi (GB/kun), eng katta papkalar (agent "top-N" ma'lumoti, F2).
  5. Runbook RB-004: `docker system df`, log tozalash, eski image'lar.

### UC-04. Zaxira muvaffaqiyatsiz bo'lganini aniqlash

- **Aktyor:** Tizim → Owner.
- **Old shart:** Agent zaxira papkasini kuzatadi.
- **Asosiy oqim:**
  1. Agent har 5 daqiqada `/opt/applications/backups/*/daily` ni skanerlaydi: eng yangi fayl vaqti, hajmi, soni.
  2. Markaziy server har loyiha uchun "oxirgi zaxira yoshi" ni hisoblaydi.
  3. Yosh > 26 soat bo'lsa — "Zaxira kechikdi" alerti.
  4. Yangi fayl hajmi oldingi 7 kun medianasining 50% dan kam bo'lsa — "Zaxira shubhali" alerti.
  5. Xabarda: loyiha, oxirgi muvaffaqiyatli zaxira, `/var/log/appctl-backup.log` dan oxirgi 10 qator.

### UC-05. Offsite zaxira tekshiruvi

- **Aktyor:** Tizim → Owner.
- **Asosiy oqim:**
  1. Offsite zaxira S-00 da (core bilan bir serverda); S-00 dagi agent ham o'rnatilgan.
  2. Agent offsite papkani (`/srv/backups/main-server`) kuzatadi.
  3. Markaziy server mahalliy (S-01) va offsite (S-00) ro'yxatlarini solishtiradi: oxirgi fayl nomi va hajmi mos kelishi kerak.
  4. Farq 26 soatdan oshsa — "Offsite sinxronlash to'xtadi" alerti.

### UC-06. Sertifikat muddati yaqinlashishi

- **Aktyor:** Tizim → Owner.
- **Asosiy oqim:**
  1. Markaziy server har 6 soatda domenga TLS ulanib `notAfter` ni o'qiydi.
  2. 21 kun qolsa — ogohlantirish; 7 kun — yuqori; 3 kun — kritik.
  3. Xabarda: domen, server, sertifikat chiqaruvchi, ehtimoliy sabab (certbot log yoki DNS o'zgargani) va runbook RB-006.

### UC-07. Domen noto'g'ri IP ga qarab qolishi

- **Aktyor:** Tizim → Owner.
- **Asosiy oqim:**
  1. Reyestrda `asadbekashurov.uz` uchun kutilgan IP: `173.249.37.152`.
  2. Tizim har 5 daqiqada bir nechta ommaviy DNS (8.8.8.8, 1.1.1.1, 9.9.9.9) va domenning o'z NS serveridan A yozuvini so'raydi.
  3. Natija kutilgan IP ga mos kelmasa va 10 daqiqadan beri o'zgarmasa — "DNS mos emas" alerti.
  4. Qisqa farq (propagation) bo'lsa, "ehtimoliy tarqalish" belgisi bilan ogohlantirishsiz.

### UC-08. Texnik ish oynasi (maintenance)

- **Aktyor:** Operator.
- **Asosiy oqim:**
  1. "Texnik ish" yaratiladi: qamrov (server yoki loyiha), boshlanish va tugash vaqti, sabab.
  2. Shu vaqt davomida qamrovdagi alertlar yuborilmaydi, lekin yig'iladi (jim rejim).
  3. Oyna tugagach, hali yechilmagan alertlar bo'lsa, ular yuboriladi.
  4. Davomiylik hisobotlarida bu vaqt "rejalashtirilgan to'xtash" deb alohida hisoblanadi.

### UC-09. Incident bilan ishlash

- **Aktyor:** Operator.
- **Asosiy oqim:**
  1. Telegram'dagi xabardagi tugma: "Qabul qilaman" (ack).
  2. Incident holati "Qabul qilingan" ga o'tadi, takroriy eslatma to'xtaydi.
  3. Operator panelda izoh yozadi ("nginx konfigini tuzatdim").
  4. Shart yo'qolgach, incident avtomatik "Yechildi" bo'ladi; tizim davomiylikni hisoblaydi.
  5. Haftalik hisobotga qo'shiladi.

### UC-10. Yangi loyiha qo'shish

- **Aktyor:** Operator / Owner.
- **Asosiy oqim:**
  1. "Loyihalar → Qo'shish": nom, server, domenlar, portlar, turi (compose/systemd/static).
  2. Tizim domen uchun avtomatik HTTP, TLS va DNS tekshiruvlarini taklif qiladi.
  3. Docker compose loyihasi bo'lsa, agent loyiha nomi bo'yicha konteynerlarni bog'laydi.
  4. Zaxira papkasi nomi ko'rsatiladi; tizim mos qoidalarni qo'llaydi.
  5. Reyestr yangilanadi; loyiha kartasi paydo bo'ladi.

### UC-11. Restore drill (tiklash testi)

- **Aktyor:** Operator.
- **Asosiy oqim:**
  1. Panel chorakda bir marta eslatadi: "BiznesRadar tiklash testi muddati o'tdi".
  2. Operator test serverda (yoki vaqtinchalik konteynerda) `appctl restore` ni sinab ko'radi.
  3. Panelda natija kiritiladi: sana, zaxira fayli, davomiyligi, jadvallar soni, qator soni, muvaffaqiyat/xato, izoh.
  4. Tizim "oxirgi muvaffaqiyatli restore" sanasini loyiha kartasida ko'rsatadi.

### UC-12. Nazoratchi'ning o'zi o'chganda

- **Aktyor:** Watchdog → Owner.
- **Asosiy oqim:**
  1. Nazoratchi har daqiqada watchdog xizmatiga "ping" yuboradi.
  2. 5 daqiqa ping kelmasa, watchdog Telegram orqali (Nazoratchi botidan boshqa botdan) xabar yuboradi.
  3. Xabarda: "Nazoratchi ishlamayapti", oxirgi ping vaqti, runbook RB-012.

## 3.5. Stsenariylarga bog'liq o'zgarmas qoidalar

1. Alert hech qachon "jimgina" yo'qolmasligi kerak: kanal xatosi bo'lsa, qayta urinish va ikkinchi kanalga o'tish (FR-NTF-004).
2. Hostning o'zi o'chganda, uning ichidagi barcha servislar uchun alohida alertlar yuborilmaydi (bostirish, "suppression"): bitta "Server o'chdi" incident'i yetarli (FR-ALR-011).
3. Agent yig'ilgan ma'lumotni yuborib bo'lmasa, 15 daqiqagacha mahalliy buferda saqlaydi va aloqa tiklangach yuboradi (FR-AGT-007).
4. Foydalanuvchi interfeysida hech qachon parol, token yoki `.env` qiymati ko'rsatilmaydi.

# 4. Funksional talablar

Har talab jadvalda quyidagi ustunlar bilan beriladi: **ID**, **Talab**, **Ust.** (M/S/C/W), **Faza** (F1/F2/F3), **Qabul mezoni** (talab bajarilganini qanday tekshirish).

Modullar: REG (reyestr), HST (host), AGT (agent), DKR (Docker), SVC (systemd va cron), HTTP (faol tekshiruvlar), TLS (sertifikat), DNS (DNS va domen), BKP (zaxira), ALR (alert), NTF (bildirishnoma), UI (interfeys), AUTH (autentifikatsiya), AUD (audit), INT (integratsiya), SYS (tizimning o'zi).

## 4.1. REG — Reyestr moduli

Reyestr — barcha serverlar, loyihalar, domenlar va portlarning yagona manbai. Nazoratchi'dagi boshqa barcha modullar reyestrga tayanadi.

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-REG-001 | **Server obyekti.** Har serverda: nom, provayder, ommaviy IP (IPv4, ixtiyoriy IPv6), OS, rol, teglar, joylashuv, narx/oy (ixtiyoriy), izoh, holat (yangi/onlayn/aloqa yo'q/pauza/arxiv). Nom noyob bo'lishi kerak | M | F1 | Server yaratilganda barcha maydonlar saqlanadi; takroriy nom 409 xato qaytaradi |
| FR-REG-002 | **Loyiha obyekti.** Har loyihada: nom, qisqa tavsif, egasi/mijoz, tur (compose / systemd / static / boshqa), asosiy server, `appctl` papka nomi, teglar, holat (ishlayapti / ko'chirilmagan / arxiv) | M | F1 | Loyiha yaratiladi va serverga bog'lanadi; bir loyiha bir nechta serverda bo'lishi mumkin (staging/production) |
| FR-REG-003 | **Domen obyekti.** Har domenda: FQDN, loyiha, kutilgan IP lar ro'yxati, DNS provayder, registrar, domen egasi/nazoratchisi (biz/mijoz), WHOIS muddati, HTTP tekshiruv sozlamalari | M | F1 | Bitta loyihaga ko'p domen biriktiriladi; FQDN noyob |
| FR-REG-004 | **Birlashtirilgan ro'yxat.** Loyihalar sahifasi har loyiha uchun: domenlar, server, port(lar), joriy holat, oxirgi zaxira yoshi, eng yaqin sertifikat muddati ustunlarini ko'rsatadi; ustun bo'yicha saralash va qidiruv | M | F1 | 10 loyiha bilan ro'yxat < 1 soniyada yuklanadi; saralash ishlaydi |
| FR-REG-005 | **Loyiha qo'shish formasi.** UC-10 bo'yicha; domen kiritilganda avtomatik HTTP/TLS/DNS tekshiruvlari qo'shiladi (rozilik bilan) | M | F1 | Yangi loyiha qo'shilgach 1 daqiqada tekshiruvlar ishga tushadi |
| FR-REG-006 | **Tahrirlash va arxivlash.** Obyektlar o'chirilmaydi, arxivga o'tadi (tarix va incident bog'lanishlari saqlanadi). Arxivlangan obyekt alert yubormaydi | M | F1 | Arxivlangan loyiha ro'yxatda ko'rinmaydi (filtr bilan ko'rinadi); tarixi saqlanadi |
| FR-REG-007 | **"Ko'chirilmagan" holati.** Loyiha hali ishlamayotgan bo'lsa ham reyestrda saqlanadi; unga tekshiruv va alert qo'llanmaydi | S | F2 | `moved=false` loyiha 2.4-bo'lim ro'yxatiga mos holda ko'rinadi, alert yubormaydi |
| FR-REG-008 | **Teglar va guruhlash.** Teglar (masalan, `django`, `client:wmg`, `prod`) bo'yicha filtr va guruhlangan ko'rinish | S | F2 | Tegli filtr ro'yxatni to'g'ri toraytiradi |
| FR-REG-009 | **Import/eksport.** Butun reyestrni YAML/JSON ga eksport va import; formatning versiyasi bor. Maqsad: zaxira va git'da saqlash | S | F2 | Eksport → yangi bo'sh tizimga import → bir xil reyestr (testda diff yo'q) |
| FR-REG-010 | **Port to'qnashuvi nazorati.** Bir serverda bir xil port ikki loyihaga berilsa, saqlashda ogohlantirish; keyingi bo'sh port taklif qilinadi (REGISTRY.md dagi qoida) | S | F2 | Port 8020 ikkinchi marta kiritilsa xato; "keyingi bo'sh: 8021" ko'rinadi |
| FR-REG-011 | **Aloqa shaxslari.** Loyiha uchun mas'ul shaxs va mijoz kontakti (ism, Telegram) saqlanadi; alertda mas'ul ko'rsatiladi | C | F3 | Alert xabarida mas'ul ism chiqadi |
| FR-REG-012 | **Havolalar va eslatmalar.** Loyihada: repo URL, deploy yo'riqnomasi, runbook havolalari, markdown izoh | C | F3 | Loyiha sahifasida havolalar ishlaydi |

## 4.2. HST — Host (server) monitoringi

Agent tomonidan yig'iladigan ko'rsatkichlar (to'liq katalog — 7-bo'lim).

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-HST-001 | **CPU.** Umumiy foiz (user, system, iowait, steal, idle bo'yicha ham), har yadro uchun; yig'ish intervali 15 soniya | M | F1 | `stress` ostida panel 30 soniyada o'sishni ko'rsatadi; `top` bilan ±3% farq |
| FR-HST-002 | **Yuklama (load average).** 1, 5, 15 daqiqalik; yadro soniga normallashtirilgan ko'rinish ham | M | F1 | `uptime` qiymatlariga mos |
| FR-HST-003 | **RAM.** Jami, band, bo'sh, mavjud (`MemAvailable`), cache/buffers; foiz va bayt | M | F1 | `free -b` bilan ±2% farq |
| FR-HST-004 | **Swap.** Jami va band; swap-in/out tezligi | S | F1 | `free` bilan mos |
| FR-HST-005 | **Disk hajmi va bashorat.** Har o'rnatilgan fayl tizimi (tmpfs, overlay va squashfs'siz) uchun jami/band/bo'sh; asosiy bo'limda chiziqli regressiya bilan "to'lishga qolgan kun" hisoblanadi | M | F1 | `df` bilan mos; sun'iy o'sish testida bashorat ±20% aniq |
| FR-HST-006 | **Disk I/O.** O'qish/yozish baytlari/s, IOPS, o'rtacha kutish vaqti (await), band foiz (util) har blok qurilma uchun | S | F2 | `fio` ostida o'sish ko'rinadi |

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-HST-007 | **Tarmoq.** Interfeys bo'yicha kiruvchi/chiquvchi bayt/s, paket/s, xato va tashlangan paketlar; `lo` va `docker*`/`veth*` ni yashirish mumkin | M | F1 | `iperf3`/`curl` yuklamasida o'sish ko'rinadi |
| FR-HST-008 | **Inode.** Fayl tizimi bo'yicha inode bandligi (kichik fayllar ko'p bo'lganda disk "bo'sh", lekin yozib bo'lmaydigan holat uchun) | S | F2 | `df -i` bilan mos |
| FR-HST-009 | **Kutilayotgan reboot va kernel.** `/var/run/reboot-required` mavjudligi; ishlayotgan va o'rnatilgan eng yangi kernel versiyasi; farq bo'lsa — "reboot kerak" belgisi va yoshi (necha kundan beri) | S | F2 | Kernel o'rnatilgan serverda belgisi paydo bo'ladi, reboot'dan keyin yo'qoladi |
| FR-HST-010 | **Uptime va boot vaqti.** Server ishga tushgan vaqt, uptime; kutilmagan reboot aniqlanadi (boot vaqti o'zgargan va ruxsat etilgan texnik ish yo'q) | M | F1 | Reboot'dan keyin 1 daqiqada "reboot aniqlandi" voqeasi yaratiladi |
| FR-HST-011 | **OS va jihoz haqida ma'lumot.** Hostname, OS nomi/versiyasi, kernel, arxitektura, CPU modeli/yadro soni, jami RAM, disk hajmi; o'zgarsa voqea yoziladi | M | F1 | Server qo'shilganda avtomatik to'ldiriladi |
| FR-HST-012 | **Jarayonlar.** Jami jarayonlar soni, zombi jarayonlar, ochiq fayl deskriptorlari (umumiy), threadlar soni | S | F2 | `ps` bilan mos |
| FR-HST-013 | **Vaqt sinxronligi.** Soat farqi (agent vaqti va markaziy vaqt); `chrony`/`timesyncd` holati. Katta farq (> 5 s) ma'lumot sifatiga ta'sir qiladi | S | F2 | Qo'lda soat siljitilsa, 1 daqiqada belgi chiqadi |
| FR-HST-014 | **SSH xavfsizlik ko'rsatkichi.** So'nggi soatdagi muvaffaqiyatsiz SSH kirishlar soni (`/var/log/auth.log` yoki `journalctl -u ssh`), noyob IP soni, `fail2ban` taqiqlagan IP soni (mavjud bo'lsa) | C | F3 | Sun'iy noto'g'ri parol bilan urinishdan keyin hisob oshadi |
| FR-HST-015 | **Paket yangilanishlari.** Kutilayotgan yangilanishlar soni va shundan xavfsizlik yangilanishlari soni (`apt`), kuniga 1 marta | C | F3 | `apt list --upgradable` bilan mos |
| FR-HST-016 | **Pauza.** Host "pauza" rejimiga o'tkazilsa, unga alert yuborilmaydi, lekin ma'lumot yig'ilaveradi; pauzani kim va qachon qo'yganini ko'rsatadi | M | F1 | Pauzadagi serverga alert kelmaydi; audit jurnalida yoziladi |
| FR-HST-017 | **Eng og'ir jarayonlar (top-N).** CPU va RAM bo'yicha 5 ta eng og'ir jarayon (nomi, PID, foiz); har 60 soniyada | C | F3 | `stress` ishga tushirilganda ro'yxatda ko'rinadi |

## 4.3. AGT — Agent moduli

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-AGT-001 | **Bir buyruqli o'rnatish.** `curl ... \| sudo bash -s -- --token <T>` shaklidagi skript: arxitekturani aniqlaydi, agent faylini yuklaydi, SHA-256 imzosini tekshiradi, systemd unit yaratadi va ishga tushiradi. Skript idempotent: qayta ishga tushirsa buzmaydi | M | F1 | Toza Ubuntu 22.04 va 24.04 da 60 soniyada o'rnatiladi; qayta ishga tushirish xatosiz |
| FR-AGT-002 | **Token almashtirish.** Bir martalik "enroll" tokeni (1 soat amal qiladi) birinchi ulanishda doimiy agent tokeniga almashtiriladi; doimiy token faqat agent faylida (rejim 0600) saqlanadi | M | F1 | Eskirgan enroll tokeni rad etiladi; doimiy token oddiy foydalanuvchi o'qiy olmaydi |
| FR-AGT-003 | **Sozlama fayli.** `/etc/nazoratchi/agent.yaml` (core manzil, intervallar, yoqilgan kollektorlar, kuzatiladigan servislar/papkalar); o'zgarishi `SIGHUP` yoki servis qayta ishga tushirilganda qo'llanadi | M | F1 | Intervalni o'zgartirib qayta yuklagach yangi interval ishlaydi |
| FR-AGT-004 | **Yig'ish intervallari.** Standart: tizim metrikalari 15 s; Docker 30 s; disk/papka skaneri 5 daqiqa; sertifikat skaneri 6 soat. Minimal interval 5 s | M | F1 | Standart qiymatlar sozlamada ko'rinadi, o'zgartirilsa ishlaydi |
| FR-AGT-005 | **Yuborish.** Ma'lumot HTTPS POST bilan paketlab (JSON, gzip), har 15 soniyada; agent faqat chiquvchi ulanish ochadi | M | F1 | Serverda hech qanday yangi inbound port ochilmaydi (`ss -ltn` farqi yo'q) |
| FR-AGT-006 | **Heartbeat.** Har 15 soniyada kichik "tirikman" paketi (versiya, uptime, bufer holati); ma'lumot paketi bilan birlashtirilishi mumkin | M | F1 | 120 soniya kelmasa host "aloqa yo'q" bo'ladi |
| FR-AGT-007 | **Mahalliy bufer.** Aloqa uzilsa, ma'lumot diskdagi navbatda (maks. 20 MB yoki 15 daqiqa) saqlanadi, aloqa tiklangach vaqt tartibida yuboriladi; bufer to'lsa eng eski ma'lumot o'chiriladi | M | F1 | Tarmoq 5 daqiqaga uzilib tiklangach grafikda bo'shliq yo'q |
| FR-AGT-008 | **O'z-o'zini yangilash.** Core yangi versiya borligini bildiradi; agent faylni yuklaydi, imzoni tekshiradi, o'zini almashtiradi va qayta ishga tushadi; muvaffaqiyatsiz bo'lsa eski versiyaga qaytadi | S | F2 | Versiya yangilanishi paneldan boshqariladi; buzuq fayl bilan rollback ishlaydi |
| FR-AGT-009 | **Resurs cheklovi.** Agent o'rtacha ≤ 1% CPU va ≤ 60 MB RAM; systemd `MemoryMax=100M`, `CPUQuota=20%` | M | F1 | 24 soatlik o'lchovda chegaradan oshmaydi |
| FR-AGT-010 | **Minimal huquq.** Agent `root` bo'lmagan `nazoratchi` foydalanuvchisi bilan ishlaydi; Docker ma'lumoti uchun socket'ga to'g'ridan-to'g'ri emas, faqat o'qish huquqli proksi (`docker-socket-proxy`) orqali; tizim fayllarini o'qish uchun `adm`/`systemd-journal` guruhlari | M | F1 | `id nazoratchi` da `docker` guruhi yo'q; agent `docker run` qila olmaydi |
| FR-AGT-011 | **Versiya hisoboti.** Agent o'z versiyasi, build vaqti va OS/arxitekturasini bildiradi; panelda eskirgan agentlar belgilanadi | S | F2 | Eski versiyali agent "yangilash kerak" deb ko'rinadi |

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-AGT-012 | **Soat farqi aniqlash.** Core javobida o'z vaqtini qaytaradi; agent farq > 5 s bo'lsa shuni bildiradi | S | F2 | Soat 10 s ga siljitilsa belgi chiqadi |
| FR-AGT-013 | **Toza o'chirish.** `nazoratchi-agent uninstall` servis, fayl, foydalanuvchi va sozlamani o'chiradi va core'ga "xayrlashuv" yuboradi (host "arxiv" holatiga tushadi) | S | F2 | O'chirishdan keyin tizimda agent izi qolmaydi |
| FR-AGT-014 | **Kollektorlar tizimi.** Har metrika guruhi alohida kollektor (`system`, `disk`, `net`, `docker`, `systemd`, `backup`, `certs`, `ssh`); kollektor xatosi boshqalarga ta'sir qilmaydi, xato `agent_collector_errors` metrikasi sifatida yuboriladi | M | F1 | Docker to'xtatilsa ham tizim metrikalari uzluksiz yuboriladi |

## 4.4. DKR — Docker monitoringi

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-DKR-001 | **Konteynerlar ro'yxati va holati.** Nom, image, holat (running/exited/restarting/paused), ishga tushgan vaqt, compose loyiha va servis nomi (labellardan) | M | F1 | `docker ps -a` bilan mos |
| FR-DKR-002 | **Health status.** Konteynerda `HEALTHCHECK` bo'lsa, `healthy/unhealthy/starting` holati; unhealthy → alert | S | F2 | Sun'iy buzilgan healthcheck 2 daqiqada alert beradi |
| FR-DKR-003 | **Konteyner resurs sarfi.** Har konteyner uchun CPU %, RAM (bayt va limitga nisbatan %), tarmoq, blok I/O | S | F2 | `docker stats` bilan ±5% |
| FR-DKR-004 | **Qayta ishga tushish va crash loop.** Restart soni; 10 daqiqada ≥ 3 marta qayta ishga tushsa — "crash loop" | M | F1 | Doim xato bilan chiqadigan konteynerda alert paydo bo'ladi |
| FR-DKR-005 | **Compose loyihaga bog'lash.** `com.docker.compose.project` labeli bo'yicha konteynerlar reyestrdagi loyihaga avtomatik biriktiriladi; biriktirilmaganlar "noma'lum konteynerlar" ro'yxatida | M | F1 | Nazar konteynerlari Nazar loyihasi kartasida ko'rinadi |
| FR-DKR-006 | **Log hajmi.** Har konteyner log fayli hajmi (`/var/lib/docker/containers/<id>/<id>-json.log*`); limitdan (standart 50 MB) oshgan konteyner belgilanadi. Bu `daemon.json` limiti qo'llanmagan eski konteynerlarni topadi | S | F2 | Limitsiz konteyner ro'yxatda "log limiti yo'q" deb chiqadi |
| FR-DKR-007 | **Docker disk sarfi.** Image, konteyner, volume, build cache hajmlari (`docker system df`) va "qaytarib olinadigan" hajm | S | F2 | `docker system df` bilan mos |
| FR-DKR-008 | **OOM-kill voqealari.** Konteyner xotira yetishmasligidan o'ldirilsa (`OOMKilled`) voqea yoziladi va alert | S | F2 | `stress` bilan limitdan oshirilganda voqea paydo bo'ladi |
| FR-DKR-009 | `daemon.json` **muvofiqligi.** Docker konfiguratsiyasi qoidalarga mosligi: log drayveri/limiti, `live-restore`. Mos kelmasa — ma'lumot belgisi | C | F3 | `daemon.json` yo'q serverda belgi chiqadi |
| FR-DKR-010 | **Image yoshi.** Konteyner image'ining yaratilgan sanasi; 180 kundan eski bo'lsa — ma'lumot (xavfsizlik yangilanishi uchun) | C | F3 | Eski image belgilanadi |

## 4.5. SVC — systemd va cron

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-SVC-001 | **Kuzatiladigan servislar ro'yxati.** Agent sozlamasida yoki panelda belgilangan systemd unit lari (`avlo-contact`, `nginx`, `docker`, `cron`, `redis` va h.k.) | M | F1 | Ro'yxatga qo'shilgan unit holati 30 soniyada ko'rinadi |
| FR-SVC-002 | **Unit holati.** `active/inactive/failed/activating`; `failed` yoki kutilmagan `inactive` → alert | M | F1 | `systemctl stop avlo-contact` dan keyin 90 soniyada alert |
| FR-SVC-003 | **Qayta ishga tushish soni.** `NRestarts` va oxirgi qayta ishga tushish vaqti; tez-tez qayta ishga tushsa alert | S | F2 | Sun'iy crash bilan alert |
| FR-SVC-004 | **Cron va davriy vazifalar heartbeat'i.** Muhim cron ishlari (zaxira, certbot renew) oxirida core'ga "men bajarildim" signali yuboradi (`curl` yoki agent komandasi); signal belgilangan oraliqda kelmasa — alert | M | F1 | `appctl backup all` signal yubormasa, 26 soatdan keyin alert |
| FR-SVC-005 | **Muvaffaqiyatsiz unitlar soni.** `systemctl --failed` soni; > 0 bo'lsa ma'lumot sifatida panelda | S | F2 | Failed unit paydo bo'lsa son ko'rinadi |
| FR-SVC-006 | **nginx konfiguratsiyasi sog'lig'i.** Agent `nginx -t` natijasini (muvaffaqiyat/xato) kuniga bir marta va konfig o'zgarganda tekshiradi; xato bo'lsa alert (reload qilinsa buzilishi mumkin) | C | F3 | Sun'iy sintaksis xatosi alert beradi |

## 4.6. HTTP — Faol tekshiruvlar

Faol tekshiruvlarni markaziy server bajaradi (agent'dan mustaqil, tashqaridan).

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-HTTP-001 | **HTTP(S) tekshiruv.** URL, metod (GET/HEAD), kutilgan status kodi (standart 200; oraliq `200-299`), timeout (standart 10 s) | M | F1 | Ishlamayotgan URL 90 soniyada "qizil" |
| FR-HTTP-002 | **Kalit so'z tekshiruvi.** Javob matnida belgilangan matn bo'lishi (yoki bo'lmasligi) kerak | S | F2 | Soxta "502 Bad Gateway" sahifasi (200 bilan) aniqlanadi |
| FR-HTTP-003 | **Maxsus sarlavhalar.** `Host`, `X-Forwarded-Proto`, `Authorization` kabi sarlavhalar (Django `ALLOWED_HOSTS` / `SECURE_SSL_REDIRECT` uchun kerak). Maxfiy qiymatlar shifrlab saqlanadi | M | F1 | Nazar `/healthz/` `Host: nazar.avlo.ai` bilan 200 beradi |
| FR-HTTP-004 | **Intervall, qayta urinish.** Standart interval 60 s; xato bo'lsa 15 s oraliq bilan 2 marta qayta tekshiriladi; faqat 3 ketma-ket xatodan keyin "tushdi" deb belgilanadi | M | F1 | Bitta tasodifiy xato alert bermaydi |
| FR-HTTP-005 | **Yo'naltirishlar.** `301/302` ni kuzatish (maks. 5) yoki bevosita javobni baholash (sozlanadi) | S | F2 | `www → asosiy` yo'naltirish to'g'ri baholanadi |
| FR-HTTP-006 | **Javob vaqti.** Umumiy vaqt, DNS, TCP ulanish, TLS qo'l siqish, birinchi bayt vaqti (TTFB) alohida o'lchanadi; chegaradan sekin bo'lsa ogohlantirish | S | F2 | `curl -w` bilan mos taqsimot |
| FR-HTTP-007 | **TCP tekshiruv.** Berilgan host:port ga ulanish (masalan, 22, 5432 — faqat ichki tarmoqdan emas, tashqi ochiq portlar uchun); kutilmagan ochiq port aniqlansa — xavfsizlik ogohlantirishi | S | F2 | Serverda 5432 tashqariga ochilsa alert |
| FR-HTTP-008 | **Ko'p lokatsiyali tekshiruv.** Ikkinchi tekshiruvchi tugun (boshqa serverdan) — "yolg'on tushdi"ni yo'qotish uchun: ikkala tugun ham xato bersa — tasdiq | C | F3 | Bitta tugun tarmoq muammosida alert chiqmaydi |
| FR-HTTP-009 | **Uptime hisobi.** Har tekshiruv uchun kun/hafta/oy/90 kun uptime %; rejalashtirilgan texnik ish vaqti alohida hisoblanadi | M | F2 | 24 soatlik uptime = (muvaffaqiyatli / jami) bilan mos |
| FR-HTTP-010 | **Tekshiruv tarixi.** Har tekshiruvning oxirgi 90 kunlik holati, javob vaqti grafigi, xato sabablari (timeout, DNS, TLS, 5xx) | S | F2 | Grafik va xato jadvali ochiladi |

## 4.7. TLS — Sertifikat monitoringi

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-TLS-001 | **Muddatni o'qish.** Domenning 443 portiga TLS ulanib, sertifikat `notAfter`, chiqaruvchi, subject, SAN larni olish. SNI ishlatiladi | M | F1 | `openssl s_client` natijasi bilan bir xil sana |
| FR-TLS-002 | **Chegaralar.** Qolgan kun: ≤ 21 — ogohlantirish, ≤ 7 — yuqori, ≤ 3 — kritik, 0 — o'tib ketgan (kritik). Chegaralar sozlanadi | M | F1 | Test sertifikati bilan har chegarada alert |
| FR-TLS-003 | **Ko'p domen.** Har domenning ro'yxatdagi barcha nomlari (`www`, subdomenlar) alohida tekshiriladi | M | F1 | `www.` uchun alohida holat ko'rinadi |
| FR-TLS-004 | **Zanjir va nom mosligi.** Zanjir to'liqligi, ishonchli CA, hostname SAN ga mosligi; mos kelmasa — xato | S | F2 | Noto'g'ri nom sertifikati aniqlanadi |
| FR-TLS-005 | **Mahalliy skaner.** Agent `/etc/letsencrypt/live/*/cert.pem` va konfigda ko'rsatilgan yo'llardagi sertifikatlarni o'qib, muddatini yuboradi — shu bilan ishlatilmayotgan, lekin yangilanayotgan sertifikatlar ham ko'rinadi | S | F2 | `appctl status` bo'limi "Sertifikatlar" bilan mos |
| FR-TLS-006 | **Yangilanish natijasi.** `certbot renew` (yoki konteynerdagi yangilanish) oxirgi natijasi va vaqti; yangilanish 2 marta ketma-ket xato bersa — alert (muddat hali yaqin bo'lmasa ham) | S | F2 | Sun'iy buzilgan renew alert beradi |
| FR-TLS-007 | **Protokol versiyalari.** TLS 1.0/1.1 yoqilmaganligi, HTTP/2 mavjudligi | C | F3 | Eski protokol yoqilgan serverda belgi |
| FR-TLS-008 | **Kutilgan chiqaruvchi.** Domen uchun kutilgan CA (Let's Encrypt); boshqasi chiqsa — ma'lumot | C | F3 | CA o'zgarishi ko'rinadi |

## 4.8. DNS — DNS va domen monitoringi

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-DNS-001 | **A yozuvini tekshirish.** Har domen uchun A (va ixtiyoriy AAAA) yozuvi so'raladi va reyestrdagi kutilgan IP lar bilan solishtiriladi | M | F1 | `asadbekashurov.uz` → `173.249.37.152` mos deb ko'rinadi |
| FR-DNS-002 | **Ko'p resolver.** So'rov kamida 3 ommaviy resolverga (Google 8.8.8.8, Cloudflare 1.1.1.1, Quad9 9.9.9.9) va domenning o'z NS serveriga alohida yuboriladi; natijalar alohida ko'rsatiladi | M | F1 | Panelda har resolver javobi ko'rinadi |
| FR-DNS-003 | **Mos kelmaslik alerti.** Natija kutilgan IP dan boshqa bo'lsa va 10 daqiqa davomida o'zgarmasa (tarqalish vaqtidan keyin) — alert. Eski IP ga qaragan domen (masalan, o'chirilgan GCP `34.173.220.247`) alohida sabab sifatida ko'rsatiladi | M | F1 | `lutfanai.uz` hali GCP IP da bo'lsa alert chiqadi |
| FR-DNS-004 | **Boshqa yozuv turlari.** CNAME, MX, TXT (masalan, SPF), NS yozuvlari o'zgarishi kuzatiladi; kutilgan qiymat bo'lsa solishtiriladi | C | F3 | MX o'zgarsa voqea yoziladi |

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-DNS-005 | **Tarqalish (propagation) rejimi.** DNS o'zgartirilgandan keyin "kutish" rejimini yoqish: har 1 daqiqada tekshiradi va barcha resolverlar yangi IP ga o'tganda xabar beradi ("o'zgarish tarqaldi") | S | F2 | Rejim yoqilgach, Telegram'ga "hamma resolver yangi IP ga o'tdi" xabari keladi |
| FR-DNS-006 | **Domen muddati (WHOIS/RDAP).** Domenning ro'yxatdan o'tgan muddati tugash sanasi (RDAP orqali; `.uz` uchun qo'lda kiritish imkoniyati, chunki RDAP har doim mavjud emas); 60, 30, 14, 7 kun qolganda ogohlantirish | S | F2 | `.com` domen uchun avtomatik, `.uz` uchun qo'lda kiritilgan sana ishlaydi |
| FR-DNS-007 | **NS kuzatuvi.** Domenning NS yozuvlari (masalan, `rdns1.ahost.uz`) o'zgarsa voqea; reyestrdagi DNS provayder maydoni bilan solishtiriladi | C | F3 | NS o'zgarsa voqea yoziladi |
| FR-DNS-008 | **DNS o'zgarishlar tarixi.** Har domen uchun yozuvlar o'zgarish tarixi (qachon, nimadan nimaga) | S | F2 | `A: 34.173.220.247 → 164.68.105.194` yozuvi vaqt bilan ko'rinadi |
| FR-DNS-009 | **Kesib o'tishdan oldingi tekshiruv (cut-over yordamchisi).** DNS o'zgartirilishidan **oldin** domenni yangi IP ga to'g'ridan-to'g'ri (`Host` va SNI saqlangan holda) HTTPS bilan tekshirish: status, sertifikat (muddati, hostname mosligi), javob vaqti; natija "tayyor / tayyor emas (sabab)". `resolve_to` parametri orqali (11.4.4, 18.4.1) | S | F2 | Sertifikati yo'q serverga tekshiruv "sertifikat yo'q" sababi bilan rad etiladi; tayyor serverga "tayyor" deydi |

## 4.9. BKP — Zaxira nusxalar monitoringi

Zaxirani `appctl` oladi; Nazoratchi uni **kuzatadi va tekshiradi** (restore drill bundan mustasno — u tiklash jarayonini qayd etadi, 10-bo'limga qarang).

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-BKP-001 | **Zaxira papkasini skanerlash.** Agent har 5 daqiqada `<BACKUP_ROOT>/<loyiha>/{daily,weekly}` ni skanerlaydi: fayllar ro'yxati, nomi, hajmi, o'zgarish vaqti, soni | M | F1 | Yangi fayl paydo bo'lgach 5 daqiqada panelda ko'rinadi |
| FR-BKP-002 | **Oxirgi muvaffaqiyatli zaxira yoshi.** Har loyiha uchun eng yangi to'liq (`.tmp` bo'lmagan) kunlik fayl yoshi (soat) hisoblanadi | M | F1 | `BiznesRadar` uchun yosh soat aniqligida ko'rinadi |
| FR-BKP-003 | **Loyiha zaxira kartasi.** Loyiha sahifasida: oxirgi zaxira vaqti/hajmi, kunlik/haftalik soni, offsite holati, oxirgi restore drill, 30 kunlik hajm grafigi | M | F1 | Karta barcha elementlarni ko'rsatadi |
| FR-BKP-004 | **Kechikish alerti.** Yosh > 26 soat bo'lsa — yuqori ahamiyatli alert; > 50 soat — kritik | M | F1 | `appctl backup` to'xtatilsa 26 soatdan keyin alert |
| FR-BKP-005 | **Hajm anomaliyasi.** Yangi zaxira hajmi oldingi 7 ta zaxira medianasining 50% dan kam yoki 300% dan ko'p bo'lsa — "shubhali zaxira" alerti. Birinchi 7 kunda faqat ma'lumot | M | F1 | Bo'sh/yarim dump alertni ishga tushiradi |
| FR-BKP-006 | **Yaxlitlik tekshiruvi.** Agent eng yangi `.sql.gz` ni `gzip -t` bilan tekshiradi (kuniga 1 marta, oxirgi fayl); natija `ok/corrupt`; buzuq bo'lsa kritik | M | F2 | Kesilgan gz fayl "corrupt" deb belgilanadi |
| FR-BKP-007 | **Saqlash siyosati tekshiruvi.** Kunlik fayllar soni 7 dan, haftalik 5 dan oshmasligi (appctl tozalashi ishlayotganini bilish); oshsa — ma'lumot | S | F2 | Qo'lda 12 ta fayl qo'shilsa belgi chiqadi |
| FR-BKP-008 | **O'sish tendensiyasi.** Zaxira hajmining 90 kunlik o'sishi va zaxira papkasi diskka sig'ish bashorati | C | F3 | Grafik va "N kunda to'ladi" bashorati |
| FR-BKP-009 | **Haftalik fayl zaxirasi.** `BACKUP_PATHS` tar.gz fayllari haftada bir marta (yakshanba) chiqishi kerak; 8 kundan eski bo'lsa — ogohlantirish | S | F2 | Haftalik fayl yo'q bo'lsa alert |
| FR-BKP-010 | **Offsite sozlanganligi.** Reyestrdagi har serverda `OFFSITE` bo'sh bo'lsa, panelda "Offsite zaxira yo'q" doimiy ogohlantirish (yopiq bo'lmaydigan, silence qilinguncha) | M | F1 | Hozirgi S-01 uchun ogohlantirish ko'rinadi |
| FR-BKP-011 | **Offsite oxirgi sinxronlash.** Offsite serverdagi agent shu papkani skanerlaydi; core har loyiha uchun offsite'dagi eng yangi fayl vaqtini biladi | M | F2 | Offsite'ga fayl yetib borgach, 15 daqiqada panelda ko'rinadi |
| FR-BKP-012 | **Mahalliy va offsite solishtirish.** Eng yangi mahalliy fayl nomi va hajmi offsite'dagi bilan mos bo'lishi kerak | M | F2 | Farq bo'lsa "offsite orqada" belgisi |
| FR-BKP-013 | **Offsite alerti.** Offsite oxirgi sinxronlash > 26 soat — yuqori; > 50 soat — kritik | M | F2 | Rsync to'xtatilsa alert |
| FR-BKP-014 | **Offsite diskida joy.** Offsite serverning bo'sh joyi va zaxiralarning o'sishi; < 15% — ogohlantirish | S | F2 | Offsite agent diskni ko'rsatadi |
| FR-BKP-015 | **Restore drill rejasi.** Har loyiha uchun tiklash testi davriyligi (standart: chorakda 1 marta); muddati yaqinlashsa eslatma | S | F2 | 90 kundan keyin eslatma chiqadi |
| FR-BKP-016 | **Restore drill natijasini qayd etish.** Forma: sana, zaxira fayli, davomiyligi, tiklangan jadval/qator soni, natija (muvaffaqiyat/qisman/xato), izoh, bajargan shaxs | S | F1 | Natija saqlanadi va kartada ko'rinadi |
| FR-BKP-017 | **Restore drill muddati o'tdi alerti.** 100 kun o'tib hali muvaffaqiyatli drill bo'lmasa — ogohlantirish | S | F2 | Muddati o'tgan loyiha ro'yxatda belgilanadi |
| FR-BKP-018 | **Zaxira logini tahlil qilish.** `/var/log/appctl-backup.log` ning oxirgi bajarilishi: `XATO` qatorlari bo'lsa alert matnida oxirgi 10 qator ko'rsatiladi | S | F2 | Sun'iy `XATO` qatori alert matniga tushadi |

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-BKP-019 | **Dump mazmuni tekshiruvi.** Dump boshi (`head`) PostgreSQL dump belgilarini o'z ichiga olishi; (ixtiyoriy) `pg_restore --list` jadvallar soni | C | F3 | Bo'sh/yaroqsiz dump aniqlanadi |
| FR-BKP-020 | **Avtomatik restore drill (`appctl drill`).** Vaqtinchalik konteynerda (bir xil PostgreSQL asosiy versiyasi) dump tiklanadi, jadval/qator soni hisoblanadi, konteyner o'chiriladi, natija API orqali yuboriladi; ishlab turgan bazaga hech qachon tegmaydi (10.8.2, 10.8.6) | C | F3 | Drill natijasi kartada "avtomatik" belgisi bilan paydo bo'ladi; ishlab turgan baza o'zgarmagan |

## 4.10. ALR — Alert tizimi

Batafsil mexanika va qoidalar katalogi — 12-bo'limda.

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-ALR-001 | **Qoida modeli.** Qoida: nom, qamrov (server/loyiha/tekshiruv/teg), shart, davomiylik (`for`), ahamiyat, kanallar, runbook, yoqilgan/o'chirilgan | M | F1 | Qoida yaratiladi va ishlaydi |
| FR-ALR-002 | **Shart turlari.** (a) chegara (`>`, `<`, `==`), (b) yo'qlik (`absent`, ma'lumot kelmayapti), (c) o'zgarish tezligi (rate), (d) bashorat (forecast), (e) holat o'tishi (healthy → unhealthy) | M | F1 | Har tur uchun test mavjud |
| FR-ALR-003 | **Davomiylik sharti.** Shart belgilangan vaqt davomida uzluksiz bajarilsa alert (masalan, CPU > 90% 10 daqiqa) | M | F1 | Qisqa sakrash alert bermaydi |
| FR-ALR-004 | **Ahamiyat darajalari.** `info`, `warning`, `high`, `critical`. Kanal va sokin soatlar ahamiyatga bog'liq | M | F1 | Har daraja o'z kanal qoidasiga amal qiladi |
| FR-ALR-005 | **"Server o'chdi" alerti.** Heartbeat 120 s kelmasa **va** tashqi TCP:22/ICMP tekshiruvi ham muvaffaqiyatsiz bo'lsa — `critical`; faqat heartbeat yo'q bo'lsa — `high` ("agent javob bermayapti") | M | F1 | Serverni o'chirib sinab ko'riladi: 2 daqiqada alert |
| FR-ALR-006 | **Standart qoidalar to'plami.** Tizim o'rnatilganda 12-bo'limdagi qoidalar (AL-001..AL-089) tayyor keladi; foydalanuvchi ularni o'chirishi yoki chegarasini o'zgartirishi mumkin | M | F1 | Toza o'rnatishda kamida F1 qoidalari yoqilgan |
| FR-ALR-007 | **Qoidalarni tahrirlash interfeysi.** Panelda qoida chegaralari va kanallarini o'zgartirish; o'zgarish audit jurnaliga yoziladi | S | F2 | Chegara o'zgartirilgach keyingi tekshiruvda qo'llanadi |
| FR-ALR-008 | **Disk to'lishi bashorati.** Bashorat < 7 kun — `warning`; < 24 soat yoki band ≥ 90% — `critical` | M | F1 | Sun'iy o'sishda alert |
| FR-ALR-009 | **Sekin javob alerti.** HTTP javob vaqti belgilangan chegaradan (standart 3 s) 5 daqiqa davomida oshsa — `warning` | S | F2 | `tc`/sun'iy kechikish bilan alert |
| FR-ALR-010 | **Crash loop alerti.** FR-DKR-004 asosida `high` | M | F1 | Sinov konteynerida alert |
| FR-ALR-011 | **Bostirish (suppression) va bog'liqlik.** Server `critical` "o'chdi" holatida bo'lsa, shu serverdagi servis/HTTP/zaxira alertlari alohida yuborilmaydi; ular incident ichida "ta'sirlangan" ro'yxati sifatida ko'rinadi | M | F1 | Server o'chganda 1 ta xabar keladi, 20 ta emas |
| FR-ALR-012 | **Jim qilish (silence).** Qamrov va muddat (15 daqiqa … 30 kun) bilan; sabab majburiy; muddati tugagach avtomatik o'chadi; audit jurnaliga yoziladi | M | F1 | Jim qilingan alert yuborilmaydi, muddatdan keyin qayta ishlaydi |
| FR-ALR-013 | **Qabul qilish (acknowledge).** Operator incident'ni qabul qiladi; takroriy eslatmalar to'xtaydi; kim va qachon qabul qilgani yoziladi | M | F1 | Telegram tugmasi va panel orqali ishlaydi |
| FR-ALR-014 | **Tiklandi bildirishnomasi.** Shart yo'qolsa incident yopiladi va "tiklandi" xabari davomiylik bilan yuboriladi | M | F1 | Xabarda "davomiyligi: 12 daqiqa" bor |
| FR-ALR-015 | **Eskalatsiya.** `critical` alert 15 daqiqa qabul qilinmasa, ikkinchi kontaktga/kanalga yuboriladi (kechikish sozlanadi) | C | F3 | Ikkinchi chat'ga xabar keladi |
| FR-ALR-016 | **Incident tarixi.** Har incident: boshlanishi, tugashi, davomiyligi, ahamiyati, ta'sirlangan obyektlar, ack, izohlar, sabab (qo'lda) | S | F2 | Ro'yxat va filtr ishlaydi |
| FR-ALR-017 | **Runbook havolasi.** Har qoidada runbook (RB-nnn) havolasi; Telegram xabarida ko'rinadi | S | F2 | Xabarda havola bor va ochiladi |
| FR-ALR-018 | **Titrashdan himoya (flapping).** Bir alert 1 soatda 5 martadan ko'p holat o'zgartirsa, "beqaror" deb belgilanadi va yig'ma xabar bilan almashadi; gisterezis (yoqish va o'chirish chegaralari farqli) | S | F2 | Titragan HTTP tekshiruvi 1 soatda ≤ 2 xabar beradi |
| FR-ALR-019 | **Guruhlash.** Bir vaqtning o'zida 3 va undan ko'p bog'liq alert bitta yig'ma xabarga birlashadi (60 soniya oynada) | S | F2 | Sinovda 10 alert 1–2 xabarga aylanadi |
| FR-ALR-020 | **Sinov alerti.** Panel tugmasi orqali kanalni tekshirish uchun sinov xabari yuborish | M | F1 | Tugma bosilganda Telegram'ga xabar keladi |

## 4.11. NTF — Bildirishnomalar

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-NTF-001 | **Telegram kanali.** Bot orqali bir yoki bir nechta chatga (shaxsiy, guruh) xabar yuborish; bot tokeni maxfiy saqlanadi (shifrlangan) | M | F1 | Sinov xabari keladi |
| FR-NTF-002 | **Chuqur havola.** Har xabarda tegishli incident/obyekt sahifasiga havola | M | F1 | Havola brauzerda to'g'ri sahifani ochadi |
| FR-NTF-003 | **Xabar formati.** Ahamiyat belgisi (🔴/🟠/🟡/🔵), qisqa sarlavha, server/loyiha, qiymat va chegara, boshlanish vaqti (mahalliy zonada, Asia/Tashkent), runbook havolasi; xabar 1000 belgidan oshmaydi | M | F1 | Namuna xabarlar 12.9-bo'limga mos |
| FR-NTF-004 | **Qayta urinish va zaxira kanal.** Yuborish muvaffaqiyatsiz bo'lsa — eksponensial kutish bilan 5 marta; hali yetmasa — ikkinchi kanal (email); kanal ishlamayotgani panelda ko'rinadi | M | F1 | Telegram bloklanganda email ketadi |
| FR-NTF-005 | **Email kanali.** SMTP orqali (matn va HTML); `critical` va kunlik hisobot uchun | S | F2 | Email keladi |
| FR-NTF-006 | **Sokin soatlar.** Belgilangan vaqtda (masalan, 23:00–07:00) faqat `critical` darrov; qolganlari ertalabgacha yig'iladi va 07:00 da bitta xabar bo'ladi | S | F2 | Kechasi `warning` kelmaydi |
| FR-NTF-007 | **Webhook kanali.** Ixtiyoriy URL ga JSON POST (HMAC imzo bilan) | C | F3 | Sinov server webhookni oladi va imzoni tekshiradi |
| FR-NTF-008 | **Davriy hisobotlar.** Kunlik qisqa (08:00) va haftalik batafsil (dushanba 09:00): incidentlar, uptime, zaxira holati, sertifikat muddatlari, disk bashorati | S | F2 | Hisobot jadvalga ko'ra keladi |
| FR-NTF-009 | **Telegram tugmalari.** Xabarda "Qabul qilaman", "1 soat jim", "Panelda ochish" inline tugmalari; bosish faqat ruxsat etilgan chat ID lardan qabul qilinadi | S | F2 | Tugma bosilganda incident ack bo'ladi |

## 4.12. UI — Interfeys

Batafsil ekranlar va wireframe — 13-bo'limda.

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-UI-001 | **Bosh sahifa (dashboard).** Umumiy holat bannerlari ("Hammasi joyida" / "N ta muammo"), serverlar va loyihalar soni, ochiq incidentlar, eng yaqin muddatlar (sertifikat, domen, drill) | M | F1 | Banner holati ochiq incidentlarga mos |
| FR-UI-002 | **Holat kartochkalari.** Har server va har loyiha uchun rangli holat (yashil/sariq/qizil/kulrang) va qisqa ko'rsatkichlar (CPU, RAM, disk) | M | F1 | Rang qoidalari 13.4 ga mos |
| FR-UI-003 | **Loyihalar ro'yxati.** FR-REG-004 ustunlari, filtr, qidiruv, saralash | M | F1 | 50 loyiha bilan tez ishlaydi |
| FR-UI-004 | **Serverlar ro'yxati.** Har server: holat, uptime, CPU/RAM/disk mini-grafiklar (sparkline), agent versiyasi, so'nggi aloqa | M | F1 | Ro'yxat 5 soniyada yangilanadi |
| FR-UI-005 | **Server sahifasi.** CPU, yuklama, RAM, swap, disk (hajm va I/O), tarmoq, konteynerlar, servislar, incidentlar bo'limlari; vaqt oralig'i (1 s, 6 s, 24 s, 7 k, 30 k, 90 k, maxsus) | M | F1 | Barcha grafiklar tanlangan oraliqda chiziladi |
| FR-UI-006 | **Loyiha sahifasi.** Domenlar, HTTP/TLS/DNS holati, konteynerlar/servislar, zaxira kartasi, incidentlar, havolalar | M | F1 | Barcha bo'limlar ko'rinadi |
| FR-UI-007 | **Eng yuklangan ro'yxati.** CPU/RAM/disk bo'yicha eng ko'p ishlatayotgan serverlar va konteynerlar | S | F2 | Ro'yxat ma'lumotga mos |
| FR-UI-008 | **Zaxiralar sahifasi.** Barcha loyihalar bo'yicha jadval: oxirgi zaxira, yoshi, hajmi, offsite holati, drill; qizil/sariq belgilash | M | F1 | `appctl backup all` dan keyin yangilanadi |
| FR-UI-009 | **Domenlar va sertifikatlar sahifasi.** Domen, qolgan kun (sertifikat, ro'yxat), DNS mosligi; muddati bo'yicha saralash | M | F1 | Eng yaqin muddat tepada |
| FR-UI-010 | **Incidentlar sahifasi.** Ochiq va yopilgan; filtr (ahamiyat, server, loyiha, vaqt); incident tafsiloti: vaqt chizig'i, ta'sirlanganlar, izohlar | M | F1 | Incident ochiladi va izoh qo'shiladi |
| FR-UI-011 | **Texnik ishlar sahifasi.** Maintenance oynalari yaratish/ko'rish (UC-08) va jim qilishlar ro'yxati | M | F1 | Oyna yaratilgach alertlar jim bo'ladi |
| FR-UI-012 | **Hisobotlar.** Oylik/chorak uptime, incident soni, MTTR, zaxira muvaffaqiyat foizi; CSV va PDF eksport | S | F2 | Hisobot sonlari ma'lumotga mos |
| FR-UI-013 | **Sozlamalar.** Kanallar, standart chegaralar, retention, foydalanuvchilar, tokenlar, vaqt zonasi, til | M | F1 | Sozlamalar saqlanadi |
| FR-UI-014 | **Qorong'i/yorug' mavzu.** Tizim sozlamasiga mos avtomatik va qo'lda almashtirish | S | F2 | Ikkala mavzuda kontrast WCAG AA |
| FR-UI-015 | **Global qidiruv.** Server, loyiha, domen, incident bo'yicha (`/` tugmasi) | C | F3 | "biznes" yozilganda BiznesRadar topiladi |
| FR-UI-016 | **Jonli yangilanish.** Dashboard va server sahifasi sahifani qayta yuklamasdan SSE orqali yangilanadi | S | F2 | Yangi ma'lumot 5 soniyada ko'rinadi |

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-UI-017 | **Maxsus oraliq va yaqinlashtirish.** Grafikda vaqt oralig'ini tanlab yaqinlashtirish, tooltip da aniq qiymat | S | F2 | Zoom ishlaydi |
| FR-UI-018 | **CSV eksport.** Har jadval va grafik ma'lumotini CSV qilish | C | F3 | Fayl to'g'ri ochiladi |
| FR-UI-019 | **Ommaviy holat sahifasi (status page).** Tanlangan loyihalar uchun avtorizatsiyasiz o'qiladigan sahifa (masalan, mijoz uchun) | C | F3 | Alohida URL, faqat tanlangan loyihalar |

## 4.13. AUTH — Autentifikatsiya va kirish

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-AUTH-001 | **Kirish.** Email/login va parol; parollar `argon2id` bilan xeshlanadi | M | F1 | Noto'g'ri parol rad etiladi |
| FR-AUTH-002 | **Parol siyosati.** Kamida 12 belgi; mashhur parollar ro'yxatida bo'lmasligi | M | F1 | Zaif parol qabul qilinmaydi |
| FR-AUTH-003 | **Ikki bosqichli tasdiqlash (TOTP).** Authenticator ilovasi bilan; zaxira kodlari | S | F2 | TOTP bilan kirish ishlaydi |
| FR-AUTH-004 | **Sessiyalar.** Xavfsiz cookie (`HttpOnly`, `Secure`, `SameSite=Lax`), 12 soat harakatsizlik muddati; faol sessiyalar ro'yxati va "hammasidan chiqish" | M | F1 | Muddatdan keyin qayta kirish talab etiladi |
| FR-AUTH-005 | **Loyiha bo'yicha cheklangan viewer.** Viewer faqat biriktirilgan loyihalarni ko'radi (boshqa server/loyiha/incident yashirin) | C | F3 | Cheklangan viewer boshqa loyihani ko'rmaydi |
| FR-AUTH-006 | **Token boshqaruvi.** Agent tokenlari va API tokenlari: yaratish, nomlash, oxirgi ishlatilgan vaqt, bekor qilish; token faqat yaratilganda bir marta ko'rsatiladi, bazada faqat xesh | M | F1 | Bekor qilingan token bilan so'rov 401 |
| FR-AUTH-007 | **Kirishga urinishlar cheklovi.** 5 ta xato → 15 daqiqa blok (IP + hisob bo'yicha); barcha urinishlar audit jurnalida | M | F1 | 6-urinish rad etiladi |
| FR-AUTH-008 | **Parolni tiklash.** Telegram bot orqali bir martalik havola (egasining ruxsat etilgan chat ID siga) | C | F3 | Havola 15 daqiqa amal qiladi |

## 4.14. AUD — Audit jurnali

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-AUD-001 | **Yoziladigan hodisalar.** Kirish/chiqish, rolga oid o'zgarish, obyekt yaratish/o'zgartirish/arxivlash, qoida o'zgarishi, silence/maintenance, token yaratish/bekor qilish, restore drill kiritish, sozlama o'zgarishi | M | F1 | Har hodisa jurnalda paydo bo'ladi |
| FR-AUD-002 | **Yozuv tarkibi.** Vaqt (UTC), foydalanuvchi, IP, amal turi, obyekt, eski va yangi qiymat (maxfiy maydonlar maskalanadi) | M | F1 | Parol/token qiymatlari jurnalda yo'q |
| FR-AUD-003 | **Ko'rish va filtr.** Faqat Owner; foydalanuvchi, amal, sana bo'yicha filtr | S | F2 | Filtr ishlaydi |
| FR-AUD-004 | **Saqlash va eksport.** Kamida 1 yil; jurnal faqat qo'shiladi (o'zgartirilmaydi/o'chirilmaydi); CSV eksport | S | F2 | UI/API orqali o'chirib bo'lmaydi |

## 4.15. INT — Integratsiyalar

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-INT-001 | `appctl list --json`. Mavjud `appctl` ga JSON chiqarish rejimi qo'shiladi: loyiha, holat, konteynerlar soni; Nazoratchi agenti shu bilan loyiha holatini oladi | M | F1 | `appctl list --json \| jq .` xatosiz ishlaydi |
| FR-INT-002 | `appctl status --json`. Disk, xotira, konteynerlar, sertifikatlar (qolgan kun), oxirgi zaxira vaqtlari JSON formatda | M | F1 | Chiqish 18-bo'limdagi sxemaga mos |
| FR-INT-003 | **Zaxira holati fayli.** `appctl backup` oxirida `/opt/applications/backups/<loyiha>/state.json` yozadi: vaqt, natija (ok/fail), fayl nomi, hajm, davomiylik, xato matni (offsite natijasi `_run.json` da, 10.3.2). Agent shuni o'qiydi; papka skaneridan ishonchliroq | M | F1 | Zaxira muvaffaqiyatsiz bo'lsa `state.json` da `status: "fail"` va sabab bor |
| FR-INT-004 | **Docker API (faqat o'qish).** Agent Docker ma'lumotini `docker-socket-proxy` orqali oladi (`CONTAINERS=1`, `INFO=1`, `IMAGES=1`, `VOLUMES=1`; yozuvchi so'rovlar taqiqlangan) | M | F1 | Proksi POST so'rovlarini 403 bilan rad etadi |
| FR-INT-005 | **Kiruvchi heartbeat endpointi.** Har qanday cron/skript `curl https://<core>/hb/<token>` bilan "bajarildim" deb xabar bera oladi (muvaffaqiyat/xato parametri bilan) | M | F1 | Endpoint 200 qaytaradi, voqea yoziladi |
| FR-INT-006 | **Deploy belgilari.** `appctl deploy` yakunida core'ga "deploy" voqeasi yuboriladi (loyiha, versiya/commit, natija); grafiklarda vertikal chiziq sifatida ko'rinadi | C | F3 | Deploy'dan keyin grafikda belgi |

## 4.16. SYS — Tizimning o'zi

| ID | Talab | Ust. | Faza | Qabul mezoni |
|---|---|---|---|---|
| FR-SYS-001 | **O'z-o'zini kuzatish.** Core'ning o'z metrikalari: qabul qilish tezligi, navbat uzunligi, bazaga yozish kechikishi, alert hisoblash vaqti, xatolar soni | M | F1 | `/metrics` va panelda ko'rinadi |
| FR-SYS-002 | **Watchdog (alohida serverda).** Core har 60 soniyada S-01 dagi watchdog'ga ping yuboradi (3.1 P-5); 5 daqiqa kelmasa, watchdog mustaqil Telegram bot orqali xabar beradi. O'zaro kuzatuv: S-01 ni core (agent heartbeat) kuzatadi | M | F1 | Core to'xtatilsa 5–6 daqiqada xabar keladi |
| FR-SYS-003 | **O'z zaxirasi (Telegram orqali).** Nazoratchi bazasi (reyestr, incident, audit, kanal va qoida sozlamalari; metrikalarsiz) va sozlamalari kunlik zaxiralanadi, `age` bilan shifrlanadi va bot orqali **Owner'ning shaxsiy Telegram chatiga** (guruhga emas) fayl sifatida yuboriladi. Fayl 45 MB dan oshsa bo'laklanadi. Yuborish natijasi qayd etiladi; yuborilmasa alert. Zaxira o'zi ham FR-BKP qoidalariga bo'ysunadi | M | F1 | Chatda shifrlangan fayl bor; undan yangi serverda tiklash sinab ko'rilgan |
| FR-SYS-004 | **Saqlash muddati vazifasi.** Kunlik ish: xom ma'lumotlarni siyraklashtirish (downsampling) va muddati o'tganini o'chirish (8-bo'lim) | M | F1 | Bazada eski xom ma'lumot qolmaydi |
| FR-SYS-005 | **Sog'liq endpointi.** `/healthz` (baza va navbat tekshiruvi bilan), alohida `/readyz`; tashqi tekshiruvchi uchun | M | F1 | Baza o'chiq bo'lsa 503 |
| FR-SYS-006 | **Xavfsiz migratsiya.** Sxema o'zgarishlari orqaga qaytariladigan migratsiyalar bilan; yangilash vaqtida ma'lumot qabul qilish to'xtamaydi (agent buferi qoplaydi) | S | F2 | Yangilash paytida ma'lumot yo'qolmaydi |

## 4.17. Talablar xulosasi (jami)

| Modul | M | S | C | Jami |
|---|---|---|---|---|
| REG | 6 | 4 | 2 | 12 |
| HST | 8 | 6 | 3 | 17 |
| AGT | 10 | 4 | 0 | 14 |
| DKR | 3 | 5 | 2 | 10 |
| SVC | 3 | 2 | 1 | 6 |
| HTTP | 4 | 5 | 1 | 10 |
| TLS | 3 | 3 | 2 | 8 |
| DNS | 3 | 4 | 2 | 9 |
| BKP | 10 | 7 | 3 | 20 |
| ALR | 13 | 6 | 1 | 20 |
| NTF | 4 | 4 | 1 | 9 |
| UI | 11 | 5 | 3 | 19 |
| AUTH | 5 | 1 | 2 | 8 |
| AUD | 2 | 2 | 0 | 4 |
| INT | 5 | 0 | 1 | 6 |
| SYS | 5 | 1 | 0 | 6 |
| **Jami FR** | **95** | **59** | **24** | **178** |

Sonlar hujjat qismlaridan skript bilan sanalgan (har talab bir marta, takrorlanish yo'q). Nofunksional talablar (5-bo'lim): 65 ta; umumiy sanoq va bosqichlar bo'yicha taqsimot — 18.7.

# 5. Nofunksional talablar

## 5.1. Hisob-kitob asoslari (o'lchamlar)

Talablardagi raqamlar quyidagi taxminiy hisoblarga asoslangan. Ular loyiha davomida haqiqiy o'lchovlar bilan tekshiriladi (17-bo'lim, xavf R-104).

### 5.1.1. Miqyos

| Parametr | Joriy (2026-10) | Maqsad (1 yil) | Zaxira (3x) |
|---|---|---|---|
| Serverlar | 3 (+1 core) | 10 | 30 |
| Loyihalar | 7 ishlayapti + ~15 ko'chirilmagan | 30 | 90 |
| Domenlar | ~20 | 50 | 150 |
| Konteynerlar (jami) | ~60 | 150 | 450 |
| Faol tekshiruvlar (HTTP/TLS/DNS) | 0 | 150 | 450 |
| Foydalanuvchilar | 1–3 | 5 | 15 |

Tizim **maqsad miqyosida** (10 server) hech qanday sozlamasiz, **zaxira miqyosida** (30 server) esa faqat resurs oshirilishi bilan ishlashi kerak. Arxitektura o'zgarishi talab qilinmasligi kerak.

### 5.1.2. Ma'lumot oqimi hajmi (zaxira miqyosi: 30 server)

Bir hostdan intervalga ko'ra namunalar soni (7-bo'limdagi katalog asosida, taxminan):

| Guruh | Qatorlar (series) | Interval | Namuna/soniya (1 host) |
|---|---|---|---|
| Tizim (CPU: 8 yadro × 4 rejim + umumiy, load, RAM, swap) | ~60 | 15 s | 4.0 |
| Disk (3 bo'lim × 5 + 2 blok qurilma × 6) | ~27 | 15 s | 1.8 |
| Tarmoq (2 interfeys × 8) | ~16 | 15 s | 1.1 |
| Docker (15 konteyner × 8) | ~120 | 30 s | 4.0 |
| Servislar, jarayonlar, boshqa | ~20 | 30 s | 0.7 |
| **Jami 1 host** | **~243** | | **~11.6** |

30 host × 11.6 ≈ **350 namuna/soniya** → kuniga ≈ **30 million** namuna, yiliga ≈ 11 milliard (xom). Shu sababli xom ma'lumot qisqa saqlanadi va siyraklashtiriladi (8-bo'lim).

### 5.1.3. Saqlash hajmi bahosi

Asos: 30 host × ~243 qator = **7 290 qator (series)**. Taxminiy yozuv hajmi (TimescaleDB siqishdan keyin): xom namuna ~12 bayt, agregat yozuv (min/avg/max/count) ~20 bayt.

| Daraja | Saqlash | Yozuv/kun (7 290 qator) | Hajm bahosi (30 host, hamma metrika) |
|---|---|---|---|
| Xom (15–30 s) | 14 kun | 30.2 mln | 30.2M × 14 × 12 B ≈ 5.1 GB |
| 1 daqiqalik agregat | 90 kun | 7 290 × 1 440 = 10.5 mln | 10.5M × 90 × 20 B ≈ 18.9 GB |
| 5 daqiqalik agregat | 1 yil | 7 290 × 288 = 2.1 mln | 2.1M × 365 × 20 B ≈ 15.3 GB |
| 1 soatlik agregat | 3 yil | 7 290 × 24 = 0.175 mln | 0.175M × 1 095 × 20 B ≈ 3.8 GB |
| **Jami (yuqori chegara)** | | | **≈ 43 GB** |

Maqsad miqyosida (10 server) yuqori chegara ≈ **14 GB**. Agregat darajalar uchun faqat **muhim metrikalar** (CPU, RAM, disk, tarmoq umumiy) saqlansa, agregat hajmi taxminan 3 baravar kamayadi va maqsad miqyosidagi kutilgan jami hajm **~6 GB** bo'ladi.

> Muhim: siqish koeffitsiyenti (12 va 20 bayt) TimescaleDB hujjatlaridagi tipik natijaga asoslangan taxmin edi. **P-02 o'lchovi** (sintetik, lekin real naqshli: 120 qator × 24 soat × 15 s = 691 200 namuna): siqilmagan 91 bayt, siqilgan **6.9 bayt/namuna** (13× siqish), ya'ni xom ma'lumot uchun taxmin konservativ. Real ma'lumot (spayklar, ko'p kasr xonali) sekinroq siqilishi mumkin; F1 oxirida haqiqiy hajm bilan qayta o'lchanadi (xavf R-104).

## 5.2. PRF — Unumdorlik

| ID | Talab | Ust. | Faza | O'lchash usuli |
|---|---|---|---|---|
| NFR-PRF-001 | **Qabul qilish kechikishi.** Ingest endpoint so'rovga 95-persentilda ≤ 300 ms, 99-persentilda ≤ 1 s javob beradi (30 host, har 15 s) | M | F1 | Yuklama testi (`k6`/`locust`), 30 virtual agent |
| NFR-PRF-002 | **Ma'lumot yangiligi (freshness).** Agent namunani yig'ganidan boshlab panelda ko'rinishigacha ≤ 30 soniya (95-persentil) | M | F1 | Test namunaning vaqt tamg'asi va paneldagi vaqtini solishtirish |
| NFR-PRF-003 | **Alert hisoblash kechikishi.** Shart paydo bo'lgandan alert yaratilgunga qadar: `for` davomiyligi + ≤ 30 s | M | F1 | Sun'iy hodisa bilan stopwatch |
| NFR-PRF-004 | **Bildirishnoma yetkazilishi.** Incident yaratilgandan Telegram'da paydo bo'lishigacha ≤ 10 soniya (95-persentil), kanal ishlaganda | M | F1 | Sinov alertlari, 50 ta o'lchov |
| NFR-PRF-005 | **Dashboard yuklanishi.** Bosh sahifa to'liq ko'rinishi ≤ 2 s (10 server, 30 loyiha, tezkor internet, kesh bilan ≤ 1 s) | M | F1 | Lighthouse / qo'lda o'lchov |
| NFR-PRF-006 | **Grafik so'rovlari.** 24 soatlik grafik ≤ 1 s; 7 kunlik ≤ 1.5 s; 30 kunlik ≤ 2 s; 90 kunlik ≤ 3 s (tegishli agregat jadvaldan) | M | F1 | So'rov vaqtini yozish, 100 ta o'lchov medianasi |
| NFR-PRF-007 | **Agent resurslari.** O'rtacha CPU ≤ 1%, maksimal RAM ≤ 60 MB, disk yozuvi ≤ 5 MB/soat (bufer ishlamaganda) | M | F1 | 24 soatlik o'lchov (`pidstat`) |
| NFR-PRF-008 | **Core resurslari.** 10 server yuklamasida core (backend + baza + redis) o'rtacha ≤ 40% CPU va ≤ 1.5 GB RAM, 2 vCPU/4 GB da | M | F1 | Yuklama testi |
| NFR-PRF-009 | **Frontend hajmi.** Dastlabki yuklanadigan JS+CSS ≤ 400 KB (gzip); grafik kutubxonasi kechiktirib yuklanadi (lazy) | S | F2 | Build hisoboti |
| NFR-PRF-010 | **Faol tekshiruvlar sig'imi.** Bitta prober jarayoni 450 tekshiruvni (60 s interval) bir vaqtda, 1 vCPU ning ≤ 30% bilan bajaradi | S | F2 | Sun'iy 450 tekshiruv |

## 5.3. CAP — Sig'im va masshtablash

| ID | Talab | Ust. | Faza | O'lchash usuli |
|---|---|---|---|---|
| NFR-CAP-001 | **Serverlar soni.** 30 serverni qo'llab-quvvatlaydi (5.1.1); 100 serverga o'tish uchun faqat gorizontal bo'lmagan masshtablash (resurs oshirish) yetarli | M | F2 | Yuklama testi 30 va 60 virtual agent bilan |
| NFR-CAP-002 | **Saqlash byudjeti.** Maqsad miqyosida (10 server) jami baza hajmi ≤ 25 GB (1 yil; kutilgan 6–14 GB, 5.1.3); 3 yilda ≤ 60 GB | M | F2 | `pg_total_relation_size` kuzatuvi |
| NFR-CAP-003 | **Indeks va so'rov rejalari.** Hech bir asosiy so'rov (7 ta kritik, 8.8-bo'lim) ketma-ket skanerlashga (seq scan) tushmaydi | S | F2 | `EXPLAIN` tekshiruvi testda |
| NFR-CAP-004 | **Namuna shakli bilan masshtab.** Yangi metrika turi qo'shish bazaga sxema o'zgartirishni talab qilmaydi (uzun/tor jadval modeli) | M | F1 | Yangi metrika kodsiz/migratsiyasiz qabul qilinadi |
| NFR-CAP-005 | **Gorizontal ingest.** Ingest xizmati holatsiz (stateless): bir nechta nusxa ishlashi mumkin | C | F3 | Ikkita nusxa bilan test |

## 5.4. REL — Ishonchlilik va chidamlilik

| ID | Talab | Ust. | Faza | O'lchash usuli |
|---|---|---|---|---|
| NFR-REL-001 | **Core mavjudligi.** Oylik ≥ 99.5% (≈ 3.6 soat/oy ruxsat etilgan to'xtash) | M | F2 | Tashqi watchdog hisobi |
| NFR-REL-002 | **Ma'lumot yo'qolmasligi.** Core 15 daqiqagacha mavjud bo'lmasa ham agent buferi tufayli grafikda bo'shliq (> 1 namuna) yo'q | M | F1 | Core to'xtatilib qayta yoqilgan test |
| NFR-REL-003 | **Idempotent qabul.** Bir paket takror yuborilsa, ma'lumot ikkilanmaydi (paket ID va namuna kaliti bo'yicha) | M | F1 | Bir paketni 3 marta yuborib hisoblash |
| NFR-REL-004 | **Yumshoq degradatsiya.** Baza vaqtincha o'chsa, core navbatga (Redis) yozadi (15 daqiqagacha), UI esa "ma'lumot kechikmoqda" ko'rsatadi; alert hisoblash to'xtamasdan kechiktiriladi | S | F2 | Bazani to'xtatib sinash |
| NFR-REL-005 | **Bildirishnoma ishonchliligi.** Telegram API vaqtincha ishlamasa, 24 soat ichida kamida 99.9% xabar yetkaziladi (qayta urinish va zaxira kanal bilan) | M | F2 | Sinov (Telegram'ni bloklab) |
| NFR-REL-006 | **Tiklash vaqti (RTO).** Nazoratchi'ni to'liq yangi serverga zaxiradan tiklash ≤ 1 soat | M | F2 | Yiliga 1 marta sinov (drill) |
| NFR-REL-007 | **Tiklash nuqtasi (RPO).** Nazoratchi sozlamalari va reyestr uchun ≤ 24 soat; vaqt qatorlari uchun yo'qotish qabul qilinadi (agent qayta yuboradi) | M | F1 | Zaxira jadvali |
| NFR-REL-008 | **Vaqt va vaqt zonalari.** Ichkarida hamma vaqt UTC (`timestamptz`); UI da foydalanuvchi zonasi (standart `Asia/Tashkent`) | M | F1 | Kod ko'rigi va test |
| NFR-REL-009 | **Qayta ishga tushishga chidamlilik.** Core konteynerlari `restart: unless-stopped`; server reboot'dan keyin 3 daqiqada hammasi ishlaydi, qo'lda aralashuvsiz | M | F1 | Reboot sinovi |

| ID | Talab | Ust. | Faza | O'lchash usuli |
|---|---|---|---|---|
| NFR-REL-010 | **Soxta alertlarni kamaytirish.** Tekshiruvlar tasdiqlash qoidasi (3 ketma-ket xato) va gisterezis bilan; oylik soxta alert ulushi ≤ 5% | M | F2 | 90 kunlik qo'lda baholash |

## 5.5. SEC — Xavfsizlik (qisqa; batafsili 14-bo'limda)

| ID | Talab | Ust. | Faza | O'lchash usuli |
|---|---|---|---|---|
| NFR-SEC-001 | **Barcha aloqa shifrlangan.** Brauzer↔core, agent↔core, core↔Telegram: TLS 1.2+; agent sertifikatni tekshiradi (o'zini-o'zi imzolagan sertifikat bo'lsa, qotirilgan (pinned) CA) | M | F1 | `testssl.sh` / agent testi |
| NFR-SEC-002 | **Maxfiy ma'lumotni saqlash.** Telegram bot tokeni, SMTP paroli, tekshiruv sarlavhalari: bazada shifrlangan (AES-256-GCM, kalit `.env`/fayl orqali, bazada emas) | M | F1 | Bazadan to'g'ridan-to'g'ri o'qib ko'rish: ochiq matn yo'q |
| NFR-SEC-003 | **Tokenlar.** Agent/API tokenlari kamida 256 bit tasodifiy; bazada faqat SHA-256 xesh | M | F1 | Kod ko'rigi |
| NFR-SEC-004 | **Eng kam huquq.** Agent `root` siz; Docker o'qish proksisi; backend bazaga cheklangan foydalanuvchi bilan | M | F1 | Konfiguratsiya tekshiruvi |
| NFR-SEC-005 | **Veb-zaifliklardan himoya.** OWASP ASVS 4.0 L1 talablari to'liq, L2 ning asosiy qismi (autentifikatsiya, sessiya, kirishni tekshirish, XSS/CSRF/SQLi) | M | F2 | Xavfsizlik tekshiruv ro'yxati va avtomatik skanerlash (ZAP) |
| NFR-SEC-006 | **Bog'liqliklar xavfsizligi.** `pip-audit`, `npm audit`, `govulncheck` CI da; kritik zaiflik bo'lsa build to'xtaydi | S | F2 | CI hisoboti |
| NFR-SEC-007 | **Tezlik cheklovi.** Ingest: token bo'yicha daqiqada ≤ 30 so'rov; login: 5 xato/15 daqiqa; API: foydalanuvchi bo'yicha 120 so'rov/daqiqa | M | F1 | Test |
| NFR-SEC-008 | **Loglarda maxfiy ma'lumot yo'q.** Token, parol, `Authorization` sarlavhalari, `.env` qiymatlari hech qachon logga yozilmaydi | M | F1 | Log namunalarini avtomatik skanerlash (regex) |
| NFR-SEC-009 | **Xavfsizlik sarlavhalari.** `Strict-Transport-Security`, `Content-Security-Policy`, `X-Content-Type-Options`, `Referrer-Policy`, `X-Frame-Options: DENY` | M | F1 | `securityheaders.com` darajasi ≥ A |
| NFR-SEC-010 | **O'z zaxirasi shifrlanishi.** Nazoratchi'ning o'z zaxirasi Telegram'ga yuborilishidan oldin `age` bilan shifrlanadi; ochiq kalit core'da, maxfiy kalit faqat Owner'da (parol menejeri) | M | F1 | Telegram'dagi fayl shifrlangan; maxfiy kalitsiz ochilmaydi |
| NFR-SEC-011 | **Ma'muriy kirishni cheklash.** `/admin` va sozlamalar uchun ixtiyoriy IP ro'yxati (allowlist) va 2FA | S | F2 | Konfiguratsiya testi |
| NFR-SEC-012 | **Imzolangan yangilanishlar.** Agent yangilanishi faqat CI maxfiy kaliti bilan imzolangan va agent binariga joylangan ochiq kalit bilan tekshirilgan bo'lsa qabul qilinadi; imzo kaliti core serverda saqlanmaydi; versiya pasaytirish (downgrade) rad etiladi (14.5.4) | M | F1 | Buzuq/imzosiz binar bilan yangilash rad etiladi (test) |

## 5.6. USA — Qulaylik va til

| ID | Talab | Ust. | Faza | O'lchash usuli |
|---|---|---|---|---|
| NFR-USA-001 | **Til.** Standart til — o'zbek (lotin, to'g'ri imlo: o', g', sh, ch, ng); qo'shimcha: rus, ingliz. Barcha matnlar tarjima fayllarida; kodda qattiq yozilgan matn yo'q | M | F1 | Kod ko'rigi, til almashtirish testi |
| NFR-USA-002 | **Mobil moslashuv.** Interfeys 360 px kenglikdan boshlab to'liq ishlaydi; asosiy ekranlar (dashboard, incidentlar) telefonda qulay | M | F1 | Qurilma/emulyator tekshiruvi |
| NFR-USA-003 | **Foydalanish imkoniyati (a11y).** WCAG 2.1 AA: kontrast, klaviatura bilan boshqarish, rang bilan birga belgi/matn (rang ko'rlik uchun) | S | F2 | `axe` skaneri, qo'lda tekshiruv |
| NFR-USA-004 | **Tushunarli xabarlar.** Har xato va alert xabari: nima bo'ldi, qayerda, nima qilish kerak — texnik jargonsiz emas, lekin aniq | M | F1 | Namuna xabarlarni ko'rib chiqish |
| NFR-USA-005 | **Vaqt formati.** UI da vaqt `Asia/Tashkent` zonasida, `2026-10-22 14:35` formatida; nisbiy vaqt ("5 daqiqa oldin") tooltip bilan aniq vaqtni ko'rsatadi | S | F2 | Ko'rik |
| NFR-USA-006 | **Birinchi sozlash vaqti.** Yangi o'rnatilgan tizimda birinchi serverni qo'shib, birinchi alertni olishgacha ≤ 15 daqiqa | S | F1 | Yangi odam bilan sinov |
| NFR-USA-007 | **Ranglar semantikasi.** Yashil — sog'lom, sariq — ogohlantirish, to'q sariq — yuqori, qizil — kritik, kulrang — noma'lum/pauza. Hammasi ikkala mavzuda | M | F1 | Dizayn tekshiruvi |

## 5.7. MNT — Qo'llab-quvvatlash va sifat

| ID | Talab | Ust. | Faza | O'lchash usuli |
|---|---|---|---|---|
| NFR-MNT-001 | **Test qamrovi.** Biznes mantiq (alert hisoblash, bashorat, bostirish, zaxira tahlili) uchun qator qamrovi ≥ 85%; umumiy ≥ 70% | M | F2 | `coverage` hisoboti CI da |

| ID | Talab | Ust. | Faza | O'lchash usuli |
|---|---|---|---|---|
| NFR-MNT-002 | **Statik tahlil.** Python: `ruff` + `mypy` (qat'iy rejim asosiy modullarda); TypeScript: `strict`; Go: `go vet` + `staticcheck`. CI da xato bo'lsa birlashtirib bo'lmaydi | M | F1 | CI |
| NFR-MNT-003 | **CI/CD.** Har commit'da: lint, test, build; main shoxda — konteyner image'lari (versiya tegi bilan) chiqariladi | M | F1 | Pipeline |
| NFR-MNT-004 | **Sozlama muhit o'zgaruvchilari orqali.** 12-factor: kod ichida qattiq yozilgan sozlama yo'q; `.env.example` to'liq | M | F1 | Kod ko'rigi |
| NFR-MNT-005 | **Strukturalangan loglar.** JSON log (`ts`, `level`, `msg`, `request_id`, `component`); sathlari sozlanadi | M | F1 | Log namunasi |
| NFR-MNT-006 | **Migratsiyalar.** Sxema o'zgarishi faqat versiyalangan migratsiya orqali; har migratsiya testda ikki marta (oldinga/orqaga) o'tadi | M | F1 | CI |
| NFR-MNT-007 | **Hujjatlar.** `README`, o'rnatish yo'riqnomasi, arxitektura qisqacha, runbook'lar, API OpenAPI sxemasi (avtomatik) | M | F2 | Hujjat ro'yxati |
| NFR-MNT-008 | **Modulli tuzilma.** Backend `apps/` bo'yicha mustaqil modullarga bo'lingan (6-bo'lim); modullar orasidagi bog'liqlik bir yo'nalishli | S | F1 | Import grafi tekshiruvi |
| NFR-MNT-009 | **Versiyalash.** Semantik versiyalash; agent va core o'rtasida protokol versiyasi (`/v1/`); eski agent kamida 2 ta minor versiya davomida ishlaydi | M | F1 | Moslik testi |

## 5.8. POR — Ko'chma va o'rnatish

| ID | Talab | Ust. | Faza | O'lchash usuli |
|---|---|---|---|---|
| NFR-POR-001 | **Core — Docker Compose.** Bitta `docker compose up -d` bilan butun core ko'tariladi | M | F1 | Toza serverda sinov |
| NFR-POR-002 | **Agent platformalari.** Linux amd64 va arm64 (Ubuntu 22.04, 24.04); statik binar, tashqi bog'liqliksiz | M | F1 | Ikkala arxitekturada sinov |
| NFR-POR-003 | **Provayderga bog'liq emas.** Hech qanday bulut provayderning xizmatiga (GCP, AWS) bog'lanmaydi; Contabo, Hetzner, boshqa VPS da bir xil ishlaydi (eski serverdagi GCP agentlari bilan bog'liq saboq) | M | F1 | Kod ko'rigi |
| NFR-POR-004 | **Minimal xost talablari.** Core: 2 vCPU, 4 GB RAM, 40 GB SSD; Ubuntu 24.04 + Docker; qo'shimcha dasturlar talab qilinmaydi | M | F1 | Spetsifikatsiya |
| NFR-POR-005 | **Oflayn o'rnatish imkoniyati.** Agent binari core'ning o'zidan yuklab olinadi (GitHub'ga bog'liq emas) | S | F2 | Tashqi internetsiz sinov |

## 5.9. OBS — Kuzatiluvchanlik (tizimning o'zi)

| ID | Talab | Ust. | Faza | O'lchash usuli |
|---|---|---|---|---|
| NFR-OBS-001 | **Ichki metrikalar.** Core Prometheus formatida `/metrics` (faqat ichki tarmoq/token bilan) beradi: so'rovlar, xatolar, navbat, alert hisoblash vaqti | S | F2 | `curl /metrics` |
| NFR-OBS-002 | **So'rov kuzatuvi.** Har HTTP so'rovga `request_id`; xato javobida u ko'rsatiladi | S | F2 | Test |
| NFR-OBS-003 | **Xato to'plash.** Kutilmagan istisnolar (exception) `ERROR` darajasida, to'liq stek bilan logga; ixtiyoriy Sentry (o'z-hosting) | C | F3 | Log namunasi |

## 5.10. DAT — Ma'lumot va maxfiylik

| ID | Talab | Ust. | Faza | O'lchash usuli |
|---|---|---|---|---|
| NFR-DAT-001 | **Yig'ilmaydigan ma'lumot.** Agent: fayl mazmuni, ilova ma'lumoti, foydalanuvchi shaxsiy ma'lumoti, buyruq argumentlari (parol bo'lishi mumkin), muhit o'zgaruvchilari — yig'maydi. Jarayon nomi faqat bajariladigan fayl nomi (argumentsiz) | M | F1 | Kod ko'rigi, paket namunasi tekshiruvi |
| NFR-DAT-002 | **Saqlash siyosati hujjatlashtirilgan.** Retention qiymatlari sozlamalarda ko'rinadi va o'zgartiriladi | S | F2 | Sozlamalar sahifasi |
| NFR-DAT-003 | **Ma'lumot eksporti.** Egasi barcha ma'lumotini (reyestr, incident, audit) standart formatda (JSON/CSV) yuklab oladi | S | F2 | Eksport sinovi |
| NFR-DAT-004 | **Mijoz ma'lumoti.** Mijozga (masalan, wmg.uz) faqat o'z loyihasi ma'lumoti ko'rsatiladi (FR-AUTH-005); serverning umumiy tuzilmasi yashirin | C | F3 | Cheklangan viewer testi |

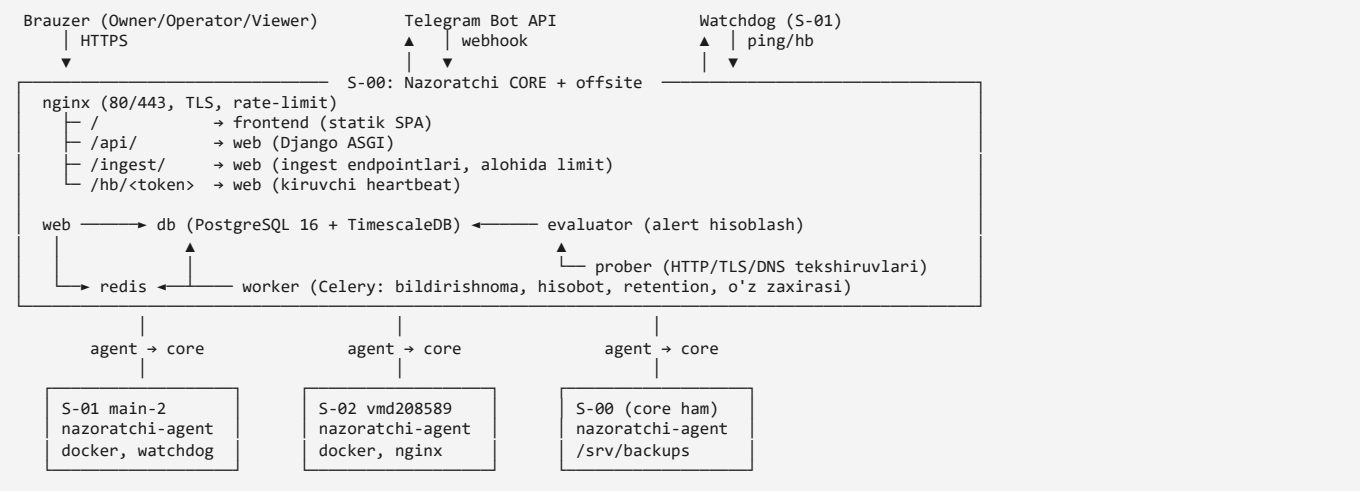
## 5.11. Nofunksional talablar xulosasi

| Toifa | Soni |
|---|---|
| PRF | 10 |
| CAP | 5 |

| Toifa | Soni |
|---|---|
| REL | 10 |
| SEC | 12 |
| USA | 7 |
| MNT | 9 |
| POR | 5 |
| OBS | 3 |
| DAT | 4 |
| **Jami NFR** | **65** |

# 6. Tizim arxitekturasi

## 6.1. Umumiy ko'rinish




Asosiy g'oyalar:

1. **Agent itaruvchi (push) modelda ishlaydi.** Kuzatiladigan serverlarda hech qanday yangi kiruvchi port ochilmaydi; bu xavfsizlikni oshiradi va NAT/firewall orqasidagi serverlar bilan ham ishlaydi.
2. **Core kuzatiladigan serverlardan alohida** joyda turadi. Aks holda bitta server o'lganda monitoring ham o'ladi va hech kim bilmaydi (eski serverdagi hodisaning saboqi).
3. **Faol tekshiruvlarni (HTTP/TLS/DNS) core bajaradi** — mijoz nuqtai nazaridan, tashqaridan. Agent esa ichkaridan ko'radi. Ikkalasi ham kerak: sayt "tashqaridan ishlamayapti, ichkaridan hammasi yaxshi" holati (masalan, nginx yoki DNS muammosi) faqat shunday topiladi.
4. **Core ham kuzatiladi**: boshqa serverdagi (S-01) watchdog uni tekshiradi (FR-SYS-002); S-01 ni esa core kuzatadi.

## 6.2. Komponentlar

| Komponent | Vazifa | Texnologiya | Ma'lumot saqlaydimi | Masshtab |
|---|---|---|---|---|
| **agent** (`nazoratchi-agent`) | Metrikalarni yig'ish va yuborish, mahalliy bufer | Go 1.23, `gopsutil`, systemd | Faqat bufer (diskda, 20 MB) | Har host 1 nusxa |
| **nginx** | TLS, marshrutlash, statik fayllar, tezlik cheklovi | nginx | Yo'q | 1 |
| **web** | REST API, ingest, autentifikatsiya, SSE, Telegram webhook | Python 3.13, Django 5, DRF, uvicorn | Yo'q (holatsiz) | 1–N |
| **evaluator** | Alert qoidalarini davriy hisoblash, incident holati mashinasi | Python (asyncio), Django ORM | Bazada holat | 1 (leader) |
| **prober** | HTTP/TCP/TLS/DNS/WHOIS faol tekshiruvlari | Python (asyncio, httpx, dnspython) | Bazaga natija yozadi | 1–2 |
| **worker** | Bildirishnoma yuborish, hisobotlar, retention, o'z zaxirasi | Celery 5 + Redis | Yo'q | 1–N |
| **beat** | Davriy vazifalar jadvali | Celery beat | Yo'q | 1 |
| **db** | Barcha ma'lumot | PostgreSQL 16 + TimescaleDB 2.x | Ha | 1 |
| **redis** | Celery navbati, kesh, rate-limit, SSE pub/sub | Redis 7 | Vaqtinchalik | 1 |
| **frontend** | Veb-panel | React 18 + TypeScript + Vite, build → statik | Yo'q | CDN-siz, nginx |
| **watchdog** | Nazoratchi tirikligini tashqaridan tekshirish | Kichik Python skript + systemd, **S-01 da** (Owner qarori; ADR-014) | Faqat oxirgi ping vaqti | 1 |
| **docker-socket-proxy** | Agentga Docker ma'lumotini faqat o'qish uchun berish | `tecnativa/docker-socket-proxy` | Yo'q | Har Docker hostda |

## 6.3. Texnologiya tanlovi — arxitektura qarorlari (ADR)

Har qaror: **Kontekst** → **Qaror** → **Sabab** → **Oqibatlar** → **Muqobillar**.

### ADR-001. Markaziy server: Django + DRF

- **Kontekst.** Buyurtmachi Django (Nazar, BiznesRadar, wmg) bilan tajribali; kod egasi bitta odam.
- **Qaror.** Django 5 + DRF, ASGI (uvicorn) da.
- **Sabab.** Tanish stek — tez ishlab chiqish va ekspluatatsiya; Django admin zaxira boshqaruv oynasi sifatida foydali; ORM, migratsiya, autentifikatsiya tayyor. Ingest yuklamasi past (30 host / 15 s ≈ 2 so'rov/soniya) — Django uchun muammo emas.
- **Oqibat.** Ingest yozuvlari `bulk_create`/`COPY` bilan paketlanadi (ORM orqali bittalab emas).
- **Muqobillar.** FastAPI (tezroq, lekin admin/autentifikatsiyani qo'lda yozish kerak); Go backend (eng yengil, lekin buyurtmachi tajribasi kam).

### ADR-002. Vaqt qatorlari bazasi: PostgreSQL + TimescaleDB

- **Qaror.** Alohida TSDB (Prometheus/VictoriaMetrics/InfluxDB) o'rniga PostgreSQL ustida TimescaleDB kengaytmasi.
- **Sabab.** Bitta baza: reyestr, incidentlar, audit va vaqt qatorlari birga (JOIN, tranzaksiya, bitta zaxira). Timescale gipertablitsalari, siqish, doimiy agregatlar (continuous aggregates) va saqlash siyosatlari tayyor — o'z downsampling kodimizni yozmaymiz.
- **Oqibat.** Docker image `timescale/timescaledb:latest-pg16`. Zaxira va tiklash `pg_dump` bilan (bizga tanish). TimescaleDB litsenziyasi (TSL) o'z-hosting uchun bepul.
- **Muqobil (agar Timescale tanlanmasa).** Oddiy PostgreSQL deklarativ partitsiyalash (kunlik) + `pg_cron`/Celery bilan agregat. Kod ikkala holatda ishlashi uchun repozitoriy (data access) qatlami orqali yoziladi (NFR-MNT-008).
- **Xavf.** Timescale tajribasi yo'q; R-102.

### ADR-003. Agent tili: Go

- **Qaror.** Agent — Go da, statik binar (`CGO_ENABLED=0`), amd64 va arm64.
- **Sabab.** Bitta fayl, interpreter va bog'liqliklarsiz, ~10 MB, ~15–25 MB RAM; Python agent har serverda virtual muhit va versiya muammolarini keltiradi (eski serverda Python 3.13, tizim Python 3.12 — bunday nomuvofiqliklar ma'lum). Cross-compile oson.
- **Oqibat.** Agent kodi alohida modul/repo; hajmi taxminan 3–5 ming qator. Buyurtmachi Go bilan tanish bo'lmasa, F0 da prototip Python (`psutil`) bilan yoziladi va protokol tasdiqlangach Go ga o'tkaziladi (17-bo'lim).
- **Muqobillar.** Python + psutil (oson, lekin og'irroq va bog'liqlik), Rust (eng yengil, lekin o'rganish qiyin).

### ADR-004. Agent → core: HTTPS push, JSON+gzip

- **Qaror.** Agent HTTPS POST yuboradi (`/ingest/v1/metrics`), JSON, `Content-Encoding: gzip`.
- **Sabab.** Eng sodda, firewall/proksi bilan muammosiz, curl bilan nosozlikni topish oson. Hajm kichik (≈ 2–4 KB siqilgan paket), shuning uchun protobuf/gRPC qulayligi katta emas.
- **Oqibat.** Protokol versiyalanadi (`/v1/`). Kelajakda yuklama oshsa protobuf qo'shish mumkin.

### ADR-005. Faol tekshiruvlar core ichida (prober)

- **Qaror.** HTTP/TLS/DNS tekshiruvlari core serverdagi alohida `prober` jarayoni bilan.
- **Sabab.** Tashqi ko'rinish; agent o'chsa ham sayt tekshiriladi.
- **Oqibat.** Core serverning chiquvchi tarmog'i barqaror bo'lishi kerak; F3 da ikkinchi tekshiruvchi tugun (FR-HTTP-008).

### ADR-006. Frontend: React + TypeScript + Vite, statik build

- **Qaror.** SPA; build `dist/` ni nginx beradi (SSR yo'q).
- **Sabab.** Panel autentifikatsiyalangan ichki ilova — SEO kerak emas; statik fayllar oddiy va tez; Next.js ga ehtiyoj yo'q. Buyurtmachi loyihalarida React/TS tajribasi bor.
- **Kutubxonalar.** TanStack Query (ma'lumot keshi), React Router, Tailwind CSS, `uPlot` (juda tez vaqt qatori grafiklari; 100+ nuqtali ko'p grafik uchun ECharts'dan yengil), `i18next` (uz/ru/en), `zod` (API javob validatsiyasi).

### ADR-007. Real vaqt: SSE (Server-Sent Events)

- **Qaror.** Dashboard yangilanishi uchun WebSocket emas, SSE.
- **Sabab.** Faqat serverdan klientga yo'nalish kerak; SSE oddiy va avtomatik qayta ulanadi; Redis pub/sub bilan oson.
- **Muqobil.** Har 5 s polling (zaxira rejimi; SSE ishlamasa).

### ADR-008. Navbat: Celery + Redis

- **Qaror.** Bildirishnoma, hisobot, retention uchun Celery.

- **Sabab.** Buyurtmachi Nazar loyihasida Celery + Redis + beat ni allaqachon ishlatadi; bir xil tajriba va monitoring.
- **Muhim istisno.** Alert hisoblash (evaluator) va faol tekshiruvlar (prober) Celery'da emas, alohida uzoq yashaydigan jarayonlar: Celery beat'ning soniyalik aniqligi va leader boshqaruvi yetarli emas.

### ADR-009. Deploy: Docker Compose, bitta server

- **Qaror.** Core Docker Compose bilan, bitta VPS da.
- **Sabab.** Miqyos kichik; Kubernetes ortiqcha. Compose buyurtmachi `appctl` standartiga mos: Nazoratchi o'zi ham `/opt/applications/projects/Nazoratchi/` da, `.appctl` bilan boshqariladi.
- **Oqibat.** Core — SPOF; shuning uchun RTO ≤ 1 soat (zaxira + qayta ko'tarish) va tashqi watchdog.

### ADR-010. Autentifikatsiya: Django sessiyalari (brauzer) + tokenlar (agent/API)

- **Qaror.** Brauzer uchun cookie-sessiya + CSRF; agent/API uchun `Authorization: Bearer`.
- **Sabab.** JWT ning murakkabligi ortiqcha; sessiyani server tomonida bekor qilish oson.

### ADR-011. Alert holati: bazada (xotirada emas)

- **Qaror.** Alert/incident holati PostgreSQL da saqlanadi; evaluator qayta ishga tushsa holatni bazadan tiklaydi.
- **Sabab.** Qayta ishga tushganda "tiklandi/yangi alert" xabarlari takrorlanmasligi kerak.

### ADR-012. Reyestr — yagona manba

- **Qaror.** Kuzatuv sozlamasi (qaysi domen tekshiriladi, qaysi konteyner qaysi loyihaga tegishli) reyestrdan hosil bo'ladi, agent sozlamasidan emas.
- **Sabab.** Bitta joyda o'zgartirish; ikki manba nomuvofiqlikka olib keladi.

### ADR-013. Ma'lumot modeli: "tor" (long) metrika jadvali

- **Qaror.** `metric_samples(time, series_id, value)` — bitta uzun jadval; `series` jadvali (metrika nomi + belgilar/labels) alohida (Prometheus modeliga o'xshash).
- **Sabab.** Yangi metrika qo'shish migratsiyasiz (NFR-CAP-004); siqish samarali.
- **Oqibat.** Qatorlar kardinalligi (cardinality) nazorat qilinadi: har host uchun `series` soni ≤ 1 000 (agent chegarasi), aks holda ogohlantirish.

### ADR-014. Watchdog: mustaqil va minimal

- **Qaror (Owner, 2026-10-26).** Watchdog — Nazoratchi koddan mustaqil kichik xizmat (≤ 100 qator), alohida jarayon va alohida Telegram bot bilan; **S-01 (main-2) da**, core (S-00) dan boshqa serverda.
- **Sabab.** Nazoratchi o'chsa, uning o'z xabarlari ham ketmaydi (kod, bot va server mustaqilligi shundan).
- **O'zaro kuzatuv.** Core S-01 ni agent heartbeat orqali kuzatadi; S-01 dagi watchdog core'dan 60 soniyada bir ping kutadi (va ixtiyoriy ravishda `/healthz` ni tekshiradi). Bitta server o'lsa, ikkinchisi xabar beradi.
- **Qoldiq xavf.** S-00 va S-01 bir provayderda (Contabo). Ikkalasi bir vaqtda tushsa, hech kim xabar bermaydi (RS-06). Oxirgi yumshatish — o'z Uptime Kuma yoki tashqi bepul xizmat (majburiy emas).

## 6.4. Backend modullari (Django ilovalari)

```
nazoratchi/
  apps/
    accounts/       # foydalanuvchi, rol, sessiya, 2FA, tokenlar
    registry/       # Server, Project, Domain, Service (reyestr), teglar
    ingest/         # /ingest/v1/* endpointlari, validatsiya, yozish
    metrics/        # Series, MetricSample, agregatlar, so'rov API
    inventory/      # host ma'lumoti (OS, kernel, jihoz), o'zgarish voqealari
    probes/         # HTTP/TCP/TLS/DNS/WHOIS tekshiruvlari, natijalar, uptime
    backups/        # zaxira holati, restore drill, offsite solishtirish
    alerts/         # qoida, holat, incident, silence, maintenance, bostirish
    notify/         # kanallar (telegram/email/webhook), shablonlar, yuborish
    reports/        # kunlik/haftalik hisobotlar, uptime hisoblash
    audit/          # audit jurnali
    realtime/       # SSE, Redis pub/sub
    core/           # umumiy: sozlamalar, vaqt, kriptografiya, yordamchilar
  workers/
    evaluator.py    # alert hisoblash jarayoni (asyncio)
    prober.py       # faol tekshiruvlar jarayoni (asyncio)
  config/           # Django sozlamalari (base/dev/prod)
```

Bog'liqlik qoidasi (bir yo'nalishli): `core ← registry ← (metrics, probes, backups, inventory) ← alerts ← notify`. `alerts` boshqa modullarni **o'qiydi**, ular esa `alerts` ni import qilmaydi (signal/hodisa orqali).

## 6.5. Asosiy oqimlar

### 6.5.1. Metrikalarni qabul qilish (ingest)

1. Agent paket tuzadi: `{agent_id, seq, sent_at, samples:[{name, labels, t, v}], facts:{...}}`.
2. `POST /ingest/v1/metrics + Authorization: Bearer <agent_token>`.
3. nginx tezlik cheklovini qo'llaydi (NFR-SEC-007), so'rovni `web` ga uzatadi.
4. `web` tokenni tekshiradi (xesh bo'yicha), host faolligini (arxiv emasligini) tekshiradi.
5. Paket tekshiriladi (sxema, `sent_at` majburiy; namuna vaqti tuzatilgandan keyin `now − 1 soat ≤ t ≤ now + 1 daqiqa`, 6.10).
6. `series` kesh (Redis/xotira) orqali `series_id` lar aniqlanadi; yo'q bo'lsa yaratiladi (kardinallik chegarasi bilan).
7. Namunalar `COPY/bulk_create` bilan `metric_samples` ga yoziladi; oxirgi qiymatlar `series_latest` jadvalida yangilanadi (UPSERT).
8. `hosts.last_seen = now`; Redis'ga `host:<id>:updated` hodisasi (SSE uchun).
9. Javob: `200 {"ok":true,"server_time":"...","next_interval":15,"commands":[]}`. Qabul qilinmagan (takroriy) paketga ham `200` (idempotent, agent navbatdan o'chiradi).

### 6.5.2. Alert hisoblash

1. `evaluator` har 15 soniyada (`tick`) yoqilgan qoidalarni yuklaydi (60 s kesh).
2. Har qoida uchun qamrovdagi obyektlar bo'yicha shart hisoblanadi: `series_latest`, oxirgi N daqiqa xom ma'lumoti yoki `probe_results` dan.
3. Natijada har (qoida, obyekt) juftligi uchun holat: `ok` → `pending` (shart bajarildi, `for` kutilmoqda) → `firing` → `resolved`.
4. `firing` ga o'tganda: bostirish (6.7.3) va silence/maintenance tekshiriladi; o'tgan bo'lsa incident yaratiladi/yangilanadi va `notify` ga topshiriq (Celery) yuboriladi.
5. Har tick oxirida `evaluator_heartbeat` bazaga yoziladi (FR-SYS-001).

### 6.5.3. Faol tekshiruv

1. `prober` har tekshiruv uchun navbatdagi vaqtni (`next_run_at`) bazadan oladi.
2. Asinxron ravishda (`httpx.AsyncClient`, `dnspython`, `ssl`) bajaradi, vaqtlarni o'lchaydi.
3. Natija `probe_results` ga yoziladi; holat o'zgargan bo'lsa (3 ketma-ket qoida) — `check_state` yangilanadi va `evaluator` ga hodisa (Redis) yuboriladi.
4. Xato turi (timeout, dns, tls, 5xx, kalit so'z yo'q) saqlanadi.

### 6.5.4. Bildirishnoma yuborish

1. `evaluator` Celery vazifasi `notify.send_incident(incident_id, event)` ni navbatga qo'yadi.
2. `worker` kanal sozlamalarini oladi, sokin soatlarni (FR-NTF-006) tekshiradi, xabarni shablondan hosil qiladi.
3. Telegram'ga yuboradi; muvaffaqiyatsiz bo'lsa eksponensial kutish (5 s, 20 s, 1 daq, 5 daq, 15 daq); hammasi yo'qolsa, ikkinchi kanal (email).
4. Yetkazilish natijasi `notifications` jadvaliga yoziladi (kim, qachon, qaysi kanal, holat).

### 6.5.5. Telegram tugmasi (callback)

1. Telegram `POST /api/v1/telegram/webhook/<secret>` ga yuboradi.
2. `web` `chat_id` ni ruxsat etilganlar ro'yxati bilan solishtiradi, `X-Telegram-Bot-Api-Secret-Token` sarlavhasini tekshiradi.
3. `ack/silence` amali bajariladi, audit jurnaliga yoziladi, xabar yangilanadi (`editMessageText`).

### 6.5.6. Panelga jonli yangilanish (SSE)

1. Brauzer `GET /api/v1/stream?scope=dashboard` ochadi (cookie bilan).
2. `web` Redis pub/sub'ga obuna bo'ladi; `ingest/evaluator` hodisalarini (host holati, incident) 1 soniyalik partiyalar bilan yuboradi.
3. Aloqa uzilsa brauzer avtomatik qayta ulanadi (`Last-Event-ID`); 3 marta muvaffaqiyatsiz bo'lsa, polling (5 s) ga o'tadi.

# 6.6. Joylashtirish (deployment) topologiyasi

## 6.6.1. Serverlar

| Server | Rol | Joylashuv | Tavsiya etilgan xususiyat |
|---|---|---|---|
| **S-00** | Nazoratchi CORE + offsite zaxira (S-01 uchun) | Contabo (`185.214.135.9`), kuzatiladigan S-01 dan **boshqa server**, lekin bir provayderda (R-110) | 200 GB SSD, Ubuntu 24.04 (qayta o'rnatiladi), kuniga snapshot |
| S-01, S-02 | Agent o'rnatiladi (S-00 da ham agent bor); S-01 da watchdog ham | Mavjud | — |
| Watchdog | Nazoratchi tirikligini tekshirish | **S-01** (Owner qarori) | Minimal |

**Qaror (Owner, 2026-10-26).** Core S-00 da, S-01 dan alohida: S-01 o'lsa, S-01 uchun xabar beradigan tizim yo'qolmaydi. S-02 aaclub bilan band va uning nginx front-proksi qoidalariga bog'liq (R-009), shuning uchun core uchun ishlatilmaydi. S-00 oldin yuklanmagan (1.3.1), shu sabab qayta o'rnatishdan keyin darhol provayder snapshoti olinadi. S-03 zaxira uchun ishlatilmaydi.

**Vazifalar taqsimoti.** Core va S-01 uchun offsite zaxira S-00 da turadi. Watchdog S-01 da turadi (ADR-014). Nazoratchi'ning **o'z zaxirasi** (FR-SYS-003) S-00 dan tashqarida — Owner'ning shaxsiy Telegram chatida, `age` bilan shifrlangan holda saqlanadi.

## 6.6.2. Docker Compose servislari (core)

| Servis | Image | Portlar (127.0.0.1) | Hajm (volume) |
|---|---|---|---|
| `db` | `timescale/timescaledb:latest-pg16` | 5432 (faqat ichki tarmoq) | `nazoratchi_pgdata` |
| `redis` | `redis:7-alpine` | — | `nazoratchi_redis` (AOF o'chiq, kesh) |
| `web` | `ghcr.io/<owner>/nazoratchi-web:<ver>` (o'zimiz quramiz) | 8040 | — |
| `evaluator` | shu image, `command: python -m workers.evaluator` | — | — |
| `prober` | shu image, `command: python -m workers.prober` | — | — |
| `worker` | shu image, `celery -A config worker` | — | — |
| `beat` | shu image, `celery -A config beat` | — | `nazoratchi_beat` |
| `frontend` | `nginx:alpine` + statik fayllar | 3040 | — |

Port tanlovi `REGISTRY.md` qoidasiga mos: keyingi bo'sh backend/frontend portlari (8040, 3040 taklif qilinadi; S-01 dagi bandlari bilan to'qnashuvni reyestr tekshiradi, FR-REG-010).

## 6.6.3. Tarmoq va xavfsizlik chegaralari

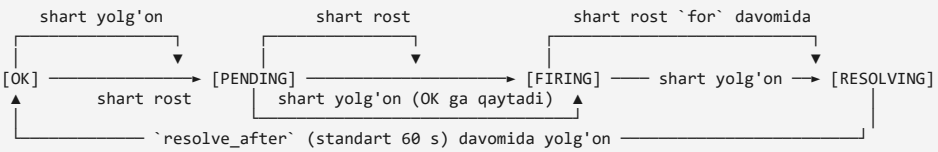
- Tashqariga faqat 80/443 (nginx). `db`, `redis` ichki Docker tarmog'ida; xostga chiqarilmaydi.
- `/ingest/*` — tashqaridan ochiq (agentlar uchun), lekin token + tezlik cheklovi bilan.
- `/api/*`, UI — autentifikatsiya talab qiladi; ixtiyoriy IP allowlist (NFR-SEC-011).
- `/metrics`, `/healthz` — faqat ichki yoki token bilan.
- Agent → core: faqat 443. Core → Telegram: 443 (chiquvchi).
- DNS: `nazoratchi.<domen>` A yozuvi core serverga; sertifikat Let's Encrypt (certbot).

## 6.6.4. Nazoratchi'ning o'zi `appctl` bilan

`/opt/applications/projects/Nazoratchi/` + `.appctl` (`DB_SERVICE=db`, `DB_USER=nazoratchi`, `DB_NAME=nazoratchi`, `HEALTH_URL=http://127.0.0.1:8040/healthz`). `appctl backup all` uni ham zaxiralaydi; xuddi shu mexanizm bilan Nazoratchi **o'z zaxirasini** ham kuzatadi (FR-SYS-003).

# 6.7. Alert dvigateli dizayni

## 6.7.1. Holat mashinasi (har qoida × obyekt)




- `PENDING`: shart rost bo'ldi, lekin `for` hali o'tmadi — xabar yuborilmaydi. Agar shu paytda ma'lumot butunlay yo'qolsa (agent jim), uzluksizlik tasdiqlanmagan hisoblanadi va holat `OK` ga qaytadi; aks holda agent bir soatdan keyin qaytganda eski `PENDING` darrov `FIRING` bo'lib ketardi. `FIRING`/`RESOLVING` ma'lumot yo'qligida o'zgarmaydi.

- `FIRING`: alert faol; incident ochiq.
- `RESOLVING`: shart yolg'on bo'ldi, `resolve_after` kutiladi (titrashdan himoya, FR-ALR-018).
- `OK`: yopildi; "tiklandi" xabari yuboriladi (agar FIRING bo'lgan bo'lsa).

### 6.7.2. Incident va alert o'rtasidagi bog'liqlik

- **Alert instance** — bitta (qoida × obyekt) ning FIRING holati.
- **Incident** — bir yoki bir nechta alert instance dan tashkil topgan, odam uchun mo'ljallangan yagona voqea. Yangi alert, agar xuddi shu **qamrov kaliti** (masalan, `host:S-01`) bo'yicha ochiq incident bo'lsa, o'shanga qo'shiladi; aks holda yangi incident ochiladi.
- Incident'ning ahamiyati = ichidagi alertlarning eng yuqorisi. Incident `started_at` — birinchi alertning shart rost bo'la boshlagan vaqti (`for` kutish davri ham kiradi), shuning uchun "Davomiyligi" haqiqiy muammo vaqtini ko'rsatadi.
- Incident yopiladi: barcha alert instance lari `OK` bo'lganda (+ 60 s).

### 6.7.3. Bostirish (suppression) qoidalari

| Sharoit | Natija |
|---|---|
| Host `down` (critical) | Shu hostdagi barcha `docker_*`, `svc_*`, `backup_age_*`, `http_*` (agar loyiha asosiy serveri shu bo'lsa) alertlari bostiriladi va incident "ta'sirlangan" ro'yxatiga qo'shiladi |
| Domen DNS mos emas | O'sha domen uchun `http_down` va `tls_*` alertlari bostiriladi (sabab bitta) |
| Maintenance oynasi | Qamrovdagi hamma alert bostiriladi, yig'iladi |
| Silence | Moslik (matcher) bo'yicha bostiriladi |
| Server `pause` | Hamma alert bostiriladi |

### 6.7.4. Bashorat (forecast)

Disk to'lishi: oxirgi 6 soatlik (`disk_used_bytes`) nuqtalariga eng kichik kvadratlar usuli bilan chiziq o'tkaziladi; `eta = (jami − hozirgi) / qiyalik`. Chiziq $R^2 < 0.5$ bo'lsa (tartibsiz), bashorat ishonchsiz deb belgilanadi va faqat chegara qoidasi ishlaydi. Qo'shimcha: 24 soatlik va 7 kunlik oynalar ham hisoblanadi, **eng yomon (eng yaqin)** natija ishlatiladi.

## 6.8. Prober dizayni

- **Rejalashtirish.** Har tekshiruvning `next_run_at` maydoni bor; jitter (±10%) bilan taqsimlanadi, shunda 450 tekshiruv bir soniyada boshlanmaydi.
- **Parallellik.** `asyncio.Semaphore(50)`; har tekshiruv uchun qattiq timeout.
- **HTTP.** `httpx` (HTTP/2 qo'llab-quvvatlanadi), DNS/TCP/TLS/TTFB vaqtlari `httpx` hodisa ilgaklari yoki alohida `socket`/`ssl` o'lchovlari bilan.
- **TLS.** `ssl.create_default_context()` + `getpeercert()`; tekshiruvsiz rejim muddatni o'qish uchun (hostname mos kelmaydigan holatni ham aniqlash uchun alohida o'tish).
- **DNS.** `dnspython`, resolverlar ro'yxati; har resolver uchun alohida natija (`probe_dns_results`).
- **Tasdiqlash qoidasi.** Muvaffaqiyatsiz natijadan keyin 15 s oraliq bilan 2 marta qayta tekshiriladi; 3 ketma-ket xato = `down`. Bitta muvaffaqiyat = `up`. *Amalga oshirish (S4):* qayta tekshiruv oqimni uxlatmaydi — `next_run_at` 15 s ga suriladi; tasdiqlanmagan xatolarda holat o'zgarmaydi (`unknown` bo'lsa `unknown` qoladi). Parallellik: `asyncio` o'rniga 50 oqimli `ThreadPoolExecutor`; vaqti kelganlar `SELECT ... FOR UPDATE SKIP LOCKED` bilan olinib, `next_run_at` avval suriladi (bir nechta prober takrorlamaydi).
- **TLS (S5).** Sertifikat avval tekshiruvsiz o'qiladi (muddati o'tgan yoki nomi mos kelmasa ham ko'rish uchun), `cryptography` bilan `notAfter`, issuer, subject, SAN olinadi; keyin alohida ulanishda zanjir ishonchliligi (`chain_ok`) va SAN ga mosligi (`hostname_ok`, bitta darajali wildcard bilan) tekshiriladi. Tekshiruv `ok` = sertifikat o'qildi; chegaralarni (21/7/3 kun) alert qoidalari qo'llaydi.
- **DNS (S5).** `dnspython`, har resolverga alohida (8.8.8.8, 1.1.1.1, 9.9.9.9 va `ns:auto` — domenning o'z NS serveri). Resolver ichki manzil bo'lsa rad etiladi. Javob bermagan resolver (timeout) hisobga olinmaydi; hamma jim bo'lsa `dns` xatosi (tarmoq muammosi), hammasi `NXDOMAIN` — `nxdomain`, aks holda kutilgan IP lar bilan nomuvofiqlik — `mismatch`. Har resolver natijasi `dns_results` ga yoziladi, javob o'zgarsa `dns_changes` ga (FR-DNS-008). WHOIS/RDAP — F2.
- **IP-pinning (S4).** Nom bir marta aniqlanadi, tekshiruvdan o'tgan IP ga ulanadi; `Host` sarlavhasi va TLS SNI/sertifikat tekshiruvi asl nomda qoladi (DNS rebinding oldini oladi, `resolve_to` shu mexanizm). Har yo'naltirish qayta aniqlanadi va tekshiriladi. Vaqtlar (`dns/connect/tls/ttfb`) `httpx` trace hodisalaridan olinadi. S4 da faqat HTTP va TCP; TLS/DNS/WHOIS — S5.

- **Ruxsat etilgan chiquvchi manzillar.** Prober ichki/loopback IP larga (SSRF xavfi) so'rov yubormaydi: manzil aniqlangach shaxsiy diapazonlar (`10/8`, `172.16/12`, `192.168/16`, `127/8`, `169.254/16`) rad etiladi (maxsus ruxsat bilan o'zgartiriladi).

## 6.9. Muvaffaqiyatsizlik rejimlari (FMEA)

| Nosozlik | Ta'siri | Aniqlash | Yumshatish |
|---|---|---|---|
| Core serveri (S-00) o'ldi | Monitoring yo'q | S-01 dagi watchdog (5 daq) | Agent buferi 15 daq; zaxiradan tiklash RTO 1 soat |
| `db` to'xtadi | Yozib bo'lmaydi | `/healthz` 503, watchdog | Redis navbati (15 daq), `restart: unless-stopped` |
| `evaluator` qotdi | Yangi alert chiqmaydi | `evaluator_heartbeat` eskirdi → watchdog | Docker healthcheck + avto-restart |
| `prober` qotdi | Faol tekshiruv natijasi yo'q | `probe_last_run` eskirdi | healthcheck, "ma'lumot eskirgan" alerti |
| Telegram API ishlamayapti | Xabar yetmaydi | `notify` xato hisoblagichi | Qayta urinish + email zaxira kanal |
| Agent to'xtadi (host tirik) | Host ma'lumoti yo'q | Heartbeat 120 s | "Agent javob bermayapti" (`high`), TCP:22 tekshiruvi |
| Host o'ldi | Hamma narsa yo'q | Heartbeat + TCP:22 | "Server o'chdi" (`critical`), bostirish |
| Core va agent o'rtasida tarmoq uzildi | Hamma host "jim" | Ko'p hostlar bir vaqtda jim bo'lsa — "core tarmog'i" gumoni | Massaviy holat: > 50% host jim bo'lsa bitta yig'ma alert, TCP:22 tekshiruvi farqlaydi |
| Soat siljishi (core) | Vaqt tamg'alari xato | NTP kuzatuvi (FR-HST-013) | chrony, agent soat farqi hisoboti |
| Disk to'ldi (core) | Yozish to'xtaydi | O'z disk alerti | Retention, hajm chegarasi, 85% da `critical` |
| Zararli/buzuq paket | Baza ifloslanadi | Validatsiya xatolari soni | Sxema tekshiruvi, qiymat diapazoni, token cheklovi |

## 6.10. Vaqt va soat

- Barcha vaqt UTC (`timestamptz`), API da ISO-8601 (`2026-10-22T09:35:00Z`).
- Namuna vaqti — agent soatidan. Soat siljishi paketdagi `sent_at` bilan aniqlanadi: `|core vaqti − sent_at|`

  > 10 daqiqa`bo'lsa, butun paket shu farqqa surilib tuzatiladi va`clock_skew` belgisi qo'yiladi. Aks holda namuna vaqti o'zgartirilmaydi: core o'chib qaytganda 15 daqiqalik bufer (FR-AGT-007) o'z vaqtlari bilan yoziladi (P-02 da aniqlangan: namuna vaqtini "core vaqti bilan almashtirish" buferni buzardi). Tuzatishdan keyin 1 soatdan eski yoki 1 daqiqadan ilgari namuna rad etiladi.

- Agent hodisa tartibi uchun monoton `seq` raqamidan ham foydalanadi.

## 6.11. Kengaytirish nuqtalari

| Kengaytma | Qanday amalga oshadi |
|---|---|
| Yangi metrika | Agentda yangi kollektor yoki `custom` kollektor; core o'zgarmaydi (ADR-013) |
| Yangi tekshiruv turi | `probes` ilovasida yangi `Probe` kichik sinfi + prober ro'yxatdan o'tkazish |
| Yangi bildirishnoma kanali | `notify.channels` da `Channel` interfeysini amalga oshirish (`send`, `validate`) |
| Yangi alert shart turi | `alerts.conditions` da `Condition` interfeysi |
| Yangi integratsiya | Kiruvchi webhook/heartbeat endpointi (FR-INT-005) |
| Ikkinchi core (HA) | F3 dan keyin: bazada replika + ikkinchi core; hozir qamrovda emas (X-6) |

# 7. Agent spetsifikatsiyasi va metrikalar katalogi

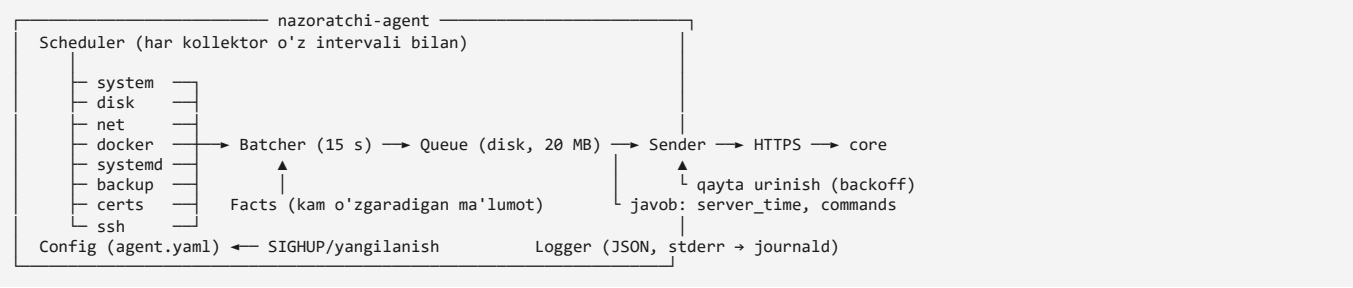
## 7.1. Agentning vazifasi va chegaralari

`nazoratchi-agent` — kuzatiladigan serverda ishlaydigan yagona, statik Go binari. U:

- tizim, disk, tarmoq, Docker, systemd, zaxira, sertifikat va xavfsizlik ma'lumotlarini yig'adi;
- ularni paketlab core'ga yuboradi;
- core'dan kelgan cheklangan buyruqlarni (faqat interval o'zgartirish, qayta yuklash, o'z-o'zini yangilash) bajaradi.

Agent **qilmaydi**: serverda ixtiyoriy buyruq bajarmaydi (masofaviy shell yo'q), fayl mazmunini yubormaydi, hech qanday kiruvchi port ochmaydi, konteynerlarni boshqarmaydi (X-8, X-3).

## 7.2. Ichki tuzilma




| Qism | Tavsifi |
|---|---|
| Scheduler | Har kollektor mustaqil goroutine; `ticker` + jitter (±10%); kollektor `timeout` (interval/2) dan oshsa to'xtatiladi va `agent_collector_errors` oshadi |
| Batcher | Oxirgi 15 soniya ichidagi namunalarni bitta paketga yig'adi; `facts` o'zgarganda yoki 10 daqiqada bir marta qo'shadi |
| Queue | Diskdagi navbat `/var/lib/nazoratchi/queue/` (har paket alohida fayl `seq.json.gz`); maksimum 20 MB yoki 15 daqiqa; to'lsa eng eski o'chadi |
| Sender | HTTPS `POST`, TLS 1.2+, timeout 10 s; xato → eksponensial kutish (1, 2, 4, 8, 16, 30 s, jitter bilan); muvaffaqiyatda navbatni vaqt tartibida bo'shatadi |
| Config | `/etc/nazoratchi/agent.yaml`; xato bo'lsa agent ishga tushmaydi va aniq xabar beradi |
| Logger | JSON, `level`, `component`; token va maxfiy qiymatlar maskalanadi |

## 7.3. Metrika modeli va nomlash

### 7.3.1. Qoidalar

- Nom: `snake_case`, birlik oxirida (`_bytes`, `_percent`, `_seconds`, `_per_sec`), bitta nom bitta ma'no.
- Turi: **gauge** (lahzalik qiymat) yoki **counter** (faqat oshadi, qayta boshlanishi mumkin). Tezlikni (`*_per_sec`) agent o'zi hisoblaydi (ikki o'lchov farqi / vaqt); sanagichlar (counter) faqat "umumiy qayta ishga tushishlar" kabi hollarda xom holda yuboriladi.
- Belgilar (labels): kardinallik kichik bo'lishi kerak. Ruxsat: `device`, `mount`, `iface`, `container`, `project`, `service`, `unit`, `core`, `mode`, `name`, `kind`. **Taqiq:** URL, foydalanuvchi ID, IP manzillar (har safar yangi qiymat chiqadigan belgilar).
- Har host uchun noyob qatorlar (series) soni ≤ 1 000 (agent o'zi cheklaydi; oshsa `agent_series_dropped` metrikasi yuboriladi).
- Vaqt tamg'asi: namuna olingan paytdagi UTC, millisekund aniqligida.

### 7.3.2. Standart intervallar

| Kollektor | Interval | Sozlanadi |
|---|---|---|
| `system` | 15 s | ha (5–60 s) |
| `disk` (fayl tizimi) | 30 s | ha |
| `diskio` | 15 s | ha |
| `net` | 15 s | ha |
| `docker` | 30 s | ha |
| `systemd` | 30 s | ha |

| Kollektor | Interval | Sozlanadi |
|---|---|---|
| `backup` | 300 s | ha |
| `certs` | 21 600 s (6 soat) | ha |
| `ssh` | 300 s | ha |
| `facts` | 600 s | ha |
| `apt` | 86 400 s (kuniga 1) | ha |

## 7.4. Metrikalar katalogi

Quyidagi jadvallarda har metrika uchun: **Nom**, **Tur**, **Birlik**, **Belgilar**, **Manba**, **Izoh**. Ustuvorlik belgisi (F1/F2/F3) — qaysi bosqichda amalga oshirilishi.

### 7.4.1. `system` kollektori

| Nom | Tur | Birlik | Belgilar | Manba | Izoh | Faza |
|---|---|---|---|---|---|---|
| `cpu_usage_percent` | gauge | % | `mode` = user, nice, system, idle, iowait, irq, softirq, steal | `/proc/stat` (ikki o'lchov farqi) | Hamma yadro bo'yicha o'rtacha | F1 |
| `cpu_busy_percent` | gauge | % | — | `/proc/stat` | `100 − idle − iowait` | F1 |
| `cpu_core_busy_percent` | gauge | % | `core` | `/proc/stat` | Har yadro (kardinallik = yadro soni) | F2 |
| `load1`, `load5`, `load15` | gauge | — | — | `/proc/loadavg` | | F1 |
| `load_per_core_1m` | gauge | — | — | `load1 / cpu_count` | 1.0 dan oshsa — yuklama yadro soniga teng | F1 |
| `mem_total_bytes` | gauge | bayt | — | `/proc/meminfo` `MemTotal` | | F1 |
| `mem_available_bytes` | gauge | bayt | — | `MemAvailable` | Haqiqiy mavjud xotira | F1 |
| `mem_used_bytes` | gauge | bayt | — | `MemTotal − MemAvailable` | | F1 |
| `mem_used_percent` | gauge | % | — | hisoblangan | Alert shu asosida | F1 |
| `mem_free_bytes` | gauge | bayt | — | `MemFree` | | F1 |
| `mem_cached_bytes`, `mem_buffers_bytes` | gauge | bayt | — | `Cached`, `Buffers` | | F2 |
| `swap_total_bytes`, `swap_used_bytes` | gauge | bayt | — | `SwapTotal`, `SwapTotal−SwapFree` | | F1 |
| `swap_used_percent` | gauge | % | — | hisoblangan | | F1 |
| `swap_in_pages_per_sec`, `swap_out_pages_per_sec` | gauge | sahifa/s | — | `/proc/vmstat` (`pswpin`, `pswpout`) | Faol swap = xotira bosimi | F2 |
| `oom_kill_total` | counter | son | — | `/proc/vmstat` (`oom_kill`) | Yadro xotira yetishmasligidan jarayon o'ldirganlari soni (host darajasi) | F2 |
| `procs_total`, `procs_running`, `procs_blocked` | gauge | son | — | `/proc/stat`, `/proc/loadavg` | | F2 |
| `procs_zombie` | gauge | son | — | `/proc/[pid]/stat` | | F2 |
| `threads_total` | gauge | son | — | `/proc/loadavg` | | F2 |
| `fd_allocated`, `fd_max` | gauge | son | — | `/proc/sys/fs/file-nr` | | F2 |
| `context_switches_per_sec` | gauge | 1/s | — | `/proc/stat` `ctxt` | | F3 |
| `uptime_seconds` | gauge | s | — | `/proc/uptime` | | F1 |

| Nom | Tur | Birlik | Belgilar | Manba | Izoh | Faza |
|---|---|---|---|---|---|---|
| `boot_time_seconds` | gauge | unix s | — | `/proc/stat` `btime` | O'zgarsa — reboot voqeasi | F1 |
| `psi_cpu_some_avg10`, `psi_memory_some_avg10`, `psi_io_some_avg10` | gauge | % | — | `/proc/pressure/*` | Resurs bosimi (agar yadro qo'llasa) | F3 |
| `clock_offset_seconds` | gauge | s | — | `chronyc tracking` yoki `timedatectl timesync-status` | Mavjud bo'lsa | F2 |
| `ntp_synchronized` | gauge | 0/1 | — | `timedatectl show -p NTPSynchronized` | | F2 |
| `top_cpu_process_percent`, `top_mem_process_bytes` | gauge | %/byt | `name`, `rank` (1..5) | `/proc/[pid]/stat` | Faqat bajariladigan fayl nomi (argumentsiz), NFR-DAT-001 | F3 |

### 7.4.2. `disk` va `diskio` kollektorlari

| Nom | Tur | Birlik | Belgilar | Manba | Izoh | Faza |
|---|---|---|---|---|---|---|
| `fs_total_bytes` | gauge | bayt | `mount`, `device`, `fstype` | `statfs()` | `tmpfs`, `overlay`, `squashfs`, `devtmpfs`, `proc`, `sysfs`, `cgroup*` e'tiborsiz | F1 |
| `fs_used_bytes` | gauge | bayt | xuddi shunday | `statfs()` | | F1 |
| `fs_avail_bytes` | gauge | bayt | xuddi shunday | `statfs()` (`f_bavail`) | Oddiy foydalanuvchi uchun bo'sh | F1 |
| `fs_used_percent` | gauge | % | xuddi shunday | hisoblangan | `used / (used + avail)` (`df` ga mos) | F1 |
| `fs_inodes_total`, `fs_inodes_used` | gauge | son | xuddi shunday | `statfs()` | | F2 |
| `fs_inodes_used_percent` | gauge | % | xuddi shunday | hisoblangan | | F2 |
| `fs_readonly` | gauge | 0/1 | `mount`, `device` | `/proc/mounts` opsiyalari (`ro`) | **Muhim:** fayl tizimi xato tufayli faqat o'qish rejimiga o'tganini aniqlaydi | F1 |
| `disk_read_bytes_per_sec`, `disk_write_bytes_per_sec` | gauge | bayt/s | `device` | `/proc/diskstats` | Faqat jismoniy/virtual blok qurilmalar (`sd*`, `vd*`, `nvme*`); `loop*`, `ram*`, `dm-*` hisoblanmaydi | F2 |
| `disk_read_iops`, `disk_write_iops` | gauge | 1/s | `device` | `/proc/diskstats` | | F2 |
| `disk_io_util_percent` | gauge | % | `device` | `/proc/diskstats` (io_ticks) | | F2 |
| `disk_await_ms` | gauge | ms | `device` | `/proc/diskstats` | O'rtacha so'rov kutish vaqti | F2 |

### 7.4.3. `net` kollektori

| Nom | Tur | Birlik | Belgilar | Manba | Izoh | Faza |
|---|---|---|---|---|---|---|
| `net_rx_bytes_per_sec`, `net_tx_bytes_per_sec` | gauge | bayt/s | `iface` | `/proc/net/dev` | Standart: `lo`, `veth*`, `br-*`, `docker0` yashirin | F1 |
| `net_rx_packets_per_sec`, `net_tx_packets_per_sec` | gauge | 1/s | `iface` | `/proc/net/dev` | | F2 |
| `net_rx_errors_per_sec`, `net_tx_errors_per_sec` | gauge | 1/s | `iface` | `/proc/net/dev` | | F2 |
| `net_rx_drops_per_sec`, `net_tx_drops_per_sec` | gauge | 1/s | `iface` | `/proc/net/dev` | | F2 |
| `tcp_established` | gauge | son | — | `/proc/net/snmp` (`CurrEstab`) | | F2 |
| `tcp_time_wait` | gauge | son | — | `/proc/net/sockstat` (`tw`) | | F3 |
| `sockets_used` | gauge | son | — | `/proc/net/sockstat` | | F3 |
| `net_listen_ports` | gauge | son | — | `/proc/net/tcp*` | Tashqariga (`0.0.0.0`/`::`) ochiq tinglovchi portlar soni; ro'yxat `facts` da (kutilmagan ochiq port aniqlash, FR-HTTP-007) | F2 |

### 7.4.4. `docker` kollektori

Docker ma'lumoti `docker-socket-proxy` orqali (faqat GET) olinadi. Konteyner nomi `/` siz. Kardinallik chegarasi: bitta hostda ≤ 100 konteyner kuzatiladi (oshsa — `docker_series_dropped`).

> **S5 da amalga oshirilgan (F1):** `docker_container_running`, `_restarts_total`, `_uptime_seconds`, `_health`, `_oom_killed`, `docker_containers_running/total` va `docker_container_expected` (quyida). Manzil: `tcp://127.0.0.1:2375` (proksi) yoki `unix:///var/run/docker.sock`. Konteyner holati `/containers/json?all=1`, qolgani har konteyner uchun `/containers/{id}/json` dan (≤ 100 so'rov/30 s). `health` faqat ishlayotgan konteynerda hisoblanadi (Docker to'xtagan konteynerda eski holatni saqlaydi). Resurs sarfi (`cpu_percent`, `mem_*`), `system/df` va log hajmi — F2. `docker_container_expected` **(0/1):** konteyner kamida 1 soat uzluksiz ishlagan va `RestartPolicy` `no` dan farqli bo'lsa 1 (11.5.1). Bu holat agent **xotirasida** saqlanadi (agent qayta ishga tushsa, qaytadan 1 soat kerak). Kutilgan konteyner ro'yxatdan yo'qolsa 10 daqiqagacha `running=0` yuboriladi, keyin ataylab olib tashlangan deb unutiladi.

| Nom | Tur | Birlik | Belgilar | Manba | Izoh | Faza |
|---|---|---|---|---|---|---|
| `docker_container_running` | gauge | 0/1 | `container`, `project`, `service` | `GET /containers/json?all=1` | `project`/`service` — `com.docker.compose.*` labellaridan | F1 |
| `docker_container_health` | gauge | kod | `container`, `project`, `service` | `State.Health.Status` | 0 — healthcheck yo'q, 1 — healthy, 2 — unhealthy, 3 — starting | F2 |
| `docker_container_restarts_total` | counter | son | `container`, `project`, `service` | `RestartCount` | | F1 |
| `docker_container_uptime_seconds` | gauge | s | `container`, … | `State.StartedAt` | | F1 |
| `docker_container_cpu_percent` | gauge | % | `container`, … | `GET /containers/{id}/stats?stream=false` | Bir yadro = 100% | F2 |
| `docker_container_mem_bytes` | gauge | bayt | `container`, … | stats (`memory_stats.usage − cache`) | | F2 |
| `docker_container_mem_limit_bytes` | gauge | bayt | `container`, … | stats (`memory_stats.limit`) | Limit xost RAM ga teng bo'lsa — "cheksiz" deb belgilanadi | F2 |
| `docker_container_mem_percent` | gauge | % | `container`, … | hisoblangan | | F2 |
| `docker_container_net_rx_bytes_per_sec`, `..._tx_...` | gauge | bayt/s | `container`, … | stats | | F3 |
| `docker_container_blk_read_bytes_per_sec`, `..._write_...` | gauge | bayt/s | `container`, … | stats | | F3 |
| `docker_container_oom_killed` | gauge | 0/1 | `container`, … | `State.OOMKilled` | | F2 |
| `docker_container_log_bytes` | gauge | bayt | `container`, … | log fayli hajmi (agent o'qiy olsa) yoki `LogPath` `stat` | `/var/lib/docker/containers` ga o'qish huquqi kerak (7.8 ga qarang) | F2 |
| `docker_containers_running`, `docker_containers_total` | gauge | son | — | `GET /info` | | F1 |
| `docker_images_bytes`, `docker_containers_bytes`, `docker_volumes_bytes`, `docker_build_cache_bytes` | gauge | bayt | — | `GET /system/df` | Kuniga ko'p marta emas, 10 daqiqada bir | F2 |
| `docker_reclaimable_bytes` | gauge | bayt | — | `GET /system/df` | Tozalab qaytariladigan | F3 |
| `docker_log_limit_configured` | gauge | 0/1 | — | `GET /info` (`LoggingDriver`) + `/etc/docker/daemon.json` (o'qiy olsa) | | F3 |

### 7.4.5. `systemd` kollektori

| Nom | Tur | Birlik | Belgilar | Manba | Izoh | Faza |
|---|---|---|---|---|---|---|
| `systemd_unit_active` | gauge | 0/1 | `unit` | D-Bus (`ListUnits`) yoki `systemctl show` | Faqat sozlamadagi unit lar | F1 |

| Nom | Tur | Birlik | Belgilar | Manba | Izoh | Faza |
|---|---|---|---|---|---|---|
| `systemd_unit_failed` | gauge | 0/1 | `unit` | `ActiveState == failed` | | F1 |
| `systemd_unit_restarts_total` | counter | son | `unit` | `NRestarts` | | F2 |
| `systemd_unit_uptime_seconds` | gauge | s | `unit` | `ActiveEnterTimestamp` | | F2 |
| `systemd_failed_units_total` | gauge | son | — | `systemctl --failed` | | F2 |
| `nginx_config_ok` | gauge | 0/1 | — | `nginx -t` (kuniga + o'zgarishda; `sudo` kerak: 7.8) | | F3 |
| `cron_job_last_success_timestamp` | gauge | unix s | `job` | Heartbeat'lar `/var/lib/nazoratchi/heartbeats/<job>` | Cron skript oxirida `nazoratchi-agent beat <job>` chaqiradi (FR-SVC-004) | F1 |

> **S6 da amalga oshirilgan (agent):** `systemd_unit_active`, `systemd_unit_failed` (`systemctl show`, shell yo'q; unit nomi `^[A-Za-z0-9:_.@\-]{1,100}$` ga mos, aks holda sozlama xato), `cron_job_last_success_timestamp{job}` (`heartbeats_dir` fayllari `mtime`; cron skript oxirida `nazoratchi-agent beat <job>`). Yo'q unit (`LoadState=not-found`) — `active=0, failed=0`. F2 metrikalari (`restarts_total`, `uptime`, `failed_units_total`) va `nginx_config_ok` hali yo'q.

### 7.4.6. `backup` kollektori

Manba — `appctl` yozgan `state.json` (FR-INT-003) **va** papka skaneri (qo'shimcha tekshiruv). Agent `BACKUP_ROOT` (standart `/opt/applications/backups`) ni ko'radi; ichidagi har papka — loyiha.

| Nom | Tur | Birlik | Belgilar | Manba | Izoh | Faza |
|---|---|---|---|---|---|---|
| `backup_last_status` | gauge | 0/1 | `project` | `state.json` (`status`) | 1 — ok, 0 — fail | F1 |
| `backup_last_success_timestamp` | gauge | unix s | `project` | `state.json` (`finished_at` agar ok) | | F1 |
| `backup_last_size_bytes` | gauge | bayt | `project` | `state.json` yoki eng yangi fayl hajmi | | F1 |
| `backup_last_duration_seconds` | gauge | s | `project` | `state.json` | | F2 |
| `backup_newest_file_age_seconds` | gauge | s | `project`, `kind` = daily, weekly, files | papka skaneri (`mtime`) | `.tmp` fayllar hisobga olinmaydi | F1 |
| `backup_files_count` | gauge | son | `project`, `kind` | papka skaneri | | F1 |
| `backup_dir_bytes` | gauge | bayt | `project` | papka skaneri (`du`) | | F2 |
| `backup_gz_ok` | gauge | 0/1 | `project` | `gzip -t` (eng yangi `.sql.gz`) | Kuniga 1 marta | F2 |
| `backup_offsite_configured` | gauge | 0/1 | — | `/opt/applications/appctl.conf` (`OFFSITE` bo'sh emasmi) | FR-BKP-010 | F1 |
| `backup_offsite_last_status` | gauge | 0/1 | — | `state.json` (`offsite.status`) | | F2 |
| `backup_offsite_last_success_timestamp` | gauge | unix s | — | `state.json` (`offsite.finished_at`) | | F2 |
| `backup_root_free_bytes` | gauge | bayt | — | `statfs(BACKUP_ROOT)` | Zaxira bo'limidagi bo'sh joy | F2 |

> **S6 da amalga oshirilgan (F1 metrikalari):** `backup_last_status`, `backup_last_success_timestamp`, `backup_last_size_bytes`, `backup_newest_file_age_seconds{kind}`, `backup_files_count{kind}`, `backup_offsite_configured`. Loyiha papkasi nomi `^[A-Za-z0-9_\-]{1,60}$`, `_` bilan boshlanganlari loyiha emas, ≤ 100 loyiha. `*.tmp` va yashirin fayllar, simvolik havolalar hisobga olinmaydi. `kind`: `daily/` — `daily`; `weekly/` da nomida `-files-` bo'lsa `files`, aks holda `weekly`. `backup_last_success_timestamp`: `state.json` `ok` bo'lsa `finished_at`, aks holda eng yangi kunlik fayl `mtime`. `state.json` yo'q bo'lsa (appctl hali yozmagan) xato emas — faqat skaner metrikalari chiqadi; buzuq bo'lsa kollektor xatosi (qolgan loyihalar davom etadi). `error` matni hech qachon metrikaga tushmaydi. `backup_offsite_configured`: avval `_run.json` `offsite.configured`, bo'lmasa `appctl.conf` dagi `OFFSITE=` (matn

> sifatida o'qiladi, `source` qilinmaydi). `backup_gz_ok`, `backup_dir_bytes`, `backup_root_free_bytes`, `backup_offsite_last_*` — F2.

Offsite zaxira S-00 da turadi; u yerdagi agent bir xil kollektor bilan, lekin `BACKUP_ROOT=/srv/backups/main-server` ko'rsatkichi bilan ishlaydi; core mahalliy va offsite natijalarni loyiha nomi bo'yicha solishtiradi (FR-BKP-012).

### 7.4.7. `certs` kollektori

| Nom | Tur | Birlik | Belgilar | Manba | Izoh | Faza |
|---|---|---|---|---|---|---|
| `cert_not_after_timestamp` | gauge | unix s | `name` (papka nomi), `path` | `/etc/letsencrypt/live/*/cert.pem` + sozlamadagi qo'shimcha yo'llar | Faylni agent o'qiy olishi kerak (7.8) | F2 |
| `cert_days_left` | gauge | kun | `name` | hisoblangan | | F2 |
| `certbot_last_renew_status` | gauge | 0/1 | — | `/var/log/letsencrypt/letsencrypt.log` oxirgi `renew` natijasi | | F3 |
| `certbot_last_renew_timestamp` | gauge | unix s | — | xuddi shu | | F3 |

### 7.4.8. `ssh` va `apt` kollektorlari

| Nom | Tur | Birlik | Belgilar | Manba | Izoh | Faza |
|---|---|---|---|---|---|---|
| `ssh_failed_logins_1h` | gauge | son | — | `journalctl -u ssh --since -1h` (`Failed password`, `Invalid user`) | IP/foydalanuvchi nomi **yuborilmaydi**, faqat son | F3 |
| `ssh_failed_unique_ips_1h` | gauge | son | — | xuddi shu | | F3 |
| `ssh_accepted_logins_1h` | gauge | son | — | `Accepted ...` | Kutilmagan kirishni sezish uchun | F3 |
| `fail2ban_banned_ips` | gauge | son | `jail` | `fail2ban-client status sshd` | Mavjud bo'lsa | F3 |
| `apt_upgradable_total`, `apt_security_upgradable_total` | gauge | son | — | `apt-get -s upgrade` | Kuniga 1 marta | F3 |
| `reboot_required` | gauge | 0/1 | — | `/var/run/reboot-required` | | F2 |
| `kernel_reboot_pending` | gauge | 0/1 | — | ishlayotgan kernel ≠ o'rnatilgan eng yangi kernel | FR-HST-009 | F2 |
| `reboot_required_age_seconds` | gauge | s | — | `/var/run/reboot-required` mtime | | F2 |

### 7.4.9. Agent o'zining metrikalari

| Nom | Tur | Birlik | Belgilar | Izoh |
|---|---|---|---|---|
| `agent_up` | gauge | 1 | — | Har paketda 1 (heartbeat) |
| `agent_version_info` | gauge | 1 | `version`, `go` | Versiya belgi sifatida |
| `agent_collector_errors` | counter | son | `collector` | Kollektor xatolari |
| `agent_collect_duration_seconds` | gauge | s | `collector` | Yig'ish vaqti |
| `agent_queue_files`, `agent_queue_bytes` | gauge | son/bayt | — | Yuborilmagan navbat |
| `agent_send_failures_total` | counter | son | — | Yuborish xatolari |
| `agent_series_dropped` | counter | son | — | Kardinallik chegarasidan tashlangan |
| `agent_clock_skew_seconds` | gauge | s | — | Core javobidagi vaqt bilan farq |
| `agent_rss_bytes`, `agent_cpu_percent` | gauge | bayt/% | — | Agentning o'z resurs sarfi |

### 7.4.10. Faktlar (`facts`) — kam o'zgaradigan ma'lumot

Metrika emas; JSON obyekt. O'zgarganda core voqea (`host_fact_changed`) yozadi.

| Kalit | Misol | Manba |
|---|---|---|
| `hostname` | `vmi3679093` | `os.Hostname()` |
| `os` | `Ubuntu 24.04.3 LTS` | `/etc/os-release` |
| `kernel_running` | `6.8.0-139-generic` | `uname -r` |

| Kalit | Misol | Manba |
|---|---|---|
| `kernel_latest_installed` | `6.8.0-147-generic` | `/boot/vmlinuz-*` (eng yuqori versiya) |
| `arch` | `amd64` | `runtime.GOARCH` |
| `cpu_model`, `cpu_count` | `AMD EPYC ...`, `4` | `/proc/cpuinfo` |
| `mem_total_bytes` | `8321499136` | `/proc/meminfo` |
| `boot_id` | UUID | `/proc/sys/kernel/random/boot_id` |
| `ips` | `["164.68.105.194"]` | interfeyslar (ommaviy IP larni, `lo`/docker siz) |
| `timezone` | `Etc/UTC` | `timedatectl` |
| `docker` | `{"version":"28.x","compose":"2.x","log_driver":"json-file"}` | Docker API |
| `listen_ports` | `[22,80,443]` | `/proc/net/tcp*` (tashqariga ochiqlari) |
| `systemd_failed` | `[]` | `systemctl --failed` |
| `nginx_domains` | `["avlo.ai","nazar.avlo.ai"]` | `nginx -T` dan `server_name` (reyestr bilan solishtirish uchun, F3) |

# 7.5. Agent ↔ core protokoli (v1)

## 7.5.1. Enroll (bir martalik token → doimiy token)

```
POST /ingest/v1/enroll
Content-Type: application/json

{
  "enroll_token": "enr_********",
  "hostname": "vmi3679093",
  "agent_version": "1.0.0",
  "facts": { "os": "Ubuntu 24.04.3 LTS", "arch": "amd64", "cpu_count": 4, "mem_total_bytes": 8321499136 }
}
```

Javob (`200`):

```
{
  "host_id": "0193c0de-7a4e-7b3a-9d41-2f6c8e5a1b90",
  "agent_token": "agt_********",
  "interval_seconds": 15,
  "server_time": "2026-10-22T09:35:00Z"
}
```

Xatolar: `401 invalid_enroll_token`, `410 enroll_token_expired`, `409 host_already_enrolled`. Xato tanasi umumiy formatda (`{"error": {"code":..., "message":..., "request_id":...}}`, 9.1.1). Doimiy token agent tomonida `0600` rejimli faylda saqlanadi (enroll token bir martalik, qayta ishga tushganda yana kerak bo'lmasligi uchun); prototip agent: `--enroll-token` + `--token-file`. IP bo'yicha chegara: 10 so'rov/soat (14.7.3).

## 7.5.2. Metrika yuborish

> Quyidagi namunadagi `//` izohlar faqat tushuntirish uchun; haqiqiy JSON da izoh bo'lmaydi. `sent_at` majburiy va **har yuborishda yangilanadi** (qayta urinishda ham): core agent soati siljishini shundan aniqlaydi (6.10). `batch_id` esa qayta urinishda o'zgarmaydi.

```
POST /ingest/v1/metrics
Authorization: Bearer agt_********
Content-Type: application/json
Content-Encoding: gzip
X-Nazoratchi-Agent: nazoratchi-agent/1.0.0

{
  "batch_id": "b-5f3c2a90",          // takrorlanmasin, idempotentlik kaliti
  "seq": 128443,                     // monoton oshuvchi
  "sent_at": "2026-10-22T09:35:15.120Z",
  "samples": [
    {"n": "cpu_busy_percent",        "t": 1761125700000, "v": 12.4},
    {"n": "mem_used_percent",        "t": 1761125700000, "v": 41.8},
    {"n": "fs_used_percent",         "t": 1761125700000, "v": 23.1, "l": {"mount": "/", "device": "/dev/sda1", "fstype": "ext4"}},
    {"n": "docker_container_running","t": 1761125700000, "v": 1,    "l": {"container": "nazar-web-1", "project": "nazar", "service": "web"}}
  ],
  "facts": { "kernel_running": "6.8.0-139-generic", "kernel_latest_installed": "6.8.0-147-generic" },
  "events": [
    {"type": "cron_beat", "job": "appctl-backup", "t": 1761125700000, "ok": true}
  ]
}
```

Javob (`200`):

```
{
  "ok": true,
  "accepted": 4,
  "duplicate": false,
  "server_time": "2026-10-22T09:35:15.200Z",
  "interval_seconds": 15,
  "commands": [ {"type": "set_interval", "collector": "docker", "seconds": 60} ]
}
```

Xatolar:

| Kod | Ma'nosi | Agent harakati |
|---|---|---|
| 400 | Sxema/qiymat xato | Paket tashlab yuboriladi (navbatda qolmaydi), `agent_send_failures_total` oshadi, logga yoziladi |
| 401 | Token noto'g'ri/bekor qilingan | 5 daqiqa kutib qayta urinadi; 10 marta ketma-ket 401 → logga "token bekor qilingan", yig'ishda davom etadi (navbatni to'ldiradi) |
| 413 | Paket juda katta (> 1 MB) | Paket bo'laklanadi |
| 429 | Tezlik chegarasi | `Retry-After` kutiladi |
| 5xx / tarmoq | Core muammosi | Navbatga qoladi, eksponensial kutish |

### 7.5.3. Core buyruqlari (cheklangan to'plam)

| Buyruq | Ta'siri |
|---|---|
| `set_interval` | Kollektor intervalini o'zgartirish (5–600 s) |
| `reload_config` | Sozlamani qayta yuklash |
| `update` | Yangi versiyaga yangilanish (7.7) |
| `collect_now` | Bir martalik zudlik bilan yig'ish (diagnostika) |

Boshqa hech qanday buyruq yo'q. Agent noma'lum buyruqni e'tiborsiz qoldiradi va logga yozadi.

> **Holat (S2):** agent hozircha hech bir buyruqni bajarmaydi, faqat logga yozadi; `set_interval`, `reload_config` va `update` keyingi sprintlarda (sozlamani qayta o'qish `SIGHUP` bilan ishlaydi).

## 7.6. Sozlama fayli

`/etc/nazoratchi/agent.yaml` (egasi `root:nazoratchi`, rejim `0640`; token alohida `/etc/nazoratchi/token`, rejim `0600`, egasi `nazoratchi`):

```
core:
  url: https://nazoratchi.example.uz        # ommaviy manzil
  ca_file: ""                               # ixtiyoriy: o'zi imzolagan CA
  timeout: 10s
host:
  name: ""                                  # bo'sh bo'lsa hostname
collectors:
  system:  { enabled: true,  interval: 15s }
  disk:    { enabled: true,  interval: 30s, ignore_fstypes: [tmpfs, overlay, squashfs, devtmpfs] }
  diskio:  { enabled: true,  interval: 15s }
  net:     { enabled: true,  interval: 15s, ignore_ifaces: ["lo", "veth*", "br-*", "docker0"] }
  docker:
    enabled: true
    interval: 30s
    endpoint: tcp://127.0.0.1:2375       # docker-socket-proxy
    max_containers: 100
  systemd:
    enabled: true
    interval: 30s
    units: [nginx, docker, cron, ssh, avlo-contact]
  backup:
    enabled: true
    interval: 300s
    root: /opt/applications/backups
    appctl_conf: /opt/applications/appctl.conf
  certs:
    enabled: true
    interval: 6h
    globs: ["/etc/letsencrypt/live/*/cert.pem"]
  ssh:     { enabled: true, interval: 5m }
  apt:     { enabled: true, interval: 24h }
queue:
  dir: /var/lib/nazoratchi/queue
  max_bytes: 20971520
  max_age: 15m
log:
  level: info
```

## 7.7. O'rnatish, yangilash, o'chirish

### 7.7.1. O'rnatish skripti (`install.sh`) bosqichlari

1. `root` ekanini, `systemd`, `curl`, `sha256sum`, `minisign` borligini tekshiradi.
2. Arxitekturani (`uname -m`) aniqlaydi → `amd64` yoki `arm64`.
3. `https://<core>/agent/v1/nazoratchi-agent-<arch>` va `SHA256SUMS`, `SHA256SUMS.minisig` ni yuklaydi.
4. **Imzoni tekshiradi** (`minisign` ochiq kaliti skript ichiga joylangan); mos kelmasa to'xtaydi.
5. `/usr/local/bin/nazoratchi-agent` ga o'rnatadi (rejim `0755`).
6. `nazoratchi` tizim foydalanuvchisini yaratadi (`--system --no-create-home --shell /usr/sbin/nologin`), `adm`, `systemd-journal` guruhlariga qo'shadi.
7. Docker mavjud bo'lsa `docker-socket-proxy` konteynerini (`127.0.0.1:2375`) ko'taradi.
8. `/etc/nazoratchi/agent.yaml` ni yaratadi (mavjud bo'lsa **o'zgartirmaydi**).
9. Enroll qiladi (`--token`) → doimiy tokenni `/etc/nazoratchi/token` ga yozadi.
10. `/etc/systemd/system/nazoratchi-agent.service` ni yozadi, `daemon-reload`, `enable --now`.
11. 30 soniya kutib, birinchi paket yuborilganini tekshiradi va natijani chop etadi ("Nazoratchi'ga ulandi: <host_name>").

Skript idempotent: qayta ishga tushirilsa, mavjud sozlama va tokenni saqlab, binarni yangilaydi.

> **S7:** `deploy/agent/install.sh`, Debian konteynerida sinalgan (`bash deploy/agent/test.sh`): imzo/SHA-256 rad etish, idempotentlik, enroll. Agent birinchi muvaffaqiyatli paketni `paket yuborildi (birinchi)` deb Info darajasida yozadi (skript 30 s shuni kutadi). `ReadOnlyPaths` yo'llari `-` prefiksli (papka yo'q bo'lsa unit yiqilmaydi). `uninstall` core'ga `goodbye` yuboradi.

### 7.7.2. systemd unit

```
[Unit]
Description=Nazoratchi monitoring agent
After=network-online.target docker.service
Wants=network-online.target

[Service]
User=nazoratchi
Group=nazoratchi
SupplementaryGroups=adm systemd-journal
ExecStart=/usr/local/bin/nazoratchi-agent run --config /etc/nazoratchi/agent.yaml
Restart=always
RestartSec=5
MemoryMax=100M
CPUQuota=20%
# Qattiqlashtirish
NoNewPrivileges=true
ProtectSystem=strict
ProtectHome=true
PrivateTmp=true
ProtectKernelModules=true
ProtectControlGroups=true
RestrictAddressFamilies=AF_INET AF_INET6 AF_UNIX
ReadWritePaths=/var/lib/nazoratchi
ReadOnlyPaths=/opt/applications/backups /etc/letsencrypt
CapabilityBoundingSet=
LockPersonality=true

[Install]
WantedBy=multi-user.target
```

> Eslatma: `/etc/letsencrypt/live` odatda `root` uchun yopiq (`0700`). Sertifikat muddatini o'qish uchun agent `cert.pem` (ommaviy, maxfiy emas) ni o'qishi kerak, lekin `privkey.pem` ni **hech qachon**. Yechim: `setfacl -m u:nazoratchi:rx /etc/letsencrypt/{live,archive}` va `privkey*.pem` ga `setfacl` **qo'llanmaydi**; yoki root'da ishlaydigan kichik `certbot` ilgagi (`--deploy-hook`) muddatlarni `/var/lib/nazoratchi/certs.json` ga yozadi. Ikkinchisi tavsiya etiladi.

### 7.7.3. O'z-o'zini yangilash

1. Core javobida `update` buyrug'i: `{"type":"update","version":"1.1.0","sha256":"...","url":"/agent/v1/..."}`.
2. Agent faylni yuklaydi, SHA-256 va imzoni tekshiradi.
3. Yangi binarni `/usr/local/bin/nazoratchi-agent.new` ga yozadi, joriyni `.old` ga nusxalaydi, `rename` bilan almashtiradi va `systemd` orqali qayta ishga tushadi (`exit 0` + `Restart=always`).
4. 60 soniyada yangi agent birinchi muvaffaqiyatli paketni yubormasa, `ExecStopPost` skripti `.old` ni qaytaradi (rollback).
5. Core versiyani qayd etadi; panelda "yangilandi" voqeasi.

### 7.7.4. O'chirish

`nazoratchi-agent uninstall`: servisni to'xtatadi va o'chiradi, binar va unit faylini o'chiradi, `/etc/nazoratchi` va `/var/lib/nazoratchi` ni o'chiradi (`--keep-config` bilan saqlash mumkin), `nazoratchi` foydalanuvchisini o'chiradi, core'ga `POST /ingest/v1/goodbye` yuboradi (host "arxiv").

## 7.8. Ruxsatlar va xavfsizlik chegaralari (agent)

| Nimaga kirish kerak | Qanday | Nima uchun xavfsiz |
|---|---|---|
| `/proc`, `/sys` | Oddiy foydalanuvchi o'qiy oladi | Tizim statistikasi |
| `journalctl` (SSH loglari) | `systemd-journal` guruhi | Faqat o'qish |
| Docker ma'lumoti | `docker-socket-proxy` (`127.0.0.1:2375`, GET faqat) | Agent konteyner yarata/to'xtata olmaydi |
| Docker log hajmi | `/var/lib/docker/containers` — `root` uchun; yechim: Docker API `LogPath` + `stat` o'rniga `docker system df` / `inspect` dan hajm; zarur bo'lsa `setfacl -m u:nazoratchi:rx` papka darajasida (**fayl mazmuni emas**, faqat `stat`) | Mazmunni o'qimaydi |
| Zaxira papkasi | `ReadOnlyPaths` + `setfacl -R -m u:nazoratchi:rX /opt/applications/backups` | Faqat nomlari/hajmi/ `gzip -t` (o'qish) |
| `appctl.conf` | O'qish (`OFFSITE` bo'sh emasmi) | Faqat bitta bayroq yuboriladi, qiymati (host/manzil) emas |
| `nginx -t` | Kam huquqli `sudo` qoidasi yoki root ilgagi natijani faylga yozadi | Agent `sudo` olmaydi; faqat natija faylini o'qiydi |
| Tarmoq | Faqat chiquvchi 443 core'ga | Kiruvchi port yo'q |

## 7.9. Agent sifat mezonlari va testlari

| Test | Tavsifi |
|---|---|
| Birlik testlari | Har kollektor uchun `/proc` namunalari (fixtures) bilan; qiymatlar `top`, `free`, `df` bilan solishtiriladi |
| Moslik testi | Ubuntu 22.04 va 24.04, amd64 va arm64 (Docker + QEMU) |
| Yuklama testi | 15 s intervalda 24 soat; CPU/RAM chegarasi (NFR-PRF-007) |
| Xato testlari | Core o'chirilgan, DNS ishlamaydi, sekin tarmoq, 401/429/500 javoblari, disk to'lgan |
| Xavfsizlik | `gosec`, `govulncheck`; token loglarda yo'qligi; imzo tekshiruvi |
| Yangilash | Normal va buzuq binar bilan rollback |

## 7.10. Versiyalash va chiqarish

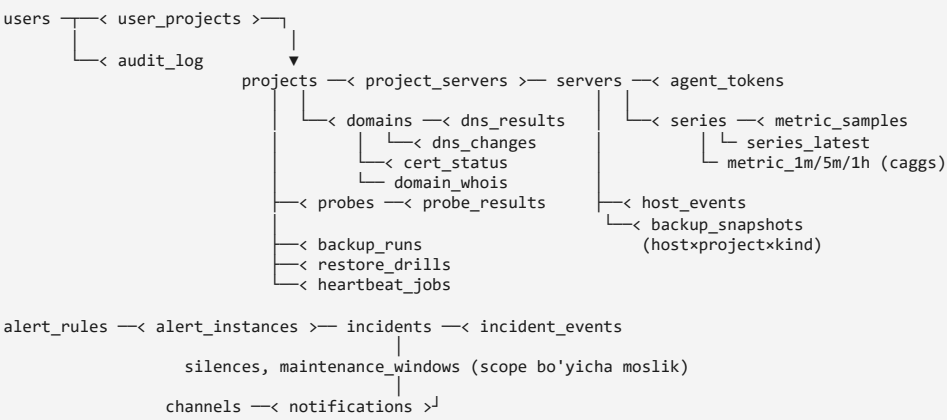
- Semantik versiya (`MAJOR.MINOR.PATCH`). Core `MAJOR` bir xil bo'lgan agentlarni qabul qiladi.
- Chiqarish artefaktlari: `nazoratchi-agent-linux-amd64`, `-arm64`, `SHA256SUMS`, `SHA256SUMS.minisig`, `install.sh`. CI da `goreleaser`; core o'z ichida bu fayllarni `/agent/v1/` orqali tarqatadi.
- Imzo kaliti (`minisign` maxfiy kaliti) CI sirlarida; ochiq kalit `install.sh` ichida.

# 8. Ma'lumotlar modeli

## 8.1. Umumiy tamoyillar

1. **Identifikatorlar.** Reyestr va holat jadvallarida `uuid` (v7, vaqt bo'yicha tartiblangan); vaqt qatorlarida ixcham `bigint`. API da UUID satr ko'rinishida.
2. **Vaqt.** Hamma vaqt `timestamptz` (UTC).
3. **Yumshoq o'chirish.** Reyestr obyektlari o'chirilmaydi: `status = 'archived'` (FR-REG-006).
4. **Qattiq cheklovlar.** `NOT NULL`, `CHECK`, `UNIQUE`, `FOREIGN KEY` bazada, faqat kodda emas.
5. **JSONB** faqat haqiqatan o'zgaruvchan tuzilma uchun (`facts`, `config`, `labels`); qidiriladigan maydonlar alohida ustun.
6. **Maxfiy maydonlar** (`*_enc`) ilova darajasida AES-256-GCM bilan shifrlangan `bytea`.
7. **Nomlash.** Jadvallar — ko'plik, `snake_case`; tashqi kalit — `<obyekt>_id`.

## 8.2. Mantiqiy (ER) diagramma




## 8.3. Reyestr jadvallari

> **Yaratish tartibi.** Quyidagi DDL o'qish qulayligi uchun mavzu bo'yicha guruhlangan. Amalda Django migratsiyalari bog'liqlik tartibini o'zi hal qiladi; qo'lda bajarilsa tartib: `users` (8.8) → `servers` → `projects` → `project_servers` → `domains` → qolganlari.

```
CREATE EXTENSION IF NOT EXISTS timescaledb;
CREATE EXTENSION IF NOT EXISTS pgcrypto;   -- gen_random_uuid()

CREATE TABLE servers (
  id               uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  name             text NOT NULL UNIQUE,
  provider         text,
  public_ipv4      inet,
  public_ipv6      inet,
  role             text,
  tags             text[] NOT NULL DEFAULT '{}',
  location         text,
  monthly_cost_usd numeric(8,2),
  notes            text NOT NULL DEFAULT '',
  status           text NOT NULL DEFAULT 'new'
                   CHECK (status IN ('new','online','offline','paused','archived')),
  last_seen_at     timestamptz,
  paused_at        timestamptz,
  paused_by        uuid REFERENCES users(id),
  agent_version    text,
  facts            jsonb NOT NULL DEFAULT '{}',
  created_at       timestamptz NOT NULL DEFAULT now(),
  updated_at       timestamptz NOT NULL DEFAULT now()
);

CREATE TABLE projects (
  id            uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  slug          text NOT NULL UNIQUE CHECK (slug ~ '^[a-zA-Z0-9_-]+$'), -- appctl papka nomi
  name          text NOT NULL,
  description   text NOT NULL DEFAULT '',
  owner_name    text,                -- mijoz/mas'ul
  owner_contact text,
  kind          text NOT NULL CHECK (kind IN ('compose','systemd','static','other')),
  moved         boolean NOT NULL DEFAULT true,   -- false = "ko'chirilmagan"
  status        text NOT NULL DEFAULT 'active' CHECK (status IN ('active','archived')),
  tags          text[] NOT NULL DEFAULT '{}',
  links         jsonb NOT NULL DEFAULT '[]',     -- [{label,url}]
  notes         text NOT NULL DEFAULT '',
  created_at    timestamptz NOT NULL DEFAULT now(),
  updated_at    timestamptz NOT NULL DEFAULT now()
);

CREATE TABLE project_servers (
  project_id  uuid NOT NULL REFERENCES projects(id),
  server_id   uuid NOT NULL REFERENCES servers(id),
  environment text NOT NULL DEFAULT 'production'
              CHECK (environment IN ('production','staging','dev')),
  ports       int[] NOT NULL DEFAULT '{}',      -- 127.0.0.1 portlar (backend/frontend)
  PRIMARY KEY (project_id, server_id, environment)
);
-- Bir serverda bir xil port ikki loyihaga berilmasligi (FR-REG-010) — ilova darajasida tekshiriladi.

CREATE TABLE domains (
  id             uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id     uuid REFERENCES projects(id),
  fqdn           text NOT NULL UNIQUE CHECK (fqdn = lower(fqdn)),
  expected_ips   inet[] NOT NULL DEFAULT '{}',  -- DNS mosligi uchun
  dns_provider   text,                          -- 'ahost.uz', 'Namecheap', 'Beget', ...
  registrar      text,
  controlled_by  text NOT NULL DEFAULT 'self' CHECK (controlled_by IN ('self','client','other')),
  tls_enabled    boolean NOT NULL DEFAULT true,
  status         text NOT NULL DEFAULT 'active' CHECK (status IN ('active','archived')),
  notes          text NOT NULL DEFAULT '',
  created_at     timestamptz NOT NULL DEFAULT now()
);
CREATE INDEX domains_project_idx ON domains(project_id);

CREATE TABLE agent_tokens (
  id            uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  server_id     uuid NOT NULL REFERENCES servers(id),
  name          text NOT NULL,
  token_hash    bytea NOT NULL UNIQUE,   -- SHA-256(token)
  created_at    timestamptz NOT NULL DEFAULT now(),
  created_by    uuid REFERENCES users(id),
  last_used_at  timestamptz,
  last_used_ip  inet,
  revoked_at    timestamptz
);
CREATE INDEX agent_tokens_server_idx ON agent_tokens(server_id) WHERE revoked_at IS NULL;

CREATE TABLE enroll_tokens (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  token_hash  bytea NOT NULL UNIQUE,
  server_id   uuid NOT NULL REFERENCES servers(id),
  expires_at  timestamptz NOT NULL,
  used_at     timestamptz,
  created_by  uuid REFERENCES users(id)
);

CREATE TABLE host_events (          -- reboot, kernel o'zgardi, agent yangilandi, ...
  id         bigserial PRIMARY KEY,
  server_id  uuid NOT NULL REFERENCES servers(id),
  time       timestamptz NOT NULL DEFAULT now(),
  type       text NOT NULL,         -- 'reboot','fact_changed','agent_updated','enrolled','paused','resumed'
  data       jsonb NOT NULL DEFAULT '{}'
);
CREATE INDEX host_events_idx ON host_events(server_id, time DESC);
```

# 8.4. Vaqt qatorlari

## 8.4.1. Series va namunalar

```
CREATE TABLE series (
  id          bigserial PRIMARY KEY,
  server_id   uuid NOT NULL REFERENCES servers(id),
  name        text NOT NULL,                -- 'cpu_busy_percent'
  labels      jsonb NOT NULL DEFAULT '{}',   -- {"mount":"/","device":"/dev/sda1"}
  labels_hash text NOT NULL,                -- md5(labels::text) tartiblangan kalit bilan
  unit        text,
  kind        text NOT NULL DEFAULT 'gauge' CHECK (kind IN ('gauge','counter')),
  rollup      boolean NOT NULL DEFAULT false, -- agregat darajalarda saqlansinmi (muhim metrika)
  first_seen  timestamptz NOT NULL DEFAULT now(),
  last_seen   timestamptz NOT NULL DEFAULT now(),
  UNIQUE (server_id, name, labels_hash)
);
CREATE INDEX series_lookup_idx ON series(server_id, name);

CREATE TABLE metric_samples (
  time      timestamptz      NOT NULL,
  series_id bigint           NOT NULL,
  value     double precision NOT NULL
);
SELECT create_hypertable('metric_samples', 'time', chunk_time_interval => INTERVAL '1 day');
CREATE UNIQUE INDEX metric_samples_uq ON metric_samples (series_id, time);

ALTER TABLE metric_samples SET (
  timescaledb.compress,
  timescaledb.compress_segmentby = 'series_id',
  timescaledb.compress_orderby   = 'time DESC'
);
SELECT add_compression_policy('metric_samples', INTERVAL '2 days');
SELECT add_retention_policy  ('metric_samples', INTERVAL '14 days');

CREATE TABLE series_latest (            -- eng oxirgi qiymat (tez so'rovlar va alert uchun)
  series_id bigint PRIMARY KEY REFERENCES series(id),
  time      timestamptz      NOT NULL,
  value     double precision NOT NULL
);
```

> **Idempotentlik chegarasi.** `ON CONFLICT DO NOTHING` (takroriy namunani o'tkazib yuborish) faqat siqilmagan bo'laklarda (so'nggi 2 kun) ishlaydi; agent buferi 15 daqiqa bilan cheklangani uchun takror paket har doim shu oraliqda keladi. **P-02 da tasdiqlandi** (TimescaleDB 2.17.2): siqilgan bo'lakka takror yozuv dublikat hosil qiladi (41 → 82 qator), UNIQUE indeks siqilgan ma'lumotda ishlamaydi. Shuning uchun ingest 1 soatdan eski namunani yozmaydi (`MAX_SAMPLE_AGE_S`). Siqilgan bo'lakdan qo'lda `DELETE` ham cheklangan (`max_tuples_decompressed_per_dml_transaction`, 100 000): eski ma'lumot faqat retention (`drop_chunks`) bilan o'chiriladi.

## 8.4.2. Agregatlar (continuous aggregates)

Agregat daraja uchun `sum` va `count` saqlanadi (o'rtacha = `sum/count`); shunda yuqori darajalar quyi darajalardan to'g'ri hisoblanadi.

```
CREATE MATERIALIZED VIEW metric_1m
WITH (timescaledb.continuous) AS
SELECT time_bucket('1 minute', time) AS bucket,
       series_id,
       min(value) AS v_min,
       max(value) AS v_max,
       sum(value) AS v_sum,
       count(*)   AS n
FROM metric_samples
GROUP BY bucket, series_id
WITH NO DATA;

SELECT add_continuous_aggregate_policy('metric_1m',
  start_offset => INTERVAL '30 minutes', end_offset => INTERVAL '1 minute',
  schedule_interval => INTERVAL '1 minute');
SELECT add_retention_policy('metric_1m', INTERVAL '90 days');

CREATE MATERIALIZED VIEW metric_5m
WITH (timescaledb.continuous) AS
SELECT time_bucket('5 minutes', bucket) AS bucket,
       series_id,
       min(v_min) AS v_min, max(v_max) AS v_max, sum(v_sum) AS v_sum, sum(n) AS n
FROM metric_1m
GROUP BY 1, series_id
WITH NO DATA;

SELECT add_continuous_aggregate_policy('metric_5m',
  start_offset => INTERVAL '2 hours', end_offset => INTERVAL '5 minutes',
  schedule_interval => INTERVAL '5 minutes');
SELECT add_retention_policy('metric_5m', INTERVAL '365 days');

CREATE MATERIALIZED VIEW metric_1h
WITH (timescaledb.continuous) AS
SELECT time_bucket('1 hour', bucket) AS bucket,
       series_id,
       min(v_min) AS v_min, max(v_max) AS v_max, sum(v_sum) AS v_sum, sum(n) AS n
FROM metric_5m
GROUP BY 1, series_id
WITH NO DATA;

SELECT add_continuous_aggregate_policy('metric_1h',
  start_offset => INTERVAL '1 day', end_offset => INTERVAL '1 hour',
  schedule_interval => INTERVAL '1 hour');
SELECT add_retention_policy('metric_1h', INTERVAL '1095 days');
```

Eslatmalar:

- Ierarxik agregatlar (agregat ustida agregat) TimescaleDB 2.9+ da mavjud; versiya shu talabga ko'ra tanlanadi.
- `series.rollup = false` bo'lgan qatorlar (har-yadro, har-konteyner tarmoq va h.k.) uchun agregat saqlanmaydi (5.1.3 dagi hajm tejashi). **P-02 da sinab ko'rildi (R-102, hal bo'ldi):** `metric_1m` agregati `JOIN series ... WHERE rollup` bilan TimescaleDB 2.17.2 da yaratiladi va ishlaydi; 5m va 1h unga ierarxik bog'lanadi (faqat `rollup` seriyalar materiallashadi). Cheklov: filtr yangilash paytida hisoblanadi, shuning uchun `rollup` bayrog'i seriya yaratilganda qo'yiladi va keyin o'zgartirilmaydi. Namuna: `backend/apps/metrics/migrations/0002_timescale.py`.
- Agregat siyosati `start_offset` dan eskisini qayta hisoblamaydi, shuning uchun xom ma'lumotni 14 kunda o'chirish agregatlarni buzmaydi.

### 8.4.3. Tarixiy ma'lumot qaysi jadvaldan olinadi (so'rov marshrutizatsiyasi)

| Vaqt oralig'i | Jadval | Nuqtalar soni (taxminan) |
|---|---|---|
| ≤ 6 soat | `metric_samples` (xom) | ≤ 1 440 |
| 6 soat — 3 kun | `metric_1m` | ≤ 4 320 → UI uchun 1 000 nuqtagacha qayta siyraklanadi |
| 3 kun — 30 kun | `metric_5m` | ≤ 8 640 → siyraklanadi |
| > 30 kun | `metric_1h` | ≤ 2 160 (90 kun) |

Backend oraliqqa qarab jadvalni tanlaydi va natijani `max_points=1000` ga keltirib (`time_bucket` bilan) qaytaradi.

## 8.5. Faol tekshiruvlar

```
CREATE TABLE probes (
  id           uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id   uuid REFERENCES projects(id),
  domain_id    uuid REFERENCES domains(id),
  type         text NOT NULL CHECK (type IN ('http','tcp','tls','dns','whois')),
  name         text NOT NULL,
  config       jsonb NOT NULL,       -- tur bo'yicha sxema (18.4)
  interval_s   int  NOT NULL DEFAULT 60 CHECK (interval_s BETWEEN 10 AND 86400),
  timeout_s    int  NOT NULL DEFAULT 10 CHECK (timeout_s BETWEEN 1 AND 60),
  enabled      boolean NOT NULL DEFAULT true,
  state        text NOT NULL DEFAULT 'unknown' CHECK (state IN ('up','down','unknown','paused')),
  state_since  timestamptz,
  consecutive_failures int NOT NULL DEFAULT 0,
  last_run_at  timestamptz,
  next_run_at  timestamptz NOT NULL DEFAULT now(),
  secrets_enc  bytea,                -- shifrlangan sarlavhalar (Authorization va h.k.)
  created_at   timestamptz NOT NULL DEFAULT now()
);
CREATE INDEX probes_due_idx ON probes(next_run_at) WHERE enabled;

CREATE TABLE probe_results (
  time        timestamptz NOT NULL,
  probe_id    uuid        NOT NULL,
  ok          boolean     NOT NULL,
  status_code int,
  total_ms    int,
  dns_ms      int,
  connect_ms  int,
  tls_ms      int,
  ttfb_ms     int,
  error_kind  text,        -- 'timeout','dns','connect','tls','status','keyword','redirect_loop'
  detail      jsonb
);
SELECT create_hypertable('probe_results', 'time', chunk_time_interval => INTERVAL '7 days');
CREATE INDEX probe_results_idx ON probe_results(probe_id, time DESC);
SELECT add_retention_policy('probe_results', INTERVAL '90 days');

CREATE TABLE probe_uptime_daily (    -- uzoq muddatli uptime hisobi (FR-HTTP-009)
  probe_id        uuid NOT NULL REFERENCES probes(id),
  day             date NOT NULL,
  checks_total    int  NOT NULL,
  checks_ok       int  NOT NULL,
  maintenance_s   int  NOT NULL DEFAULT 0,   -- rejalashtirilgan to'xtash vaqti
  avg_total_ms    int,
  p95_total_ms    int,
  PRIMARY KEY (probe_id, day)
);

-- S5: `cert_status` ORM da; UNIQUE (name, source, server) NULL larni teng hisoblaydi (agentsiz yozuvda server NULL).
-- Yangi: `settings(key, value jsonb)` — tizim sozlamalari (hozircha `retired_ips`, Q-12; `seed_registry` yozadi).
CREATE TABLE cert_status (           -- eng oxirgi sertifikat holati (TLS probe yoki agent)
  id          bigserial PRIMARY KEY,
  domain_id   uuid REFERENCES domains(id),
  server_id   uuid REFERENCES servers(id),
  name        text NOT NULL,               -- 'avlo.ai' yoki papka nomi
  source      text NOT NULL CHECK (source IN ('probe','agent')),
  not_after   timestamptz NOT NULL,
  issuer      text,
  subject     text,
  san         text[] NOT NULL DEFAULT '{}',
  chain_ok    boolean,
  hostname_ok boolean,
  checked_at  timestamptz NOT NULL DEFAULT now(),
  UNIQUE (name, source, server_id)
);

CREATE TABLE dns_results (
  time       timestamptz NOT NULL,
  domain_id  uuid        NOT NULL,
  probe_id   uuid        NOT NULL,   -- S5: qaysi DNS tekshiruvi (bir domenda bir nechta bo'lishi mumkin)
  resolver   text        NOT NULL,     -- '8.8.8.8','1.1.1.1','9.9.9.9','ns:rdns1.ahost.uz'
  rtype      text        NOT NULL DEFAULT 'A',
  answers    text[]      NOT NULL DEFAULT '{}',
  matches    boolean     NOT NULL,     -- kutilgan IP lar bilan mosmi
  rcode      text,
  latency_ms int
);
SELECT create_hypertable('dns_results', 'time', chunk_time_interval => INTERVAL '7 days');
CREATE INDEX dns_results_idx ON dns_results(domain_id, time DESC);
SELECT add_retention_policy('dns_results', INTERVAL '90 days');

CREATE TABLE dns_changes (
  id         bigserial PRIMARY KEY,
  domain_id  uuid NOT NULL REFERENCES domains(id),
  resolver   text NOT NULL,
  rtype      text NOT NULL,
  old_answers text[] NOT NULL,
  new_answers text[] NOT NULL,
  changed_at timestamptz NOT NULL DEFAULT now()
);

CREATE TABLE domain_whois (
  domain_id   uuid PRIMARY KEY REFERENCES domains(id),
  expires_at  timestamptz,
```

```
  registrar   text,
  source      text NOT NULL CHECK (source IN ('rdap','manual')),
  checked_at  timestamptz NOT NULL DEFAULT now()
);
```

## 8.6. Zaxira nusxalar

```
-- S6: loyiha zaxira siyosati `projects.backup jsonb` ({expect_daily, expect_weekly_files, require_offsite, db}; seed `backup:` dan).
-- `backup_runs` agent yuborgan state.json dan emas, `backup_*` METRIKALARIDAN hosil bo'ladi: har safar `backup_last_success_timestamp` yangil
-- bitta qator yozadi (faqat kind='daily', server_id, project_slug, finished_at, size_bytes; UNIQUE shu to'rtlikda). Qolgan ustunlar (status,
-- F2 (restore drill bilan birga). Hajm anomaliyasi (AL-071/072) va 30 kunlik grafik shu tarixdan.
CREATE TABLE backup_runs (            -- appctl state.json dan (agent yuboradi)
  id            bigserial PRIMARY KEY,
  server_id     uuid NOT NULL REFERENCES servers(id),
  project_id    uuid REFERENCES projects(id),
  project_slug  text NOT NULL,
  kind          text NOT NULL CHECK (kind IN ('daily','weekly','files')),
  status        text NOT NULL CHECK (status IN ('ok','fail')),
  started_at    timestamptz,
  finished_at   timestamptz NOT NULL,
  duration_s    int,
  file_name     text,
  size_bytes    bigint,
  error         text,
  offsite_status text CHECK (offsite_status IN ('ok','fail','skipped')),
  offsite_finished_at timestamptz,
  UNIQUE (server_id, project_slug, kind, finished_at)
);
CREATE INDEX backup_runs_idx ON backup_runs(project_slug, finished_at DESC);

CREATE TABLE backup_snapshots (       -- papka skaneri natijasi (har host × loyiha × tur uchun 1 qator)
  server_id       uuid NOT NULL REFERENCES servers(id),
  project_slug    text NOT NULL,
  kind            text NOT NULL,
  newest_file     text,
  newest_mtime    timestamptz,
  newest_size     bigint,
  files_count     int,
  dir_bytes       bigint,
  gz_ok           boolean,
  checked_at      timestamptz NOT NULL,
  PRIMARY KEY (server_id, project_slug, kind)
);

CREATE TABLE restore_drills (
  id            uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id    uuid NOT NULL REFERENCES projects(id),
  performed_at  timestamptz NOT NULL,
  performed_by  uuid REFERENCES users(id),
  backup_file   text NOT NULL,
  backup_age_h  int,
  duration_s    int,
  tables_count  int,
  rows_count    bigint,
  result        text NOT NULL CHECK (result IN ('success','partial','failed')),
  notes         text NOT NULL DEFAULT ''
);
CREATE INDEX restore_drills_idx ON restore_drills(project_id, performed_at DESC);

CREATE TABLE heartbeat_jobs (          -- cron/skript "bajarildim" signallari (FR-SVC-004, FR-INT-005)
  id             uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  name           text NOT NULL UNIQUE,   -- 'appctl-backup@main-2'
  server_id      uuid REFERENCES servers(id),
  project_id     uuid REFERENCES projects(id),
  token_hash     bytea NOT NULL UNIQUE,
  expected_every_s int NOT NULL,
  grace_s        int NOT NULL DEFAULT 3600,
  last_beat_at   timestamptz,
  last_ok        boolean,
  enabled        boolean NOT NULL DEFAULT true
);
```

## 8.7. Alert tizimi

```
CREATE TABLE alert_rules (
  id           uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  key          text NOT NULL UNIQUE,        -- 'AL-003'
  name         text NOT NULL,
  description  text NOT NULL DEFAULT '',
  scope        jsonb NOT NULL,              -- {"type":"host"|"project"|"probe"|"domain", "tags":[...], "ids":[...]}
  condition    jsonb NOT NULL,              -- 12.3 sxemasi
  for_seconds  int  NOT NULL DEFAULT 0,
  resolve_after_seconds int NOT NULL DEFAULT 60,
  severity     text NOT NULL CHECK (severity IN ('info','warning','high','critical')),
  channels     uuid[] NOT NULL DEFAULT '{}',   -- bo'sh bo'lsa standart kanallar
  runbook_key  text,                        -- 'RB-004'
  enabled      boolean NOT NULL DEFAULT true,
  built_in     boolean NOT NULL DEFAULT false,
  updated_at   timestamptz NOT NULL DEFAULT now()
);

CREATE TABLE incidents (
  id            uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  scope_key     text NOT NULL,              -- 'host:<server_id>' | 'project:<id>' | 'domain:<id>'
  title         text NOT NULL,
  severity      text NOT NULL CHECK (severity IN ('info','warning','high','critical')),
  status        text NOT NULL DEFAULT 'open' CHECK (status IN ('open','acknowledged','resolved')),
  started_at    timestamptz NOT NULL DEFAULT now(),
  acked_at      timestamptz,
  acked_by      uuid REFERENCES users(id),
  resolved_at   timestamptz,
  suppressed    boolean NOT NULL DEFAULT false,   -- silence/maintenance ichida ochilgan
  root_cause    text,
  summary       jsonb NOT NULL DEFAULT '{}'
);
CREATE UNIQUE INDEX incidents_one_open_per_scope
  ON incidents(scope_key) WHERE status <> 'resolved';
CREATE INDEX incidents_time_idx ON incidents(started_at DESC);

CREATE TABLE alert_instances (
  id            uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  rule_id       uuid NOT NULL REFERENCES alert_rules(id),
  subject_type  text NOT NULL,            -- 'host','probe','domain','project','container','unit'
  subject_id    text NOT NULL,            -- UUID yoki 'host:container' kabi kalit
  fingerprint   text NOT NULL,            -- hash(rule_id, subject_type, subject_id, labels)
  state         text NOT NULL CHECK (state IN ('ok','pending','firing','resolving')),
  pending_since timestamptz,
  firing_since  timestamptz,
  resolving_since timestamptz,
  last_eval_at  timestamptz NOT NULL DEFAULT now(),
  last_value    double precision,
  labels        jsonb NOT NULL DEFAULT '{}',
  incident_id   uuid REFERENCES incidents(id),
  UNIQUE (fingerprint)
);
CREATE INDEX alert_instances_state_idx ON alert_instances(state) WHERE state <> 'ok';

-- Amalga oshirishda qo'shilgan ustunlar (S4): servers.paused_until, servers.pause_reason (pauza muddati va sababi, FR-HST-016);
-- probe_results — ORM modeli emas, migratsiyada SQL bilan yaratilgan gipertablitsa; probes.secrets_enc hozircha ishlatilmaydi.
-- Amalga oshirishda qo'shilgan ustunlar (S3):
--   alert_rules.runbook_hint text         -- xabardagi "Nima qilish kerak" matni
--   incidents.last_notified_at, reminders_sent   -- takroriy eslatmalar (12.4.1)
--   alert_instances.subject_name, detail, suppressed, notified_at   -- xabar matni va bostirish holati
-- `evaluator_state` (id=1, last_run_at, last_duration_ms): dvigatel ishlayotganini bildiradi (AL-084).
-- notifications.incident_id -> incidents(id) ON DELETE SET NULL (migratsiya bilan bog'langan).

CREATE TABLE incident_events (
  id          bigserial PRIMARY KEY,
  incident_id uuid NOT NULL REFERENCES incidents(id),
  time        timestamptz NOT NULL DEFAULT now(),
  type        text NOT NULL,   -- opened, alert_added, alert_resolved, acked, comment, notified, escalated, resolved, severity_changed
  user_id     uuid REFERENCES users(id),
  data        jsonb NOT NULL DEFAULT '{}'
);
CREATE INDEX incident_events_idx ON incident_events(incident_id, time);

CREATE TABLE silences (
  id         uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  matchers   jsonb NOT NULL,      -- [{"label":"server","op":"=","value":"main-2"}, ...]
  starts_at  timestamptz NOT NULL DEFAULT now(),
  ends_at    timestamptz NOT NULL CHECK (ends_at > starts_at),
  reason     text NOT NULL CHECK (length(reason) > 0),
  created_by uuid REFERENCES users(id),
  created_at timestamptz NOT NULL DEFAULT now(),
  cancelled_at timestamptz
);

CREATE TABLE maintenance_windows (
  id         uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  scope      jsonb NOT NULL,      -- {"servers":[...], "projects":[...]}
  starts_at  timestamptz NOT NULL,
  ends_at    timestamptz NOT NULL CHECK (ends_at > starts_at),
  reason     text NOT NULL,
  created_by uuid REFERENCES users(id),
  cancelled_at timestamptz
);
```

## 8.8. Bildirishnomalar, foydalanuvchilar, audit

```
CREATE TABLE channels (
  id            uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  type          text NOT NULL CHECK (type IN ('telegram','email','webhook')),
  name          text NOT NULL UNIQUE,
  config_enc    bytea NOT NULL,        -- shifrlangan JSON: bot_token, chat_id / smtp / url, secret
  min_severity  text NOT NULL DEFAULT 'warning'
                CHECK (min_severity IN ('info','warning','high','critical')),
  quiet_hours   jsonb,                 -- {"from":"23:00","to":"07:00","tz":"Asia/Tashkent","pass":["critical"]}
  is_default    boolean NOT NULL DEFAULT false,
  enabled       boolean NOT NULL DEFAULT true,
  last_ok_at    timestamptz,
  last_error    text
);

CREATE TABLE notifications (
  id           bigserial PRIMARY KEY,
  incident_id  uuid REFERENCES incidents(id),
  channel_id   uuid NOT NULL REFERENCES channels(id),
  event        text NOT NULL,         -- 'opened','escalated','resolved','reminder','digest','test'
  status       text NOT NULL CHECK (status IN ('queued','sent','failed','suppressed')),
  attempts     int  NOT NULL DEFAULT 0,
  created_at   timestamptz NOT NULL DEFAULT now(),
  sent_at      timestamptz,
  external_id  text,                  -- Telegram message_id (tahrirlash uchun)
  error        text,
  payload      jsonb NOT NULL,        -- yuboriladigan xabar {"text","buttons"} (sirlarsiz); qayta urinish uchun
  next_attempt_at timestamptz NOT NULL DEFAULT now()   -- status='queued' bo'lganda keyingi urinish vaqti
);
-- incident_id: incidents jadvali S3 da yaratiladi; shu paytgacha FK emas (keyin migratsiya bilan bog'lanadi).
CREATE INDEX notifications_due_idx ON notifications(next_attempt_at) WHERE status = 'queued';
CREATE INDEX notifications_idx ON notifications(incident_id, created_at);

CREATE TABLE users (
  id            uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  login         text NOT NULL UNIQUE,
  email         text UNIQUE,
  password_hash text NOT NULL,          -- argon2id
  role          text NOT NULL CHECK (role IN ('owner','operator','viewer')),
  totp_secret_enc bytea,
  telegram_chat_id bigint,
  language      text NOT NULL DEFAULT 'uz' CHECK (language IN ('uz','ru','en')),
  timezone      text NOT NULL DEFAULT 'Asia/Tashkent',
  is_active     boolean NOT NULL DEFAULT true,
  failed_logins int NOT NULL DEFAULT 0,
  locked_until  timestamptz,
  last_login_at timestamptz,
  created_at    timestamptz NOT NULL DEFAULT now()
);

CREATE TABLE user_projects (          -- viewer cheklovi (FR-AUTH-005)
  user_id    uuid NOT NULL REFERENCES users(id),
  project_id uuid NOT NULL REFERENCES projects(id),
  PRIMARY KEY (user_id, project_id)
);

CREATE TABLE user_sessions (          -- Django sessiya saqlagichi + egasi (FR-AUTH-004); amalda Django nomi: accounts_usersession
  session_key  varchar(40) PRIMARY KEY,
  session_data text NOT NULL,
  expire_date  timestamptz NOT NULL,
  user_id      uuid,                          -- users.id; FK emas (Django sessiyani ro'yxatdan o'tkazmasdan yozadi); NULL: kirilmagan sessiy
  public_id    uuid NOT NULL DEFAULT gen_random_uuid(),  -- API'da ko'rinadigan id; session_key hech qachon chiqmaydi
  ip           inet,
  user_agent   varchar(300) NOT NULL DEFAULT '',
  created_at   timestamptz NOT NULL DEFAULT now()
);
CREATE INDEX user_sessions_expire_idx ON user_sessions(expire_date);
CREATE INDEX user_sessions_user_idx ON user_sessions(user_id);
CREATE INDEX user_sessions_public_idx ON user_sessions(public_id);   -- UNIQUE emas, pastga qarang
-- `public_id` ataylab UNIQUE emas: Django `cycle_key()` yangi qatorni eskisi o'chirilishidan OLDIN
-- yaratadi, UNIQUE bo'lsa `create()` cheksiz qayta uriniladi (S1 da topilgan). Ro'yxat/o'chirish
-- so'rovlari `user_id` bilan birga filtrlanadi, shuning uchun to'qnashuv xavfsizlikka ta'sir qilmaydi.

CREATE TABLE audit_log (
  id          bigserial PRIMARY KEY,
  time        timestamptz NOT NULL DEFAULT now(),
  user_id     uuid REFERENCES users(id),
  ip          inet,
  action      text NOT NULL,          -- 'login','logout','server.create','rule.update','silence.create',...
  object_type text,
  object_id   text,
  before      jsonb,
  after       jsonb
);
CREATE INDEX audit_log_idx ON audit_log(time DESC);
-- Audit jurnali faqat qo'shiladi: ilova foydalanuvchisiga UPDATE/DELETE huquqi berilmaydi.
REVOKE UPDATE, DELETE ON audit_log FROM nazoratchi_app;
```

## 8.9. Kritik so'rovlar va kutilgan rejalar

| № | So'rov | Maqsad | Indeks / yo'l | Kutilgan vaqt |
|---|---|---|---|---|
| Q1 | Host sahifasidagi "hozirgi" qiymatlar | Eng oxirgi qiymatlar | `series (server_id)` → `series_latest` PK | < 20 ms |
| Q2 | Grafik: 1 qator, 24 soat | UI grafigi | `metric_1m` (series_id, bucket) | < 100 ms |
| Q3 | Barcha loyihalar uchun zaxira yoshi | Zaxiralar sahifasi | `backup_runs (project_slug, finished_at desc)` + `backup_snapshots` PK | < 50 ms |
| Q4 | Ochiq incidentlar | Dashboard | `incidents` qisman indeks (status <> 'resolved') | < 10 ms |
| Q5 | Probe uptime 30 kun | Uptime foizi | `probe_uptime_daily` PK | < 30 ms |
| Q6 | Disk bashorati (regressiya) | Alert/forecast | `metric_samples (series_id, time)` oxirgi 6 soat | < 100 ms |
| Q7 | "Jim" hostlar | Server o'chdi | `servers (last_seen_at)` | < 10 ms |

Namuna (Q1):

```
SELECT s.name, s.labels, l.value, l.time
FROM series s
JOIN series_latest l ON l.series_id = s.id
WHERE s.server_id = $1;
```

Namuna (Q2, 24 soatlik CPU):

```
SELECT bucket AS t, v_sum / n AS avg, v_min, v_max
FROM metric_1m
WHERE series_id = $1 AND bucket >= now() - INTERVAL '24 hours'
ORDER BY bucket;
```

Namuna (Q6, to'lishga qolgan vaqt):

```
WITH pts AS (
  SELECT extract(epoch FROM time) AS x, value AS y
  FROM metric_samples
  WHERE series_id = $1 AND time >= now() - INTERVAL '6 hours'
), fit AS (
  SELECT regr_slope(y, x) AS slope, regr_intercept(y, x) AS icpt, regr_r2(y, x) AS r2
  FROM pts
)
SELECT slope, r2,
       CASE WHEN slope > 0 THEN ($2 - (slope * extract(epoch FROM now()) + icpt)) / slope END AS seconds_to_full
FROM fit;   -- $2 = fs_total_bytes
```

## 8.10. Saqlash siyosati (retention) xulosasi

| Ma'lumot | Saqlash | Mexanizm |
|---|---|---|
| `metric_samples` (xom) | 14 kun (2 kundan keyin siqiladi) | Timescale retention + compression |
| `metric_1m` | 90 kun | CAgg retention |
| `metric_5m` | 365 kun | CAgg retention |
| `metric_1h` | 3 yil | CAgg retention |
| `probe_results` | 90 kun | Timescale retention |
| `probe_uptime_daily` | cheksiz (kichik) | — |
| `dns_results` | 90 kun | Timescale retention |
| `dns_changes`, `host_events` | cheksiz | — |
| `incidents`, `incident_events` | cheksiz (yiliga ~ yuzlab) | — |
| `notifications` | 1 yil | Celery vazifasi |
| `audit_log` | ≥ 1 yil (tavsiya: cheksiz) | Faqat qo'shiladi |
| `backup_runs` | 2 yil | Celery vazifasi |

Sozlamalar paneldan o'zgartiriladi (NFR-DAT-002), o'zgarish audit jurnaliga yoziladi.

## 8.11. Boshlang'ich ma'lumot (seed)

Birinchi o'rnatishda reyestr 2-bo'limdagi inventardan to'ldiriladi (`seed/inventory.yaml`, 18-bo'limdagi format): S-00, S-01, S-02 (S-03 arxivda), 7 ishlayotgan loyiha, ~20 domen, kutilgan IP lar, DNS provayderlar va "ko'chirilmagan" loyihalar. Shu bilan 90-kunlik KPI (G-6) dastlabki kundan boshlanadi.

## 8.12. Migratsiya qoidalari

- Django migratsiyalari + Timescale maxsus buyruqlari (`RunSQL`) bilan; har migratsiya qaytariladigan (`reverse_sql`).
- Gipertablitsa va agregatlar migratsiyasi tranzaksiyadan tashqarida bajarilishi kerak (`atomic = False`; `CREATE MATERIALIZED VIEW ... WITH (timescaledb.continuous)` talabi).
- Katta jadvallarga ustun qo'shish `DEFAULT` siz va `NULL` ruxsat bilan (qulflanishni oldini olish), keyin to'ldirish va cheklov qo'shish.
- Sxema o'zgarishi paytida ingest to'xtamaydi; agentlar buferi bilan qoplaydi (FR-SYS-006).

# 9. API spetsifikatsiyasi

## 9.1. Umumiy qoidalar

| Mavzu | Qoida |
|---|---|
| Bazaviy manzil | `https://<core>/api/v1/` (brauzer/API), `https://<core>/ingest/v1/` (agent), `https://<core>/hb/` (heartbeat) |
| Format | JSON (`application/json; charset=utf-8`), maydon nomlari `snake_case` |
| Vaqt | ISO-8601 UTC: `2026-10-22T09:35:00Z`. Vaqt oralig'i parametrlari ham shu formatda |
| Identifikator | UUID satr |
| Autentifikatsiya | Brauzer: cookie-sessiya + `X-CSRFToken` (o'zgartiruvchi so'rovlarda). API tokeni: `Authorization: Bearer nzr_...`. Agent: `Authorization: Bearer agt_...` |
| Ruxsat | 3.2-bo'limdagi rollar matritsasi; ruxsat yo'q — `403`, kirmagan — `401` |
| Versiyalash | URL da `/v1/`; buzuvchi o'zgarish yangi `/v2/` ni talab qiladi; eski versiya kamida 6 oy ishlaydi |
| Sahifalash | Kursor bo'yicha: `?limit=50&cursor=<opaque>`; javobda `next_cursor` (oxirida `null`). `limit` ≤ 200 |
| Filtr va saralash | `?filter[status]=open&filter[severity]=critical&sort=-started_at` |
| Maydonlarni tanlash | `?fields=id,name,status` (ixtiyoriy) |
| Idempotentlik | O'zgartiruvchi `POST` da `Idempotency-Key: <uuid>` sarlavhasi qo'llab-quvvatlanadi (24 soat) |
| Tezlik cheklovi | Foydalanuvchi bo'yicha 120 so'rov/daqiqa; javob sarlavhalari `X-RateLimit-Limit`, `X-RateLimit-Remaining`, `Retry-After` |
| Hujjat | OpenAPI 3.1 sxemasi `/api/v1/schema/` da, Swagger UI `/api/v1/docs/` (faqat Owner/Operator) |
| Kodlash | `UTF-8`; matnlar `NFC` normallashtirilgan |
| Til | `Accept-Language: uz`, `ru` yoki `en` — xato va shablon matnlari uchun |

### 9.1.1. Xato formati

Barcha xatolar bir xil shaklda:

```
{
  "error": {
    "code": "validation_error",
    "message": "Ma'lumot noto'g'ri kiritilgan.",
    "details": [
      {"field": "fqdn", "code": "invalid", "message": "Domen nomi noto'g'ri."}
    ],
    "request_id": "req_01JA4E9Q3T7"
  }
}
```

| HTTP | `code` | Ma'nosi |
|---|---|---|
| 400 | `bad_request`, `validation_error` | So'rov noto'g'ri / maydon xato |
| 401 | `unauthenticated`, `invalid_token`, `token_revoked` | Kirmagan yoki token noto'g'ri |
| 403 | `forbidden`, `csrf_failed` | Ruxsat yo'q |
| 404 | `not_found` | Obyekt yo'q yoki ko'rishga ruxsat yo'q (mavjudligini oshkor qilmaydi); noma'lum `/api/`, `/ingest/`, `/agent/` yo'llari ham shu formatda |
| 409 | `conflict`, `duplicate` | Takroriy nom, joriy holat bilan nomuvofiqlik |
| 410 | `gone`, `enroll_token_expired` | Muddati o'tgan |
| 413 | `payload_too_large` | Hajm chegarasi (ingest: 1 MB) |
| 415 | `unsupported_media_type` | JSON emas |
| 422 | `unprocessable` | Mantiqiy xato (masalan, `ends_at < starts_at`) |
| 423 | `account_locked` | Ko'p xato kirishdan keyin |
| 429 | `rate_limited` | Tezlik chegarasi; `Retry-After` |
| 500 | `internal_error` | Kutilmagan xato; `request_id` loglarda qidiriladi |
| 503 | `unavailable` | Baza/navbat vaqtincha mavjud emas |

## 9.2. Autentifikatsiya va foydalanuvchi

| Metod | Yo'l | Rol | Tavsif |
|---|---|---|---|
| POST | `/auth/login` | — | `{login, password}` → sessiya cookie; 2FA yoqilgan bo'lsa `202 {"mfa_required":true,"mfa_token":"..."}` |

| Metod | Yo'l | Rol | Tavsif |
|---|---|---|---|
| POST | `/auth/mfa` | — | `{mfa_token, code}` → sessiya |
| POST | `/auth/logout` | hamma | Joriy sessiyani tugatadi |
| POST | `/auth/logout-all` | hamma | Barcha sessiyalardan chiqish |
| GET | `/auth/me` | hamma | Joriy foydalanuvchi, rol, til, vaqt zonasi, ruxsat etilgan loyihalar |
| PATCH | `/auth/me` | hamma | Til, vaqt zonasi, parol almashtirish (`current_password` bilan) |
| POST | `/auth/mfa/setup` | hamma | TOTP sirini va QR (`otpauth://`) ni qaytaradi |
| POST | `/auth/mfa/confirm` | hamma | `{code}` → 2FA yoqiladi, 10 ta zaxira kod qaytariladi (faqat bir marta) |
| GET | `/auth/sessions` | hamma | Faol sessiyalar |
| DELETE | `/auth/sessions/{id}` | hamma | Sessiyani tugatish |
| GET/POST | `/users/` | owner | Foydalanuvchilar ro'yxati / yaratish (`role`, `projects[]`) |
| GET/PATCH/DELETE | `/users/{id}` | owner | Ko'rish / o'zgartirish (rol, faollik) / o'chirish (aslida o'chirib qo'yish, `is_active=false`) |
| POST | `/users/{id}/unlock` | owner | Bloklangan hisobni ochish |

Namuna:

```
POST /api/v1/auth/login
Content-Type: application/json

{"login": "jasur", "password": "********"}
```

```
HTTP/1.1 200 OK
Set-Cookie: __Host-nzr_sid=...; HttpOnly; Secure; SameSite=Lax; Path=/

{"user": {"id": "0193...", "login": "jasur", "role": "owner", "language": "uz",
          "timezone": "Asia/Tashkent", "mfa_enabled": true}}
```

## 9.3. Serverlar

| Metod | Yo'l | Rol | Tavsif |
|---|---|---|---|
| GET | `/servers/` | hamma | Ro'yxat (filtr: `status`, `tag`, `q`) |
| POST | `/servers/` | owner, operator | Yangi server yaratish → enroll tokeni va o'rnatish buyrug'ini qaytaradi |
| GET | `/servers/{id}` | hamma | Tafsilot: reyestr maydonlari, `facts`, so'nggi qiymatlar, ochiq incidentlar soni |
| PATCH | `/servers/{id}` | owner, operator | Maydonlarni o'zgartirish |
| POST | `/servers/{id}/pause` | owner, operator | `{reason, until?}` → pauza (FR-HST-016) |
| POST | `/servers/{id}/resume` | owner, operator | Pauzani bekor qilish |
| POST | `/servers/{id}/archive` | owner | Arxivga o'tkazish |
| GET | `/servers/{id}/events` | hamma | Reboot, fakt o'zgarishi, yangilanish tarixi |
| GET | `/servers/{id}/containers` | hamma | Konteynerlar va oxirgi qiymatlari |
| GET | `/servers/{id}/services` | hamma | systemd servislar |
| GET | `/servers/{id}/filesystems` | hamma | Fayl tizimlari, bandlik, bashorat (to'lishga qolgan kun) |
| POST | `/servers/{id}/enroll-token` | owner | Yangi bir martalik enroll tokeni (1 soat amal qiladi) |
| GET | `/servers/{id}/tokens` | owner | Agent tokenlari ro'yxati (token qiymatisiz) |
| DELETE | `/servers/{id}/tokens/{token_id}` | owner | Tokenni bekor qilish |

Qoidalar (S4): `GET /servers/` — `status` (arxivlangan faqat `?status=archived` bilan), `tag`, `q` (nom, provayder, IP boshi). `PATCH` da `status` va `name` dan boshqa tizim maydonlari o'zgarmaydi; nom takrorlansa `409`. `pause`: `{reason (majburiy), until? (≤ 90 kun)}`, muddat tugagach evaluator o'zi davom ettiradi; allaqachon pauzada bo'lsa `409`. `archive`: tokenlar bekor qilinadi, server alert bermaydi, tarix saqlanadi. `tokens` javobida token qiymati yo'q (faqat xesh saqlanadi). `GET /servers/{id}` ga `open_incidents` qo'shilgan.

Namuna — server yaratish:

```
POST /api/v1/servers/
{"name": "main-2", "provider": "Contabo", "public_ipv4": "164.68.105.194",
 "role": "production", "tags": ["contabo", "docker"]}
```

```
201 Created
{
  "id": "0193c0de-7a4e-7b3a-9d41-2f6c8e5a1b90",
  "name": "main-2",
  "status": "new",
  "enroll": {
    "token": "enr_Zk3...",
    "expires_at": "2026-10-22T10:35:00Z",
    "install_command": "curl -fsSL https://nazoratchi.example.uz/install.sh | sudo bash -s -- --token enr_Zk3..."
  }
}
```

## 9.4. Loyihalar, domenlar, servislar

| Metod | Yo'l | Rol | Tavsif |
|---|---|---|---|
| GET | `/projects/` | hamma (L) | Birlashtirilgan ro'yxat (FR-REG-004): domenlar, serverlar, holat, oxirgi zaxira yoshi, eng yaqin sertifikat muddati |
| POST | `/projects/` | owner, operator | Loyiha yaratish (`domains[]` bilan birga — avtomatik tekshiruvlar yaratiladi) |
| GET | `/projects/{id}` | hamma (L) | Tafsilot |
| PATCH | `/projects/{id}` | owner, operator | O'zgartirish |
| POST | `/projects/{id}/archive` | owner | Arxivlash |
| GET | `/projects/{id}/status` | hamma (L) | Umumiy holat: domenlar (HTTP/TLS/DNS), konteynerlar, servislar, zaxira, incidentlar |
| PUT | `/projects/{id}/servers` | owner, operator | Serverlarga biriktirish (`[{server_id, environment, ports}]`); port to'qnashuvida `409` |
| GET | `/projects/ports/next` | owner, operator | `?server_id=` uchun keyingi bo'sh port juftligi |
| GET | `/domains/` | hamma (L) | Domenlar ro'yxati (`filter[days_left_lt]=21`, `filter[dns_ok]=false`) |
| POST | `/domains/` | owner, operator | Domen yaratish; `auto_probes=true` bo'lsa HTTP, TLS, DNS tekshiruvlari yaratiladi |
| GET/PATCH | `/domains/{id}` | hamma (L) / owner, operator | Ko'rish / o'zgartirish |
| GET | `/domains/{id}/dns` | hamma (L) | Resolverlar bo'yicha oxirgi natija va o'zgarishlar tarixi |
| POST | `/domains/{id}/dns/watch` | owner, operator | Tarqalish rejimi (FR-DNS-005): `{expect_ips:[...], until}` |
| PUT | `/domains/{id}/whois` | owner, operator | Domen muddatini qo'lda kiritish (`.uz` uchun) |
| GET | `/registry/export` | owner | Butun reyestr YAML/JSON (FR-REG-009) |
| POST | `/registry/import` | owner | Reyestrni import qilish (`?dry_run=true` bilan farqni ko'rsatadi) |

Qoidalar (S4):

- `POST /projects/` tanasi: loyiha maydonlari + ixtiyoriy `servers: [{server_id, environment, ports}]`, `domains: [{fqdn, ...}]`, `auto_probes` (standart `true`: ko'chirilgan loyihaning har domeni uchun HTTP tekshiruv). Hammasi bitta tranzaksiyada: biror domen takrorlansa loyiha ham yaratilmaydi (`409`). `slug` yaratilgandan keyin o'zgarmaydi.
- **Port to'qnashuvi:** faol loyihalar orasida bir serverda bir port ikki marta berilsa `409 port_conflict`, `details[]` da band qilgan loyiha va keyingi bo'sh port taklifi. Arxivlangan loyiha portni bo'shatadi. Portlar 1024–65535, bir biriktirishda ≤ 10 ta; berilgan tartib saqlanadi.
- **Keyingi bo'sh port** (`GET /projects/ports/next?server_id=`): `[A, B]`, A — 8000–8999 dagi eng kichik bo'sh port, B — `3000 + (A − 8000)`, u band bo'lsa 3000–3999 dagi eng kichik bo'sh. (REGISTRY.md qoidasining amalga oshirilgan talqini; hujjat bilan farq qilsa, shu yerni yangilang.)
- `domains`: `fqdn` kichik harfga keltiriladi, oxirgi nuqta olib tashlanadi, IP va bir bo'g'inli nomlar rad etiladi. Domen `archived` bo'lsa uning tekshiruvlari o'chadi. Loyiha arxivlansa, uning tekshiruvlari ham o'chadi (alert yo'q, tarix qoladi).
- `GET /projects/` standart holatda arxivlanganlarni ko'rsatmaydi; filtrlar: `status`, `q` (nom, slug, domen), `tag`, `server_id`, `moved`.
- S5 da qo'shilgan: `GET /domains/{id}/dns` (har resolver bo'yicha oxirgi natija va `changes[]` o'zgarishlar tarixi). Yaratilmagan: `/projects/{id}/status`, `/domains/{id}/dns/watch`, `/domains/{id}/whois` (F2), `/registry/export|import` (F2).

## 9.5. Metrikalar so'rovi

| Metod | Yo'l | Rol | Tavsif |
|---|---|---|---|
| GET | `/metrics/query` | hamma (L) | Vaqt qatorini olish |
| GET | `/metrics/latest` | hamma (L) | Server uchun oxirgi qiymatlar: `?server_id=&names=cpu_busy_percent,mem_used_percent` |
| GET | `/metrics/names` | hamma | Mavjud metrika nomlari va belgilar |
| GET | `/metrics/top` | hamma (L) | Eng yuklangan (CPU/RAM/disk) serverlar va konteynerlar |

`/metrics/query` parametrlari:

| Parametr | Majburiy | Tavsif |
|---|---|---|
| `server_id` | ha | Server |
| `name` | ha | Metrika nomi (`cpu_busy_percent`) |
| `labels` | yo'q | `labels[mount]=/` |
| `from`, `to` | ha | ISO vaqt; `to` ≤ hozir |
| `step` | yo'q | Soniya; ko'rsatilmasa avtomatik (`max_points=1000`) |
| `agg` | yo'q | `avg` (standart), `min`, `max`, `last` |
| `max_points` | yo'q | 100–2000, standart 1000 |

```
GET /api/v1/metrics/query?server_id=0193c0de-...&name=cpu_busy_percent&from=2026-10-22T08:00:00Z&to=2026-10-22T09:00:00Z
```

```
{
  "name": "cpu_busy_percent",
  "unit": "%",
  "step_seconds": 15,
  "source": "raw",
  "series": [
    {"labels": {}, "points": [[1761120000000, 11.8], [1761120015000, 12.4], [1761120030000, 13.0]]}
  ]
}
```

Qoidalar (S4-S5): `step` (soniya) berilsa nuqtalar soni `ceil(oraliq/step)` (≤ 2000); aks holda `max_points` (100–2000, standart 1000). Oraliq ≤ 400 kun, `to` kelajakda bo'lmasligi kerak (≤ 1 daqiqa tolerantlik), `from`/`to` vaqt zonasi bilan. `agg`: `avg|min|max|last`. Ko'pi bilan 50 seriya. **Bo'shliq:** qo'shni nuqtalar orasi `max(3 qadam, 2 daqiqa)` dan uzoq bo'lsa, orasiga `null` nuqta qo'yiladi (agent jim bo'lgan davr grafikda uziladi, oddiy 15-30 s namuna oralig'i bo'shliq emas). `GET /metrics/latest?server_id=&names=a,b` — har seriyaning oxirgi qiymati. `GET /metrics/names[?server_id=]` — nom, birlik, tur, belgi kalitlari. `GET /metrics/top?metric=&limit=` — oxirgi 5 daqiqadagi qiymat bo'yicha kamayish tartibida (arxivlangan serverlar yashirin). `GET /servers/{id}/containers` — konteynerlar va oxirgi qiymatlari (5 daqiqadan eskisi `stale: true`); `GET /servers/{id}/filesystems` — bandlik va `days_until_full` (oxirgi 6 soat, $R^2 \geq 0.5$ bo'lsa).

`points` — `[vaqt_ms, qiymat]` juftliklari (ixcham). `source` — qaysi daraja ishlatilgani: `raw`, `1m`, `5m`, `1h` (8.4.3). Ma'lumot bo'lmagan oraliqlar `null` qiymat bilan beriladi (grafikda bo'shliq).

## 9.6. Faol tekshiruvlar

| Metod | Yo'l | Rol | Tavsif |
|---|---|---|---|
| GET | `/probes/` | hamma (L) | Ro'yxat (filtr: `type`, `state`, `project_id`) |
| POST | `/probes/` | owner, operator | Yaratish |
| GET | `/probes/{id}` | hamma (L) | Tafsilot va oxirgi natija |
| PATCH | `/probes/{id}` | owner, operator | O'zgartirish (interval, chegara, yoqish/o'chirish) |
| DELETE | `/probes/{id}` | owner | O'chirish (aslida arxiv) |
| POST | `/probes/{id}/run` | owner, operator | Zudlik bilan bir marta bajarish (natija javobda) |
| GET | `/probes/{id}/results` | hamma (L) | Natijalar tarixi (`from`, `to`, `limit`) |
| GET | `/probes/{id}/uptime` | hamma (L) | Uptime %: 24 soat, 7, 30, 90 kun |
| GET | `/certs/` | hamma (L) | Barcha sertifikatlar va qolgan kun |

Namuna — HTTP tekshiruv yaratish:

```
POST /api/v1/probes/
{
  "type": "http",
  "project_id": "0193...",
  "domain_id": "0193...",
  "name": "Nazar healthz",
  "interval_s": 60,
  "timeout_s": 10,
  "config": {
    "url": "https://nazar.avlo.ai/healthz/",
    "method": "GET",
    "expected_status": [200],
    "keyword": null,
    "follow_redirects": true,
    "max_redirects": 5,
    "headers": {"X-Check": "nazoratchi"},
    "latency_warn_ms": 3000
  }
}
```

Konfiguratsiya sxemalari (tur bo'yicha) 18.4-bo'limda.

Qoidalar (S4): hozircha `http` va `tcp` turlari (qolganlari `400`). `DELETE` tekshiruvni o'chirmaydi: `enabled=false`, `state=paused` (tarix saqlanadi); qayta yoqilganda holat `unknown` dan boshlanadi. `config.allow_private=true` (ichki manzillarga ruxsat, SSRF) faqat Owner tomonidan o'rnatiladi. `POST /probes/{id}/run` — bir marta bajaradi, natijani qayd etadi va tasdiqlash qoidasini qo'llaydi. `GET /probes/{id}/results`: `from`, `to`, `limit` (≤ 500). `GET /probes/{id}/uptime`: 1, 7, 30, 90 kun. `GET /certs/` — sertifikatlar va `days_left` (`?days_left_lt=21`, `?project_id=`), muddati yaqinlari birinchi. TLS va DNS turlari endi qo'llab-quvvatlanadi (`tls`: 18.4.3, `dns`: 18.4.4; DNS uchun `domain` majburiy emas, lekin tarix faqat domenli tekshiruvlar uchun yoziladi); `whois` — F2.

## 9.7. Zaxira nusxalar

| Metod | Yo'l | Rol | Tavsif |
|---|---|---|---|
| GET | `/backups/overview` | hamma (L) | Barcha loyihalar bo'yicha jadval: oxirgi zaxira, yoshi, hajmi, offsite, oxirgi drill |
| GET | `/backups/projects/{project_id}` | hamma (L) | Loyiha zaxira kartasi: oxirgi 30 ta run, hajm grafigi, holat |
| GET | `/backups/runs` | hamma (L) | Run'lar ro'yxati (`project`, `status`, `from`, `to`) |
| GET | `/backups/offsite` | hamma (L) | Mahalliy va offsite solishtirish |
| GET | `/backups/drills` | hamma (L) | Restore drill tarixi |
| POST | `/backups/drills` | owner, operator | Drill natijasini kiritish |
| PATCH | `/backups/drills/{id}` | owner, operator | Tuzatish |
| GET/PUT | `/backups/policy` | owner | Har loyiha uchun siyosat: kutilgan yosh (26 s), hajm chegarasi, drill davriyligi (90 kun) |

> **F1 S10 da qo'shildi:** `GET/POST /backups/drills` (filtr `project`, `result`), `GET/PATCH /backups/drills/{id}`; `drills` yo'li `{slug}` dan oldin. Tanasi: `project` (slug), `performed_at`, `backup_file`, `backup_age_h`, `duration_s`, `tables_count`, `rows_count`, `result` (`success|partial|failed`), `notes`; kelajak sanasi rad etiladi, loyihani PATCH bilan o'zgartirib bo'lmaydi, har yozuv audit'ga tushadi. `GET /backups/` va `GET /backups/{slug}` javobida `drill` bloki: `every_days` (siyosatdagi `drill_every_days`, standart 90), `last`, `last_success_days`, `never`, `overdue`; karta qo'shimcha `drills` (oxirgi 20). AL-081…083 hali F2. **S6 da amalga oshirilgan (soddalashtirilgan yo'llar):** `GET /backups/` (jadval) va `GET /backups/{slug}` (karta). Yuqoridagi `/backups/overview`, `/projects/{id}`, `/runs`, `/offsite`, `/drills`, `/policy` — F2. Siyosat hozircha `PATCH /projects/{id}` dagi `backup` maydoni orqali. `GET /backups/` elementi: `project, project_id, name, server, server_id, expect_daily, expect_weekly_files, require_offsite, state, age_hours, last_success_at, last_size_bytes, last_status (ok|fail|null), daily_count, weekly_count, files_count, newest_age_seconds{kind}, offsite_configured (server darajasida), data_seen_at`. `state`: `ok`, `late` (> 26 s), `critical` (> 50 s), `failed` (oxirgi zaxira xato), `stale` (agent 15 daqiqadan beri jim: ma'lumot ishonchsiz), `missing` (kutilgan, lekin hech qachon olinmagan, loyiha ≥ 26 soatlik), `unknown` (yangi loyiha), `none` (zaxira kutilmaydi). Ro'yxatga faqat faol, ko'chirilgan va siyosati bor yoki agent ma'lumot yuborayotgan loyihalar kiradi. Karta qo'shimcha: `runs[]` (oxirgi 30, eskidan yangiga: `finished_at, size_bytes`), `size_median_7`, `size_ratio`, `size_anomaly`.

Namuna — `GET /backups/overview` javobi:

```
{
  "generated_at": "2026-10-22T09:35:00Z",
  "items": [
    {
      "project": {"id": "0193...", "slug": "BiznesRadar", "name": "BiznesRadar"},
      "server": "main-2",
      "last_success_at": "2026-10-22T03:15:41Z",
      "age_hours": 6.3,
      "size_bytes": 48273114,
      "gz_ok": true,
      "daily_count": 7, "weekly_count": 5,
      "offsite": {"configured": true, "last_success_at": "2026-10-22T03:16:55Z", "in_sync": true},
      "last_drill": {"performed_at": "2026-09-30T10:00:00Z", "result": "success", "days_ago": 22},
      "status": "ok"
    },
    {
      "project": {"slug": "lutfan_ai"},
      "age_hours": 31.2,
      "status": "stale",
      "problems": ["backup_too_old"]
    }
  ]
}
```

`status` qiymatlari: `ok`, `warning` (offsite yo'q, drill muddati o'tgan), `stale` (yosh > chegara), `failed` (oxirgi run `fail`), `suspicious` (hajm anomaliyasi), `unknown`.

## 9.8. Alertlar va incidentlar

| Metod | Yo'l | Rol | Tavsif |
|---|---|---|---|
| GET | `/alert-rules/` | hamma | Qoidalar (filtr: `enabled`, `severity`, `built_in`) |
| POST | `/alert-rules/` | owner, operator | Yangi qoida |
| GET/PATCH | `/alert-rules/{id}` | hamma / owner, operator | Ko'rish / o'zgartirish |
| POST | `/alert-rules/{id}/test` | owner, operator | Qoidani hozirgi ma'lumotda quruq hisoblash (kim "firing" bo'lardi), alert yubormasdan |
| GET | `/alerts/` | hamma (L) | Faol alert instance lar (`pending`/`firing`/`resolving`) |
| GET | `/incidents/` | hamma (L) | Incidentlar (`filter[status]`, `severity`, `server_id`, `project_id`, `from`, `to`) |
| GET | `/incidents/{id}` | hamma (L) | Tafsilot: alertlar, ta'sirlanganlar, vaqt chizig'i |
| POST | `/incidents/{id}/ack` | owner, operator | Qabul qilish (`{comment?}`) |
| POST | `/incidents/{id}/comments` | owner, operator | Izoh |
| PATCH | `/incidents/{id}` | owner, operator | `root_cause` ni kiritish |
| POST | `/incidents/{id}/resolve` | owner, operator | Qo'lda yopish (faqat shart allaqachon yo'qolgan bo'lsa; aks holda `409`) |
| GET/POST | `/silences/` | hamma / owner, operator | Ro'yxat / yaratish (`matchers`, `ends_at`, `reason`) |
| DELETE | `/silences/{id}` | owner, operator | Bekor qilish |
| GET/POST | `/maintenance/` | hamma / owner, operator | Ro'yxat / yaratish |
| DELETE | `/maintenance/{id}` | owner, operator | Bekor qilish |

Namuna — silence:

```
POST /api/v1/silences/
{
  "matchers": [{"label": "server", "op": "=", "value": "main-2"},
               {"label": "severity", "op": "!=", "value": "critical"}],
  "ends_at": "2026-10-22T12:00:00Z",
  "reason": "Kernel yangilanishi va reboot"
}
```

## 9.9. Bildirishnoma kanallari

| Metod | Yo'l | Rol | Tavsif |
|---|---|---|---|
| GET | `/channels/` | owner | Kanallar (maxfiy maydonlarsiz: `bot_token: "••••"`) |
| POST | `/channels/` | owner | Yaratish |
| PATCH/DELETE | `/channels/{id}` | owner | O'zgartirish / o'chirish |
| POST | `/channels/{id}/test` | owner | Sinov xabari (FR-ALR-020) |

| Metod | Yo'l | Rol | Tavsif |
|---|---|---|---|
| GET | `/notifications/` | owner, operator | Yuborilgan xabarlar tarixi (muvaffaqiyat/xato); `?channel=`, `?status=`, `?before=<id>` |
| POST | `/telegram/webhook/{secret}` | Telegram | Bot callback (9.12) |

Qoidalar (F1 S2):

- Yaratishda `config` yoziladi: `{"bot_token": "123456:ABC...", "chat_id": -1001234567890}` (`chat_id` — butun son yoki `@kanal_nomi`). Javobda `config` **hech qachon** qaytmaydi, o'rniga `config_masked`: `{"bot_token": "••••", "chat_id": ...}`. Audit jurnaliga ham sir tushmaydi.
- `PATCH` da `config` qisman bo'lishi mumkin: berilmagan kalit (masalan, `bot_token`) eskicha qoladi. Kanal turini o'zgartirib bo'lmaydi. Hozircha faqat `telegram` (email F2, webhook F3) — qolganlari `400`.
- `POST /channels/{id}/test`: bitta urinish, qayta urinishsiz. Muvaffaqiyatda `200 {"ok": true}`, Telegram xatosida `502 {"ok": false, "error": "..."}` (xato matnida token bo'lmaydi).
- `GET /notifications/`: 50 tadan, yangisidan eskisiga; keyingi sahifa uchun `next_before` qiymatini `?before=` ga bering (`null` — oxiri). Xabar matni va sirlar tarixda qaytmaydi.

## 9.10. Hisobotlar, audit, tizim

| Metod | Yo'l | Rol | Tavsif |
|---|---|---|---|
| GET | `/reports/uptime` | hamma (L) | Oy/chorak uptime (`from`, `to`, `group=project` yoki `group=domain`) |
| GET | `/reports/incidents` | hamma (L) | Incident statistikasi: soni, MTTR, ahamiyat bo'yicha |
| GET | `/reports/backups` | hamma (L) | Zaxira muvaffaqiyat foizi |
| GET | `/reports/weekly` | owner, operator | Haftalik hisobot (JSON/PDF: `Accept: application/pdf`) |
| GET | `/audit/` | owner | Audit jurnali (`user`, `action`, `from`, `to`) |
| GET | `/settings/` | owner | Tizim sozlamalari (retention, standart kanallar, chegaralar) |
| PATCH | `/settings/` | owner | O'zgartirish |
| GET | `/system/status` | owner, operator | Nazoratchi o'zining holati: evaluator, prober, navbat, baza hajmi, oxirgi zaxira |
| GET | `/dashboard/summary` | hamma (L) | Bosh sahifa ma'lumoti: banner holati, sonlar, eng yaqin muddatlar |
| GET | `/stream` | hamma (L) | SSE oqimi (9.11) |

## 9.11. SSE oqimi

```
GET /api/v1/stream?scope=dashboard
Accept: text/event-stream
```

Voqealar (`event:` turi va `data:` JSON):

| Voqea | Qachon | `data` |
|---|---|---|
| `host.status` | Host holati o'zgarganda | `{"server_id":"...","status":"offline","since":"..."}` |
| `host.metrics` | Har 5 s (partiya) | `{"server_id":"...","cpu":12.4,"mem":41.8,"disk":23.1}` |
| `incident.opened` / `incident.updated` / `incident.resolved` | Incident o'zgarganda | Incident qisqa obyekti |
| `probe.state` | Tekshiruv holati o'zgarganda | `{"probe_id":"...","state":"down"}` |
| `backup.updated` | Zaxira holati yangilanganda | `{"project":"BiznesRadar","status":"ok"}` |
| `ping` | Har 25 s | `{}` (aloqani ushlab turish) |

`scope`: `dashboard`, `server:<id>`, `project:<id>`, `incidents`. Cheklangan viewer faqat o'z loyihalariga tegishli voqealarni oladi. `Last-Event-ID` bilan qayta ulanganda o'tkazib yuborilgan voqealar (5 daqiqagacha) qayta yuboriladi.

## 9.12. Telegram webhook

`POST /api/v1/telegram/webhook/{secret}` — Telegram yuboradi. Talablar:

1. `X-Telegram-Bot-Api-Secret-Token` sarlavhasi sozlamadagi qiymatga teng bo'lishi kerak.
2. `callback_query.from.id` / `message.chat.id` ruxsat etilgan chat ID lar ro'yxatida bo'lishi kerak.

3. `callback_data` formati: `ack:<incident_id>:<sig>`, `snz:<incident_id>:<daqiqa>:<sig>`. `sig` — HMAC-SHA256 ning dastlabki 12 hex belgisi (kalit `SECRET_KEY` dan olingan; butun qiymat Telegram chegarasi 64 baytga sig'adi). Muddat 15 daqiqadan 24 soatgacha. Chat ID kanallardan birining `chat_id` siga teng bo'lishi shart. Bosgan odam `users.telegram_chat_id` bilan bog'langan bo'lsa, uning roli tekshiriladi (`viewer` rad etiladi, audit shu foydalanuvchi nomidan); bog'lanmagan bo'lsa chat ruxsati yetarli va amal `telegram:<username>` deb yoziladi. "1 soat jim" qamrovi: host incident'i — `server`, loyiha incident'i — `project`, qolgani — `rule`. Maxfiy kalit `TELEGRAM_WEBHOOK_SECRET`: URL segmenti VA sarlavha ikkalasi ham shunga teng bo'lishi kerak; bo'sh bo'lsa webhook o'chiq. O'rnatish: `setWebhook(url=https://<core>/api/v1/telegram/webhook/<secret>, secret_token=<secret>)`.
4. Javob har doim `200` (Telegram qayta yubormasligi uchun); ruxsatsiz so'rov faqat loglanadi.

Qo'llab-quvvatlanadigan bot buyruqlari (F2):

| Buyruq | Natija |
|---|---|
| `/status` | Qisqa umumiy holat (serverlar, ochiq incidentlar) |
| `/incidents` | Ochiq incidentlar |
| `/silence <server> <daqiqa> <sabab>` | Jim qilish |
| `/backups` | Zaxira holati |
| `/help` | Yordam |

## 9.13. Agent (ingest) endpointlari

Protokol tafsiloti 7.5-bo'limda. Qisqa jadval:

| Metod | Yo'l | Autentifikatsiya | Tavsif |
|---|---|---|---|
| POST | `/ingest/v1/enroll` | `enroll_token` (tanada) | Enroll |
| POST | `/ingest/v1/metrics` | Bearer `agt_` | Metrika paketi |
| POST | `/ingest/v1/goodbye` | Bearer `agt_` | Agent o'chirildi |
| GET | `/agent/v1/latest` | — | `{version, sha256, url}` (oxirgi agent versiyasi) |
| GET | `/agent/v1/nazoratchi-agent-{arch}` | — | Agent binari |
| GET | `/install.sh` | — | O'rnatish skripti |

## 9.14. Kiruvchi heartbeat (cron va skriptlar uchun)

```
GET|POST https://<core>/hb/<token>[?status=ok|fail][&msg=<qisqa matn>][&duration=<s>]
```

- `token` — `heartbeat_jobs` yaratilganda berilgan maxfiy kalit (URL da; shuning uchun alohida, tor ruxsatli).
- `status` ko'rsatilmasa `ok`. `fail` — darhol alert (`AL-048`).
- `msg` 200 belgigacha; maxfiy ma'lumot yubormaslik ogohlantiriladi.
- Javob: `200 ok` (matn), tezkor (< 50 ms), tezlik cheklovi: token bo'yicha daqiqada 10.

Namuna — `appctl-backup` uchun (E-03): cron qatori o'zgarmaydi, heartbeat'ni `appctl` o'zi yuboradi (10.3.5). Token `appctl.secret` ga yoziladi:

```
sudo bash setup.sh --heartbeat-url=https://nazoratchi.example.uz/hb/<TOKEN>   # yoki: appctl heartbeat-set <url>
appctl heartbeat ok                                                           # qo'lda sinash
```

```
15 3 * * * root /usr/local/bin/appctl backup all >> /var/log/appctl-backup.log 2>&1
```

Eski namunadagi \ bilan ko'p qatorli `&& curl ... || curl ...` zanjiri `/etc/cron.d` da ishlamaydi (qator bo'linmaydi) va faqat `curl` yiqilganda ham noto'g'ri `fail` yuboradi, shuning uchun almashtirildi.

## 9.15. Tizim endpointlari

| Metod | Yo'l | Kirish | Javob |
|---|---|---|---|
| GET | `/healthz` | ichki/tashqi watchdog | Liveness: `200 {"status":"ok"}` yoki `503` (baza/navbat) |
| GET | `/readyz` | ichki | Tayyorlik: baza, Redis, notifier va evaluator yurak urishi (tsikl yangi) tekshiriladi; biri yiqilsa `503` |
| GET | `/metrics` | `METRICS_TOKEN` yoki faqat loopback | Prometheus matni (NFR-OBS-001); nginx tashqariga bermaydi |
| GET | `/api/v1/version` | hamma | `{"version":"1.0.0","build":"...","agent_latest":"1.0.0"}` |

> **S7 da amalga oshirilgan:** evaluator har tsikl muvaffaqiyatli tugagach (kamida 60 s oralig'ida) `WATCHDOG_URL` (`https://<host>/ping/<id>`) ga ping yuboradi; tsikl yiqilsa ping to'xtaydi va watchdog 5 daqiqada xabar beradi (AL-084). `WATCHDOG_URL` `https://` bo'lishi shart (maxfiy id ochiq ketmasligi uchun; loopback `http` mumkin). Tarqatish: `GET /install.sh` (`__CORE_URL__`, `__MINISIGN_PUBKEY__` almashtiriladi; kalit yo'q bo'lsa `503`), `GET /agent/v1/<nazoratchi-agent-amd64|arm64|SHA256SUMS|SHA256SUMS.minisig>` (oq ro'yxat).

## 9.16. Moslik va o'zgarishlar siyosati

- Javobga yangi maydon qo'shish — buzuvchi emas (klient noma'lum maydonlarni e'tiborsiz qoldirishi kerak).
- Maydonni o'chirish yoki ma'nosini o'zgartirish — buzuvchi, yangi versiya.
- Agent protokoli alohida versiyalanadi (`/ingest/v1/`); eski agentlar kamida 2 minor versiya davomida qo'llab-quvvatlanadi (NFR-MNT-009).
- Barcha o'zgartiruvchi so'rovlar audit jurnaliga yoziladi (FR-AUD-001).

# 10. Zaxira nusxalar (backup) monitoringi

## 10.1. Nima uchun alohida bo'lim

Zaxira — Nazoratchi'ning eng muhim qiymat beradigan qismi, chunki bu yerda xato **jim** bo'ladi: zaxira ishlamayotgani faqat kerak bo'lgan kuni, tiklash paytida ma'lum bo'ladi. 2026-10 hodisasida aynan shu xavf amalga oshishiga oz qoldi: zaxiralar faqat o'sha serverning o'zida edi.

Bu bo'limning printsipi: **"tekshirilmagan zaxira — zaxira emas".** Nazoratchi to'rt savolga doim javob berishi kerak:

1. **Bormi?** Har loyiha uchun yaqinda zaxira olinganmi? (yosh)
2. **To'g'rimi?** Fayl butun va mazmunli ko'rinadimi? (hajm, yaxlitlik)
3. **Boshqa joydami?** Server o'lsa, nusxa boshqa serverda bormi? (offsite)
4. **Tiklasa bo'ladimi?** Oxirgi marta qachon haqiqatan tiklab ko'rilgan? (restore drill)

## 10.2. Chegaralar: kim nima qiladi

| Vazifa | Mas'ul |
|---|---|
| Zaxira olish (`pg_dump`, `tar`), saqlash muddati, offsite `rsync` | `appctl` (mavjud) |
| Zaxira fayllarini va holatini kuzatish, anomaliyani aniqlash, alert | **Nazoratchi** |
| Tiklash testining natijasini qayd etish, eslatish | **Nazoratchi** |
| Tiklash testining o'zini bajarish | Operator (qo'lda) yoki `appctl drill` (F3, 10.8) |
| Haqiqiy ma'lumotni tiklash (avariya) | Operator (`appctl restore`, 10.10) |

Nazoratchi zaxirani **o'zi olmaydi** va hech qachon `appctl restore` ni avtomatik ishga tushirmaydi (`restore` mavjud bazani o'chiradi; X-7).

## 10.3. `appctl` ga kerakli o'zgartirishlar

Hozirgi `appctl backup` natijani faqat terminal chiqishi va `/var/log/appctl-backup.log` ga yozadi. Nazoratchi uchun mashinada o'qiladigan natija kerak. Quyidagi ishlar Nazoratchi loyihasining **F1 oldi shartidir** (17-bo'lim, ish E-01..E-03).

### 10.3.1. Loyiha holati fayli: `state.json`

Har `appctl backup <loyiha>` oxirida (muvaffaqiyat ham, xato ham) yoziladi: `/opt/applications/backups/<loyiha>/state.json`. Yozish atomik (`.tmp` → `mv`).

```
{
  "schema": 1,
  "project": "BiznesRadar",
  "host": "vmi3679093",
  "status": "ok",
  "started_at": "2026-10-22T03:15:01Z",
  "finished_at": "2026-10-22T03:15:41Z",
  "duration_s": 40,
  "daily": {
    "status": "ok",
    "file": "BiznesRadar-20261022-031501.sql.gz",
    "size_bytes": 48273114,
    "error": null
  },
  "weekly_files": {
    "status": "ok",
    "file": "BiznesRadar-files-20261018-031501.tar.gz",
    "size_bytes": 1204331520,
    "error": null
  }
}
```

Qoidalar:

- `status` — `ok` faqat barcha bajarilgan qadamlar muvaffaqiyatli bo'lsa; aks holda `fail` va `error` da bir qatorli sabab (maks. 300 belgi, yangi qatorsiz).
- `weekly_files` — haftalik fayl zaxirasi (yakshanba) bo'lmagan kunlarda `null`.
- `daily` — loyihada `DB_SERVICE` yo'q bo'lsa (baza dumpi olinmaydi) `null`. Dump xato bo'lsa `status: "fail"`, `file: null`, `size_bytes: 0`, `error` to'ldiriladi.
- Offsite natijasi `state.json` da **emas**, `_run.json` da (`rsync` barcha loyihalar uchun bir marta bajariladi).
- Haftalik `tar` xatosi ham `status: "fail"` beradi va `appctl backup` chiqish kodi 1 bo'ladi (asl skript bu holda jimgina 0 qaytarardi; E-01 da aniqlangan).

- `error`: faqat bosma ASCII, maks. 300 belgi, `token|secret|password|authorization` qiymatlari `***` bilan yopiladi.
- Vaqt — UTC, ISO-8601.
- `error` matnida parol/token bo'lishi mumkin emas (`pg_dump` xato matni odatda bunday ma'lumot bermaydi; baribir agent yuborishdan oldin `NFR-SEC-008` regex'i bilan maskalaydi).

### 10.3.2. Umumiy ishga tushirish holati: `_run.json`

`appctl backup all` (yoki cron) oxirida `/opt/applications/backups/_run.json`:

```
{
  "schema": 1,
  "host": "vmi3679093",
  "run_id": "20261022-031501",
  "started_at": "2026-10-22T03:15:01Z",
  "finished_at": "2026-10-22T03:17:12Z",
  "projects": [
    {"name": "BiznesRadar", "status": "ok"},
    {"name": "Nazar", "status": "ok"},
    {"name": "lutfan_ai", "status": "ok"}
  ],
  "offsite": {
    "configured": true,
    "status": "ok",
    "target_host": "185.214.135.9",
    "exit_code": 0,
    "finished_at": "2026-10-22T03:17:12Z",
    "bytes_sent": 51237892
  }
}
```

`offsite.configured=false` bo'lsa, `status` yo'q (`null`) — bu AL-075 ni ishga tushiradi.

### 10.3.3. Yozish kodi (appctl ga qo'shiladigan eskiz)

```
json_escape() {
  local s=${1//\\/\\\\}
  s=${s//\"/\\\"}; s=${s//$'\n'/ }; s=${s//$'\t'/ }
  printf '%s' "${s:0:300}"
}

# write_state <loyiha> <status> <started> <kundalik_fayl> <hajm> <xato>
write_state() {
  local dir="$BACKUPS/$1" err="null"
  [ -n "${6:-}" ] && err="\"$(json_escape "$6")\""
  printf '{"schema":1,"project":"%s","host":"%s","status":"%s","started_at":"%s","finished_at":"%s","daily":{"status":"%s","file":"%s","size_b
    "$1" "$(hostname)" "$2" "$3" "$(date -u +%FT%TZ)" "$2" "${4:-}" "${5:-0}" "$err" \
    > "$dir/state.json.tmp" && mv "$dir/state.json.tmp" "$dir/state.json"
}
```

> Bu eskiz yo'nalishni ko'rsatadi; haftalik fayl bloki va davomiylik hisobi amalga oshirishda qo'shiladi. `jq` ga bog'liq bo'lmaslik uchun (serverlarda bo'lmasligi mumkin) JSON `printf` bilan yoziladi.
>
> **Amalga oshirildi (E-01):** `deploy/appctl-patches/appctl` (README va 20 ta sinov bilan). Eskizdagi `json_escape` tuzatildi: 300 belgida kesishdan *keyin* qochirish `\"` ni bo'lib, yaroqsiz JSON berishi mumkin edi (avval kesiladi, keyin qochiriladi); ASCII bo'lmagan baytlar va boshqaruv belgilari yaroqsiz JSON/UTF-8 hosil qilardi (bosma ASCII ga keltiriladi); maxfiy qiymatlar yopiladi.

### 10.3.4. `appctl list --json` va `appctl status --json`

Ikkala buyruq ham `--json` bayrog'ini qo'llab-quvvatlaydi (sxema 18.1-bo'limda). Nazoratchi agenti `appctl status --json` ni 5 daqiqada bir marta chaqiradi (yoki zaxira/sertifikat ma'lumotini fayllardan o'zi o'qiydi; ikkala usul ham F1 da mumkin, ustuvor — fayllar, chunki ular `appctl` bajarilishiga bog'liq emas).

### 10.3.5. Cron heartbeat

`appctl backup all` oxirida Nazoratchi heartbeat'i yuboradi (9.14-bo'lim). Bu `state.json` dan mustaqil ikkinchi signal: cron umuman ishlamasa (masalan, `crond` o'chiq yoki fayl o'chirilgan), `state.json` yangilanmaydi va heartbeat ham kelmaydi — AL-047 va AL-068 birgalikda aniqlaydi.

**Amalga oshirildi (E-03), TZ dagi dastlabki dizayndan farqli.** Heartbeat cron qatoriga emas, `appctl` ichiga qo'yildi (`/etc/cron.d/appctl-backup` o'zgarmaydi). Sabablari:

1. `/etc/cron.d` qatorini `\` bilan bir necha qatorga bo'lib bo'lmaydi (cron buni qo'llab-quvvatlamaydi), 9.14 dagi eski namuna cron'da ishlamasdi.
2. `A && curl ok || curl fail` zanjiri zaxira muvaffaqiyatli bo'lib, faqat birinchi `curl` yiqilsa, noto'g'ri "fail" yuborardi.
3. Token URL ichida; `/etc/cron.d` hamma o'qiy oladigan (0644) fayl, `appctl.conf` ni esa agent o'qiydi (7.8). Token `/opt/applications/appctl.secret` da (0600, `HEARTBEAT_URL`).

Xatti-harakat (barchasi sinalgan, `deploy/appctl-patches/`):

- Faqat `backup all` (cron ishi) yuboradi; bitta loyiha zaxirasi yubormaydi. Muvaffaqiyatda `status=ok&duration=<s>`, xatoda `status=fail&duration=<s>&msg=fail: <loyihalar>[,offsite]` (loyiha nomlari, maxfiy ma'lumotsiz).
- **R-109:** heartbeat zaxira natijasini, fayllarni va chiqish kodini hech qachon o'zgartirmaydi. Xato bo'lsa (core yetib bo'lmaydi, 500, `curl` yo'q) faqat stderr'ga ogohlantirish; URL (token) chop etilmaydi. `curl --connect-timeout 5 -m 10` (`HEARTBEAT_TIMEOUT`).
- Kutilmagan to'xtash (SIGTERM/SIGINT/SIGHUP) EXIT tuzog'i orqali bitta `fail` yuboradi (`... to'xtatildi (kod 143)`): jimlik (AL-047) o'rniga tezroq AL-048.
- `appctl heartbeat [ok|fail] [xabar]` — sozlamani qo'lda sinash; `appctl heartbeat-set <https-url>` URL ni qat'iy tekshirib (bash o'qiydigan fayl!) `appctl.secret` ga yozadi; `setup.sh --heartbeat-url=...` (yoki `NAZORATCHI_HEARTBEAT_URL`) shuni chaqiradi, noto'g'ri URL bo'lsa nginx o'zgartirilmasdan to'xtaydi.

E-03 da topilgan: asl skriptda buzuq `.appctl backup all` ichida jimgina o'tib ketardi (dump o'tkazib yuborilib, loyiha `ok`). Endi shu loyiha `fail` bo'ladi, qolganlari zaxiralanadi.

## 10.4. Loyiha zaxira holati: qoidalar

Har loyiha uchun holat quyidagi tartibda (yuqoridan pastga, birinchi mos kelgani) hisoblanadi:

| Tartib | Holat | Shart | Rang |
|---|---|---|---|
| 1 | `failed` | oxirgi `state.json.status = fail` | qizil |
| 2 | `stale` | eng yangi muvaffaqiyatli kunlik zaxira yoshi > `max_age_h` (standart 26) | qizil |
| 3 | `suspicious` | hajm anomaliyasi (10.5.2) yoki `gz_ok = false` | qizil/to'q sariq |
| 4 | `warning` | offsite sozlanmagan yoki orqada; drill muddati o'tgan; haftalik fayl yo'q | sariq |
| 5 | `ok` | yuqoridagilarning hech biri | yashil |
| — | `unknown` | agent jim yoki loyiha uchun zaxira siyosati belgilanmagan | kulrang |

Eslatma: `failed` ham, `stale` ham bo'lishi mumkin; UI barcha muammolarni ro'yxat qilib ko'rsatadi, eng og'iri belgilangan rangni beradi.

### 10.4.1. Siyosat (har loyiha uchun sozlanadi)

| Parametr | Standart | Izoh |
|---|---|---|
| `expect_daily` | `true` | Kunlik DB dump kutiladi (statik saytlar uchun `false`) |
| `max_age_h` | 26 | 24 soat + 2 soat zaxira (cron kechikishi, katta dump) |
| `critical_age_h` | 50 | Ikki kun o'tkazib yuborilgan |
| `expect_weekly_files` | `BACKUP_PATHS` bor bo'lsa `true` | Yakshanba tar.gz |
| `weekly_max_age_h` | 192 | 8 kun |
| `size_floor_bytes` | 1 024 | `appctl` ning o'z chegarasi |
| `size_min_ratio` | 0.5 | Median'ning 50% idan kichik — shubhali |
| `size_max_ratio` | 3.0 | Median'ning 300% idan katta — shubhali |
| `require_offsite` | `true` | Offsite sinxron kutiladi |
| `drill_every_days` | 90 | Tiklash testi davriyligi |
| `retention_daily` / `retention_weekly` | 7 / 5 | `appctl` bilan mos |

## 10.5. Hisoblash qoidalari (aniq)

### 10.5.1. Zaxira yoshi

`age_h = (now − finished_at_of_latest_successful_daily) / 3600`.

"Eng yangi muvaffaqiyatli" manbalari ustuvorlik tartibida:

1. `state.json` (`daily.status = ok` → `finished_at`);
2. agar `state.json` yo'q yoki eskirgan — papka skaneri: eng yangi `*.sql.gz` (`.tmp` siz) `mtime` si.

Ikkala manba farq qilsa (masalan, `state.json` "ok", lekin fayl yo'q), **fayl** haqiqat manbai: `backup_state_mismatch` belgisi qo'yiladi va holat `suspicious`.

**Soat zonasi.** Serverlar `UTC` da; cron `03:15` = `08:15 Asia/Tashkent`. UI vaqtni foydalanuvchi zonasida ko'rsatadi; hisoblash UTC da.

### 10.5.2. Hajm anomaliyasi

- Asos: oxirgi 7 ta **muvaffaqiyatli kunlik** zaxira hajmlarining medianasi (`M`).
- Yangi zaxira hajmi `S` bo'lsa:
  - `S < size_floor_bytes` → `failed` (appctl o'zi qabul qilmaydi, ammo tekshiriladi);
  - `S < M × size_min_ratio` → "shubhali (juda kichik)" (AL-071);
  - `S > M × size_max_ratio` → "shubhali (juda katta)" (AL-072, `info`).
- 7 tadan kam tarix bo'lsa (yangi loyiha) — faqat `size_floor_bytes` tekshiriladi.
- Qo'lda "bu normal" deb belgilash mumkin (`baseline_reset`): bazada katta ma'lumot o'chirilgan/qo'shilgan bo'lsa, tarix qayta boshlanadi.

### 10.5.3. Yaxlitlik

Agent eng yangi `*.sql.gz` ga kuniga 1 marta `gzip -t` (o'qish orqali) bajaradi (CPU: 50 MB fayl ≈ 0.3 s). Natija `backup_gz_ok`. Fayl o'zgarish vaqti oxirgi tekshiruvdan keyin o'zgarmagan bo'lsa, qayta tekshirilmaydi. `gzip -t` xato bersa — `suspicious` + AL-073 (`critical`).

### 10.5.4. Haftalik fayl zaxirasi

`BACKUP_PATHS` bor loyihalar (BiznesRadar media, lutfan_ai media+minio, wmg sqlite+media) uchun yakshanba kuni `...-files-*.tar.gz` chiqishi kerak. 8 kundan eski bo'lsa — AL-074. `tar` fayli ham `gzip -t` bilan tekshiriladi (haftada 1 marta; hajmi katta bo'lsa — faqat `gzip -t` ning birinchi 64 MB si, F3).

### 10.5.5. Saqlash siyosati

Kunlik fayllar soni ≤ `retention_daily + 2`, haftalik ≤ `retention_weekly + 2` bo'lishi kerak; oshsa — `info` belgi (appctl tozalamayapti, disk to'lishi mumkin).

## 10.6. Offsite monitoringi

### 10.6.1. Ikki darajali tekshiruv

| Daraja | Manba | Nimani isbotlaydi | Ishonchlilik |
|---|---|---|---|
| 1 | `_run.json.offsite` (asosiy serverdagi `appctl` natijasi) | `rsync` xatosiz tugadi | O'rtacha (rsync "muvaffaqiyat" qaytarsa ham mazmun to'liq bo'lmasligi mumkin) |
| 2 | Offsite papkadagi agent skaneri (S-00) | Fayl **haqiqatan** u yerda, hajmi mos | Yuqori |

Nazoratchi ikkalasini ham ishlatadi. Offsite serverda agent bo'lmasa, faqat 1-daraja ishlaydi va panelda "offsite tasdiqlanmagan (agent yo'q)" belgisi ko'rinadi.

### 10.6.2. Solishtirish algoritmi

Har loyiha uchun, har 15 daqiqada:

1. Mahalliy eng yangi kunlik fayl: `(nom, hajm, mtime)`.
2. Offsite'dagi shu loyiha papkasi (`<offsite_root>/<loyiha>/daily/`) eng yangi fayli: `(nom, hajm, mtime)`.
3. Agar `nom` bir xil va `hajm` bir xil → `in_sync = true`.
4. Aks holda `lag_h = (mahalliy.mtime − offsite.mtime)`; `lag_h` > 26 (AL-076) yoki > 50 (AL-077) bo'lsa alert.
5. Nom farqi, lekin hajm mos kelmasa: `out_of_sync` (AL-078) — rsync oraliq holatda bo'lishi mumkinligi uchun 30 daqiqa kutiladi.

Offsite'da `--delete` bo'lmagani sababli, u yerda mahalliydan ko'ra **ko'p** fayl bo'lishi normal.

### 10.6.3. Offsite serverni tayyorlash (spetsifikatsiya)

| Parametr | Qiymat |
|---|---|
| Joy | **S-00** (Owner qarori): asosiy S-01 dan boshqa server, core bilan bir serverda; bir provayder (Contabo) — R-110. S-03 ishlatilmaydi |
| Papka | `/srv/backups/main-server`, alohida foydalanuvchi (`appctl.conf` `OFFSITE` shunga moslanadi) |
| Kirish | Faqat SSH-kalit; asosiy serverning kalitiga `command="rrsync /srv/backups/main-server"` cheklovi (faqat shu papkaga yozish) |
| Agent | S-00 dagi `nazoratchi-agent`, `backup.root` shu papkaga ko'rsatilgan |
| Disk | 200 GB SSD S-00 da core bazasi (≈ 6–14 GB), Docker va offsite uchun; offsite papkasi alohida kvota/bo'lim bilan cheklanadi, 25% bo'sh |
| Tozalash | Offsite'da alohida cron: 30 kundan eski kunlik, 12 haftadan eski haftalik fayllarni o'chirish |

| Parametr | Qiymat |
|---|---|
| Xavfsizlik | Asosiy server buzilsa, hujumchi offsite zaxirani o'chira olmasligi uchun: `rrsync` yozuv-faqat (`--ro` emas, lekin `--delete` yo'q) yoki offsite'da `chattr +a` / kunlik snapshot |

> Bu talablar `rsync -a` (`--delete` siz) tanlovi bilan bog'liq: hujumchi yoki xato asosiy serverda fayllarni o'chirsa ham, offsite'dagi nusxalar o'chmaydi.

## 10.7. Zaxira hajmi va disk

- `backup_dir_bytes` (har loyiha) va `backup_root_free_bytes` kuzatiladi.
- 90 kunlik o'sish chizig'i bo'yicha "zaxira bo'limi to'lishiga N kun" bashorati (FR-BKP-008).
- `backup_root_free_bytes` < 15% (yoki < 5 GB) → AL-079 (`warning`); < 5% → `critical` (rejalashtirilgan zaxira yozilmay qolishi mumkin).
- Offsite uchun ham xuddi shu (AL-080).

## 10.8. Restore drill (tiklash testi)

### 10.8.1. Maqsad

Zaxiradan haqiqatan ma'lumotni tiklab ko'rish, shuni isbotlash: (a) fayl o'qiladi, (b) bazaga yuklanadi, (c) ma'lumot mazmunli (jadvallar va qatorlar bor), (d) ilova shu bilan ishlay oladi (F3), (e) qancha vaqt ketishini bilish (RTO).

### 10.8.2. Xavfsizlik qoidasi (juda muhim)

> `appctl restore` **ni ishlab turgan (production) bazaga drill uchun ishlatish qat'iyan taqiqlanadi** — u bazani `DROP DATABASE ... WITH (FORCE)` bilan o'chiradi. Drill faqat **alohida (vaqtinchalik) bazada** bajariladi.

### 10.8.3. Qo'lda drill tartibi (PostgreSQL loyihalar, F2)

1. Eng yangi (yoki tanlangan) `*.sql.gz` ni aniqlash.
2. Shu loyihaning **PostgreSQL asosiy versiyasi bilan bir xil** vaqtinchalik konteyner ko'tarish (masalan, BiznesRadar — 17, Nazar — 16, lutfan_ai — 15):

```
docker run -d --rm --name drill-pg -e POSTGRES_PASSWORD=drill -p 127.0.0.1:55432:5432 postgres:17
```

3. Bazani va rolni yaratish (dump rol nomiga bog'liq bo'lsa):

```
docker exec drill-pg psql -U postgres -c "CREATE ROLE biznesradar LOGIN; CREATE DATABASE biznesradar OWNER biznesradar;"
```

4. Yuklash va vaqtni o'lchash:

```
time gunzip -c BiznesRadar-20261022-031501.sql.gz | docker exec -i drill-pg psql -U postgres -d biznesradar -v ON_ERROR_STOP=1 -q
```

5. Tekshirish: jadvallar soni va asosiy jadvallardagi qatorlar soni; ularni ishlab turgan bazadagi qiymat bilan (taxminan, zaxira vaqtidan keyingi o'sishni hisobga olib) solishtirish:

```
docker exec drill-pg psql -U postgres -d biznesradar -Atc "SELECT count(*) FROM information_schema.tables WHERE table_schema='public'"
docker exec drill-pg psql -U postgres -d biznesradar -Atc "SELECT relname, n_live_tup FROM pg_stat_user_tables ORDER BY n_live_tup DESC L
```

6. Konteynerni to'xtatish (`docker stop drill-pg`) — `--rm` tufayli o'chadi.
7. Natijani panelda qayd etish (FR-BKP-016): sana, fayl, davomiylik, jadval/qator soni, natija, izoh.

**Muvaffaqiyat mezonlari:** xatosiz yuklandi; jadvallar soni ishlab turgan bazadagi bilan bir xil (yoki kamida shu); eng katta 3 jadvalning qator soni ishlab turgan bazadagidan 5% dan ko'p kam emas.

### 10.8.4. Boshqa turdagi loyihalar

| Tur | Drill usuli | Tekshiruv |
|---|---|---|
| SQLite (wmg) | `.sqlite3` nusxasini vaqtinchalik papkaga ochish, `sqlite3 db 'PRAGMA integrity_check'` va `SELECT count(*)` | `integrity_check = ok`; asosiy jadvallar bo'sh emas |
| Media (tar.gz) | `tar -tzf` ro'yxat; tasodifiy 20 fayl ni vaqtinchalik papkaga chiqarib solishtirish (`sha256sum`) | Xatosiz; fayllar soni kutilganga yaqin |
| MinIO ma'lumoti | Vaqtinchalik MinIO konteyneriga tiklab, bucket ro'yxatini ko'rish | Bucketlar bor, ob'ektlar ochiladi |

| Tur | Drill usuli | Tekshiruv |
|---|---|---|
| Nazoratchi'ning o'zi | Yangi bo'sh serverda tiklash yo'riqnomasi (15.9) bo'yicha | Panel ochiladi, reyestr to'liq |

### 10.8.5. Davriylik va eslatma

- Har loyiha uchun `drill_every_days` (standart 90). Eslatma: muddatdan 14 kun oldin `info`, muddati o'tganda `warning` (AL-082), 100 kun — yuqoriroq.
- Birinchi drill — Nazoratchi F1 ishga tushishi bilan, **hammasi uchun** (hozirgi zaxiralar hech qachon tiklab ko'rilmagan).
- Drill xato bilan tugasa (`failed`) — AL-083 (`high`) va zaxira holati `warning` → `failed` (agar oxirgi muvaffaqiyatli drill 100 kundan eski bo'lsa).

### 10.8.6. Avtomatik drill (`appctl drill`, F3, ixtiyoriy)

`appctl drill <loyiha> [fayl]` — yuqoridagi 1–6 qadamlarni o'zi bajaradi (vaqtinchalik konteyner, yuklash, hisoblash, tozalash) va natijani API orqali yuboradi (`POST /backups/drills`, API token bilan). Haftada bir marta bitta tasodifiy loyiha uchun cron orqali. Bu talab FR-BKP-020 (4-bo'lim).

## 10.9. Zaxira sahifasi va alertlar bilan bog'lanish

| Ko'rsatkich / hodisa | Alert (12-bo'lim) | UI joyi |
|---|---|---|
| Yosh > 26 s / > 50 s | AL-068 / AL-069 | Zaxiralar jadvali, loyiha kartasi |
| `status = fail` | AL-070 | Loyiha kartasi, incident |
| Hajm kichik / katta | AL-071 / AL-072 | Hajm grafigi (anomaliya nuqtasi) |
| `gz_ok = false` | AL-073 | Kartada qizil belgi |
| Haftalik fayl yo'q | AL-074 | Kartada |
| Offsite sozlanmagan | AL-075 | Dashboard va jadvalda doimiy belgi |
| Offsite eskirdi / farq | AL-076, AL-077, AL-078 | "Offsite" ustuni |
| Zaxira bo'limi to'ldi | AL-079, AL-080 | Server sahifasi |
| Drill muddati / xato | AL-081, AL-082, AL-083 | Kartadagi "Oxirgi drill" |

## 10.10. Avariya holatida tiklash (ma'lumot uchun)

Nazoratchi bu jarayonni bajarmaydi, lekin panel zarur ma'lumotni bir joyda ko'rsatadi ("Tiklash yo'riqnomasi" tugmasi → runbook RB-030):

1. Eng yangi **tekshirilgan** zaxira (`gz_ok = true`, hajm normal) — mahalliy yoki offsite.
2. Zaxira fayli joyi va serveri.
3. `appctl restore <loyiha> <fayl.sql.gz> --yes` buyrug'i (ogohlantirish: mavjud bazani o'chiradi).
4. Oxirgi drill natijasi: davomiylik (RTO bahosi) va muvaffaqiyat.
5. Loyiha `.appctl` konfigi va `docker-compose` fayli joylashuvi.

## 10.11. Qabul mezonlari (zaxira moduli)

| № | Sinov | Kutilgan natija |
|---|---|---|
| T-BKP-01 | `appctl backup BiznesRadar` muvaffaqiyatli | `state.json` yozildi; panel 5 daqiqada yangilandi |
| T-BKP-02 | Cron'ni o'chirib 27 soat kutish (yoki vaqtni siljitish) | AL-068 `high` alerti |
| T-BKP-03 | Dump faylini kesib qo'yish (`truncate`) | AL-073 `critical` |
| T-BKP-04 | `pg_dump` xatosini keltirib chiqarish (noto'g'ri `DB_USER`) | `state.json: fail`, AL-070, xato matni xabarda |
| T-BKP-05 | `OFFSITE` bo'sh | AL-075 doimiy ogohlantirish |
| T-BKP-06 | Offsite'ga rsync'ni to'xtatish | 26 soatdan keyin AL-076 |
| T-BKP-07 | Offsite'da faylni o'chirish/kichraytirish | AL-078 |
| T-BKP-08 | Bo'sh (1 KB) dump | `appctl` rad etadi; Nazoratchi `failed` ko'rsatadi |
| T-BKP-09 | Drill natijasini kiritish | Kartada yangilandi, eslatma muddati qayta boshlandi |
| T-BKP-10 | `state.json` ni o'chirib, fayl qoldirish | Skaner manbasi ishlaydi, `unknown` emas |

# 11. Loyiha, servis, domen va sertifikat monitoringi

Bu bo'lim ikki xil kuzatuvni birlashtiradi:

- **Tashqaridan (faol)** — core'dagi `prober` mijoz nuqtai nazaridan tekshiradi: sayt ochiladimi, sertifikat yaroqlimi, DNS to'g'ri joyga qaraydimi.
- **Ichkaridan (passiv)** — agent serverning ichidan ko'radi: konteyner, systemd servis, cron.

Ikkalasi ham kerak: "tashqaridan ishlamayapti, ichkaridan hammasi yaxshi" holati (masalan, DNS yoki nginx muammosi) va "tashqaridan ishlayapti, lekin ichkarida worker o'lgan" holati faqat shu birlashma bilan topiladi.

## 11.1. Loyiha holati qanday hisoblanadi

Loyiha kartasidagi umumiy rang quyidagi tarkibiy qismlardan **eng yomoni** bo'yicha olinadi:

| Tarkibiy qism | Manba | Qizil bo'ladi, agar | Sariq bo'ladi, agar |
|---|---|---|---|
| **Domenlar (HTTP)** | prober | Hech bo'lmaganda bitta asosiy domen `down` | Sekin javob yoki bitta ikkilamchi domen `down` |
| **Sertifikat** | prober, agent | ≤ 3 kun yoki yaroqsiz | ≤ 21 kun |
| **DNS** | prober | Kutilgan IP ga qaramaydi (10 daq. dan ko'p) | Resolverlar bir-biriga mos kelmaydi |
| **Konteynerlar/servislar** | agent | Kutilgan konteyner ishlamayapti / `unhealthy` / crash loop | Qayta ishga tushishlar ko'p, xotira limitga yaqin |
| **Zaxira** | agent | `failed`, `stale` | Offsite yo'q, drill muddati o'tgan |
| **Server** | agent | Server o'chiq | Disk/RAM ogohlantirish |
| **Incident** | alert tizimi | Ochiq `critical`/`high` incident | Ochiq `warning` |

Kulrang — "noma'lum": ma'lumot kelmayapti yoki loyiha pauzada/ko'chirilmagan.

Har tarkibiy qismning holati loyiha sahifasida alohida ko'rinadi; umumiy rang faqat bosh sahifadagi xulosa uchun.

## 11.2. HTTP(S) tekshiruvlari

### 11.2.1. Tekshiruv bajarilishi (bir urinish)

1. DNS so'rov (tizim resolveri, vaqt o'lchanadi).
2. TCP ulanish (`connect_ms`).
3. TLS qo'l siqish (`tls_ms`), sertifikat tekshiruvi (zanjir, hostname, muddat).
4. HTTP so'rov (`GET` yoki `HEAD`), sarlavhalar: `User-Agent: Nazoratchi/1.0 (+https://<core>/about)`, foydalanuvchi sarlavhalari.
5. Birinchi bayt (`ttfb_ms`) va to'liq javob (`total_ms`) vaqti; javob tanasi birinchi 64 KB gacha o'qiladi (kalit so'z tekshiruvi uchun).
6. Baholash: status kodi kutilgan oraliqda; kalit so'z (bo'lsa); vaqt chegarasi.

### 11.2.2. Tasdiqlash qoidasi

Soxta (bir martalik) xatoni alert qilmaslik uchun:

```
urinish 1 muvaffaqiyatsiz → 15 s kutish → urinish 2 → 15 s kutish → urinish 3
3 ta ketma-ket muvaffaqiyatsiz  → holat = DOWN   (alert hisoblash `for` bilan davom etadi)
1 ta muvaffaqiyat               → holat = UP
```

Shu sababli eng tez aniqlash vaqti ≈ 30 s (qayta urinishlar) + interval (60 s) kechikishi: **≤ 90 soniya** (G-2).

### 11.2.3. Xato turlari

| `error_kind` | Sabab | Odatdagi ma'nosi |
|---|---|---|
| `dns` | Nom hal qilinmadi | DNS yozuvi yo'q/xato |
| `connect` | TCP ulanmadi / timeout | Server o'chiq, firewall, port yopiq |
| `tls` | Qo'l siqish/sertifikat xatosi | Sertifikat tugagan, hostname mos emas |
| `timeout` | Javob vaqti oshdi | Ilova qotgan, server haddan tashqari band |
| `status` | Kutilmagan status kodi | 5xx — ilova xatosi; 4xx — sozlama |
| `keyword` | Kalit so'z topilmadi/topildi (teskari rejim) | Soxta "xizmat ishlamayapti" sahifasi |

| error_kind | Sabab | Odatdagi ma'nosi |
|---|---|---|
| redirect | Yo'naltirish halqasi / ko'p yo'naltirish | nginx/ilova sozlamasi |

Alert xabarida shu turdan kelib chiqib **birinchi tekshirish qadami** ko'rsatiladi (masalan, `connect` → "serverga ulanish: RB-001", `tls` → "RB-006").

### 11.2.4. Django loyihalar uchun maxsus sozlama

Nazar kabi Django loyihalar tashqi so'rovni `ALLOWED_HOSTS` va `SECURE_SSL_REDIRECT` tufayli `400`/`301` bilan rad etishi mumkin (bu 2026-10 da bevosita uchragan). Nazoratchi'da:

- Odatiy: haqiqiy domen orqali `https://` bilan tekshirish — Django uchun hech qanday maxsus sozlama kerak emas, chunki so'rov to'g'ri `Host` va `X-Forwarded-Proto` bilan keladi.
- Ichki (loopback) tekshiruv kerak bo'lsa (agent ichidan), `headers` bilan `Host` va `X-Forwarded-Proto: https` beriladi (`appctl` dagi `HEALTH_CURL_OPTS` bilan bir xil g'oya).
- Maxsus `healthz` yo'li bo'lsa (Nazar: `/healthz/`), shu ishlatiladi; bo'lmasa — bosh sahifa.

### 11.2.5. Standart tekshiruv to'plami (boshlang'ich seed)

Quyidagi tekshiruvlar Nazoratchi birinchi ishga tushganda reyestrdan avtomatik taklif qilinadi. "Kutilgan" ustuni 2026-10-22 holatidagi real xatti-harakatga asoslangan; belgi (?) — aniqlash kerak.

| Loyiha | URL | Kutilgan status | Izoh |
|---|---|---|---|
| Nazar | `https://nazar.avlo.ai/healthz/` | 200 | `healthz` mavjud |
| Nazar | `https://nazar.avlo.ai/` | 200–399 | Til bo'yicha yo'naltirish (307) normal |
| Nazar (Beyondlinks UI) | `https://app.beyondlinks.uz/` | 200–399 | Xuddi shunday |
| BiznesRadar | `https://biznesradar.uz/` | 200 | Bosh sahifa |
| BiznesRadar | `https://api.biznesradar.uz/` | 404 (?) | Ildiz yo'li 404 normal; haqiqiy health yo'li (?) aniqlash |
| BiznesRadar | `https://cabinet.biznesradar.uz/` | 404 (?) | Xuddi shunday |
| BiznesRadar | `https://admin.biznesradar.uz/` | 200–399 (?) | |
| beyondlinks_landing | `https://beyondlinks.uz/` | 200 | Statik |
| freighthubllc | `https://freighthubllc.co/` va `.com` | 200 | Statik |
| avlo.ai | `https://avlo.ai/` | 200 | Statik sayt |
| avlo.ai | `https://avlo.ai/api/` | 404/405 (?) | `GET` ga backend javobi; aniq kodni birinchi sinovda aniqlash |
| asadbekashurov.uz | `https://asadbekashurov.uz/` | 200 | S-02 da |
| asadbekashurov.uz | `https://www.asadbekashurov.uz/` | 301 → asosiy | Yo'naltirishni tekshirish (`follow_redirects=false`) |
| lutfan_ai | `https://lutfanai.uz/` | 200 (?) | DNS GCP da; ko'chirilgach tekshiruv yoqiladi |
| wmg | `https://moliyaviyerkinlik.wmg.uz/`, `https://investitsiya.wmg.uz/` | 200 (?) | DNS mijozda |

### 11.2.6. TCP tekshiruvlari (xavfsizlik va mavjudlik)

| Maqsad | Host:port | Kutilgan |
|---|---|---|
| SSH mavjud (server tirikligi ikkilamchi signal) | `164.68.105.194:22`, `173.249.37.152:22` | ochiq |
| HTTPS ochiq | `...:443` | ochiq |
| **Yopiq bo'lishi kerak** (xavfsizlik) | `...:5432`, `:6379`, `:9000`, `:8020`, `:3017`, `:2375` | yopiq (ulanish rad etiladi) |

Yopiq bo'lishi kerak bo'lgan portlar tekshiruvi (`expect: closed`) — Docker port nashri `ufw` ni chetlab o'tishi (2026-10 da ko'rilgan) kabi xatolarni topadi (AL-030).

# 11.3. Sertifikat (TLS) monitoringi

## 11.3.1. Ikki manba

| Manba | Nima ko'radi | Qachon foydali |
|---|---|---|
| **Prober** (TLS probe) | Tashqaridan haqiqatan taqdim etilayotgan sertifikat | Ishlayotgan sertifikat — foydalanuvchi ko'radigan narsa |
| **Agent** (`certs`) | Serverdagi `/etc/letsencrypt/live/*` fayllari | Yangilangan, lekin nginx hali o'qimagan (reload qilinmagan) sertifikatni topish |

Ikkalasi taqqoslanadi: agent'dagi sertifikat yangi (`not_after` kattaroq), prober hali eskisini ko'rsatsa — **"sertifikat yangilangan, lekin nginx reload qilinmagan"** (AL-061 ning alohida holati). Bu `certbot` deploy-hook muammosining klassik belgisi.

## 11.3.2. Qaysi nomlar tekshiriladi

Har faol domenning FQDN va (agar reyestrda `www` bor bo'lsa) `www.` varianti. Sertifikat SAN ro'yxatida `FQDN` yo'q bo'lsa — `hostname_ok = false` (AL-060).

## 11.3.3. Chegaralar va ahamiyat

| Qolgan kun | Holat | Alert | Ahamiyat |
|---|---|---|---|
| > 21 | OK | — | — |
| 8–21 | Yaqinlashmoqda | AL-057 | `warning` |
| 4–7 | Yuqori | AL-058 | `high` |
| 1–3 | Kritik | AL-059 | `critical` |
| ≤ 0 | O'tib ketgan | AL-059 (`critical`, "tugadi") | `critical` |

Let's Encrypt sertifikatlari certbot tomonidan **30 kun qolganda** avtomatik yangilanadi; shuning uchun 21 kunlik ogohlantirish "yangilanish ishlamayapti" degan ishonchli signal (30 kundan 21 kunga tushgan bo'lsa, kamida ~9 kun yangilash urinishlari muvaffaqiyatsiz bo'lgan).

## 11.3.4. Yangilanish natijasi

- **S-01 (certbot, nginx authenticator):** agent `/var/log/letsencrypt/letsencrypt.log` dan oxirgi `renew` natijasini o'qiydi (`certbot_last_renew_status`).
- **S-02 (`asadbekashurov.uz`, konteyner certbot, kunlik cron):** cron natijasini yozuvchi heartbeat (`/hb/<token>`) — `cron: asadbekashurov-certbot` (FR-SVC-004).
- Ikki ketma-ket muvaffaqiyatsiz yangilash → AL-061 (`high`), sertifikat muddati yaqin bo'lmasa ham.

## 11.3.5. Maxsus holatlar

| Holat | Nazoratchi qanday ishlaydi |
|---|---|
| Domen DNS i boshqa serverga qaraydi | Sertifikat o'sha server taqdim etganini tekshiradi (Nazoratchi domenning haqiqiy egasini emas, ko'rinayotgan holatni baholaydi); DNS mos emasligi alohida alert (AL-062) |
| Sertifikat yo'q (HTTP faqat) | `tls_enabled = false` bo'lsa tekshirilmaydi |
| Wildcard sertifikat | `SAN` da `*.domen` bo'lsa, subdomenlar mos deb hisoblanadi |
| Front-proksi orqali (S-02: SNI bo'yicha) | Prober `SNI` bilan ulanadi, shuning uchun to'g'ri stekning sertifikatini ko'radi |

# 11.4. DNS va domen monitoringi

## 11.4.1. Reyestrdagi DNS ma'lumoti

Har domen uchun saqlanadi:

- `expected_ips` — qaysi IP lar kutiladi (masalan, `164.68.105.194`);
- `dns_provider` — DNS qayerda boshqariladi (`ahost.uz`, Namecheap, Google Cloud DNS, Beget);
- `controlled_by` — `self` yoki `client` (mijoz o'zgartiradi: wmg.uz);
- `registrar`, domen muddati (WHOIS yoki qo'lda).

### 11.4.2. Tekshirish

Har 5 daqiqada (o'zgartirish rejimida — 1 daqiqada):

| Resolver | Maqsad |
|---|---|
| 8.8.8.8 (Google) | Eng keng tarqalgan; o'zgarishni sekin oladi |
| 1.1.1.1 (Cloudflare) | Tez yangilanadi |
| 9.9.9.9 (Quad9) | Qo'shimcha nuqta |
| Domenning o'z NS serveri (`rdns1.ahost.uz` va h.k.) | **Haqiqat manbai** — o'zgartirish aynan shu yerda ko'rinadi |

**Holatlar:**

| Holat | Ma'nosi | Alert |
|---|---|---|
| `match` | Hammasi kutilgan IP | — |
| `propagating` | NS yangi IP da, ba'zi resolverlar hali eski | Ma'lumot (alert yo'q) |
| `mismatch` | NS ham kutilgan IP ni bermaydi | AL-062 (`high`, 10 daqiqadan keyin) |
| `retired_ip` | Domen **eski/yopilgan** IP ga qaraydi (masalan, o'chirilgan GCP `34.173.220.247`) | AL-063 (`high`) — ko'chirishdan keyin unutilgan yozuv |
| `resolver_disagree` | NS kutilgan, lekin 30+ daqiqadan beri resolverlar mos emas | AL-065 (`warning`) — TTL/kesh muammosi |
| `nxdomain` | Domen mavjud emas | AL-062 (`critical`) |

"Yopilgan IP ro'yxati" sozlamada (`retired_ips = [34.173.220.247]`, o'chirilgan GCP) — Nazoratchi shu IP ga qaragan har bir domenni darrov belgilaydi. Bu `lutfanai.uz` va `*.wmg.uz` kabi DNS yangilanmagan domenlarni **avtomatik** topadi. `185.214.135.9` endi core serveri; unga qaragan boshqa loyiha domeni kutilgan IP ga mos kelmasligi uchun baribir AL-062 (`mismatch`) beradi.

### 11.4.3. Domen muddati

- `.com`, `.ai` va boshqa RDAP qo'llaydigan zonalar uchun avtomatik (RDAP, kuniga 1 marta).
- `.uz` uchun RDAP har doim mavjud emas — muddat **qo'lda kiritiladi** (FR-DNS-006), tizim eslatadi.
- Chegaralar: 60 / 30 / 14 / 7 kun (AL-066: ≤ 30 kun `warning`, AL-067: ≤ 7 kun `critical`).

### 11.4.4. Kesib o'tish (cut-over) yordamchisi

Domen bir serverdan boshqasiga ko'chirilayotganda (2026-10 da ko'p marta bajarilgan jarayon) Nazoratchi quyidagi ishni bajaradi:

**Oldindan tekshiruv (DNS o'zgartirilishidan oldin):**

1. Operator: "domen X ni IP Y ga ko'chiraman".
2. Nazoratchi `prober` orqali **to'g'ridan-to'g'ri Y IP ga**, `Host: X` va `SNI: X` bilan HTTPS so'rov yuboradi (`curl --resolve X:443:Y` ning ekvivalenti).
3. Natija: HTTP status, sertifikat (muddati, hostname mosligi), javob vaqti.
4. Hammasi yaxshi bo'lsa — "tayyor, DNS ni o'zgartirishingiz mumkin"; aks holda sabab ("sertifikat yo'q", "502", ...) ko'rsatiladi. Bu aynan ko'chirish jarayonida qo'lda qilingan `curl --resolve` tekshiruvini avtomatlashtiradi.

**Kuzatish (DNS o'zgartirilgandan keyin):**

1. Tarqalish rejimi yoqiladi (FR-DNS-005): har 1 daqiqada barcha resolverlar so'raladi.
2. Panelda real vaqtda: qaysi resolver yangi IP ga o'tdi, qaysisi hali eski.
3. Hammasi o'tganda Telegram'ga "domen X yangi serverga to'liq o'tdi" xabari va avtomatik HTTP + TLS tekshiruv natijasi.
4. 2 soat o'tib ham ba'zi resolverlar eski bo'lsa — maslahat: "Google DNS keshini tozalash: dns.google/cache", TTL qiymati.

Bu funksiya talab sifatida FR-DNS-009 (4-bo'lim).

## 11.5. Konteynerlar (Docker)

### 11.5.1. Qaysi konteyner "kutilgan"

Nazoratchi konteyner o'chib qolganini ("kutilgan, lekin yo'q") qanday biladi — qo'lda ro'yxat yuritmasdan?

1. Agent har 30 soniyada barcha konteynerlarni (`all=1`) ko'radi.
2. Konteyner **kutilgan** deb belgilanadi, agar: (a) kamida 1 soat uzluksiz `running` bo'lgan **va** (b) `RestartPolicy` `no` dan farqli (`always`/`unless-stopped`/`on-failure`).

3. Kutilgan konteyner yo'qolsa yoki `exited` bo'lsa → AL-035 (`high`), 90 soniyadan keyin. *(S5)* "Kutilgan" belgisi agent xotirasida (`docker_container_expected`), core'da emas; agent qayta ishga tushsa 1 soat qaytadan kutiladi. Ro'yxatdan butunlay yo'qolgan konteyner 10 daqiqa alert beradi, keyin unutiladi: ataylab olib tashlash uchun alohida amal kerak emas (F3 da paneldagi tugma bilan almashadi).
4. **Ataylab olib tashlash:** operator panelda "kutilgan ro'yxatdan chiqarish" tugmasini bosadi yoki `appctl down <loyiha>` natijasida loyiha uchun avtomatik maintenance (F3). Shu holda alert yo'q.
5. Compose loyihada konteyner nomi (`nazar-web-1`) o'zgarmaydi; yangi image bilan qayta yaratilganda (`--build`) 1 daqiqagacha yo'q bo'lishi normal — shuning uchun 90 s `for`.

### 11.5.2. Crash loop

- Mezon: 10 daqiqada `docker_container_restarts_total` ≥ 3 ga oshdi (AL-036).
- Xabarda: konteyner, loyiha, oxirgi `exit code`, `OOMKilled` belgisi, runbook RB-020 (`docker logs --tail 100 <konteyner>`).

### 11.5.3. Resurslar

| Mezon | Alert |
|---|---|
| RAM ≥ 90% limitdan 10 daqiqa (limit belgilangan bo'lsa) | AL-039 (`warning`) |
| `OOMKilled = true` | AL-038 (`high`) |
| Konteyner CPU ≥ 95% (bir yadro) 30 daqiqa | AL-040 (`info`) |
| Log fayli > 100 MB yoki log limiti yo'q | AL-041 (`warning`) — eski konteynerlarga `daemon.json` limiti qo'llanmaganini topadi |

### 11.5.4. Noma'lum konteynerlar

Reyestrdagi hech bir loyihaga bog'lanmagan konteynerlar "noma'lum" ro'yxatida; AL-042 (`info`): "Reyestrda yo'q konteyner ishlayapti" — yangi ko'tarilgan loyiha reyestrga qo'shilmagani yoki ruxsatsiz konteyner (xavfsizlik) belgisi.

## 11.6. systemd servislari va cron

| Mezon | Alert |
|---|---|
| Sozlamadagi unit `failed` yoki kutilmagan `inactive` (90 s) | AL-043 (`high`) |
| `NRestarts` 10 daqiqada ≥ 3 | AL-044 (`high`) |
| `systemctl --failed` soni > 0 (1 soat) | AL-045 (`warning`) |
| `nginx -t` xato | AL-046 (`high`) |
| Heartbeat kutilgan oraliqda kelmadi | AL-047 (`high`) |
| Heartbeat `status=fail` | AL-048 (`high`) |

Boshlang'ich kuzatiladigan unit lar (S-01): `nginx`, `docker`, `cron`, `ssh`, `avlo-contact` (`fail2ban` o'rnatilgach qo'shiladi). S-02: `nginx`, `docker`, `ssh`, `fail2ban`.

Boshlang'ich cron heartbeat'lar:

| Nom | Server | Kutilgan oraliq | Grace |
|---|---|---|---|
| `appctl-backup` | S-01 | 24 soat | 2 soat |
| `certbot-renew` | S-01 | 24 soat (certbot timer: kuniga 2 marta) | 12 soat |
| `asadbekashurov-certbot` | S-02 | 24 soat | 3 soat |
| `nazoratchi-self-backup` | S-00 | 24 soat | 2 soat |

## 11.7. Server darajasidagi ichki-tashqi mosligi

Quyidagi kombinatsiyalar diagnostika uchun alohida ko'rsatiladi (incident xulosasida):

| HTTP tashqaridan | Konteyner/servis ichkaridan | Taxminiy sabab | Birinchi qadam |
|---|---|---|---|
| ❌ | ✅ (hammasi ishlayapti) | nginx/DNS/firewall/sertifikat | `nginx -t`, DNS holati, `ufw status` |
| ❌ | ❌ (konteyner yo'q) | Ilova yiqilgan | `docker ps -a`, `docker logs` |
| ✅ | ❌ (worker yo'q) | Fon vazifalari to'xtagan (Celery worker/beat) | `appctl ps <loyiha>` |
| ❌ | ❌ + host jim | Server o'lgan | RB-001 |
| ✅ | ✅ | Hammasi joyida | — |

## 11.8. Qabul sinovlari (servis monitoringi)

| № | Sinov | Kutilgan natija |
|---|---|---|
| T-SVC-01 | Nginx'ni to'xtatish | 90 s ichida `http_down`, ichki servislar sog'lom ekani ko'rsatiladi |
| T-SVC-02 | Sinov konteynerini doimiy xato bilan ishga tushirish | AL-036 crash loop |
| T-SVC-03 | `systemctl stop avlo-contact` | AL-043 90 s ichida |
| T-SVC-04 | Sinov sertifikati (3 kunlik) | AL-059 `critical` |
| T-SVC-05 | A yozuvini boshqa IP ga o'zgartirish (sinov domen) | AL-062/AL-063 10 daqiqadan keyin |
| T-SVC-06 | Cut-over oldindan tekshiruv: sertifikati yo'q serverga | "sertifikat yo'q" aniq sababi bilan rad |
| T-SVC-07 | Xost reboot qilish | `reboot aniqlandi` voqeasi, `Server o'chdi` bo'lmasa — qisqa uzilish alerti yo'q (120 s dan oz) |
| T-SVC-08 | Yopiq bo'lishi kerak port (5432) ni ochish | AL-030 (`high`) |
| T-SVC-09 | `docker compose up -d --build` paytida konteyner 40 s yo'qolishi | Alert **yo'q** (90 s `for`) |

# 12. Alert (ogohlantirish) tizimi

## 12.1. Tamoyillar

Yomon alert tizimi monitoringning o'zidan ham zararliroq: soxta xabarlar odamni charchatadi va u haqiqiy xabarni ham e'tiborsiz qoldiradi. Shuning uchun quyidagi tamoyillar qat'iy:

1. **Har alert harakat talab qilishi kerak.** "Qiziq, lekin hech narsa qilmaydigan" ma'lumot alert emas — u panelda yoki haftalik hisobotda ko'rinadi (`info`).
2. **Alert simptomni emas, ta'sirni ko'rsatishi kerak.** Foydalanuvchiga ta'sir qiladigan holat (sayt ochilmayapti) yuqori ahamiyat; sababchi holat (CPU 92%) pastroq.
3. **Bir muammo — bitta xabar.** Server o'lsa, 40 ta alert emas, bitta incident (suppression).
4. **Xabar o'zi yetarli bo'lishi kerak:** nima, qayerda, qancha, qachondan, nima qilish kerak.
5. **Chegaralar sinab tanlanadi.** Standart qiymatlar boshlang'ich; 4 haftalik kuzatuvdan keyin sozlanadi (12.11).
6. **Tiklanish ham xabar qilinadi,** shunda odam "hal bo'ldimi?" deb tekshirib o'tirmaydi.
7. **Jimlik — signal.** Ma'lumotning yo'qligi (heartbeat, cron) ham alert.

## 12.2. Ahamiyat darajalari

| Daraja | Ma'nosi | Misol | Kutilgan reaksiya | Belgi |
|---|---|---|---|---|
| `critical` | Xizmat to'xtagan yoki darhol to'xtash xavfi, ma'lumot yo'qolishi xavfi | Server o'chdi, sayt ishlamayapti, disk 95%, sertifikat 3 kun, zaxira buzuq | **Zudlik bilan** (kechasi ham) | 🔴 |
| `high` | Jiddiy muammo; tez orada kritikka aylanishi mumkin yoki qisman ta'sir | Konteyner o'chdi, zaxira kechikdi, DNS mos emas | Bir necha soat ichida (ish vaqtida darrov) | 🟠 |
| `warning` | Diqqat talab qiladigan, lekin shoshilinch emas | CPU 90%, disk 80%, sertifikat 21 kun | Bugun-ertaga | 🟡 |
| `info` | Ma'lumot | Reboot kerak, agent eskirgan | Hisobotda ko'rib chiqiladi | 🔵 |

## 12.3. Qoida sxemasi

Qoida `alert_rules.condition` (JSONB) maydonida saqlanadi. Beshta shart turi:

### 12.3.1. `threshold` — chegara

```
{
  "type": "threshold",
  "metric": "cpu_busy_percent",
  "labels": {},
  "op": ">",
  "value": 90,
  "aggregate": "avg",
  "window_s": 60
}
```

`aggregate`: `avg|min|max|last` — `window_s` oralig'ida; `for_seconds` shart shu holda uzluksiz bajarilishi kerakligini bildiradi (qoida darajasida).

Ixtiyoriy `when`: `{"metric": "docker_container_expected", "op": "==", "value": 1}` — shart faqat shu seriyaning (xuddi shu server va belgilar) hamroh metrikasi `when` ni qanoatlantirsa qo'llanadi, aks holda natija `yolg'on`. AL-035 shuni ishlatadi (faqat "kutilgan" konteyner).

### 12.3.2. `absent` — ma'lumot yo'qligi

```
{"type": "absent", "metric": "agent_up", "max_age_s": 120}
```

Berilgan qamrovdagi obyekt uchun metrikaning oxirgi namunasi `max_age_s` dan eski bo'lsa rost.

### 12.3.3. `rate` — o'zgarish tezligi

```
{"type": "rate", "metric": "docker_container_restarts_total", "increase": 3, "window_s": 600}
```

`window_s` ichida qiymat `increase` dan ko'proqqa oshgan bo'lsa rost (counter qayta boshlanishi hisobga olinadi).

### 12.3.4. `forecast` — bashorat

```
{
  "type": "forecast",
  "metric": "fs_used_bytes",
  "labels": {"mount": "/"},
  "limit_metric": "fs_total_bytes",
  "horizon_s": 604800,
  "fit_window_s": 21600,
  "min_r2": 0.5,
  "min_used_percent": 50
}
```

Chiziqli regressiya (6.7.4) bo'yicha `horizon_s` ichida limitga yetsa rost. `min_r2` dan past bo'lsa yoki bandlik `min_used_percent` dan kam bo'lsa — hisoblanmaydi (soxta bashorat oldini olish).

### 12.3.5. `state` — holat

```
{"type": "state", "source": "probe", "field": "state", "equals": "down"}
```

`source`: `probe` (tekshiruv holati), `backup` (zaxira holati: `status`, `age_h`), `cert` (`days_left`), `dns` (`status`), `heartbeat`. Boshqa manbalar uchun maxsus maydonlar 18.3-bo'limda.

Amalga oshirilgan manbalar (S3):

| source | field | equals | Ma'nosi |
|---|---|---|---|
| reachability | tcp22 | "up" / "down" | Serverning public_ipv4 ga TCP:22 ulanish (3 s, natija 20 s keshlanadi). Loopback va link-local manzillar tekshirilmaydi. Bu to'liq prober (S4/S5) kelguncha AL-001/AL-002 uchun minimal tashqi tekshiruv. IP kiritilmagan bo'lsa natija "noma'lum" |
| heartbeat_job | overdue | true | now - last_beat_at > expected_every_s + grace_s (hali bir marta ham xabar bermagan bo'lsa — noma'lum) |
| heartbeat_job | failed | true | Oxirgi heartbeat status=fail |
| channel | failing | true | Kanalga oxirgi 3 yuborish ketma-ket failed |
| fleet | mass_silent | true | Hostlarning ≥ 50% (kamida 2 ta) 120 s dan beri jim (AL-008) |
| probe | state | "down" / "up" | Faol tekshiruv holati (S4). Ixtiyoriy primary: true/false filtri (config.primary). unknown/paused holat — noma'lum. Ko'chirilmagan (moved=false) va arxivlangan loyiha tekshiruvlari hisoblanmaydi. Incident qamrovi — project:<id> |
| prober | stalled | true | Yoqilgan tekshiruvlar bor, lekin hech biri 180 s da bajarilmagan (AL-085) |
| backup | age_hours_gt | butun son (soat) | Loyihaning oxirgi muvaffaqiyatli zaxirasi yoshi > equals soat. Hech qachon olinmagan bo'lsa — loyiha shu soatdan eski bo'lgandagina |
| backup | last_failed | true | backup_last_status = 0 (appctl state.json) |
| backup | size_small / size_large | true | Eng yangi zaxira hajmi oldingi 7 tasi medianasining < 0.5 / > 3 baravari; tarix < 8 ta bo'lsa — noma'lum (FR-BKP-005) |
| cert | days_left_lte | butun son | cert_status (TLS tekshiruvi): floor((not_after - hozir)/1 kun) ≤ equals; manfiy = muddati o'tgan. Holat 2 kundan eski bo'lsa — noma'lum |
| cert | invalid | true | chain_ok = false yoki hostname_ok = false |
| dns | mismatch | true | DNS tekshiruvi down (3 marta tasdiqlangan) VA config.mismatch_after_min (standart 10) davomida shunday |
| dns | retired | true | Oxirgi natijada javob config.retired_ips ichida (domen o'chirilgan IP ga qaraydi) |

**"Ma'lumot yo'q" (uch qiymatli mantiq).** Shart hisoblanayotgan metrika yangi bo'lmasa (`last` uchun `max(window_s, 120)` s dan eski) yoki hisoblash uchun nuqta yetmasa, natija `rost`/`yolg'on` emas, *noma'lum*. Holat mashinasi noma'lumda `FIRING`/`RESOLVING` ni o'zgartirmaydi (6.7.1): agent jim bo'lganda CPU alerti soxta "tiklandi" bo'lib qolmaydi; jimlikni AL-001/AL-002 ushlaydi. Birlashtirishda: `and` da bitta `yolg'on` yetarli (qolganlari hisoblanmaydi: qimmat TCP tekshiruvi faqat agent jim bo'lganda bajariladi), `or` da bitta `rost`.

**Obyektlar (fan-out).** `threshold`, `rate`, `forecast` har mos seriya uchun alohida obyekt beradi (host × mount), `absent`, `reachability` va birlashtirishlar — har host uchun bitta, `heartbeat_job` — har job, `channel` — har kanal. Qoida `scope.tags`/`scope.ids` bilan serverlarga cheklanadi.

### 12.3.6. Birlashtirish

`and` / `or` / `not` (chuqurligi ≤ 3; operandlar `of` ro'yxatida, `not` uchun aynan bitta, `and`/`or` uchun 2..5; faqat host darajasidagi shartlar: `threshold`, `absent`, `rate`, `forecast`, `reachability`):

```
{"type": "and", "of": [
  {"type": "absent", "metric": "agent_up", "max_age_s": 120},
  {"type": "state", "source": "probe", "field": "tcp22", "equals": "down"}
]}
```

## 12.4. Yetkazish siyosati

### 12.4.1. Kanal va vaqt

| Ahamiyat | Telegram | Email | Sokin soatlarda (23:00–07:00) | Takroriy eslatma |
|---|---|---|---|---|
| `critical` | darhol | darhol | **darhol** | 30 daqiqada bir, ack bo'lguncha (6 marta, keyin 2 soatda bir) |
| `high` | darhol | 07:00 yig'ma | 07:00 da | 2 soatda bir, ack bo'lguncha (3 marta) |
| `warning` | darhol (kunduzi), ertalabki yig'ma (kechasi) | kunlik hisobot | 07:00 yig'ma | Bir marta; 24 soatdan keyin hali ochiq bo'lsa qayta |
| `info` | faqat kunlik hisobotda | haftalik | — | — |

Qiymatlar sozlamada kanal darajasida o'zgartiriladi (`min_severity`, `quiet_hours`).

### 12.4.2. Eskalatsiya (F3)

`critical` 15 daqiqa qabul qilinmasa — ikkinchi kontakt/kanalga yuboriladi; 45 daqiqada ham qabul qilinmasa — "ikkala kontaktga ham" xabar.

### 12.4.3. Takroriy eslatma to'xtashi

- Incident `ack` qilinsa — eslatmalar to'xtaydi (lekin ahamiyat oshsa, qayta xabar).
- `silence` / `maintenance` — to'xtaydi.
- Incident yopilsa — "tiklandi".

## 12.5. Bostirish, guruhlash, titrashdan himoya

### 12.5.1. Bog'liqlik bo'yicha bostirish (suppression)

6.7.3 dagi qoidalarga qo'shimcha: bostirilgan alert "yo'qolmaydi" — incident ichida "ta'sirlangan" ro'yxatida ko'rinadi va asosiy sabab hal bo'lganda, shart hamon rost bo'lsa, ularning o'zi yangidan hisoblanadi (masalan, server qaytdi, lekin konteyner hamon ishlamayapti — yangi xabar).

### 12.5.1a. Ahamiyat bo'yicha bostirish (amalga oshirilgan)

Bir obyekt va bir metrikada bir nechta chegara qoidasi bo'lsa (AL-019/020/021), ularning eng yuqori ahamiyatlisi `FIRING` bo'lganda pastroqlari bostiriladi (80% → 90% → 95% da uchta xabar o'rniga bittasi, eng og'iri). Yuqorisi yopilib pastrog'i hamon rost bo'lsa, pastrog'i xabar beradi. Bostirish sabablari: `pause`, `maintenance`, `silence`, `host_down` (AL-001 dan keyin shu hostning boshqa alertlari), `mass_silence` (AL-008 dan keyin AL-001/002), `superseded`. Bostirilgan alert incident'da qoladi.

> **S3 holati.** Guruhlash (12.5.2), titrash/beqarorlik va gisterezis (12.5.3), eskalatsiya (12.4.2), sokin soatlar va email hozircha yo'q (F2/F3). Tugma bilan `ack` (Telegram) S4 da; `ack` hozir API orqali.

### 12.5.2. Guruhlash

60 soniyalik oyna ichida bir qamrov (`scope_key`) bo'yicha 3 va undan ko'p alert bitta xabarga birlashadi: sarlavha — eng og'ir alert, tanasida qisqa ro'yxat (maks. 10 ta, qolgani "+N ta").

### 12.5.3. Titrash (flapping)

- 1 soat ichida 5 martadan ko'p `FIRING ⇔ OK` o'tishi — "beqaror" belgisi.
- Beqaror alert uchun har o'tishda xabar yuborilmaydi; o'rniga bitta "beqaror: N marta o'zgardi" xabari va 30 daqiqalik tinch davr kutiladi.
- Gisterezis: ko'p qoidada yoqish chegarasi va o'chirish chegarasi farqli (masalan, disk: yoqish 80%, o'chirish 78%).

### 12.5.4. Massaviy jimlik

Agar ≥ 2 host va ≥ 50% hostlar 2 daqiqa ichida jim bo'lsa — AL-008 (core tarmog'i yoki core muammosi gumoni) bitta incident; alohida AL-001/AL-002 lar bostiriladi, chunki ularning sababi Nazoratchi tomonida bo'lishi ehtimoli yuqori.

## 12.6. Silence va maintenance

### 12.6.1. Silence

- Moslash (matchers): `server`, `project`, `rule`, `severity`, `domain`, `container`, `tag` bo'yicha (`=`, `!=`, `=~` regex).
- Muddat: 15 daqiqa – 30 kun. Sabab majburiy.
- **Silence alertni o'chirmaydi:** holat hisoblanaveradi, incident ochiladi (`suppressed=true`), faqat bildirishnoma yuborilmaydi. Silence tugagach ham shart rost bo'lsa — xabar yuboriladi.
- Silence tugashiga 15 daqiqa qolganda `info` eslatma (agar o'sha qamrovda ochiq incident bo'lsa).

### 12.6.2. Maintenance oynasi

- Qamrov: server(lar) yoki loyiha(lar); boshlanish/tugash vaqti.
- Oyna ichida: bildirishnoma yuborilmaydi; uptime hisobida "rejalashtirilgan to'xtash" sifatida alohida; boshqa incidentlar kabi yig'iladi.
- Oyna tugagach: ochiq `firing` alertlar yuboriladi ("texnik ish tugadi, hali hal bo'lmagan: ...").
- **Reboot oldidan nazorat ro'yxati** (L-4 saboq): oyna yaratilganda panel eslatadi: "Snapshot olindimi? Zaxira yangimi? Offsite'dagi nusxa tekshirildimi?" (12.9-bo'lim).

### 12.6.3. Ustuvorlik

`pause` (server) > `maintenance` > `silence` > oddiy yuborish. Hammasi audit jurnalida.

## 12.7. Qoidalar katalogi (AL-001 … AL-089)

Ustunlar: **ID**; **Nom**; **Qamrov**; **Shart**; `for` — shart shuncha vaqt uzluksiz rost bo'lishi; **Ahamiyat**; **Faza**; **Runbook**. Metrika nomlari 7-bo'limdagi katalogga mos. `*` — chegara sozlanadi.

### 12.7.1. Server va agent (A)

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-001 | Server o'chdi | host | Heartbeat ≥ 120 s yo'q **va** TCP:22/ICMP tashqi tekshiruv muvaffaqiyatsiz | 0 | critical | F1 | RB-001 |
| AL-002 | Agent javob bermayapti | host | Heartbeat ≥ 120 s yo'q, lekin TCP:22 javob beradi | 3 daq | high | F1 | RB-002 |
| AL-003 | Kutilmagan reboot | host | `boot_time_seconds` o'zgardi va maintenance oynasi yo'q | 0 | high | F1 | RB-003 |
| AL-004 | Agent eskirgan | host | Agent versiyasi core'dagi so'nggi versiyadan ≥ 2 minor orqada | 1 soat | info | F2 | RB-043 |
| AL-005 | Agent navbati o'smoqda | host | `agent_queue_files` > 10 | 10 daq | warning | F2 | RB-002 |
| AL-006 | Agent kollektor xatolari | host | `agent_collector_errors` o'sish tezligi > 0 | 15 daq | warning | F2 | RB-002 |
| AL-007 | Soat siljishi | host | `abs(agent_clock_skew_seconds)` > 5 | 10 daq | warning | F2 | RB-016 |
| AL-008 | Ko'p host jim (core tarmog'i) | global | ≥ 50% hostlar (kamida 2) heartbeat'siz | 2 daq | critical | F1 | RB-012 |

### 12.7.2. CPU, RAM (B)

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-009 | CPU yuqori | host | `cpu_busy_percent` > 90* | 10 daq | warning | F1 | RB-009 |
| AL-010 | CPU to'yingan | host | `cpu_busy_percent` > 97* | 15 daq | high | F2 | RB-009 |
| AL-011 | Yuklama yuqori | host | `load_per_core_1m` > 2* | 15 daq | warning | F2 | RB-009 |
| AL-012 | Disk kutish (iowait) | host | `cpu_usage_percent{mode=iowait}` > 30* | 10 daq | warning | F2 | RB-011 |

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-013 | CPU steal yuqori | host | `cpu_usage_percent{mode=steal}` > 15* (provayder resursi yetmayapti) | 15 daq | warning | F2 | RB-009 |
| AL-014 | RAM yuqori | host | `mem_used_percent` > 90* | 10 daq | warning | F1 | RB-010 |
| AL-015 | RAM kritik | host | `mem_used_percent` > 96* | 5 daq | critical | F1 | RB-010 |
| AL-016 | Swap faol ishlatilmoqda | host | `swap_used_percent` > 50 **va** `swap_in_pages_per_sec` > 100 | 10 daq | warning | F2 | RB-010 |
| AL-017 | Yadro OOM o'ldirdi | host | `oom_kill_total` oshdi (rate, 10 daq) | 0 | high | F2 | RB-010 |
| AL-018 | Zombi jarayonlar ko'p | host | `procs_zombie` > 20 | 30 daq | info | F3 | RB-009 |

### 12.7.3. Disk va fayl tizimi (C)

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-019 | Disk 80% | host × mount | `fs_used_percent` > 80* | 5 daq | warning | F1 | RB-004 |
| AL-020 | Disk 90% | host × mount | `fs_used_percent` > 90* | 5 daq | high | F1 | RB-004 |
| AL-021 | Disk 95% | host × mount | `fs_used_percent` > 95* | 2 daq | critical | F1 | RB-004 |
| AL-022 | Disk 7 kunda to'ladi (bashorat) | host × mount | Bashorat < 7 kun ($R^2$ ≥ 0.5, band ≥ 50%) | 30 daq | warning | F1 | RB-004 |
| AL-023 | Disk 24 soatda to'ladi (bashorat) | host × mount | Bashorat < 24 soat | 15 daq | critical | F1 | RB-004 |
| AL-024 | Inode yuqori | host × mount | `fs_inodes_used_percent` > 85* | 10 daq | warning | F2 | RB-004 |
| AL-025 | Fayl tizimi faqat o'qish | host × mount | `fs_readonly = 1` (real FS) | 1 daq | critical | F1 | RB-005 |
| AL-026 | Disk I/O to'yingan | host × device | `disk_io_util_percent` > 90* | 15 daq | warning | F2 | RB-011 |
| AL-027 | Disk kechikishi yuqori | host × device | `disk_await_ms` > 100* | 10 daq | warning | F2 | RB-011 |
| AL-028 | Docker tozalab bo'ladigan hajm katta | host | `docker_reclaimable_bytes` > 20 GB yoki > 25% disk | 1 soat | info | F3 | RB-004 |

### 12.7.4. Tarmoq, OS, xavfsizlik (D)

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-029 | Tarmoq xatolari | host × iface | `net_rx_errors_per_sec + net_tx_drops_per_sec` > 10* | 10 daq | warning | F2 | RB-013 |
| AL-030 | Kutilmagan ochiq port | host | TCP tekshiruvi `expect: closed` portlardan biri ochiq **yoki** `facts.listen_ports` kutilgan ro'yxatdan tashqari | 2 daq | high | F2 | RB-027 |
| AL-031 | SSH hujumi (bruteforce) | host | `ssh_failed_logins_1h` > 200* | 0 | warning | F3 | RB-014 |
| AL-032 | Reboot uzoq kutilmoqda | host | `reboot_required_age_seconds` > 14 kun | 0 | warning | F2 | RB-015 |
| AL-033 | Yangi kernel kutilmoqda | host | `kernel_reboot_pending = 1` | 24 soat | info | F2 | RB-015 |
| AL-034 | Xavfsizlik yangilanishlari | host | `apt_security_upgradable_total` > 0 | 7 kun | warning | F3 | RB-015 |

### 12.7.5. Docker (E)

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-035 | Konteyner ishlamayapti | container | Kutilgan konteyner `running` emas (11.5.1) | 90 s | high | F1 | RB-020 |
| AL-036 | Konteyner crash loop | container | `docker_container_restarts_total` 10 daq ≥ 3 ga oshdi | 0 | high | F1 | RB-020 |
| AL-037 | Konteyner sog'lom emas | container | `docker_container_health = 2` | 2 daq | high | F2 | RB-020 |
| AL-038 | Konteyner OOM bilan o'ldi | container | `docker_container_oom_killed = 1` | 0 | high | F2 | RB-021 |
| AL-039 | Konteyner xotirasi limitga yaqin | container | `docker_container_mem_percent` > 90* (limit bor bo'lsa) | 10 daq | warning | F2 | RB-021 |

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-040 | Konteyner CPU yuqori | container | `docker_container_cpu_percent` > 95* | 30 daq | info | F3 | RB-009 |
| AL-041 | Konteyner logi katta / limitsiz | container | `docker_container_log_bytes` > 100 MB* | 1 soat | warning | F2 | RB-021 |
| AL-042 | Reyestrda yo'q konteyner | container | Hech bir loyihaga bog'lanmagan konteyner ishlayapti | 1 soat | info | F2 | RB-020 |

### 12.7.6. systemd va cron (F)

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-043 | Servis ishlamayapti | unit | `systemd_unit_active = 0` yoki `systemd_unit_failed = 1` | 90 s | high | F1 | RB-022 |
| AL-044 | Servis qayta ishga tushish halqasi | unit | `systemd_unit_restarts_total` 10 daq ≥ 3 ga oshdi | 0 | high | F2 | RB-022 |
| AL-045 | Muvaffaqiyatsiz unitlar bor | host | `systemd_failed_units_total` > 0 | 1 soat | warning | F2 | RB-022 |
| AL-046 | nginx konfigi yaroqsiz | host | `nginx_config_ok = 0` | 5 daq | high | F3 | RB-023 |
| AL-047 | Heartbeat kelmadi | job | `now − last_beat_at` > `expected_every_s + grace_s` | 0 | high | F1 | RB-024 |
| AL-048 | Heartbeat "xato" dedi | job | Oxirgi heartbeat `status=fail` | 0 | high | F1 | RB-024 |

### 12.7.7. HTTP va TCP tekshiruvlari (G)

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-049 | Sayt ishlamayapti | probe (http) | Holat `down` (3 ketma-ket muvaffaqiyatsiz) | 0 | critical (teg `primary`) / high | F1 | RB-025 |
| AL-050 | Sayt sekin | probe (http) | `total_ms` > `latency_warn_ms` (3000*) | 5 daq | warning | F2 | RB-026 |
| AL-051 | Kalit so'z mos emas | probe (http) | Kalit so'z tekshiruvi 3 marta xato | 0 | high | F2 | RB-025 |
| AL-052 | TTFB yuqori | probe (http) | `ttfb_ms` > 2000* | 10 daq | warning | F3 | RB-026 |
| AL-053 | TCP port ishlamayapti | probe (tcp) | `expect: open` bo'lgan port 3 marta ulanmadi | 0 | high | F2 | RB-025 |
| AL-054 | Uptime past | probe | 30 kunlik uptime < 99.5%* | 1 soat | info | F3 | RB-025 |
| AL-055 | Tekshiruv bajarilmayapti | probe | `now − last_run_at` > 3 × interval | 0 | high | F2 | RB-040 |
| AL-056 | Tekshiruv titrayapti | probe | 1 soatda > 5 holat o'zgarishi | 0 | warning | F2 | RB-025 |

### 12.7.8. Sertifikat (H)

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-057 | Sertifikat 21 kunda tugaydi | domain | `days_left` ≤ 21* | 0 | warning | F1 | RB-006 |
| AL-058 | Sertifikat 7 kunda tugaydi | domain | `days_left` ≤ 7* | 0 | high | F1 | RB-006 |
| AL-059 | Sertifikat 3 kunda tugaydi / tugadi | domain | `days_left` ≤ 3* | 0 | critical | F1 | RB-006 |
| AL-060 | Sertifikat yaroqsiz | domain | Zanjir to'liq emas yoki hostname SAN da yo'q | 10 daq | high | F2 | RB-006 |
| AL-061 | Sertifikat yangilanmayapti | domain/host | Yangilash ketma-ket 2 marta xato; yoki agentdagi sertifikat yangi, tashqaridagi eski (nginx reload qilinmagan) | 0 | high | F2 | RB-006 |

### 12.7.9. DNS va domen (I)

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-062 | DNS kutilgan IP ga qaramaydi | domain | NS serveri kutilgan IP ni bermaydi (yoki NXDOMAIN) | 10 daq | high (NXDOMAIN: critical) | F1 | RB-007 |
| AL-063 | Domen yopilgan IP ga qaraydi | domain | A yozuvi `retired_ips` ro'yxatida | 10 daq | high | F1 | RB-007 |
| AL-064 | NS yozuvi o'zgardi | domain | NS to'plami o'zgardi | 0 | info | F3 | RB-007 |
| AL-065 | Resolverlar bir-biriga mos emas | domain | NS to'g'ri, lekin ≥ 1 ommaviy resolver boshqa javob beradi | 30 daq | warning | F2 | RB-007 |

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-066 | Domen muddati 30 kunda tugaydi | domain | `whois.expires_at − now` ≤ 30 kun | 0 | warning | F2 | RB-008 |
| AL-067 | Domen muddati 7 kunda tugaydi | domain | ≤ 7 kun | 0 | critical | F2 | RB-008 |

### 12.7.10. Zaxira (J)

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-068 | Zaxira kechikdi | project | Eng yangi muvaffaqiyatli zaxira yoshi > 26 soat* | 0 | high | F1 | RB-031 |
| AL-069 | Zaxira uzoq vaqt olinmadi | project | Yosh > 50 soat* | 0 | critical | F1 | RB-031 |
| AL-070 | Zaxira muvaffaqiyatsiz | project | Oxirgi `state.json.status = fail` | 0 | high | F1 | RB-031 |
| AL-071 | Zaxira shubhali: juda kichik | project | Hajm < median(7) × 0.5* | 0 | high | F1 | RB-032 |
| AL-072 | Zaxira shubhali: juda katta | project | Hajm > median(7) × 3* | 0 | info | F2 | RB-032 |
| AL-073 | Zaxira fayli buzuq | project | `backup_gz_ok = 0` | 0 | critical | F2 | RB-032 |
| AL-074 | Haftalik fayl zaxirasi yo'q | project | Haftalik fayl yoshi > 192 soat* | 0 | warning | F2 | RB-031 |
| AL-075 | Offsite zaxira sozlanmagan | host | `backup_offsite_configured = 0` | 0 | warning (doimiy) | F1 | RB-033 |
| AL-076 | Offsite eskirdi | project | Offsite'dagi eng yangi fayl orqada > 26 soat* | 0 | high | F2 | RB-033 |
| AL-077 | Offsite uzoq vaqt yangilanmadi | project | Orqada > 50 soat* | 0 | critical | F2 | RB-033 |
| AL-078 | Offsite nusxa mos emas | project | Nom/hajm mos kelmaydi 30 daqiqadan ortiq | 30 daq | warning | F2 | RB-033 |
| AL-079 | Zaxira bo'limida joy kam | host | `backup_root_free_bytes` < 15% yoki < 5 GB* | 10 daq | warning (< 5%: critical) | F2 | RB-004 |
| AL-080 | Offsite serverda joy kam | host (offsite) | Xuddi shu, offsite serverda | 10 daq | warning | F2 | RB-033 |
| AL-081 | Restore drill muddati yaqin | project | Oxirgi muvaffaqiyatli drill > `drill_every_days − 14` kun | 0 | info | F1 | RB-034 |
| AL-082 | Restore drill muddati o'tdi | project | Oxirgi muvaffaqiyatli drill > `drill_every_days` kun (yoki hech qachon) | 0 | warning | F1 | RB-034 |
| AL-083 | Restore drill muvaffaqiyatsiz | project | Oxirgi drill natijasi `failed` | 0 | high | F1 | RB-034 |

### 12.7.11. Nazoratchi'ning o'zi (K)

| ID | Nom | Qamrov | Shart | for | Ahamiyat | Faza | RB |
|---|---|---|---|---|---|---|---|
| AL-084 | Alert dvigateli qotdi | system | `evaluator_heartbeat` > 60 s eskirgan | 0 | critical | F1 | RB-040 |
| AL-085 | Prober qotdi | system | Hech bir tekshiruv 3 daqiqada bajarilmadi | 0 | high | F1 | RB-040 |
| AL-086 | Bildirishnoma kanali ishlamayapti | channel | Kanalga yuborish ketma-ket 3 marta xato | 0 | high | F1 | RB-041 |
| AL-087 | Nazoratchi bazasi katta | system | Baza hajmi > 80% byudjetdan (NFR-CAP-002) | 1 soat | warning | F2 | RB-042 |
| AL-088 | Qabul qilishda xatolar ko'p | system | Ingest xatolari > 5% (10 daq) | 10 daq | warning | F2 | RB-042 |
| AL-089 | Saqlash vazifasi bajarilmadi | system | Retention/agregat vazifasi 26 soatdan beri muvaffaqiyatsiz | 0 | warning | F2 | RB-042 |

> **Eslatma (AL-086).** Telegram kanali ishlamayotgan bo'lsa, bu alert o'sha kanalga yuborila olmaydi. Shuning uchun u har doim ikkinchi kanal (email) va panel orqali ko'rsatiladi, tashqi watchdog ham uni qayd etadi (FR-NTF-004).

### 12.7.12. Faza bo'yicha xulosa

| Faza | Alertlar soni |
|---|---|
| F1 (MVP) | 35 ta (muhim: server o'chdi, disk, RAM, konteyner, sayt, sertifikat, DNS, zaxira, heartbeat, o'zini kuzatish) |
| F2 | 48 ta |

| Faza | Alertlar soni |
|---|---|
| F3 | 9 ta |
| **Jami** | **89 ta** |

Ahamiyat bo'yicha (asosiy daraja): `critical` 13, `high` 30, `warning` 36, `info` 10.

> **S3 da yoqilgan standart qoidalar (16 ta):** AL-001, 002, 003, 008, 009, 014, 015, 019, 020, 021, 022, 023, 025, 047, 048, 086. **S7 da qo'shilgan (2 ta):** AL-090 (o'z zaxirasi 30 soatdan eski), AL-091 (o'z zaxirasi yuborilmadi), `self_backup` state manbasi (`SelfBackup` yozuvlari). AL-084 (evaluator qotdi) core ichida baholanmaydi: `Pinger` ping yubormay qo'yadi va mustaqil watchdog (S-01) Telegram'ga xabar beradi. **S6 da qo'shilgan (6 ta):** AL-068 (zaxira kechikdi, 26 s), AL-069 (50 s, `critical`), AL-070 (muvaffaqiyatsiz), AL-071 (juda kichik), AL-072 (juda katta, `info`), AL-075 (offsite sozlanmagan, `warning`). `backup` manbasi faqat `expect_daily` siyosatli, faol va ko'chirilgan loyihalarni baholaydi; zaxira serveri — loyihaning `production` biriktirishi (bo'lmasa loyiha o'tkazib yuboriladi). Agent 15 daqiqadan beri jim bo'lsa natija noma'lum. AL-068/069 zinapoyasida eng og'iri xabar beradi. AL-075 doimiy ogohlantirish (yopilmaydi, `OFFSITE` to'ldirilguncha yoki silence qilinguncha). AL-073/074/076…083 (F2) yo'q. **S5 da qo'shilgan (8 ta):** AL-035 (kutilgan konteyner), AL-036 (crash loop: 10 daqiqada > 2 ta qayta ishga tushish), AL-057/058/059 (sertifikat 21/7/3 kun), AL-060 (zanjir/nom yaroqsiz), AL-062 (DNS mos emas), AL-063 (yopilgan IP). Sertifikat zinapoyasida eng og'iri xabar beradi (`superseded`); domen DNS si mos emas bo'lsa shu domenning HTTP va sertifikat alertlari bostiriladi (`dns_mismatch`, 6.7.3). AL-062 da NXDOMAIN uchun alohida `critical` hozircha yo'q. **S4 da qo'shilgan (3 ta):** AL-049 (asosiy sayt, `critical`), AL-049H (asosiy bo'lmagan, `high`: katalogdagi "critical (teg `primary`) / high" ikki qoidaga bo'lingan), AL-085 (prober qotdi). Tekshiruv alertlari loyihaning HAMMA serveri pauzada/texnik ishda yoki o'chgan bo'lsa bostiriladi (6.7.3). Qolgan F1 qoidalari (docker, tekshiruvlar, sertifikat, DNS, zaxira) ularning ma'lumoti paydo bo'lgan sprintda qo'shiladi. AL-084/085 tashqi kuzatuvchi (watchdog, S7) bilan. Standart qoidalar bazada yo'q bo'lsagina yaratiladi: foydalanuvchi o'zgartirgan chegaraga tegilmaydi.

## 12.8. Qoidani sozlash

- Har qoida uchun: yoqish/o'chirish, chegara, `for`, ahamiyat, qamrov (teg/ID), kanallar.
- **Qoida testi** (`POST /alert-rules/{id}/test`): joriy ma'lumotda quruq hisoblash va "hozir firing bo'ladigan obyektlar" ro'yxatini ko'rsatadi; alert yuborilmaydi. Yangi chegarani qo'llashdan oldin shu bilan tekshiriladi.
- Qoida tarixi: har o'zgarish audit jurnalida (kim, eski/yangi qiymat).
- O'rnatilgan qoidalar (`built_in`) o'chirilmaydi, faqat o'chirib qo'yiladi; yangi versiyada standart qiymatlar yangilanganda, foydalanuvchi o'zgartirgan qiymatlar saqlanadi.

## 12.9. Xabar shablonlari (Telegram, o'zbek tilida)

### 12.9.0. Yuborish va qayta urinish (FR-NTF-004)

Xabar avval `notifications` ga `queued` bo'lib yoziladi, `notifier` jarayoni yuboradi. Vaqtinchalik xatoda (tarmoq, `5xx`, `429`) jami 5 urinish: 1-dan keyin 5 s, 2-dan 15 s, 3-dan 60 s, 4-dan 300 s kutiladi; `429` da Telegram bergan `retry_after` ishlatiladi. Doimiy xatoda (`400`, `401`, `403`: noto'g'ri token yoki chat, bot bloklangan) qayta urinilmaydi: darhol `failed`, sabab `channels.last_error` da (panelda ko'rinadi). Telegram HTML rejimida yuboriladi, o'zgaruvchan matn (server nomi, log satri) ekranlanadi; sarlavha 150 belgigacha, butun xabar 1000 belgigacha (uzun bo'lsa o'rtadagi satrlar qisqartiriladi, runbook va havola qoladi). Email zaxira kanali F2 da (FR-NTF-005); hozircha kanal ishlamasa faqat panelda ko'rinadi.

Umumiy tuzilma: **ahamiyat belgisi + sarlavha**, bo'sh qator, **qayerda**, **nima bo'ldi** (qiymat va chegara), **qachondan**, **ta'sir**, **nima qilish kerak** (runbook), tugmalar.

### 12.9.1. Server o'chdi (AL-001)

```
🔴 Server o'chdi: main-2 (164.68.105.194)

Aloqa yo'q: 2 daqiqa (oxirgi marta 09:33 da ko'rilgan)
TCP:22 — javob yo'q, ICMP — javob yo'q

Ta'sirlangan loyihalar (6): Nazar, BiznesRadar, beyondlinks.uz, avlo.ai,
freighthubllc, wmg

Nima qilish kerak: RB-001 → provayder panelidan holatni tekshiring (VNC/Rescue)
Incident: https://nazoratchi.example.uz/incidents/0193...

[ Qabul qilaman ]  [ 1 soat jim ]  [ Panelda ochish ]
```

### 12.9.2. Disk to'lmoqda (AL-022)

```
🟡 Disk 7 kunda to'ladi: main-2 ( / )

Hozir: 78% (62.4 GB / 80 GB)
O'sish: +1.9 GB/kun → to'lishga ≈ 9.3 kun (7 kundan kam bashorat: oxirgi 24 soat bo'yicha 6.1 kun)

Eng katta: /var/lib/docker (41 GB), /opt/applications/backups (12 GB)

Nima qilish kerak: RB-004 → `docker system df`, eski image/log tozalash

[ Qabul qilaman ]  [ 24 soat jim ]  [ Panelda ochish ]
```

### 12.9.3. Zaxira kechikdi (AL-068)

```
🟠 Zaxira kechikdi: BiznesRadar

Oxirgi muvaffaqiyatli: 2 kun 3 soat oldin (2026-10-20 03:15 UTC)
Kutilgan: har 24 soatda (chegara 26 soat)
Oxirgi log (appctl-backup.log):
  [BiznesRadar] XATO: dump muvaffaqiyatsiz

Nima qilish kerak: RB-031 → `appctl backup BiznesRadar` ni qo'lda ishga tushiring

[ Qabul qilaman ]  [ Panelda ochish ]
```

### 12.9.4. Tiklandi

```
✅ Tiklandi: Server o'chdi — main-2

Davomiyligi: 12 daqiqa (09:33 – 09:45)
Sabab: (kiritilmagan)  ← panelda izoh qo'shing

[ Panelda ochish ]
```

### 12.9.5. Sertifikat (AL-058)

```
🟠 Sertifikat 7 kunda tugaydi: nazar.avlo.ai

Tugash: 2026-10-29 (qolgan: 7 kun)
Chiqaruvchi: Let's Encrypt (R11)
Server: main-2. Oxirgi yangilash urinishi: 2026-10-21 04:12 — muvaffaqiyatsiz

Nima qilish kerak: RB-006 → `certbot renew --dry-run`, `journalctl -u certbot`

[ Qabul qilaman ]  [ Panelda ochish ]
```

### 12.9.6. DNS (AL-063)

```
🟠 Domen yopilgan IP ga qaraydi: lutfanai.uz

Hozir: 34.173.220.247 (o'chirilgan GCP serveri)
Kutilgan: 164.68.105.194
DNS provayder: (reyestrda ko'rsatiladi)

Nima qilish kerak: RB-007 → DNS A yozuvini yangilang

[ Qabul qilaman ]  [ Panelda ochish ]
```

### 12.9.7. Yig'ma (guruhlangan) xabar

```
🔴 main-2: 7 ta muammo (1 incident)

• Server o'chdi (asosiy sabab)
• Sayt ishlamayapti: nazar.avlo.ai, biznesradar.uz, avlo.ai (+3)
• Konteyner ishlamayapti: 14 ta
• Zaxira holati noma'lum

[ Qabul qilaman ]  [ Panelda ochish ]
```

### 12.9.8. Texnik ish oldidan nazorat ro'yxati (maintenance yaratilganda)

```
🛠 Texnik ish rejalashtirildi: main-2, 22 okt 14:00–15:00

Reboot oldidan tekshiring:
 ☐ Snapshot olindi (Contabo panel)
 ☐ Oxirgi zaxira yangi (BiznesRadar 6 soat, Nazar 6 soat) — avto-tekshiruv ✅
 ☐ Offsite nusxa mos — ❌ OFFSITE sozlanmagan
 ☐ DNS TTL pasaytirildi (agar kesib o'tish bo'lsa)
```

(Avtomatik belgilanadigan bandlar Nazoratchi ma'lumotidan hisoblanadi; qo'lda bandlarni operator belgilaydi.)

# 12.10. Davriy hisobotlar

## 12.10.1. Kunlik (08:00 Asia/Tashkent, Telegram)

```
📊 Kunlik xulosa — 22 oktyabr

Serverlar: 3/3 onlayn   Loyihalar: 9 ta (hammasi sog'lom)
Ochiq incident: 1 (🟡 disk 80%)
Zaxira: 8/8 yangi (eng eskisi 6 soat), offsite: ❌ sozlanmagan
Muddatlar: sertifikat ≥ 41 kun, domen ≥ 118 kun, drill muddati o'tgan: 6
Kecha: 2 incident (jami 14 daqiqa)
```

## 12.10.2. Haftalik (dushanba 09:00, email + Telegram qisqa)

Tarkibi: uptime (har loyiha, 7 kun), incidentlar jadvali (ahamiyat, davomiyligi, sabab), MTTR, eng ko'p alert bergan qoidalar (shovqin nazorati), disk/RAM tendensiyalari (7 kunlik o'zgarish), zaxira va drill holati, yaqin muddatlar (sertifikat, domen), hal qilinmagan "doimiy" ogohlantirishlar (masalan, offsite yo'q).

# 12.11. Alert sifatini boshqarish

| Ko'rsatkich | Maqsad | Qanday o'lchanadi |
|---|---|---|
| Haftalik "qo'ng'iroq" (high+critical) alertlar soni | ≤ 10 (barqaror holatda) | Panel hisoboti |
| Soxta alertlar ulushi | ≤ 5% | Operator incidentni "soxta" deb belgilaydi (maxsus yopish sababi) |
| Aniqlanmay qolgan haqiqiy muammo | 0 (har post-mortemda tekshiriladi) | Incident tahlili |
| Ack vaqti (critical) | median ≤ 10 daqiqa | Incident hodisalaridan |

**Oylik ko'rib chiqish (30 daqiqa):** eng ko'p ishga tushgan 5 qoida → chegarasini oshirish/`for` ni uzaytirish/o'chirish; soxta deb belgilangan incidentlarni tahlil qilish; yangi post-mortemdan yangi qoida qo'shish.

**Post-mortem qoidasi:** har `critical` incidentdan keyin "Bu muammoni oldinroq aniqlay olardikmi?" savoliga javob yoziladi va kerak bo'lsa yangi qoida (yoki chegara) qo'shiladi.

# 12.12. Alert tizimi testlari

| № | Sinov | Kutilgan natija |
|---|---|---|
| T-ALR-01 | Sun'iy "agent_up" to'xtatish (agentni stop) | 120 s dan keyin AL-002 yoki AL-001 (TCP:22 ga qarab) |
| T-ALR-02 | `for` davomiyligi: 4 daqiqa CPU 95% (chegara 10 daq) | Alert yo'q |
| T-ALR-03 | `for` davomiyligi: 11 daqiqa CPU 95% | AL-009 |
| T-ALR-04 | Bostirish: serverni o'chirish | Faqat bitta incident, ta'sirlanganlar ro'yxati bilan |
| T-ALR-05 | Titrash: 10 marta tez holat o'zgarishi | "Beqaror" xabari, ≤ 2 xabar |
| T-ALR-06 | Silence: 1 soatga | Xabar yo'q; incident `suppressed`; tugagach xabar |
| T-ALR-07 | Maintenance oynasi | Xabar yo'q; oyna tugagach hal bo'lmagan alertlar yuboriladi |
| T-ALR-08 | Ack tugmasi (Telegram) | Incident `acknowledged`, eslatmalar to'xtaydi |
| T-ALR-09 | Telegram'ni bloklash | Email zaxira kanal, AL-086 |
| T-ALR-10 | Evaluator qayta ishga tushishi | Ochiq incidentlar saqlanadi, takroriy "yangi alert" xabarlari yo'q |
| T-ALR-11 | Sokin soatlar | Kechasi `warning` kelmaydi; `critical` keladi |
| T-ALR-12 | Disk forecast: sun'iy o'sish | AL-022/AL-023 to'g'ri vaqtda |
| T-ALR-13 | Takroriy eslatma | `critical` 30 daqiqada bir (ack gacha) |

# 13. Foydalanuvchi interfeysi

## 13.1. Dizayn tamoyillari

1. **5 soniya qoidasi.** Bosh sahifaga qarab, 5 soniyada "hammasi joyidami?" ga javob olinishi kerak. Birinchi ekran — raqamlar emas, **xulosa**: "Hammasi joyida" yoki "2 ta muammo".
2. **Muammo tepada.** Sog'lom narsalar jim; muammolar yuqorida, rangli, bosiladigan.
3. **Bir bosishda sababga.** Dashboard → incident → server/loyiha → grafik: har bosqich bir bosish.
4. **Rang yagona signal emas.** Har holat rang + belgi + matn (rang ko'rlik va qora-oq chop uchun).
5. **Zichlik.** DevOps foydalanuvchi uchun ma'lumot zich, lekin o'qiladigan: bir ekranda 10 ta serverni sparkline'lar bilan sig'dirish.
6. **Ishonch.** Ma'lumot yangiligini doimo ko'rsatish ("yangilandi: 3 soniya oldin"); ma'lumot kelmayotgan bo'lsa — "eskirgan", hech qachon eski raqamni "hozirgi" deb ko'rsatmaslik.
7. **Xavfsiz amallar.** Qaytarib bo'lmaydigan amallar (token bekor qilish, arxivlash) tasdiqlanadi va nima bo'lishi yozib ko'rsatiladi.
8. **Tezlik.** Skeleton yuklash, kesh, `stale-while-revalidate`; hech qachon bo'sh oq ekran.

## 13.2. Axborot arxitekturasi (sitemap)

```
/                       Bosh sahifa (Dashboard)
/servers                Serverlar ro'yxati
/servers/:id            Server sahifasi  (Umumiy | Resurslar | Konteynerlar | Servislar | Tarix)
/projects               Loyihalar ro'yxati
/projects/:id           Loyiha sahifasi  (Holat | Domenlar | Konteynerlar | Zaxira | Incidentlar | Sozlama)
/domains                Domenlar va sertifikatlar
/domains/:id            Domen sahifasi (DNS, TLS, HTTP tarixi)
/backups                Zaxiralar (umumiy jadval)
/backups/:project       Loyiha zaxira kartasi, restore drill
/incidents              Incidentlar (ochiq / yopilgan)
/incidents/:id          Incident tafsiloti
/alerts/rules           Alert qoidalari
/alerts/silences        Jim qilishlar va texnik ishlar (maintenance)
/reports                Hisobotlar
/settings               Sozlamalar (Kanallar | Foydalanuvchilar | Tokenlar | Saqlash | Tizim)
/audit                  Audit jurnali
/status/:slug           Ommaviy holat sahifasi (F3, avtorizatsiyasiz)
/login                  Kirish
```

**Asosiy navigatsiya** (chap panel desktopda, pastki panel mobilda): Bosh sahifa, Serverlar, Loyihalar, Zaxiralar, Incidentlar (ochiq soni nishonchasi bilan), Yana (Domenlar, Qoidalar, Hisobotlar, Sozlamalar). Yuqorida: global qidiruv (`/`), vaqt zonasi/til, foydalanuvchi menyusi, "yangilandi: N soniya oldin" ko'rsatkichi.

## 13.3. Ekranlar ro'yxati

| ID | Ekran | Asosiy maqsad | Talablar |
|---|---|---|---|
| SCR-01 | Dashboard | Umumiy holat, muammolar, muddatlar | FR-UI-001, 002 |
| SCR-02 | Serverlar ro'yxati | Hamma serverni bir qarashda | FR-UI-004 |
| SCR-03 | Server sahifasi | Resurslar, konteynerlar, servislar, tarix | FR-UI-005 |
| SCR-04 | Loyihalar ro'yxati | Birlashtirilgan reyestr | FR-UI-003 |
| SCR-05 | Loyiha sahifasi | Loyiha holati va barcha tarkibiy qismlari | FR-UI-006 |
| SCR-06 | Zaxiralar | Hamma loyiha zaxirasi jadvali | FR-UI-008 |
| SCR-07 | Zaxira kartasi + drill | Bitta loyiha zaxirasi, drill kiritish | FR-BKP-003, 016 |
| SCR-08 | Domenlar va sertifikatlar | Muddatlar, DNS mosligi | FR-UI-009 |
| SCR-09 | Domen sahifasi (+ cut-over) | DNS resolverlar, tarixi, kesib o'tish yordamchisi | FR-DNS-002, 005, 009 |
| SCR-10 | Incidentlar ro'yxati | Ochiq va yopilgan incidentlar | FR-UI-010 |
| SCR-11 | Incident tafsiloti | Vaqt chizig'i, ta'sirlanganlar, izohlar | FR-UI-010, FR-ALR-016 |
| SCR-12 | Alert qoidalari | Ko'rish, tahrirlash, sinash | FR-ALR-007 |
| SCR-13 | Jim qilish va texnik ishlar | Silence, maintenance | FR-UI-011 |
| SCR-14 | Hisobotlar | Uptime, incident, zaxira statistikasi | FR-UI-012 |
| SCR-15 | Sozlamalar | Kanallar, foydalanuvchilar, tokenlar, saqlash | FR-UI-013 |
| SCR-16 | Audit jurnali | O'zgarishlar tarixi | FR-AUD-003 |
| SCR-17 | Server qo'shish | O'rnatish buyrug'i, birinchi aloqa kutish | UC-01 |

| ID | Ekran | Asosiy maqsad | Talablar |
|---|---|---|---|
| SCR-18 | Loyiha qo'shish | Forma, avtomatik tekshiruvlar taklifi | UC-10 |
| SCR-19 | Kirish va 2FA | Autentifikatsiya | FR-AUTH-* |
| SCR-20 | Ommaviy holat sahifasi (F3) | Mijozlar uchun | FR-UI-019 |

# 13.4. Rang va holat qoidalari

## 13.4.1. Holatlar (yagona lug'at)

| Holat | Rang (semantik) | Belgi | Matn (uz) | Qachon |
|---|---|---|---|---|
| `ok` | Yashil | ● / ✓ | Joyida | Hammasi me'yorda |
| `warning` | Sariq | ▲ | Diqqat | Ogohlantirish ochiq |
| `high` | To'q sariq | ▲▲ | Muhim | `high` incident ochiq |
| `critical` | Qizil | ✕ / ! | Kritik | `critical` incident ochiq yoki xizmat to'xtagan |
| `unknown` | Kulrang | ? | Noma'lum | Ma'lumot kelmayapti |
| `paused` | Kulrang, shtrixli chegara | ⏸ | Pauza | Pauza/maintenance |
| `info` | Ko'k | i | Ma'lumot | Faqat ma'lumot |

## 13.4.2. Aggregatsiya

Obyektning umumiy holati = tarkibiy qismlar holatlarining **eng yomoni** (`critical > high > warning > unknown > ok`). Istisno: `unknown` faqat qolganlar `ok` bo'lsa ko'rinadi (aniq muammo noma'lumdan muhimroq). Pauza/maintenance — alohida "pauza" holati, lekin ichidagi muammolar ko'rinadi.

## 13.4.3. Rang tokenlari

Ranglar CSS o'zgaruvchilari (`:root`) sifatida; yorug' va qorong'i mavzu uchun alohida qiymatlar. Mavzu: tizim sozlamasi (`prefers-color-scheme`) + qo'lda almashtirish (`data-theme`).

```
:root {
  --bg: #f7f8fa;      --surface: #ffffff;   --text: #1b1f24;   --muted: #5c6670;  --border: #dde1e6;
  --ok: #1f9d55;      --warning: #c28400;   --high: #d9622b;   --critical: #cf2e2e;
  --unknown: #7a828c; --info: #2b6fd6;
  --ok-bg: #e7f6ed;   --warning-bg: #fdf3d9; --high-bg: #fde9df; --critical-bg: #fbe5e5;
}
:root[data-theme="dark"] {
  --bg: #0f1216;      --surface: #171b21;   --text: #e6e9ed;   --muted: #98a2ad;  --border: #2a313a;
  --ok: #3ccf7a;      --warning: #f0b429;   --high: #ff8a50;   --critical: #ff5c5c;
  --unknown: #8b95a1; --info: #5c9dff;
  --ok-bg: #12301f;   --warning-bg: #3a2e0c; --high-bg: #3d2112; --critical-bg: #3d1616;
}
```

> Qiymatlar boshlang'ich; amalga oshirishda kontrast (matn ≥ 4.5:1, katta matn va grafik elementlari ≥ 3:1) avtomatik tekshiriladi (NFR-USA-003).

## 13.4.4. Grafik ranglari

- Bir grafikda ≤ 6 qator; ranglar rang ko'rlikka chidamli palitradan, qator nomi bilan legend.
- Chegara chiziqlari (warning/critical) — shtrixli, tegishli holat rangida, 40% shaffoflik.
- Ma'lumot yo'q oraliq (`null`) — ko'rinadigan bo'shliq (kulrang fon bilan "ma'lumot yo'q"), chiziq **tutashtirilmaydi**.

# 13.5. Maketlar (wireframe)

Quyidagi maketlar tuzilmani ko'rsatadi, vizual dizayn emas.

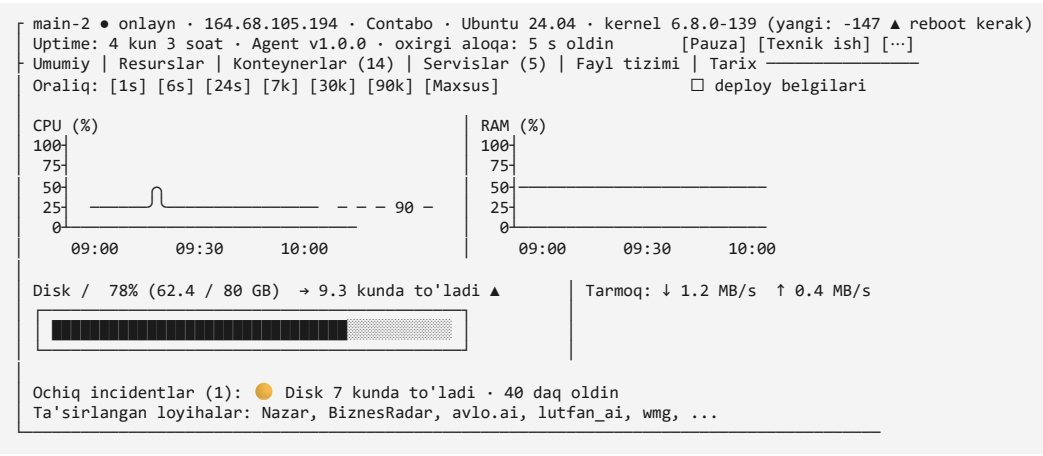
### 13.5.1. SCR-01 Dashboard (desktop)




Elementlar:

- **Banner** — butun sahifaning holati ("Hammasi joyida" yashil / "N ta muammo" ahamiyat rangida).
- **Xulosa kartochkalari** — serverlar, loyihalar, zaxira, muddatlar; har biri tegishli sahifaga havola.
- **Ochiq incidentlar** — eng og'ir tepada; tugmalar: Qabul, Ochish.
- **Serverlar jadvali** — qisqa grafiklar (sparkline, oxirgi 1 soat).
- **Loyiha kartochkalari** — holat rangi; muammo bo'lsa sababi bir qatorda.

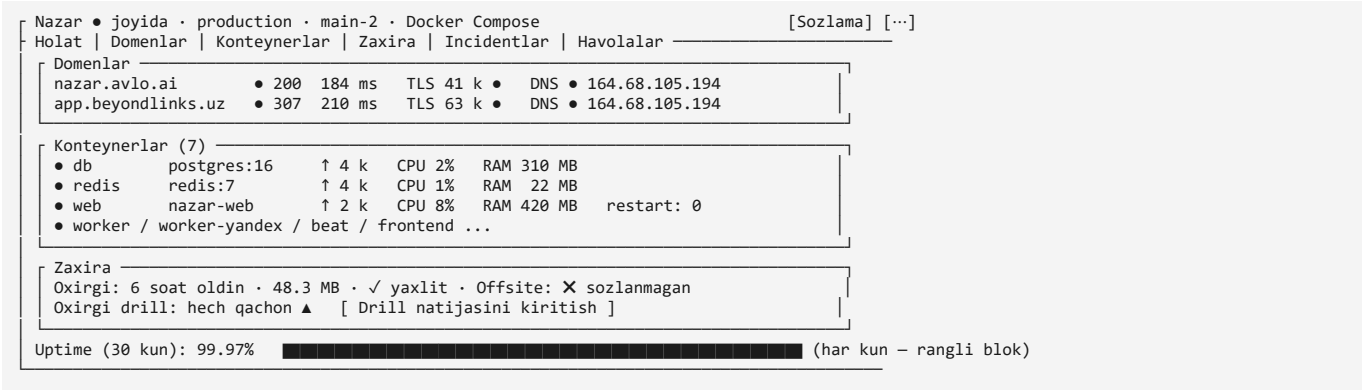
### 13.5.2. SCR-03 Server sahifasi



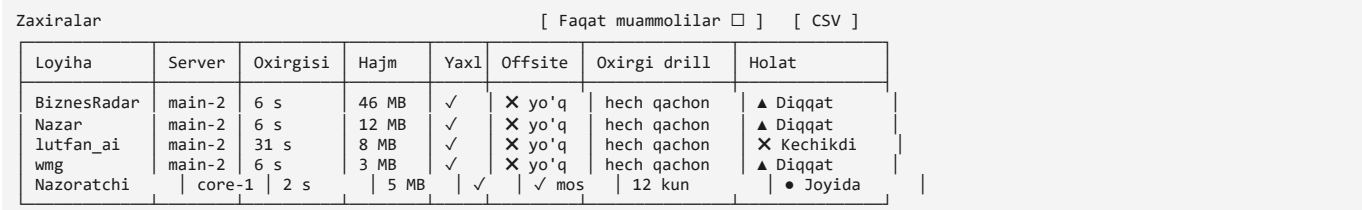

Tablar:

- **Umumiy** — "hozirgi" ko'rsatkichlar, ochiq incidentlar, ta'sirlangan loyihalar, faktlar (OS, kernel, CPU, RAM).
- **Resurslar** — CPU (rejimlar bo'yicha), yuklama, RAM/swap, disk I/O, tarmoq grafiklari.
- **Konteynerlar** — jadval: nom, loyiha, holat, uptime, CPU, RAM, qayta ishga tushishlar, log hajmi; qatorni bosish → konteyner grafiklari.
- **Servislar** — systemd unitlar holati, qayta ishga tushishlar.
- **Fayl tizimi** — har mount: hajm, band, inode, bashorat.
- **Tarix** — host voqealari (reboot, kernel o'zgardi, agent yangilandi), incidentlar.

### 13.5.3. SCR-05 Loyiha sahifasi

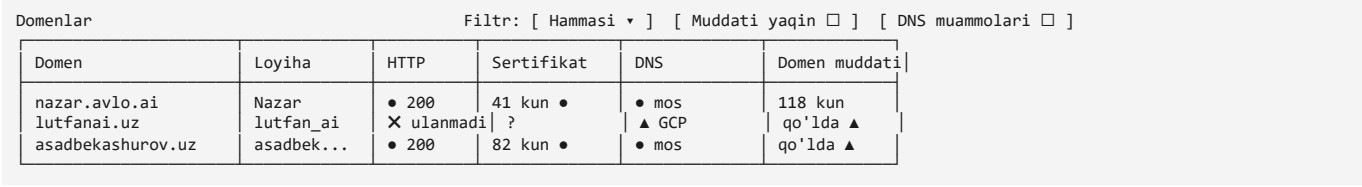



### 13.5.4. SCR-06 Zaxiralar jadvali

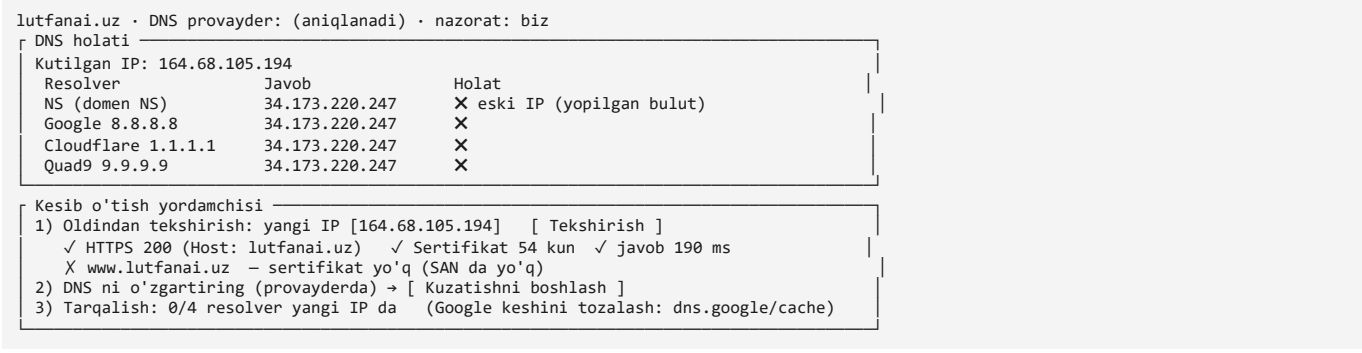



Qatorni bosish → loyiha zaxira kartasi (SCR-07): so'nggi 30 run, hajm grafigi (anomaliya nuqtalari), fayllar ro'yxati, offsite solishtirish, drill tarixi va "Drill natijasini kiritish" formasi.

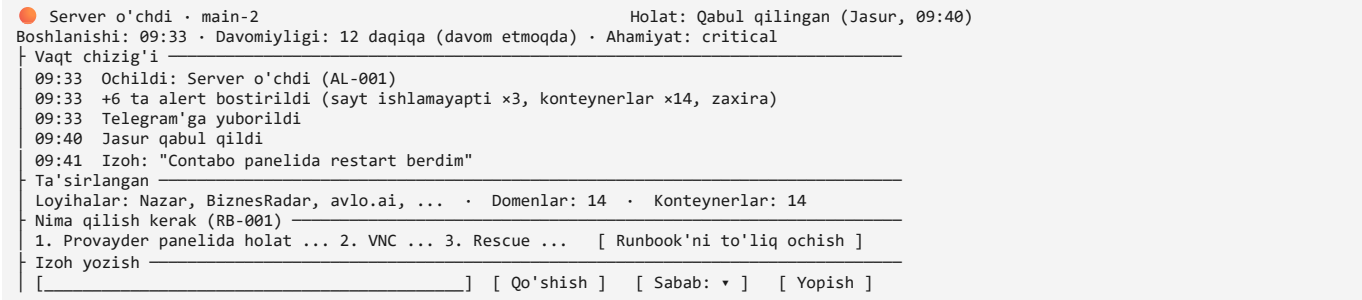
### 13.5.5. SCR-08 Domenlar va sertifikatlar



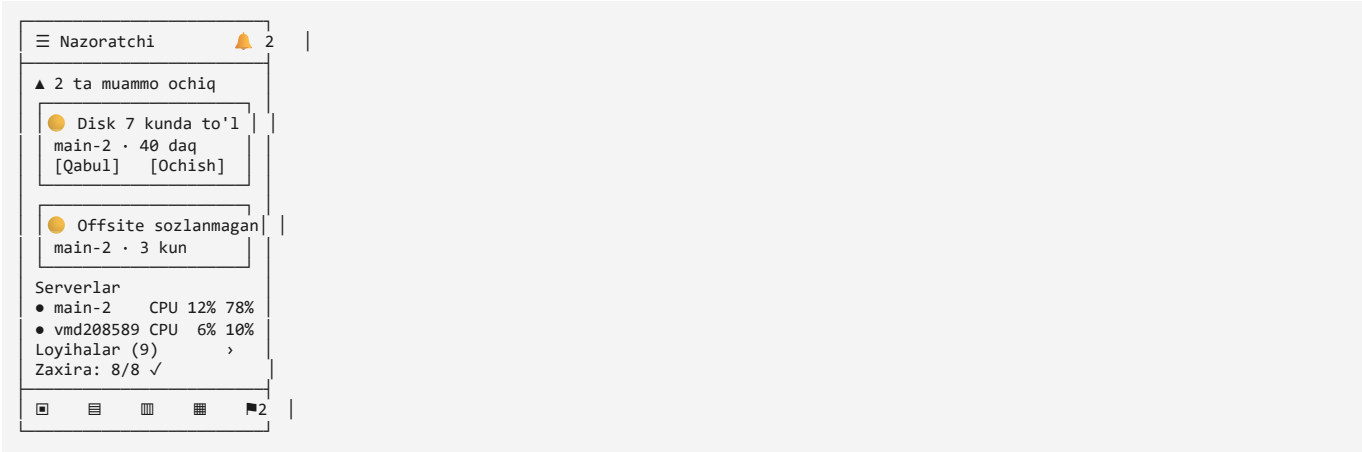

### 13.5.6. SCR-09 Domen sahifasi va kesib o'tish yordamchisi




### 13.5.7. SCR-11 Incident tafsiloti



### 13.5.8. Mobil dashboard (360–414 px)




## 13.6. Komponentlar kutubxonasi

| Komponent | Vazifa | Asosiy xususiyatlar |
|---|---|---|
| `StatusBadge` | Holat belgisi | `status` (ok/warning/high/critical/unknown/paused/info), belgi + matn, `size` |
| `StatusDot` | Kichik nuqta | Rang + `aria-label` |
| `Sparkline` | Kichik grafik | 60 nuqtagacha, rang holatga qarab, tooltip yo'q (yengil) |
| `MetricChart` | Asosiy grafik | uPlot, vaqt oralig'i, chegara chiziqlari, deploy/incident belgilari, sinxron kursor |
| `TimeRangePicker` | Oraliq tanlash | Tezkor tugmalar + maxsus; URL bilan sinxron (`?range=24h`) |
| `DataTable` | Jadval | Saralash, filtr, sahifalash/virtualizatsiya, ustunlarni yashirish, CSV |
| `IncidentCard` | Incident kartochkasi | Ahamiyat, sarlavha, vaqt, tugmalar (Qabul, Ochish) |
| `ProjectCard` | Loyiha kartochkasi | Holat, sabab, domenlar soni |
| `UptimeBar` | Kunlik uptime chizig'i | 30/90 ta rangli blok, hover → kun, foiz |
| `GaugeBar` | Foizli ko'rsatkich | Disk/RAM; chegara belgilari; bashorat yozuvi |
| `CertBadge` | Sertifikat muddati | Qolgan kun, rang chegaralari 21/7/3 |
| `BackupStatus` | Zaxira holati | Yosh, hajm, yaxlitlik, offsite, drill |
| `FreshnessLabel` | Yangilik belgisi | "3 s oldin"; > 60 s bo'lsa sariq "eskirgan" |
| `ConfirmDialog` | Tasdiqlash | Xavfli amallar uchun, nima bo'lishini yozadi |
| `CommandBlock` | Buyruq ko'rsatish | Nusxalash tugmasi (o'rnatish buyrug'i, runbook buyruqlari) |
| `EmptyState` | Bo'sh holat | Tushuntirish + keyingi qadam tugmasi |

## 13.7. Grafiklar spetsifikatsiyasi

| Mavzu | Talab |
|---|---|
| Kutubxona | uPlot (tez, kichik); ko'p grafik (20+) bir sahifada ham tez |
| Vaqt o'qi | Foydalanuvchi vaqt zonasi; oraliqqa qarab belgilar (`09:00`, `22 okt`) |
| Qiymat o'qi | Birlik bilan (`%`, `GB`, `MB/s`); bayt birliklari avtomatik (KB/MB/GB, 1024 asosida) |
| Chegara chiziqlari | Tegishli alert qoidasi chegarasi (masalan, 90%) shtrixli |
| Bo'shliqlar | `null` nuqta — chiziq uziladi, "ma'lumot yo'q" fon |
| Hodisa belgilari | Incident (rangli vertikal polosa), reboot, deploy (F3) — vertikal chiziq + tooltip |
| Tooltip | Kursor ostidagi aniq vaqt va qiymat; birdan ko'p qator bo'lsa hammasi |
| Sinxron kursor | Bir sahifadagi grafiklarda kursor vaqti sinxron |
| Zoom | Sichqoncha bilan oraliq tanlash → yaqinlashtirish; "Qaytarish" tugmasi |
| Nuqtalar soni | ≤ 1 000 (API `max_points`); undan ko'p bo'lsa backend siyraklaydi |
| Ranglar | 13.4.4 |
| Aksessibilitet | Grafik ostida "ma'lumotlar jadvali" ochiladigan blok (ekran o'quvchi uchun) |
| Yangilanish | SSE/polling bilan so'nggi nuqta qo'shiladi, butun grafik qayta chizilmaydi |
| Yuklash holati | Skeleton; xato bo'lsa "Qayta urinish" tugmasi |

## 13.8. Interaktivlik va jonli yangilanish

- **SSE** (`/api/v1/stream`) — dashboard va server sahifasi uchun. Uzilsa, avtomatik qayta ulanish (eksponensial, 1→30 s); 3 xatodan keyin polling (5 s) va "Jonli yangilanish o'chiq" belgisi.
- **Ma'lumot keshi:** TanStack Query; `staleTime` 10 s (ro'yxatlar), 30 s (grafiklar); ekran qayta fokus olganda yangilanadi.
- **Optimistik amallar:** "Qabul qilaman" — darhol UI da o'zgaradi, xato bo'lsa qaytariladi va sababi ko'rsatiladi.
- **Klaviatura:** `/` — qidiruv; `g d` — dashboard; `g s` — serverlar; `g p` — loyihalar; `g i` — incidentlar; `?` — yorliqlar ro'yxati. Matn maydonida ishlamaydi.
- **URL — holat manbai:** filtrlar, vaqt oralig'i, tab URL'da (havolani ulashish mumkin).

## 13.9. Mobil moslashuv (NFR-USA-002)

> **S9:** mobil pastki panel 5 ustun: Bosh sahifa, Serverlar, Loyihalar, Incidentlar va "Yana" menyusi (Zaxiralar, Domenlar, Texnik ish, Sozlamalar); 8 bo'lim 360 px da 3 ustunga sig'maydi.

| Element | Desktop | Mobil (< 768 px) |
|---|---|---|
| Navigatsiya | Chap panel | Pastki panel (5 band) |
| Jadvallar | To'liq ustunlar | Kartochka ko'rinishi (asosiy 3–4 maydon), qolgani "Batafsil" |
| Grafiklar | 2 ustun | 1 ustun, kamroq nuqta, tooltip bosish bilan |
| Tugmalar | Oddiy | Kamida 44×44 px |
| Incident amallari | Satr ichida | Katta tugmalar, eng muhim — "Qabul qilaman" |
| Formalar | Ko'p ustunli | 1 ustun, to'g'ri klaviatura turi (`inputmode`) |

Gorizontal sahifa skroli bo'lmasligi shart (faqat keng jadvallar ichki skroll bilan).

## 13.10. Til va formatlash (i18n)

- Tillar: `uz` (standart, lotin), `ru`, `en`. Kalitlar `i18next` JSON fayllarida: `dashboard.banner.ok`, `incident.status.acknowledged`, ...
- Ko'plik shakllari: o'zbekchada son bilan birga birlik birlik shaklida ("3 server", "1 daqiqa"); rus tilida to'liq plural qoidalari (`one/few/many`).
- **Son formati:** `uz` — bo'sh joy minglik ajratgich, vergul kasr (`12 345,6`); `en` — vergul/nuqta. `Intl.NumberFormat` ishlatiladi; qo'lda formatlash yo'q.
- **Sana/vaqt:** `Intl.DateTimeFormat`, zona `Asia/Tashkent` (sozlanadi); mutlaq format `2026-10-22 14:35`; nisbiy ("5 daqiqa oldin") — tooltip'da aniq vaqt.
- **Hajm birliklari:** `bayt, KB, MB, GB, TB` (1024). **Davomiylik:** "4 kun 3 soat", "12 daqiqa".
- **Imlo:** o'zbek lotin imlosining to'g'ri harflari (`o'`, `g'`, `sh`, `ch`, `ng`); tarjimalar lingvistik tekshiruvdan o'tadi. ASCII almashtirishlar (`o'` o'rniga `o`) qabul qilinmaydi.

## 13.11. Aksessibilitet (NFR-USA-003)

- Klaviatura bilan to'liq boshqarish; ko'rinadigan fokus halqasi.
- ARIA: holat belgilari `role="img"` + `aria-label` ("Holat: kritik"); jonli hududlar (`aria-live="polite"`) incident paydo bo'lganda.
- Rang yagona signal emas (13.4.1).
- Kontrast: matn ≥ 4.5:1; katta matn/grafik ≥ 3:1 (ikkala mavzuda).
- `prefers-reduced-motion` — animatsiyalar o'chiriladi.
- Matn kattalashtirish 200% gacha tartib buzilmaydi.

## 13.12. Bo'sh, yuklanish va xato holatlari

| Holat | Ko'rsatish |
|---|---|
| Birinchi ishga tushirish (serverlar yo'q) | "Hali server qo'shilmagan" + katta "Server qo'shish" tugmasi (UC-01) |
| Ma'lumot yuklanmoqda | Skeleton (kartochka/jadval shaklida) |
| Server yangi (hali ma'lumot yo'q) | "Birinchi ma'lumot kutilmoqda..." + o'rnatish buyrug'i qayta ko'rsatiladi |
| Ma'lumot eskirgan | Sariq "Ma'lumot N daqiqa oldin yangilangan" |
| API xatosi | Qisqa xabar + "Qayta urinish" + `request_id` (nusxalash) |

| Holat | Ko'rsatish |
|---|---|
| Ruxsat yo'q (403) | "Bu bo'limni ko'rish uchun ruxsat yo'q" (rolni ko'rsatadi) |
| Filtr natijasiz | "Hech narsa topilmadi" + "Filtrni tozalash" |
| Aloqa yo'q (offline) | Yuqorida banner: "Internet aloqasi yo'q"; oxirgi keshdagi ma'lumot ko'rinadi |

## 13.13. Frontend texnik tafsilotlari

```
frontend/
  src/
    app/            # router, providerlar, layout
    features/       # dashboard, servers, projects, backups, domains, incidents, alerts, settings, auth
    components/     # umumiy komponentlar (13.6)
    api/            # OpenAPI dan avtomatik yaratilgan klient va tiplar (openapi-typescript)
    lib/            # format (Intl), vaqt, yordamchilar
    i18n/           # uz.json, ru.json, en.json
    styles/         # tokens.css (13.4.3), global
  tests/            # Vitest (birlik), Playwright (E2E)
```

| Mavzu | Qaror |
|---|---|
| Til | TypeScript (`strict`) |
| Router | React Router |
| Server holati | TanStack Query |
| UI holati | React konteksti / Zustand (kichik); Redux ishlatilmaydi |
| Stil | Tailwind CSS + CSS tokenlari |
| Grafiklar | uPlot (qoplama: `MetricChart`) |
| Formalar | React Hook Form + `zod` |
| API klient | `openapi-typescript` + `openapi-fetch`; sxema o'zgarsa CI tipni qayta yaratadi |
| Test | Vitest + Testing Library; Playwright E2E (asosiy oqimlar) |
| Byudjet | Boshlang'ich JS ≤ 400 KB (gzip), LCP ≤ 2.5 s (4G), CLS < 0.1 |
| Xavfsizlik | CSP bilan mos (inline skriptsiz); XSS: React escaping, `dangerouslySetInnerHTML` taqiqlangan |

## 13.14. Forma spetsifikatsiyalari (asosiylari)

### 13.14.1. Server qo'shish (SCR-17)

| Maydon | Tur | Majburiy | Tekshiruv |
|---|---|---|---|
| Nom | matn | ha | 2–40 belgi, noyob, harf/raqam/ `-` / `_` |
| Provayder | tanlov/matn | yo'q | Contabo, Hetzner, GCP, ... |
| Ommaviy IPv4 | matn | yo'q (agent aniqlaydi) | IPv4 formati |
| Rol | tanlov | yo'q | production, staging, backup, core |
| Teglar | teglar | yo'q | kichik harf, `-` |
| Izoh | matn | yo'q | ≤ 500 belgi |

Yuborilgandan keyin: o'rnatish buyrug'i (`CommandBlock`), "Birinchi aloqa kutilmoqda..." (avtomatik, SSE) → ulangach "Ulandi ✓" va server sahifasiga o'tish tugmasi.

### 13.14.2. Loyiha qo'shish (SCR-18)

| Maydon | Tur | Majburiy | Tekshiruv |
|---|---|---|---|
| Nom | matn | ha | 2–60 |
| `slug` (`appctl` papka nomi) | matn | ha | `^[A-Za-z0-9_-]+$`, noyob |
| Tur | tanlov | ha | compose / systemd / static / other |
| Server(lar) + muhit | ko'p tanlov | ha | production/staging |
| Portlar | raqamlar | yo'q | 1024–65535, serverda band emas (FR-REG-010); "Keyingi bo'sh" tugmasi |
| Domenlar | ro'yxat | yo'q | FQDN; har biri uchun kutilgan IP (standart: serverning IP si), `primary` belgisi |
| Zaxira siyosati | blok | yo'q | Kunlik DB? haftalik fayl? offsite? (standart 10.4.1) |
| Avtomatik tekshiruvlar | belgi | — | HTTP + TLS + DNS (yoqilgan) |

### 13.14.3. Silence va texnik ish

Silence: moslash (server/loyiha/qoida/...), muddat (tezkor: 1 s, 4 s, 24 s, 7 k, maxsus), sabab (majburiy). Maintenance: qamrov, boshlanish/tugash, sabab, "Nazorat ro'yxati" (12.9.8).

## 13.15. Mikromatn qo'llanmasi (o'zbekcha)

- Qisqa, aniq, buyruq maylida: "Saqlash", "Bekor qilish", "Qabul qilaman".
- Texnik atamalar o'zgarishsiz: `Docker`, `nginx`, `DNS`, `TLS`, `appctl`; qolgan so'zlar o'zbekcha: "zaxira nusxa" (backup), "sertifikat", "holat", "ogohlantirish".
- Xato xabari: nima bo'ldi + nima qilish ("Domen noto'g'ri kiritilgan. Misol: `example.uz`").
- Raqam va birlik orasida bo'shliq: "12 daqiqa", "46 MB", "78 %" emas — "78%" (foiz yopishiq).
- Tasdiqlash xabarida oqibat: "Token bekor qilinadi, agent ulana olmaydi. Davom etasizmi?"
- Salbiy emas, yo'naltiruvchi: "Server qo'shilmagan" emas, "Birinchi serveringizni qo'shing".

## 13.16. Qabul sinovlari (UI)

| № | Sinov | Kutilgan natija |
|---|---|---|
| T-UI-01 | Dashboard 5 soniya qoidasi: yangi odamga ko'rsatib, "hammasi joyidami?" deb so'rash | To'g'ri javob ≤ 5 s |
| T-UI-02 | 360 px kenglikda barcha asosiy ekranlar | Gorizontal skroll yo'q, tugmalar bosiladi |
| T-UI-03 | Qorong'i va yorug' mavzu kontrasti (axe) | Jiddiy xato yo'q |
| T-UI-04 | Klaviatura bilan Dashboard → Incident → Qabul qilish | Sichqonchasiz bajariladi |
| T-UI-05 | SSE uzish | Polling'ga o'tadi, belgi chiqadi, tiklanganda qaytadi |
| T-UI-06 | 20 grafik bir sahifada | Kadr tezligi ≥ 30 fps, o'lchov ≥ 2 s ichida yuklanadi |
| T-UI-07 | Til almashtirish uz/ru/en | Hamma matn o'zgaradi, qattiq yozilgan matn yo'q |
| T-UI-08 | Ma'lumot yo'q oraliq (agent o'chgan) | Grafikda bo'shliq, chiziq tutashmaydi |
| T-UI-09 | Cheklangan viewer | Faqat o'z loyihasi ko'rinadi (F3) |
| T-UI-10 | Eski ma'lumot (agent jim) | "Eskirgan" belgisi, qiymat "hozirgi" deb ko'rsatilmaydi |

# 14. Xavfsizlik

## 14.1. Nima himoya qilinadi

Monitoring tizimi o'zi "kalit" emas, lekin **xarita**: u barcha serverlar, loyihalar, domenlar va zaxiralar joylashuvini bir joyga to'playdi. Hujumchi uchun bu razvedka. Undan tashqari, tizimda bir nechta maxfiy qiymat bor.

| Aktiv | Ahamiyati | Buzilsa nima bo'ladi |
|---|---|---|
| **Infratuzilma xaritasi** (reyestr: IP, domen, port, loyiha) | Yuqori | Hujum yuzasi ochiladi; maqsadli hujum osonlashadi |
| **Telegram bot tokeni** | Yuqori | Soxta alertlar yuborish, xabarlarni o'qish |
| **Agent tokenlari** | O'rta | Soxta metrika yuborish (monitoringni aldash: "hammasi joyida" deb ko'rsatish) |
| **Agent yangilanish kanali** | **Juda yuqori** | Hujumchi barcha serverlarda o'z kodini bajara oladi |
| **Foydalanuvchi hisoblari** | Yuqori | Sozlamalarni o'zgartirish, alertlarni o'chirish |
| **Tekshiruv sarlavhalari** (`Authorization`, API kalitlar) | O'rta | Tekshirilayotgan xizmatga kirish |
| **Zaxira holati ma'lumoti** | O'rta | Qaysi zaxira zaif ekanini bilish |
| **Audit jurnali** | O'rta | Izlarni yashirish |
| **Core serveri** | Yuqori | Yuqoridagilarning hammasi |

## 14.2. Xavf modeli (STRIDE)

| № | Tahdid | Turi | Ehtimol | Ta'sir | Chora (talab) |
|---|---|---|---|---|---|
| TH-01 | Kirish sahifasiga parol terib hujum (bruteforce, credential stuffing) | Spoofing | Yuqori | Yuqori | Argon2id, blok (5/15 daq), 2FA, tezlik chegarasi (FR-AUTH-001/003/007) |
| TH-02 | Sessiya o'g'irlanishi (XSS, cookie) | Spoofing | O'rta | Yuqori | `HttpOnly/Secure/SameSite`, CSP, XSS himoyasi, qisqa muddat (14.8) |
| TH-03 | CSRF orqali amal bajartirish | Tampering | O'rta | O'rta | CSRF token, `SameSite=Lax`, `Origin` tekshiruvi |
| TH-04 | Soxta agent (token taxmin qilish/o'g'irlash) | Spoofing | Past | O'rta | 256-bit token, xesh saqlash, tezlik chegarasi, bekor qilish (FR-AUTH-006) |
| TH-05 | Buzilgan agent hostdan soxta ma'lumot (alertni yashirish) | Tampering | Past | O'rta | Tashqi faol tekshiruvlar mustaqil; heartbeat + tashqi TCP:22; ma'lumot sifat tekshiruvlari |
| TH-06 | Core buzilib, agentlarga zararli yangilanish yuborish | Elevation | Past | **Juda yuqori** | Imzolangan yangilanish; imzo kaliti core'da **yo'q** (14.5.4) |
| TH-07 | Core buzilib, agent tokenlarini yig'ish | Info disclosure | Past | O'rta | Tokenlar bazada faqat xesh; agent faqat ma'lumot yuboradi, buyruq olmaydi (14.5.3) |
| TH-08 | SSRF: prober orqali ichki tarmoqqa so'rov | Info disclosure / Elevation | O'rta | Yuqori | Shaxsiy diapazonlarni rad etish (6.8), yo'naltirishdan keyin ham tekshirish |
| TH-09 | SQL injection | Tampering | Past | Yuqori | ORM/parametrli so'rovlar; xom SQL faqat tasdiqlangan joylarda |
| TH-10 | XSS orqali saqlangan ma'lumot (loyiha nomi, izoh, konteyner nomi) | Tampering | O'rta | Yuqori | React escaping; markdown izohlarni tozalash (`DOMPurify`); `dangerouslySetInnerHTML` taqiq |
| TH-11 | Webhook/Telegram callback soxtalashtirish | Spoofing | O'rta | O'rta | Maxfiy URL segmenti, `X-Telegram-Bot-Api-Secret-Token`, chat ID ro'yxati, HMAC imzoli callback |
| TH-12 | Maxfiy qiymatlar loglarga tushishi | Info disclosure | O'rta | O'rta | Log filtrlari, `NFR-SEC-008`, avtomatik skaner |
| TH-13 | Core DoS: ingest'ni to'ldirish | DoS | O'rta | O'rta | Tezlik chegarasi, paket hajmi 1 MB, token bo'yicha cheklov, nginx `limit_req` |
| TH-14 | Disk to'ldirish (metrika xaosi, yuqori kardinallik) | DoS | O'rta | O'rta | Kardinallik chegarasi (≤ 1 000 series/host), retention, disk alerti |
| TH-15 | Ichki odam: operator rolidan ortiqcha amal | Elevation | Past | O'rta | RBAC, obyekt darajasida tekshiruv, audit jurnali |
| TH-16 | Zaxira (Nazoratchi bazasi) ma'lumoti sizib chiqishi | Info disclosure | Past | Yuqori | Telegram'ga yuboriladigan zaxira `age` bilan shifrlangan, maxfiy kalit alohida (Owner'da) |
| TH-17 | Ta'minot zanjiri: zararli kutubxona | Tampering | O'rta | Yuqori | Versiyalarni qotirish, `pip-audit` / `npm audit` / `govulncheck`, minimal bog'liqlik |
| TH-18 | Man-in-the-middle (agent ↔ core) | Tampering / Info | Past | Yuqori | TLS 1.2+, sertifikat tekshiruvi (CA qotirish ixtiyoriy) |

| № | Tahdid | Turi | Ehtimol | Ta'sir | Chora (talab) |
|---|---|---|---|---|---|
| TH-19 | Docker socket orqali root huquqi (agent) | Elevation | O'rta | **Juda yuqori** | `docker-socket-proxy` (faqat GET), agent root emas, `docker` guruhida emas |
| TH-20 | Monitoringni "ko'r" qilish: Nazoratchi o'zi o'chirilsa | DoS | O'rta | Yuqori | Alohida serverdagi watchdog (S-01, FR-SYS-002) |

## 14.3. Autentifikatsiya va sessiya

### 14.3.1. Parol

| Parametr | Qiymat |
|---|---|
| Algoritm | Argon2id; `time_cost=3`, `memory_cost=64 MiB`, `parallelism=2` (o'lchov bilan sozlanadi: ≤ 300 ms) |
| Minimal uzunlik | 12; maksimal 128 |
| Tekshiruv | Eng ko'p ishlatiladigan 10 000 ta parol ro'yxati bilan taqqoslash; login/email bilan o'xshashlik |
| Saqlash | Faqat xesh (tuz ichida); parol hech qayerda saqlanmaydi/loglanmaydi |
| O'zgartirish | Joriy parol talab qilinadi; boshqa sessiyalar tugatiladi |
| Muddat | Majburiy muddatli almashtirish yo'q (NIST 800-63B); buzilish gumonida majburiy almashtirish |

### 14.3.2. Kirishga urinishlar

- 5 ta ketma-ket xato (login + IP bo'yicha ikkalasi ham hisoblanadi) → 15 daqiqa blok (`423`), har keyingi blok 2 baravar uzayadi (maks. 24 soat).
- Xato javoblari "login yoki parol noto'g'ri" bir xil matn va **bir xil vaqt** (timing farqini yashirish uchun soxta xesh solishtirish).
- Barcha urinishlar audit jurnalida (muvaffaqiyatli va muvaffaqiyatsiz).
- Egasi (Owner) uchun muvaffaqiyatsiz urinishlar > 10/soat → Telegram ogohlantirish.

### 14.3.3. Ikki bosqichli tasdiqlash (TOTP)

- RFC 6238; 6 raqam, 30 s, ±1 oyna. Sir 160 bit, bazada shifrlangan (`totp_secret_enc`).
- Bir marta ishlatiladigan zaxira kodlar (10 ta, xeshlangan).
- Owner roli uchun 2FA **majburiy** (F2 dan); boshqalar uchun tavsiya.
- Qayta ishlatishdan himoya: bir kod bir marta (oxirgi ishlatilgan vaqt oynasi saqlanadi).

### 14.3.4. Sessiya

| Parametr | Qiymat |
|---|---|
| Saqlash | Server tomonida (baza/Redis); cookie faqat tasodifiy identifikator |
| Cookie | `HttpOnly; Secure; SameSite=Lax; Path=/`, nomi `__Host-nzr_sid` (faqat prod; dev'da http bo'lgani uchun `nzr_sid`, `Secure`/`__Host-` brauzer tomonidan rad etilardi) |
| Muddat | Harakatsizlik 12 soat; mutlaq 7 kun |
| Aylantirish | Kirishdan keyin va rol/parol o'zgarganda yangi identifikator |
| Chiqish | Serverda sessiya o'chiriladi; "barcha qurilmalardan chiqish" |
| Bir vaqtdagi sessiyalar | Cheklanmagan, lekin ro'yxatda ko'rinadi (qurilma, IP, vaqt) |

### 14.3.5. Parolni tiklash

Email orqali emas (email hisobi alohida zaif nuqta) — Telegram bot orqali: faqat oldindan bog'langan `telegram_chat_id` ga 15 daqiqalik bir martalik havola. Havola ishlatilgach, barcha sessiyalar tugatiladi, 2FA qayta tasdiqlanadi (yoqilgan bo'lsa).

## 14.4. Avtorizatsiya

- **RBAC** (3.2 matritsasi) markaziy dekorator/ruxsat sinfi bilan; har endpoint o'z ruxsat talabini e'lon qiladi; e'lon qilinmagan endpoint **rad etiladi** (xavfsiz standart).
- **Obyekt darajasidagi tekshiruv:** viewer faqat `user_projects` dagi loyihalar va ularga bog'liq obyektlarni ko'radi; ID taxmin qilish `404` beradi (mavjudligini oshkor qilmaydi).
- **Maxfiy maydonlar** (kanal sozlamasi, tekshiruv sarlavhalari) API javobida **hech qachon** to'liq qaytmaydi (`••••` bilan maskalanadi); o'zgartirish uchun yangi qiymat yuboriladi.
- **Eng kam huquq:** Operator foydalanuvchi, token, kanal, tizim sozlamalarini o'zgartira olmaydi.
- **Audit:** barcha o'zgartiruvchi amallar (FR-AUD-001).

# 14.5. Agent xavfsizligi

## 14.5.1. Enroll va token hayoti




- `agent_token` faqat agent fayli `/etc/nazoratchi/token` (rejim `0600`, egasi `nazoratchi`) da.
- Token **host bilan bog'langan**: boshqa hostdan ishlatilsa (ma'lumot `host_id` mos emas yoki IP o'zgargan bo'lsa) — ogohlantirish (`info`), lekin rad etilmaydi (dinamik IP lar bo'lishi mumkin).
- Rotatsiya: Owner yangi token yaratadi → agentga `reload_config` orqali yetkaziladi (eski token 24 soat parallel ishlaydi) → eski bekor qilinadi. Tavsiya: yiliga 1 marta yoki buzilish gumonida.
- Bekor qilish: darhol (`revoked_at`); agent 401 oladi va 5 daqiqada qayta urinadi (14.5.3).

## 14.5.2. Agent nima qila olmaydi (cheklov kafolati)

- Masofaviy buyruq bajarmaydi (shell yo'q) — core faqat 4 turdagi buyruq beradi (7.5.3).
- Konteyner yaratmaydi/to'xtatmaydi/o'chirmaydi (Docker proksi — faqat GET).
- Fayl mazmunini o'qimaydi (faqat nom, hajm, `mtime`, `gzip -t` natijasi).
- Hech qanday kiruvchi port ochmaydi.
- `root` emas; `NoNewPrivileges`, `ProtectSystem=strict`, bo'sh `CapabilityBoundingSet`.

## 14.5.3. Core buzilsa agentga ta'siri

Agent core'dan faqat quyidagilarni qabul qiladi: `set_interval`, `reload_config`, `collect_now`, `update`. Birinchi uchtasi zararsiz (resurs sarfini oshirishdan tashqari, intervalda 5 s pastki chegara bor). `update` **xavfli** — shu sabab 14.5.4.

## 14.5.4. Imzolangan yangilanishlar (TH-06)

Talab (NFR-SEC-012, majburiy):

1. Agent binarlari CI muhitida `minisign` **maxfiy kaliti** bilan imzolanadi. Maxfiy kalit faqat CI sirlarida (yoki offlayn, egasida) turadi; **core serverda yo'q**.
2. Ochiq kalit agent binariga **joylangan** (compile-time) — yangilanish orqali ham, sozlama orqali ham o'zgartirib bo'lmaydi (kalit almashtirish uchun yangi qo'lda o'rnatish yoki ikkita kalitli "o'tish" davri).
3. Agent `update` buyrug'ini qabul qilganda: faylni yuklaydi → imzoni tekshiradi → **faqat shundan keyin** almashtiradi. Imzo mos kelmasa — rad etadi, `agent_update_rejected` voqeasini yuboradi va logga yozadi.
4. Eskiga qaytarishdan himoya (rollback attack): yangilanish versiyasi hozirgidan **katta** bo'lishi kerak (yoki aniq `--allow-downgrade` bilan).
5. `install.sh` (birinchi o'rnatish) core'dan yuklanadi — bu **trust-on-first-use** (TOFU). Yumshatish: skript ichidagi minisign ochiq kaliti va SHA-256 qiymati hujjatda (README) alohida ham e'lon qilinadi; operator o'rnatishdan oldin `sha256sum install.sh` ni solishtira oladi.

## 14.5.5. Soxta ma'lumotdan himoya (TH-05)

Buzilgan agent "hammasi joyida" deb yolg'on yuborishi mumkin. Cheklov (to'liq himoya emas):

- Faol tekshiruvlar (HTTP/TLS/DNS/TCP) mustaqil va agentga bog'liq emas.
- Heartbeat tashqi TCP:22/ICMP bilan birga baholanadi.
- Statistik tekshiruv: qiymatlar diapazondan tashqarida (masalan, `cpu_busy_percent > 100`) yoki namunalar vaqti kelajakda bo'lsa — rad etiladi va `ingest_rejected` hisoblanadi.
- Agent versiyasi/`boot_id` o'zgarishi voqea sifatida yoziladi.

# 14.6. Maxfiy ma'lumotlarni saqlash

| Maxfiy qiymat | Saqlash | Himoya |
|---|---|---|
| Foydalanuvchi parollari | `users.password_hash` | Argon2id |
| Agent/API/heartbeat tokenlari | `*.token_hash` | SHA-256 (256-bit tasodifiy tokenlar uchun yetarli); token faqat yaratilganda ko'rsatiladi |
| Telegram bot tokeni, SMTP paroli, webhook siri | `channels.config_enc` | AES-256-GCM, har yozuv uchun yangi `nonce`, `aad = channel_id` |

| Maxfiy qiymat | Saqlash | Himoya |
|---|---|---|
| Tekshiruv sarlavhalari | `probes.secrets_enc` | AES-256-GCM |
| TOTP siri | `users.totp_secret_enc` | AES-256-GCM |
| Django `SECRET_KEY` | Muhit o'zgaruvchisi / fayl | Repoda yo'q |
| Shifrlash asosiy kaliti (`NAZORATCHI_MASTER_KEY`, 32 bayt) | `/opt/applications/projects/Nazoratchi/.env` yoki `/etc/nazoratchi/master.key` | Fayl rejimi `0400`, egasi — servis foydalanuvchisi; bazadan alohida; zaxirada **alohida** saqlanadi (parol menejeri) |
| Imzo kaliti (minisign) | CI sirlari / oflayn | Core'da yo'q |

**Kalit rotatsiyasi:** `master.key` almashtirish — `nazoratchi rotate-key` buyrug'i (barcha `*_enc` maydonlarni eski kalit bilan ochib, yangisi bilan qayta shifrlaydi; tranzaksiyada). Kalitni yo'qotish = shifrlangan maydonlarni yo'qotish (kanal sozlamalarini qayta kiritish kerak) — shuning uchun kalit nusxasi parol menejerida saqlanishi shart.

# 14.7. Tarmoq va infratuzilma xavfsizligi

## 14.7.1. Core serveri

| Chora | Tafsilot |
|---|---|
| Firewall (`ufw`) | Faqat 22 (kalit bilan), 80, 443 |
| SSH | Faqat kalit (`PasswordAuthentication no`), `root` kirishi taqiqlangan yoki `prohibit-password`; `fail2ban` |
| Docker | Portlar `127.0.0.1` ga bog'langan (Docker nashr qilingan portlar `ufw` ni chetlab o'tadi); `db`, `redis` xostga chiqarilmaydi |
| nginx | `server_tokens off`; TLS 1.2+; zamonaviy shifrlar; HSTS; `limit_req` (ingest, login) |
| Yangilanishlar | `unattended-upgrades` (xavfsizlik); **reboot rejalashtirilgan** oyna bilan (L-4 saboqi) |
| Snapshot | Kunlik provayder snapshot; reboot/kernel oldidan qo'lda |
| Core ni kuzatish | Watchdog (S-01); core'ga ham agent o'rnatiladi (disk/RAM) |

## 14.7.2. SSRF himoyasi (prober)

1. Domen/URL kiritilganda va **har so'rovda** (yo'naltirishdan keyin ham) manzil aniqlanadi.
2. Rad etiladigan diapazonlar: `127.0.0.0/8`, `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`, `169.254.0.0/16` (bulut metadata), `100.64.0.0/10`, `::1`, `fc00::/7`, `fe80::/10`.
3. DNS rebinding: aniqlangan IP bilan ulaniladi (qayta aniqlanmaydi).
4. Faqat `http`/`https` protokollari; `file://`, `gopher://` va h.k. taqiq.
5. Maxsus istisno (loopback tekshiruvi uchun) faqat Owner tomonidan, har tekshiruv uchun alohida `allow_private=true` bilan va audit bilan.
6. Javob tanasi ≤ 64 KB o'qiladi; timeout qattiq.

## 14.7.3. Tezlik chegaralari (nginx va ilova)

| Endpoint | Chegara |
|---|---|
| `/auth/login` | IP bo'yicha 10/daqiqa; hisob bo'yicha 5 xato/15 daq |
| `/ingest/v1/metrics` | Token bo'yicha 30/daqiqa (normal ≈ 4/daqiqa) |
| `/ingest/v1/enroll` | IP bo'yicha 10/soat |
| `/hb/<token>` | Token bo'yicha 10/daqiqa |
| `/api/v1/*` | Foydalanuvchi bo'yicha 120/daqiqa |
| Telegram webhook | Telegram IP diapazonlari (ixtiyoriy) + maxfiy segment |

# 14.8. Ilova xavfsizligi

## 14.8.1. Kirishni tekshirish va chiqishni kodlash

- Barcha kirish DRF serializerlari bilan (turi, uzunligi, formati); noma'lum maydonlar rad etiladi.
- ORM parametrlari; xom SQL (agregatlar) faqat parametr bog'lash bilan (`%s`).
- JSON javoblar `Content-Type: application/json` + `X-Content-Type-Options: nosniff`.
- Markdown izohlar (incident, loyiha) — cheklangan (qalin, havola, ro'yxat), `DOMPurify` bilan tozalash, havolalar `rel="noopener noreferrer"`.

### 14.8.2. Sarlavhalar

```
Strict-Transport-Security: max-age=31536000; includeSubDomains
Content-Security-Policy: default-src 'self'; script-src 'self'; style-src 'self'; img-src 'self' data:;
                         connect-src 'self'; frame-ancestors 'none'; base-uri 'none'; form-action 'self'
X-Content-Type-Options: nosniff
Referrer-Policy: same-origin
X-Frame-Options: DENY
Permissions-Policy: camera=(), microphone=(), geolocation=()
Cross-Origin-Opener-Policy: same-origin
```

CORS: ruxsat yo'q (panel va API bir xil origin). Agent API (`/ingest`) CORS talab qilmaydi.

### 14.8.3. CSRF

Cookie-sessiyali so'rovlarda `X-CSRFToken` majburiy; `Origin`/`Referer` tekshiruvi; API tokenli so'rovlarda (cookie yo'q) CSRF talab qilinmaydi.

### 14.8.4. Fayl yuklash

Tizimda fayl yuklash **yo'q** (F1–F3); import (reyestr YAML) — hajm ≤ 1 MB, xavfsiz YAML yuklovchi (`safe_load`), sxema validatsiyasi.

## 14.9. Loglar va audit (xavfsizlik jihati)

- Loglarda taqiqlanganlar: parol, token, `Authorization`, `Cookie`, `Set-Cookie`, `.env` qiymatlari, TOTP kodlari, Telegram bot tokeni (URL ichida bo'lishi mumkin: `api.telegram.org/bot<TOKEN>/...` — `httpx` loglari filtrlanadi).
- Maxfiy qiymatlarni maskalash filtri (regex: `(?i)((?:token|secret|password|authorization)\S*\s*[=:]\s*(?:(?:bearer|basic)\s+)?)\S+`, qiymat `***` bilan almashadi; `Authorization: Bearer <token>` da "Bearer" dan keyingi token ham yopilishi shart — P-02 testi eski regexda shu xatoni topdi) barcha log yozuvlariga qo'llanadi; test: sun'iy token yuborilganda logda ko'rinmasligi.
- Audit jurnali faqat qo'shiladi; ilova DB roli `UPDATE/DELETE` huquqiga ega emas (8.8).
- Xavfsizlik hodisalari alohida belgilanadi: `login_failed`, `account_locked`, `token_revoked`, `agent_update_rejected`, `permission_denied`. F1–F2 da ular kunlik/haftalik hisobotda ko'rsatiladi; alohida alert qoidasi F3 da qo'shilishi mumkin (hozir katalogda yo'q).

## 14.10. Nazoratchi'ning o'z zaxirasi va tiklash

- Kunlik mantiqiy `pg_dump` (reyestr, incident, audit, sozlamalar; metrikalar kirmaydi) + `.env` (shifrlangan) + `master.key` **emas** (u alohida saqlanadi).
- `age` bilan shifrlanadi (NFR-SEC-010): ochiq kalit core'da, maxfiy kalit — faqat Owner'da (parol menejeri).
- **Saqlash joyi (Q-13):** Owner'ning shaxsiy Telegram chati (bot orqali, `sendDocument`; guruhga emas). Bot API fayl chegarasi 50 MB: katta bo'lsa bo'laklanadi yoki metrikasiz sxema kichraytiriladi. Telegram faqat shifrlangan faylni ko'radi. Owner botni bir marta `/start` qilgan bo'lishi kerak.
- Yuborish natijasi qayd etiladi; ketma-ket 2 kun yuborilmasa alert. Owner chatni o'chirib yubormasligi kerak.
- Tiklash tartibi sinovdan o'tkaziladi (yiliga 1 marta; NFR-REL-006), 15.9-bo'lim.

## 14.11. Ta'minot zanjiri

| Chora | Tafsilot |
|---|---|
| Versiyalarni qotirish | `uv.lock`/`requirements.txt` (hash bilan), `pnpm-lock.yaml`, `go.sum`; Docker image'lar versiya tegi bilan (digest tavsiya) |
| Skanerlash | CI da `pip-audit`, `npm audit`, `govulncheck`, `trivy` (image); jiddiy zaiflikda build to'xtaydi |
| Minimal bog'liqlik | Yangi kutubxona qo'shish sababi PR da yoziladi |
| SBOM | CI har relizda (`syft`) SBOM yaratadi |
| Image kelib chiqishi | Image'lar faqat o'zimizning CI da quriladi; `ghcr.io` yoki o'z registry; imzo (`cosign`) — F3 |
| Dependabot/Renovate | Haftalik yangilanish PR lari |
| Imzo kaliti | CI sirlarida, 2FA bilan himoyalangan hisob; kamida 2 kishi (yoki egasi + zaxira) ruxsat |

> Eslatma (2026-10 tajribasidan): MinIO ommaviy image'lari endi tortib bo'lmadi va manbadan qurish kerak bo'ldi. Bu "uchinchi tomon image'iga bog'liqlik" xavfini ko'rsatdi: Nazoratchi faqat o'z image'lari va keng qo'llab-quvvatlanadigan rasmiy image'lardan (`postgres`, `redis`, `nginx`, `timescale/timescaledb`) foydalanadi va ularning nusxasini o'z registry'ga qo'yish mumkin bo'lishi uchun versiyalarni qotiradi.

## 14.12. Hodisaga javob (Nazoratchi uchun)

### 14.12.1. Agent tokeni sizdi (yoki hostda buzilish gumoni)

1. Panelda tokenni bekor qilish (Sozlamalar → Tokenlar).
2. Hostda: agentni to'xtatish, `token` faylini o'chirish, hostni tekshirish (RB-014 dagi qadamlar).
3. Yangi enroll tokeni → agentni qayta o'rnatish.
4. Audit jurnalida o'sha token bilan so'nggi 7 kundagi kirishlar (`last_used_ip`) ko'rib chiqiladi.

### 14.12.2. Core buzilishi gumoni

1. Core'ni tarmoqdan ajratish (firewall/provayder panel).
2. **Agentlarni to'xtatish shart emas:** `update` imzosiz ishlamaydi (14.5.4), shuning uchun agentlar xavfsiz. Ular core'ga ma'lumot yuborishda davom etadi (buferga yig'adi) — core tiklanguncha tashqi watchdog va provayder monitoringiga tayaning.
3. Yangi serverda zaxiradan tiklash; **barcha maxfiy qiymatlarni almashtirish**: `master.key`, Django `SECRET_KEY`, Telegram bot tokeni (BotFather orqali yangi), SMTP paroli, barcha foydalanuvchi parollari va sessiyalar, agent tokenlari, tekshiruv sarlavhalari.
4. Hodisa tahlili: kirish vektori, davomiyligi, nima o'qilgan.

### 14.12.3. Telegram bot tokeni sizdi

BotFather'da `/revoke`, yangi token → `channels` da yangilash → eski tokenli xabarlar yuborilmaydi. Webhook maxfiy segmentini ham almashtirish.

## 14.13. Xavfsizlik tekshiruv ro'yxati (reliz oldidan)

| № | Tekshiruv | Qanday |
|---|---|---|
| S-01 | OWASP ASVS L1 (barcha) va L2 (autentifikatsiya, sessiya, kirish tekshiruvi) bandlari | Ro'yxat bo'yicha qo'lda + ZAP |
| S-02 | `testssl.sh` — A daraja | Avtomatik |
| S-03 | `securityheaders.com` — A | Avtomatik |
| S-04 | Maxfiy qiymat loglarda yo'q | Loglarni skaner qilish testi |
| S-05 | Bazada ochiq matnli maxfiy qiymat yo'q | `SELECT` bilan tekshirish |
| S-06 | Agent: root emas, `docker` guruhida emas, yangilanish imzosi | Qo'lda + test (buzuq imzo bilan rad etish) |
| S-07 | Prober: ichki IP ga so'rov rad etiladi | Birlik va E2E test |
| S-08 | Tezlik chegaralari ishlaydi | `k6` bilan |
| S-09 | RBAC: viewer/operator ruxsatsiz endpointlarga `403` | Avtomatik test (matritsa bo'yicha) |
| S-10 | CSRF: tokensiz POST rad etiladi | Avtomatik |
| S-11 | `db`/`redis` portlari tashqaridan yopiq | `nmap` tashqaridan |
| S-12 | Bog'liqlik skanerlari toza (jiddiy zaiflik yo'q) | CI |
| S-13 | Zaxira tiklash sinovi muvaffaqiyatli | Qo'lda |

## 14.14. Qoldiq xavflar (qabul qilingan)

| № | Xavf | Sabab | Qabul sharti |
|---|---|---|---|
| RS-01 | TOFU: `install.sh` birinchi yuklanishda core'ga ishonadi | Soddalik; hozircha bitta egasi | Ochiq kalit va skript xeshi README da e'lon qilinadi |
| RS-02 | Watchdog va Nazoratchi bir egasi hisoblari ostida (Telegram bot ham) | Kichik jamoa | Alohida bot; server bir (S-00, ADR-014 istisnosi, RS-06) |
| RS-03 | Buzilgan agent soxta ma'lumot yuborishi mumkin | Push modelining tabiati | Faol tekshiruvlar mustaqil (14.5.5) |
| RS-04 | Bitta core (SPOF) | Byudjet | RTO ≤ 1 soat, watchdog |
| RS-05 | Parol bilan SSH (core uchun tavsiya etilmaydi; kalitga o'tish rejalashtirilgan) | Hozirgi serverlarda shunday | Core uchun kalit majburiy; S-01 uchun alohida ish |
| RS-06 | S-00 (core, offsite) va S-01 (watchdog, ishchi loyihalar) bir provayderda; ikkalasi bir vaqtda tushsa hech kim xabar bermaydi | Owner qarori (2026-10-26), byudjet | Aniqlash: o'zaro kuzatuv. Yumshatish ixtiyoriy: tashqi bepul xizmat; provayder snapshoti; S-01 zaxiralari S-01 da ham qoladi |

| № | Xavf | Sabab | Qabul sharti |
|---|---|---|---|
| RS-07 | Nazoratchi o'z zaxirasi Telegram'da: Telegram yoki Owner hisobi yo'qolsa zaxira yo'qoladi; Bot API fayl chegarasi 50 MB | Q-13 qarori (2026-10-26) | `age` shifrlash; chatni o'chirmaslik; vaqti-vaqti bilan nusxani qo'lda saqlash; tiklash mashqi (15.9) |

# 15. O'rnatish, deploy va ekspluatatsiya

## 15.1. Muhitlar

| Muhit | Maqsad | Qayerda | Ma'lumot |
|---|---|---|---|
| **dev** | Ishlab chiqish | Dasturchi kompyuteri (Docker Compose, `docker-compose.dev.yml`) | Sun'iy (generator) |
| **ci** | Avtomatik testlar | GitHub Actions ishchisi | Vaqtinchalik |
| **staging** (ixtiyoriy, F2) | Reliz oldi sinov, yuklama testi | Core bilan bir xil konfiguratsiyada kichik VPS yoki vaqtincha | Sinov agentlari + generator |
| **prod** | Haqiqiy monitoring | S-00 (15.2) | Haqiqiy |

Tavsiya: staging alohida VPS sifatida doimiy bo'lmasligi mumkin; yuklama testi vaqtida vaqtincha ko'tarib, tugagach o'chirish yetarli (xarajatni tejash).

## 15.2. Core serverini tayyorlash (S-00)

> **Owner qarori (2026-10-26).** Core S-00 da (`185.214.135.9`, Contabo, 200 GB SSD). Server tozalanib Ubuntu 24.04 bilan yangidan o'rnatiladi; u oldin yuklanmagan (1.3.1), shu sabab o'rnatishdan keyin darhol provayder snapshoti olinadi. S-01 uchun offsite zaxira ham shu serverda; watchdog esa S-01 da. `retired_ips` dan bu IP olib tashlanadi; unga qaragan eski DNS yozuvlari (masalan, `avloai.uz`) ko'chiriladi, nginx'da `default_server` `return 444` bo'ladi.

### 15.2.1. Talablar

| Parametr | Minimal | Tavsiya |
|---|---|---|
| OS | Ubuntu 24.04 LTS | Ubuntu 24.04 LTS |
| CPU / RAM | 2 vCPU / 4 GB | 2–4 vCPU / 4–8 GB |
| Disk | 40 GB SSD | 80 GB SSD |
| Tarmoq | Statik ommaviy IPv4 | + IPv6 |
| Joylashuv | Kuzatiladigan serverlardan **boshqa server**, imkon bo'lsa boshqa provayder | — |
| Domen | `nazoratchi.<domen>` (A yozuvi shu serverga) | — |

### 15.2.2. Boshlang'ich sozlash (bir marta)

```
# 1. Tizim
apt-get update && apt-get -y upgrade
timedatectl set-timezone UTC          # server vaqti UTC da
apt-get install -y ufw fail2ban unattended-upgrades curl ca-certificates gnupg nginx certbot python3-certbot-nginx

# 2. Foydalanuvchi va SSH (parolsiz, kalit bilan)
adduser --disabled-password --gecos "" ops
usermod -aG sudo ops
mkdir -p /home/ops/.ssh && cp ~/.ssh/authorized_keys /home/ops/.ssh/ && chown -R ops:ops /home/ops/.ssh
printf 'PasswordAuthentication no\nPermitRootLogin no\n' > /etc/ssh/sshd_config.d/00-hardening.conf
sshd -t && systemctl reload ssh      # YANGI terminalda ops bilan kirib tekshiring, keyin sessiyani yoping

# 3. Firewall
ufw allow 22/tcp && ufw allow 80/tcp && ufw allow 443/tcp && ufw --force enable

# 4. Docker
curl -fsSL https://get.docker.com | sh
cat > /etc/docker/daemon.json <<'EOF'
{ "log-driver": "json-file", "log-opts": { "max-size": "10m", "max-file": "3" }, "live-restore": true }
EOF
systemctl restart docker

# 5. appctl va papkalar (mavjud deploy/server/setup.sh bilan)
bash setup.sh --fail2ban
```

> **Muhim (L-4 saboqi):** `unattended-upgrades` yangi kernel o'rnatishi mumkin. Avtomatik reboot **o'chiq** bo'lishi kerak (`Unattended-Upgrade::Automatic-Reboot "false"`); reboot rejalashtirilgan oynada, snapshotdan keyin qo'lda bajariladi. Nazoratchi `reboot_required` ni alert qiladi (AL-032, AL-033).

### 15.2.3. Papkalar

```
/opt/applications/
  projects/Nazoratchi/            # compose, .env, .appctl
    docker-compose.yml
    .env                        # maxfiy (0600)
    .appctl
    nginx/                      # (ixtiyoriy) konteyner ichidagi konfiglar
  nginx/nazoratchi.<domen>.conf   # host nginx konfigi
  backups/Nazoratchi/             # appctl zaxiralari
/etc/nazoratchi/master.key        # 0400 (yoki .env ichida)
```

## 15.3. Repozitoriy tuzilmasi

```
nazoratchi/                       # monorepo
  backend/                    # Django loyihasi (6.4-bo'lim)
    apps/ workers/ config/ tests/
    pyproject.toml uv.lock
  frontend/                   # React (13.13)
  agent/                      # Go agenti
    cmd/nazoratchi-agent/ internal/collectors/ internal/sender/ ...
    go.mod
  deploy/
    docker-compose.yml        # prod
    docker-compose.dev.yml
    nginx/nazoratchi.conf.template
    install.sh                # agent o'rnatuvchi (shablon)
    appctl-patches/           # appctl ga o'zgarishlar (10.3)
  docs/
    TZ/                       # shu hujjat
    runbooks/                 # RB-001 ... (18.2)
    adr/                      # arxitektura qarorlari
  seed/inventory.yaml         # boshlang'ich reyestr (8.11, 18.5)
  .github/workflows/          # CI/CD
```

## 15.4. Docker Compose (prod)

`/opt/applications/projects/Nazoratchi/docker-compose.yml`:

```yaml
name: nazoratchi

x-app: &app
  image: ghcr.io/${GHCR_OWNER}/nazoratchi-backend:${NAZORATCHI_VERSION}
  env_file: .env
  restart: unless-stopped
  depends_on:
    db:    { condition: service_healthy }
    redis: { condition: service_healthy }
  logging:
    driver: json-file
    options: { max-size: "10m", max-file: "3" }

services:
  db:
    image: timescale/timescaledb:2.17.2-pg16        # versiya reliz oldidan tekshiriladi
    restart: unless-stopped
    environment:
      POSTGRES_DB: ${DB_NAME}
      POSTGRES_USER: ${DB_USER}
      POSTGRES_PASSWORD: ${DB_PASSWORD}
    volumes:
      - nazoratchi_pgdata:/var/lib/postgresql/data
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U ${DB_USER} -d ${DB_NAME}"]
      interval: 10s
      timeout: 5s
      retries: 10
    shm_size: 256mb
    command: >
      postgres -c shared_buffers=512MB -c effective_cache_size=1536MB
               -c max_connections=60 -c work_mem=16MB
               -c timescaledb.max_background_workers=8
               -c shared_preload_libraries=timescaledb

  redis:
    image: redis:7-alpine
    restart: unless-stopped
    command: ["redis-server", "--save", "", "--appendonly", "no", "--maxmemory", "256mb", "--maxmemory-policy", "allkeys-lru"]
    healthcheck:
      test: ["CMD", "redis-cli", "ping"]
      interval: 10s
      timeout: 3s
      retries: 10

  web:
    <<: *app
    command: ["uvicorn", "config.asgi:application", "--host", "0.0.0.0", "--port", "8000", "--workers", "2", "--proxy-headers"]
    ports: ["127.0.0.1:8040:8000"]
    healthcheck:
      test: ["CMD", "python", "-c", "import urllib.request,sys; sys.exit(0 if urllib.request.urlopen('http://127.0.0.1:8000/healthz').status==
      interval: 15s
      timeout: 5s
      retries: 5

  evaluator:                            # alert dvigateli (har 15 s); S3 da: manage.py buyrug'i
    <<: *app
    command: ["python", "manage.py", "run_evaluator", "--interval", "15"]
    healthcheck: { disable: true }      # image'dagi HEALTHCHECK web uchun; ishchilarda :8000 yo'q

  prober:
    <<: *app
    command: ["python", "manage.py", "run_prober"]
    healthcheck: { disable: true }

  notifier:                             # navbatdagi bildirishnomalarni yuboradi (S3 da evaluator ichiga o'tishi mumkin)
    <<: *app
    command: ["python", "manage.py", "notify_worker"]
    healthcheck: { disable: true }

  worker:
    <<: *app
    command: ["celery", "-A", "config", "worker", "-l", "info", "--concurrency", "2"]

  beat:
    <<: *app
    command: ["celery", "-A", "config", "beat", "-l", "info", "--schedule", "/data/celerybeat-schedule"]
    volumes:
      - nazoratchi_beat:/data

  frontend:
    image: ghcr.io/${GHCR_OWNER}/nazoratchi-frontend:${NAZORATCHI_VERSION}
    restart: unless-stopped
    ports: ["127.0.0.1:3040:80"]

volumes:
  nazoratchi_pgdata:
  nazoratchi_beat:
```

> **S9 da amalga oshirilgan:** `deploy/docker-compose.yml`. Celery (`worker`/`beat`) yo'q: kodda ishlatilmaydi. Qo'shimcha servis `self-backup` (`manage.py self_backup --loop --hour 3`). `web` agent fayllari uchun `AGENT_DIST_HOST_DIR` ni `/data/agent` ga o'qish rejimida ulaydi. Backend image repo ildizidan quriladi (`docker build -f backend/Dockerfile`

> `.`): `install.sh` va `seed/` image ichida. Frontend image `nginxinc/nginx-unprivileged` (konteyner porti 8080, host `127.0.0.1:3040`).

Eslatmalar:

- Portlar faqat `127.0.0.1` ga (Docker nashri `ufw` ni chetlab o'tadi).
- `evaluator` va `prober` **bitta nusxada** ishlaydi (`replicas` yo'q). Ikkinchi nusxa tasodifan ishga tushsa, takroriy alert bo'lmasligi uchun baza darajasidagi `advisory lock` bilan "leader" tanlanadi (6.2). Amalda: `apps/core/leader.py`, alohida ulanishda `pg_try_advisory_lock`; leader bo'lmagan nusxa har davrda qayta urinadi, leader o'lsa qulf o'zi bo'shaydi. Django ulanishi `connect_timeout=5` bilan: baza javob bermasa so'rovlar osilib qolmaydi.
- `web`, `worker`, `evaluator`, `prober` bir image'dan, buyruqlari farqli.
- Migratsiyalar `deploy` paytida alohida bir martalik buyruq bilan (15.7).

## 15.5. Muhit o'zgaruvchilari (`.env`)

`.env.example` (qiymatlar o'rniga izoh; **haqiqiy qiymatlar repoda yo'q**):

| O'zgaruvchi | Misol / format | Izoh |
|---|---|---|
| `NAZORATCHI_VERSION` | `1.0.0` | Image tegi |
| `GHCR_OWNER` | `<tashkilot>` | Registry egasi |
| `DJANGO_SETTINGS_MODULE` | `config.settings.prod` | |
| `SECRET_KEY` | 50+ belgi, tasodifiy | Django |
| `ALLOWED_HOSTS` | `nazoratchi.example.uz` | Vergul bilan. Prod sozlamasi `127.0.0.1` va `localhost` ni o'zi qo'shadi: konteyner healthcheck va host watchdog `/healthz` ga shu orqali kiradi (aks holda 400) |
| `CSRF_TRUSTED_ORIGINS` | `https://nazoratchi.example.uz` | |
| `PUBLIC_URL` | `https://nazoratchi.example.uz` | Alert havolalari, install buyrug'i |
| `DB_NAME`, `DB_USER`, `DB_PASSWORD`, `DB_HOST=db`, `DB_PORT=5432` | | Bazaga ulanish |
| `REDIS_URL` | `redis://redis:6379/0` | |
| `NAZORATCHI_MASTER_KEY` | base64, 32 bayt | Maxfiy maydonlarni shifrlash (14.6). **Majburiy**: yo'q yoki 32 bayt emas bo'lsa core ishga tushmaydi |
| `TELEGRAM_API_BASE` | `https://api.telegram.org` | Ixtiyoriy; faqat sinov uchun (soxta Bot API) |
| `TELEGRAM_WEBHOOK_SECRET` | tasodifiy, `[A-Za-z0-9_-]`, 32+ belgi | Tugmalar (ack, jim) uchun webhook siri; bo'sh = webhook o'chiq (9.12) |
| `TZ_DEFAULT` | `Asia/Tashkent` | UI uchun standart zona |
| `TELEGRAM_BOT_TOKEN` | `<TOKEN>` | **Faqat boshlang'ich**; keyin panel orqali kanalda saqlanadi (shifrlangan) |
| `TELEGRAM_WEBHOOK_SECRET` | tasodifiy | Webhook yo'li va sarlavha |
| `SMTP_HOST`, `SMTP_PORT`, `SMTP_USER`, `SMTP_PASSWORD`, `SMTP_FROM` | | Email kanal (ixtiyoriy) |
| `WATCHDOG_URL` | `https://watchdog.example.uz/ping/<id>` | S-01 dagi watchdog (FR-SYS-002) |
| `RETIRED_IPS` | `34.173.220.247` | DNS "yopilgan IP" ro'yxati (11.4.2) |
| `RETENTION_RAW_DAYS` | `14` | 8.10 |
| `RETENTION_1M_DAYS` | `90` | |
| `RETENTION_5M_DAYS` | `365` | |
| `RETENTION_1H_DAYS` | `1095` | |
| `INGEST_MAX_BODY_BYTES` | `1048576` | |
| `LOG_LEVEL` | `INFO` | |
| `SENTRY_DSN` | bo'sh | Ixtiyoriy |

Qoida: `.env` fayli rejimi `0600`, egasi — deploy foydalanuvchisi; repoda yo'q; zaxirada shifrlangan holda.

## 15.6. Host nginx (core)

`/opt/applications/nginx/nazoratchi.example.uz.conf` (HTTP qismini certbot to'ldiradi):

```
limit_req_zone $binary_remote_addr zone=nzr_login:10m rate=10r/m;
limit_req_zone $http_authorization zone=nzr_ingest:10m rate=30r/m;
limit_req_zone $binary_remote_addr zone=nzr_api:10m rate=20r/s;

server {
    server_name nazoratchi.example.uz;
    server_tokens off;

    # Xavfsizlik sarlavhalari (14.8.2)
    add_header Strict-Transport-Security "max-age=31536000; includeSubDomains" always;
    add_header X-Content-Type-Options nosniff always;
    add_header X-Frame-Options DENY always;
    add_header Referrer-Policy same-origin always;
    add_header Content-Security-Policy "default-src 'self'; img-src 'self' data:; frame-ancestors 'none'; base-uri 'none'; form-action 'self'"

    client_max_body_size 2m;

    location /ingest/ {
        limit_req zone=nzr_ingest burst=10 nodelay;
        proxy_pass http://127.0.0.1:8040;
        include /etc/nginx/proxy_params;
    }
    location /hb/ {
        limit_req zone=nzr_ingest burst=10 nodelay;
        proxy_pass http://127.0.0.1:8040;
        include /etc/nginx/proxy_params;
    }
    location /api/v1/auth/login {
        limit_req zone=nzr_login burst=5 nodelay;
        proxy_pass http://127.0.0.1:8040;
        include /etc/nginx/proxy_params;
    }
    location /api/ {
        limit_req zone=nzr_api burst=40 nodelay;
        proxy_pass http://127.0.0.1:8040;
        include /etc/nginx/proxy_params;
    }
    location ~ ^/(healthz|install\.sh|agent/) {
        proxy_pass http://127.0.0.1:8040;
        include /etc/nginx/proxy_params;
    }
    location ~ ^/(metrics|readyz) { return 404; }   # ichki endpointlar tashqariga berilmaydi

    location / {                                    # SPA
        proxy_pass http://127.0.0.1:3040;
        include /etc/nginx/proxy_params;
    }

    listen 80;
}
# certbot --nginx -d nazoratchi.example.uz  → 443 va yo'naltirishni o'zi qo'shadi
```

`/etc/nginx/proxy_params` standart: `Host`, `X-Real-IP`, `X-Forwarded-For`, `X-Forwarded-Proto`. Eslatma: `limit_req` kaliti `$http_authorization` — agent tokeni bo'yicha chegara.

## 15.7. Birinchi ishga tushirish (bootstrap)

1. DNS: `nazoratchi.example.uz` A → S-00 IP (`185.214.135.9`); `appctl nginx`; `certbot --nginx -d ...`.
2. `docker compose pull && docker compose up -d db redis`.
3. Migratsiya (bir martalik): `docker compose run --rm web python manage.py migrate`.
4. Birinchi Owner yaratish: `docker compose run --rm web python manage.py createowner --login jasur` (parol terminalda so'raladi; skriptda saqlanmaydi).
5. `docker compose up -d` (hamma servis).
6. Boshlang'ich reyestr: `manage.py seed_inventory seed/inventory.yaml --dry-run`, so'ng haqiqiy.
7. Panelga kirish (HTTPS), 2FA yoqish (Owner uchun majburiy).
8. Telegram kanalini sozlash va **sinov xabari** (FR-ALR-020).
9. Watchdog S-01 da o'rnatiladi (alohida bot, systemd, nginx qoidasi va subdomen; `appctl` ga tegilmaydi); `WATCHDOG_URL` shunga qaratiladi; sinov: core'ni to'xtatib, 5–6 daqiqada xabar kelishini tekshirish. Telegram'da zaxira uchun `age` kalit jufti yaratiladi (maxfiy kalit parol menejeriga) va birinchi sinov zaxira yuboriladi.
10. Birinchi agent (S-00, core'ning o'zi va offsite papka; keyin S-02) — UC-01; panelda "Onlayn" ni tekshirish.
11. `appctl` ga Nazoratchi `.appctl` yozish va `appctl backup Nazoratchi`.

> **S9:** 3-5-qadamlar `deploy/bootstrap.sh <login>` bilan bajariladi (idempotent). Reyestr seed buyrug'i `manage.py seed_registry --file seed/inventory.yaml [--dry-run]`.

## 15.8. Agentlarni mavjud serverlarga tarqatish

Tartib — kamroq xavfdan ko'proqqa:

| Qadam | Server | Sabab | Tekshiruv |
|---|---|---|---|
| 1 | S-00 (core + offsite) | Yangi o'rnatilgan, ishlab turgan loyiha yo'q | Agent ishlaydi, zaxira skaneri bo'sh papkada xatosiz; disk/RAM ko'rinadi |
| 2 | S-02 (vmd208589) | Aaclub staging bor; Docker bor; resurs katta | Docker proksi ishlaydi; `nginx.conf` ga tegilmadi |
| 3 | S-01 (main-2) | Asosiy server | Yuk oshmadi (CPU ≤ +1%); 24 soat kuzatuv |

S-01 uchun qo'shimcha oldi shartlar:

- `appctl` ga `state.json`/`_run.json`/`--json` o'zgartirishlari o'rnatiladi (10.3) va bir marta `appctl backup all` bilan tekshiriladi.
- `docker-socket-proxy` konteyneri `127.0.0.1:2375` ga (faqat GET).
- ACL: `setfacl -R -m u:nazoratchi:rX /opt/applications/backups` (zaxira papkasi); `/etc/letsencrypt` uchun deploy-hook (7.7.2 eslatmasi).
- `nazoratchi` foydalanuvchisi `docker` guruhida **emas**.
- Cron'larga heartbeat qo'shiladi (`appctl-backup`, `certbot-renew`).

S-02 uchun ehtiyot choralari:

- `/etc/nginx/nginx.conf` front-proksiga **tegilmaydi**. Agent faqat o'qiydi.
- `asadbekashurov-certbot` cron'iga heartbeat qo'shish — `/etc/cron.d/asadbekashurov-certbot` ni tahrirlash (faqat shu fayl).

## 15.9. Nazoratchi'ni tiklash (yangi serverda) — RB-044

Maqsad: RTO ≤ 1 soat (NFR-REL-006).

> **S9:** haqiqiy buyruqlar `docs/runbooks/RB-044.md` da (`manage.py restore_self_backup <fayl> --identity <kalit> --yes`).

1. Yangi server (15.2) — 20 daqiqa.
2. `docker` va `appctl` tayyor; `Nazoratchi` papkasini yaratish; eng yangi zaxirani Owner'ning Telegram chatidan yuklab olish va `age` maxfiy kaliti (parol menejeridan) bilan ochish.
3. `.env` ni tiklash (shifrlangan zaxiradan) **va** `master.key` ni parol menejeridan.
4. `docker compose up -d db`, bazani tiklash: `appctl restore Nazoratchi <fayl.sql.gz> --yes` (yangi bo'sh serverda — xavfsiz).
5. `docker compose up -d` (hammasi).
6. DNS: `nazoratchi.example.uz` yangi IP ga (TTL kichik bo'lishi uchun doimiy 300 s).
7. Agentlar yangi core'ga ulanadi (domen o'zgarmagani uchun avtomatik); buferdagi ma'lumot yuboriladi.
8. Sinov: panel ochiladi, reyestr to'liq, sinov alerti keladi.
9. Natija va vaqt — drill jurnaliga.

## 15.10. CI/CD (GitHub Actions)

### 15.10.1. Pipeline bosqichlari

| Bosqich | Qachon | Nima qiladi | Muvaffaqiyatsiz bo'lsa |
|---|---|---|---|
| `lint` | Har PR/push | `ruff`, `mypy`, `eslint`, `tsc --noEmit`, `go vet`, `staticcheck` | Birlashtirish bloklanadi |
| `test-backend` | Har PR/push | `pytest` (Timescale konteyneri bilan), qamrov hisoboti | Bloklanadi |
| `test-frontend` | Har PR/push | `vitest`; `playwright` (asosiy oqimlar) | Bloklanadi |
| `test-agent` | Har PR/push | `go test ./...` (fixtures bilan), amd64+arm64 cross-compile | Bloklanadi |
| `security` | Har PR/push + haftalik | `pip-audit`, `npm audit`, `govulncheck`, `trivy fs` | Jiddiy zaiflikda bloklanadi |
| `build` | `main` ga birlashganda | Docker image'lar (backend, frontend), `nazoratchi-agent` binarlari, SBOM | — |
| `sign` | Reliz tegida (`v*`) | Agent binarlarini `minisign` bilan imzolash (maxfiy kalit CI sirida) | Reliz to'xtaydi |
| `release` | Reliz tegida | GitHub Release, image'larni `ghcr.io` ga, agent artefaktlarini core'ga yuklash uchun paket | — |

| Bosqich | Qachon | Nima qiladi | Muvaffaqiyatsiz bo'lsa |
|---|---|---|---|
| `deploy-prod` | Qo'lda (`workflow_dispatch`) yoki reliz tasdiqlangach | `ssh` orqali: `appctl deploy Nazoratchi` (15.11) | Avtomatik rollback (15.11.3) |

> **S9:** `release.yml` (teg `v*`): backend/frontend image'lari ghcr.io ga (SBOM, provenance), agent amd64/arm64 binarlari, `SHA256SUMS` va `SHA256SUMS.minisig` (sirlar `MINISIGN_SECRET_KEY`, `MINISIGN_PASSWORD`), GitHub Release. CI ga shellcheck, nginx `-t`, prod compose, frontend image va `install.sh` funksional testi qo'shildi.

### 15.10.2. Namuna workflow qismi (test-backend)

```
name: ci
on: [push, pull_request]
jobs:
  test-backend:
    runs-on: ubuntu-24.04
    services:
      db:
        image: timescale/timescaledb:2.17.2-pg16
        env: { POSTGRES_PASSWORD: ci, POSTGRES_DB: nazoratchi }
        ports: ["5432:5432"]
        options: >-
          --health-cmd "pg_isready -U postgres" --health-interval 5s --health-retries 10
      redis:
        image: redis:7-alpine
        ports: ["6379:6379"]
    steps:
      - uses: actions/checkout@v4
      - uses: astral-sh/setup-uv@v5
      - run: uv sync --frozen
        working-directory: backend
      - run: uv run ruff check . && uv run mypy apps
        working-directory: backend
      - run: uv run pytest --cov=apps --cov-fail-under=70
        working-directory: backend
        env:
          DB_HOST: localhost
          DB_PASSWORD: ci
          REDIS_URL: redis://localhost:6379/0
```

Sirlar (GitHub Secrets): `MINISIGN_KEY`, `GHCR_TOKEN`, `DEPLOY_SSH_KEY` (faqat deploy foydalanuvchisi, cheklangan `command=`), `DEPLOY_HOST`. Maxfiy qiymatlar loglarga chiqmaydi.

## 15.11. Yangilash (deploy) tartibi

### 15.11.1. Oddiy yangilash (backend/frontend)

1. Oldindan: oxirgi zaxira yangi (`appctl backup Nazoratchi`), ochiq `critical` incident yo'q.
2. `NAZORATCHI_VERSION` ni yangilash (`.env`), `docker compose pull`.
3. Migratsiyalar: `docker compose run --rm web python manage.py migrate --noinput` (orqaga qaytariladigan; katta jadvallarda `CONCURRENTLY` va `atomic = False`).
4. `docker compose up -d --remove-orphans` (`appctl deploy Nazoratchi` shuni bajaradi).
5. Health check: `/healthz` 200, `evaluator_heartbeat` yangilanyapti, sinov ingest 200.
6. Kuzatuv: 15 daqiqa xatolar yo'qligini ko'rish (panel → "Tizim holati").

**Uzilish:** `web` qayta ishga tushganda ingest 5–10 soniya javob bermaydi; agent buferi buni qoplaydi (NFR-REL-002). Alert hisoblash 15–30 soniya to'xtaydi (bir martalik `for` siljishi).

### 15.11.2. Agent reliz tarqatish

1. Yangi agent binari core'ga `/agent/v1/` ga yuklanadi (CI imzolagan).
2. Core `latest` ni yangilaydi; hostlar `update` buyrug'ini **bosqichma-bosqich** oladi: avval 1 ta (S-00), 1 soat kutish, so'ng qolganlari.
3. Agent o'zini yangilaydi, imzoni tekshiradi, rollback himoyasi bilan (7.7.3).

### 15.11.3. Rollback

- Backend: oldingi `NAZORATCHI_VERSION` ga qaytish va `docker compose up -d`. Migratsiya orqaga qaytariladigan bo'lsa, `migrate <app> <oldingi_raqam>`; aks holda (nisbatan kam hollarda) — bazani oldindan olingan zaxiradan tiklash (RTO ichida).
- Agent: `.old` binarga avtomatik qaytish (7.7.3), yoki core'da `latest` ni oldingi versiyaga o'rnatib `update` yuborish (versiya pasaytirish uchun maxsus bayroq kerak — 14.5.4).

## 15.12. Ekspluatatsiya: davriy vazifalar

| Davriylik | Vazifa | Kim |
|---|---|---|
| Doimiy | Alertlarga javob | Operator/Owner |
| Kunlik (avtomatik) | Zaxira (03:15), retention, kunlik hisobot (08:00) | Tizim |
| Haftalik | Haftalik hisobotni ko'rish; ochiq doimiy ogohlantirishlarni tahlil qilish; `docker image prune` kerakmi | Operator |
| Oylik | Alert sifatini ko'rib chiqish (12.11); bog'liqlik yangilanishlari; disk/baza hajmi tendensiyasi; sertifikat va domen muddatlari | Operator |
| Chorak | Restore drill (har loyiha); Nazoratchi'ni tiklash mashqi (15.9); agent token rotatsiyasini ko'rib chiqish | Operator |
| Yillik | `master.key` va tokenlar rotatsiyasi; xavfsizlik tekshiruvi (14.13); Timescale/Postgres/Docker katta yangilanishlari rejasi | Owner |

## 15.13. Monitoring monitoringi

| Nimani | Kim kuzatadi | Qanday |
|---|---|---|
| Core ishlayaptimi | Watchdog (S-01) | Core har 60 s ping; 5 daq. kelmasa Telegram |
| Core host (disk/RAM) | Core'dagi agent (o'zini kuzatadi) + watchdog (to'liq uzilish uchun) | AL-019..AL-023, AL-014/015 |
| Alert dvigateli | `evaluator_heartbeat` | AL-084 |
| Bildirishnoma | `notifications` xato soni | AL-086 |
| Zaxira | Nazoratchi'ning o'z `BACKUP` qoidalari | AL-068.. |

**Muhim cheklov:** Core to'liq o'lsa, alertlar uni kuzatuvchi watchdog'dan boshqa joydan kelmaydi. Shu sababli watchdog (a) core'dan alohida serverda (S-01), (b) alohida Telegram botida bo'lishi shart (ADR-014). O'zaro kuzatuv: S-01 ni core kuzatadi. Ikkalasi bir vaqtda tushsa (bir provayder) xabar bo'lmaydi (RS-06).

## 15.14. Masshtablash bosqichlari

| Belgi | Choralar |
|---|---|
| Baza hajmi > 80% byudjetdan | Retention qisqartirish (xom 14 → 7 kun), `rollup=false` metrikalarni agregatlamaslik, diskni oshirish |
| Ingest p95 > 300 ms | `web` workerlarini 2 → 4; `db` `shared_buffers`; indeks tekshiruvi |
| 30+ host | Core: 4 vCPU / 8 GB; Timescale chunk intervalini ko'rib chiqish |
| 100+ host | Ingest'ni alohida `ingest` servisga ajratish (gorizontal), agent paketlarini Redis navbatiga; alohida DB serveri |
| Core ishonchliligi talabi oshsa | Ikkinchi core (replika, X-6 qamrovdan chiqarilgan: alohida loyiha) |

## 15.15. Tizimni to'xtatish (decommission)

1. Barcha agentlarda `nazoratchi-agent uninstall`.
2. Ma'lumotni eksport (reyestr YAML, incident tarixi, audit CSV) va arxivlash.
3. Core'ni to'xtatish; DNS yozuvini o'chirish; `master.key` va maxfiy qiymatlarni yo'q qilish.
4. Telegram botini o'chirish yoki tokenni bekor qilish.
5. Watchdog'ni to'xtatish.

## 15.16. Qabul sinovlari (o'rnatish)

| № | Sinov | Kutilgan natija |
|---|---|---|
| T-DEP-01 | Toza Ubuntu 24.04 da 15.2 + 15.7 | 60 daqiqada ishlaydigan panel |
| T-DEP-02 | Server reboot | 3 daqiqada barcha servis ko'tariladi, qo'lda aralashuvsiz (NFR-REL-009) |
| T-DEP-03 | Yangilash (15.11.1) | Ma'lumot yo'qolmaydi; agent buferi qoplaydi |
| T-DEP-04 | Migratsiya xato bilan to'xtasa | Eski versiyaga qaytish ishlaydi |
| T-DEP-05 | Tiklash (15.9) | RTO ≤ 60 daqiqa, reyestr to'liq |
| T-DEP-06 | Agent o'rnatish 3 serverda | Hammasi onlayn; S-01 da yuk ortishi ≤ 1% |
| T-DEP-07 | CI pipeline: PR → build → release → imzolangan agent | Imzo tekshiruvidan o'tadi |
| T-DEP-08 | Core uzoq o'chiq (30 daq) → qayta yoqish | Agent buferi bo'shliqsiz yuboriladi |

# 16. Test rejasi va qabul mezonlari

## 16.1. Strategiya

Monitoring tizimida eng xavfli nuqson — **"jim nosozlik"**: tizim ishlayotgandek ko'rinadi, lekin alert chiqarmaydi. Shu sababli test strategiyasi oddiy funksional tekshiruvdan tashqari quyidagilarga alohida e'tibor beradi:

1. **Alert to'g'riligi** (to'g'ri vaqtda, to'g'ri qamrovda, takrorlanmasdan).
2. **Ma'lumot to'liqligi** (uzilishda yo'qolmaslik, takrorlanmaslik).
3. **Nosozlik holatida xatti-harakat** (baza, Redis, tarmoq, Telegram, soat).
4. **Xavfsizlik chegaralari** (SSRF, RBAC, token, imzo).

### 16.1.1. Test piramidasi

| Daraja | Ulushi | Vositalar | Qachon ishga tushadi |
|---|---|---|---|
| Birlik (unit) | ~65% | `pytest`, `vitest`, `go test` | Har commit (CI) |
| Integratsiya | ~25% | `pytest` + haqiqiy Timescale/Redis (Docker), `testcontainers` | Har PR (CI) |
| E2E (brauzer + agent + core) | ~8% | Playwright; agent simulyatori | Har PR (asosiy oqimlar), kechasi (to'liq) |
| Yuklama, chaos, xavfsizlik | ~2% (lekin muhim) | `k6`, qo'lda stsenariylar, ZAP | Har reliz oldidan |

### 16.1.2. Qamrov maqsadlari

| Komponent | Qator qamrovi |
|---|---|
| Alert dvigateli (`alerts`) | ≥ 90% |
| Zaxira tahlili (`backups`), DNS holat tasnifi (`probes`) | ≥ 90% |
| Ingest (`ingest`), Kripto (`core.crypto`) | ≥ 90% |
| Qolgan backend | ≥ 70% |
| Agent kollektorlari | ≥ 75% (fixtures bilan) |
| Frontend (mantiq va komponentlar) | ≥ 60% |
| Umumiy backend | ≥ 80% (NFR-MNT-001: mantiq uchun ≥ 85%) |

## 16.2. Birlik testlari (majburiy ro'yxat)

### 16.2.1. Alert dvigateli

| № | Test | Kutilgan |
|---|---|---|
| T-U-ALR-01 | `threshold`: qiymat chegaradan o'tdi, `for` o'tmadi | holat `pending`, xabar yo'q |
| T-U-ALR-02 | `threshold`: `for` o'tdi | `firing`, incident ochildi |
| T-U-ALR-03 | `firing` → shart yolg'on → `resolve_after` o'tmaguncha | `resolving`, xabar yo'q |
| T-U-ALR-04 | `resolving` ichida shart yana rost | `firing` ga qaytadi, yangi xabar yo'q |
| T-U-ALR-05 | `absent`: oxirgi namuna `max_age_s` dan eski | Rost |
| T-U-ALR-06 | `rate`: counter qayta boshlandi (qiymat kamaydi) | Noto'g'ri musbat yo'q (reset hisobga olinadi) |
| T-U-ALR-07 | `forecast`: chiziqli o'sish | `eta` ±5% aniqlikda |
| T-U-ALR-08 | `forecast`: $R^2 < 0.5$ | Hisoblanmaydi (alert yo'q) |
| T-U-ALR-09 | `forecast`: band < 50% | Hisoblanmaydi |
| T-U-ALR-10 | `and` / `or` / `not` birlashmalari (chuqurlik 3) | To'g'ri mantiq jadvali |
| T-U-ALR-11 | Bostirish: host `down` → shu hostdagi konteyner alerti | Alohida xabar yo'q, incident ro'yxatida |
| T-U-ALR-12 | Bostirish tugagach shart hamon rost | Yangi alert chiqadi |
| T-U-ALR-13 | Silence matcher: `server="main-2"`, `severity!="critical"` | Faqat mos kelganlar bostiriladi |
| T-U-ALR-14 | Maintenance oynasi tugashi | Hal bo'lmagan alertlar yuboriladi |
| T-U-ALR-15 | Titrash: 1 soatda 6 o'tish | "Beqaror" belgisi, xabarlar kamaytirildi |
| T-U-ALR-16 | Guruhlash: 60 s ichida 5 alert | 1 xabar |
| T-U-ALR-17 | Incident: xuddi shu `scope_key` bo'yicha ikkinchi alert | Mavjud incidentga qo'shiladi |

| № | Test | Kutilgan |
|---|---|---|
| T-U-ALR-18 | Incident yopilishi: hamma alert `ok` + 60 s | `resolved` |
| T-U-ALR-19 | Qayta ishga tushish: holat bazadan tiklanadi | Takroriy "yangi" xabar yo'q |
| T-U-ALR-20 | Takroriy eslatma jadvali (`critical`) | 30 daq, 6 marta, keyin 2 soat |
| T-U-ALR-21 | Sokin soatlar: `warning` | Xabar kechiktirildi, 07:00 yig'ma |
| T-U-ALR-22 | Vaqt zonasi: sokin soat Asia/Tashkent bo'yicha (UTC+5) | To'g'ri chegara |
| T-U-ALR-23 | Ikkita evaluator nusxasi (advisory lock) | Faqat bittasi ishlaydi |

### 16.2.2. Zaxira tahlili

| № | Test | Kutilgan |
|---|---|---|
| T-U-BKP-01 | `state.json` ok, fayl mos | `ok` |
| T-U-BKP-02 | `state.json` ok, fayl yo'q | `suspicious` + `backup_state_mismatch` |
| T-U-BKP-03 | `state.json` yo'q, fayl bor (skaner) | Skaner manbasi bilan `ok`/`stale` |
| T-U-BKP-04 | Yosh 25.9 s / 26.1 s chegara | `ok` / `stale` |
| T-U-BKP-05 | Hajm median×0.49 / ×0.51 | `suspicious` / `ok` |
| T-U-BKP-06 | Tarix 7 dan kam | Faqat `size_floor_bytes` tekshiriladi |
| T-U-BKP-07 | `.tmp` fayl | Hisobga olinmaydi |
| T-U-BKP-08 | Offsite: nom mos, hajm mos | `in_sync` |
| T-U-BKP-09 | Offsite: hajm farqli, 29 daq / 31 daq | Alert yo'q / AL-078 |
| T-U-BKP-10 | Offsite'da ko'proq fayl (`--delete` yo'q) | Normal |
| T-U-BKP-11 | Drill muddati: 75 / 91 / 101 kun | `ok` / `due` / `overdue` |
| T-U-BKP-12 | Bir nechta muammo birga | Hammasi ro'yxat, rang — eng og'iri |
| T-U-BKP-13 | `baseline_reset` | Median tarixi qayta boshlanadi |

### 16.2.3. DNS va TLS tasnifi

| № | Test | Kutilgan |
|---|---|---|
| T-U-DNS-01 | NS ham, resolverlar ham kutilgan IP | `match` |
| T-U-DNS-02 | NS yangi, resolverlar eski | `propagating`, alert yo'q |
| T-U-DNS-03 | NS eski (kutilmagan) 10 daq+ | `mismatch` |
| T-U-DNS-04 | A yozuvi `retired_ips` ichida | `retired_ip` (AL-063) |
| T-U-DNS-05 | NXDOMAIN | `nxdomain`, `critical` |
| T-U-DNS-06 | NS to'g'ri, resolver 30 daq+ boshqa | `resolver_disagree` |
| T-U-TLS-01 | Qolgan kun 22 / 21 / 8 / 7 / 4 / 3 / 0 / −1 | Chegaralar bo'yicha to'g'ri daraja |
| T-U-TLS-02 | SAN da hostname yo'q | `hostname_ok = false` |
| T-U-TLS-03 | Wildcard SAN (`*.avlo.ai`) `nazar.avlo.ai` ga | Mos |
| T-U-TLS-04 | Agent yangi sertifikat, prober eski | "reload qilinmagan" (AL-061) |

### 16.2.4. Ingest va ma'lumot

| № | Test | Kutilgan |
|---|---|---|
| T-U-ING-01 | To'g'ri paket | `200`, namunalar yozildi, `series_latest` yangilandi |
| T-U-ING-02 | Takroriy `batch_id` | `200 duplicate=true`, ikkilanmagan |
| T-U-ING-03 | Bir xil `(series_id, time)` ikki marta | `ON CONFLICT DO NOTHING`, xato yo'q |
| T-U-ING-04 | Agent soati 11 daqiqa orqada (`sent_at` va namunalar birga) | Butun paket farqqa surildi, `clock_skew` |
| T-U-ING-05 | Qiymat diapazondan tashqari (`cpu_busy_percent = 140`) | Namuna rad etildi, hisoblagich oshdi |
| T-U-ING-06 | `NaN`/`Inf` qiymat | Rad etildi |
| T-U-ING-07 | Kardinallik: 1 001-series | Rad etildi, `agent_series_dropped` signali |
| T-U-ING-08 | Token noto'g'ri / bekor | `401` |
| T-U-ING-09 | Paket > 1 MB | `413` |
| T-U-ING-10 | gzip buzuq | `400` |
| T-U-ING-11 | `labels` da taqiqlangan kalit (masalan, `url`) | Rad etildi |
| T-U-ING-12 | Core 14 daqiqa o'chib qaytdi: bufer namunasi 14 daqiqa eski | O'z vaqti bilan yoziladi, `clock_skew` yo'q |
| T-U-ING-13 | 1 soatdan eski yoki 5 daqiqa kelajakdagi namuna | Rad etildi |

| № | Test | Kutilgan |
|---|---|---|
| T-U-ING-14 | `sent_at` yo'q yoki noto'g'ri | `400 validation_error` |
| T-U-DAT-01 | Agregat: `metric_1m` yig'indisi xom bilan mos | `sum/n` teng |
| T-U-DAT-02 | Ierarxik agregat 5m = 5 ta 1m dan | Mos |
| T-U-DAT-03 | So'rov marshrutizatsiyasi (oraliq → jadval) | 8.4.3 jadvaliga mos |
| T-U-DAT-04 | Retention: 15 kunlik xom | O'chirilgan, agregat saqlangan |

### 16.2.5. Kripto va xavfsizlik birlik testlari

| № | Test | Kutilgan |
|---|---|---|
| T-U-SEC-01 | AES-GCM: shifrlash → ochish | Teng; `nonce` har safar farqli |
| T-U-SEC-02 | AES-GCM: `aad` boshqa kanal ID | Ochish xato |
| T-U-SEC-03 | Kalit rotatsiyasi | Hamma maydon yangi kalit bilan ochiladi |
| T-U-SEC-04 | SSRF filtri: `127.0.0.1`, `10.x`, `169.254.169.254`, `::1`, `localhost` | Rad etildi |
| T-U-SEC-05 | SSRF: DNS rebinding (birinchi ochiq IP, ikkinchi ichki) | Aniqlangan IP ga ulaniladi; ichki rad |
| T-U-SEC-06 | SSRF: yo'naltirish ichki IP ga | Rad etildi |
| T-U-SEC-07 | Login timing: mavjud/mavjud emas login | Vaqt farqi ≤ 20 ms (o'lchovda) |
| T-U-SEC-08 | Log maskalash: token, parol | Logda ko'rinmaydi |
| T-U-SEC-09 | RBAC matritsasi (3.2) barcha endpointlar | Mos `200/403` |
| T-U-SEC-10 | Viewer: boshqa loyiha ID | `404` |
| T-U-SEC-11 | Agent yangilanishi: imzo mos emas | Rad etiladi |
| T-U-SEC-12 | Agent yangilanishi: versiya pasayishi | Rad etiladi |

### 16.2.6. Agent birlik testlari

| № | Test | Kutilgan |
|---|---|---|
| T-U-AGT-01 | `/proc/stat` fixtures bo'yicha CPU foizi | Ma'lum javobga teng |
| T-U-AGT-02 | `/proc/meminfo` bo'yicha `mem_used_percent` | `free` formulasiga mos |
| T-U-AGT-03 | `statfs` bo'yicha `fs_used_percent` | `df` ga mos (±0.1%) |
| T-U-AGT-04 | `tmpfs/overlay/squashfs` e'tiborsiz | Metrika yaratilmaydi |
| T-U-AGT-05 | `/proc/diskstats` farqlari → IOPS, await | To'g'ri |
| T-U-AGT-06 | `/proc/net/dev` → bayt/s; counter reset | Manfiy tezlik yo'q |
| T-U-AGT-07 | Docker API javobi (fixtures) → metrikalar | To'g'ri; `compose` labellari |
| T-U-AGT-08 | `state.json` o'qish: ok / fail / buzuq JSON | Xato bilan yiqilmaydi, `unknown` |
| T-U-AGT-09 | Kollektor panic | Qolgan kollektorlar ishlaydi (FR-AGT-014) |
| T-U-AGT-10 | Bufer: to'lganda eng eskisini o'chirish | 20 MB chegarasi |
| T-U-AGT-11 | Sender: 5xx → backoff jadvali | 1,2,4,8,16,30 s ± jitter |
| T-U-AGT-12 | Konfig xato (noto'g'ri YAML) | Ishga tushmaydi, aniq xabar |
| T-U-AGT-13 | Token faylidagi ruxsat 0644 | Ogohlantirish / rad |

## 16.3. Integratsiya testlari

| № | Test | Kutilgan |
|---|---|---|
| T-I-01 | Agent simulyatori → `ingest` → bazada metrika → API `query` | Qiymatlar mos, vaqt mos |
| T-I-02 | `heartbeat` to'xtashi → evaluator → incident → Telegram (soxta server) | Xabar 10 s ichida (NFR-PRF-004) |
| T-I-03 | HTTP probe soxta serverga: 200 → 500 → 200 | `up → down → up`, tasdiqlash qoidasi (3 xato) |
| T-I-04 | TLS probe: sinov sertifikatlari (1/3/7/21/40 kun) | Mos alertlar |
| T-I-05 | DNS probe: sinov DNS server (A yozuvini o'zgartirish) | Holatlar 11.4.2 ga mos |
| T-I-06 | Zaxira: `appctl` bilan haqiqiy `backup` (test loyiha) → agent → core | Kartada ko'rinadi |
| T-I-07 | Telegram webhook: `ack` callback | Incident `acknowledged`, audit yozuvi |
| T-I-08 | Telegram callback noto'g'ri imzo / begona chat | Hech narsa bo'lmaydi, `200` qaytadi |
| T-I-09 | SSE: yangi incident → brauzer | ≤ 5 s ichida UI'da |
| T-I-10 | Sokin soatlar + `critical` | Darrov keladi |
| T-I-11 | Retention vazifasi | Eski ma'lumot o'chirildi, agregat saqlandi |

| № | Test | Kutilgan |
|---|---|---|
| T-I-12 | Reyestr import/eksport aylanishi | Bir xil natija (diff yo'q) |
| T-I-13 | Heartbeat endpoint (`/hb/<token>`) ok/fail/yo'q | AL-048 / AL-047 |
| T-I-14 | Maintenance oynasi + incident | Xabar yo'q → oyna tugagach yuboriladi |
| T-I-15 | Kanal xatosi → email zaxira | Email yuborildi, AL-086 |

## 16.4. End-to-end (E2E) stsenariylar

Brauzer (Playwright) va haqiqiy agent (test konteynerida):

| № | Stsenariy | Qadamlar | Kutilgan |
|---|---|---|---|
| T-E-01 | UC-01: yangi server qo'shish | Panelda yaratish → buyruqni test konteynerida ishga tushirish | Server "Onlayn" ≤ 60 s |
| T-E-02 | UC-02: server o'chdi | Test agent konteynerini to'xtatish | 2–3 daq ichida incident; Telegram (soxta) xabari |
| T-E-03 | UC-03: disk bashorati | Sun'iy o'sish (generator) | AL-022 keyin AL-023 |
| T-E-04 | UC-04: zaxira muvaffaqiyatsiz | `state.json: fail` | AL-070; xabarda xato matni |
| T-E-05 | UC-07/11.4.4: kesib o'tish | Soxta DNS'da A yozuvini o'zgartirish | "tarqaldi" xabari |
| T-E-06 | UC-08: maintenance | Oyna yaratish → alert chiqarish | Xabar yo'q → oyna tugagach keladi |
| T-E-07 | UC-09: incident bilan ishlash | Ack → izoh → hal bo'lishi | Vaqt chizig'i to'liq |
| T-E-08 | UC-10: yangi loyiha qo'shish | Forma → avtomatik tekshiruvlar | 1 daqiqada tekshiruvlar ishlaydi |
| T-E-09 | UC-11: restore drill kiritish | Forma | Kartada yangilandi, eslatma qayta boshlandi |
| T-E-10 | UC-12: watchdog | Core'ni to'xtatish | 5–6 daqiqada watchdog xabari |
| T-E-11 | Kirish + 2FA + chiqish | | Sessiya xavfsizligi (cookie xususiyatlari) |
| T-E-12 | Viewer cheklovi | | Faqat o'z loyihasi |

## 16.5. Unumdorlik (yuklama) testi

### 16.5.1. Vositalar va ma'lumot

- Agent simulyatori (Go yoki `k6`): har virtual agent haqiqiy paket hajmi/ketma-ketligi bilan (7.4 katalogi, ~243 qator/host) har 15 s yuboradi.
- Ma'lumot generatori (16.9): real naqshlar (kunlik tsikl, spayk, sekin o'sish).

### 16.5.2. Stsenariylar

| № | Stsenariy | Parametrlar | Muvaffaqiyat mezoni |
|---|---|---|---|
| T-P-01 | Barqaror yuklama (maqsad miqyosi) | 10 agent, 1 soat | Ingest p95 ≤ 300 ms (NFR-PRF-001), xato 0%, CPU ≤ 40% |
| T-P-02 | Barqaror yuklama (zaxira miqyosi) | 30 agent, 4 soat | p95 ≤ 300 ms, p99 ≤ 1 s, navbat o'smaydi |
| T-P-03 | Chegara: 60 agent | 60 agent, 30 daqiqa | Qachon buzilishini aniqlash (bilish uchun, mezon yo'q) |
| T-P-04 | Spayk: 30 agent bir vaqtda qayta ulandi (core qaytdi, buferdan yuboradi) | 30 × 15 daq. bufer | Hammasi 5 daqiqada qabul qilindi, bo'shliq yo'q |
| T-P-05 | 7 kunlik tarix grafigi so'rovi | 30 host, 90 kunlik ma'lumot | p95 ≤ 2 s (NFR-PRF-006) |
| T-P-06 | Dashboard (50 loyiha, 10 server) | 20 parallel foydalanuvchi | LCP ≤ 2 s, API p95 ≤ 500 ms |

| № | Stsenariy | Parametrlar | Muvaffaqiyat mezoni |
|---|---|---|---|
| T-P-07 | 450 faol tekshiruv | 60 s interval, 24 soat | Prober CPU ≤ 30% (NFR-PRF-010); kechikish ≤ 5 s |
| T-P-08 | Alert hisoblash vaqti | 89 qoida × 30 host | Bir tick ≤ 3 s (interval 15 s) |
| T-P-09 | Baza hajmi o'sishi | 30 agent × 3 kun | Bashorat (5.1.3) bilan ±30% |
| T-P-10 | Agent resurslari | 24 soat | CPU ≤ 1%, RAM ≤ 60 MB (NFR-PRF-007) |

T-P-09 natijasi bilan 5.1.3 dagi siqish koeffitsiyentlari tuzatiladi (xavf R-104).

## 16.6. Chaos va nosozlik testlari

Maqsad: nosozlikda tizim yolg'on xabar bermasligi va ma'lumotni yo'qotmasligi.

| № | Nosozlik | Kutilgan xatti-harakat |
|---|---|---|
| T-C-01 | `db` konteynerini 5 daqiqaga to'xtatish | `/healthz` 503; ingest navbatga yoziladi; tiklangach ma'lumot yo'qolmaydi; UI "kechikmoqda" |
| T-C-02 | `redis` ni to'xtatish | Ingest ishlaydi (kesh yo'qolsa ham); SSE → polling; Celery vazifalari kechikadi, tiklangach bajariladi |
| T-C-03 | `evaluator` jarayonini `kill -9` | Docker avto-restart; holat bazadan tiklandi; takroriy xabar yo'q; AL-084 vaqtincha |
| T-C-04 | `prober` ni to'xtatish | AL-085 3 daqiqada; boshqa funksiyalar ishlaydi |
| T-C-05 | Telegram API'ga chiqishni bloklash | Qayta urinish; email zaxira; AL-086 |
| T-C-06 | Core ↔ agent tarmog'ini uzish (15 daq) | Agent buferlaydi; tiklangach yuboradi; core "aloqa yo'q" belgilaydi |
| T-C-07 | Core soatini 10 daqiqa siljitish | Namunalar `clock_skew` bilan; qoidalar buzilmaydi |
| T-C-08 | Agent soatini 1 soat siljitish | Namunalar core vaqti bilan almashtiriladi; AL-007 |
| T-C-09 | Core diskini to'ldirish (95%) | AL-021; yozishda xato — tizim yiqilmaydi, retention yordam beradi |
| T-C-10 | Agentga `SIGKILL` | systemd 5 s da qayta ishga tushiradi; AL-002 kechikish < 2 daq bo'lsa chiqmaydi |
| T-C-11 | Agent diskini to'ldirish | Bufer yozolmaydi, xotira buferi; yiqilmaydi |
| T-C-12 | Docker daemon to'xtashi (kuzatiladigan hostda) | `docker` kollektori xato, boshqalar ishlaydi (FR-AGT-014) |
| T-C-13 | Chiqish tarmog'i sekin (3 s kechikish, 5% yo'qotish) | Agent backoff; ma'lumot yo'qolmaydi |
| T-C-14 | 2 ta evaluator nusxasi ataylab | Bittasi "leader" (advisory lock) |
| T-C-15 | Retention paytida yuklama | Ingest p95 buzilmaydi |
| T-C-16 | Core reboot (to'liq server) | 3 daqiqada hammasi tiklandi (T-DEP-02) |
| T-C-17 | Buzuq paket oqimi (fuzz) | `400`; yiqilish yo'q; log hajmi o'smaydi |

## 16.7. Xavfsizlik testlari

14.13 dagi ro'yxatdan tashqari:

| № | Test | Kutilgan |
|---|---|---|
| T-S-01 | OWASP ZAP (faol skaner) — autentifikatsiyalangan | Jiddiy/yuqori topilma yo'q |
| T-S-02 | Token taxmin qilish (tezlik chegarasi ostida 10 000 urinish) | Hammasi `401`, chegara ishladi |
| T-S-03 | Bruteforce login | 5 urinishdan keyin blok |

| № | Test | Kutilgan |
|---|---|---|
| T-S-04 | Sessiya: cookie xususiyatlari | `HttpOnly; Secure; SameSite=Lax; __Host-` |
| T-S-05 | CSRF (cookie bilan POST, token yo'q) | `403` |
| T-S-06 | XSS: loyiha nomiga `<script>`, izohga `<img onerror>` | Escaped / tozalangan |
| T-S-07 | SQLi: filtr parametrlariga payload | Xato yoki bo'sh natija; so'rov buzilmaydi |
| T-S-08 | IDOR: boshqa viewer loyihasi ID si | `404` |
| T-S-09 | Agent: buzuq imzoli yangilanish | Rad etildi, voqea yuborildi |
| T-S-10 | Agent: `docker run` urinishi (proksi orqali) | `403` |
| T-S-11 | `db` / `redis` tashqaridan | `nmap`: yopiq |
| T-S-12 | TLS: `testssl.sh` | A daraja |
| T-S-13 | Zaxira fayllarida maxfiy ma'lumot | Nazoratchi zaxirasida `master.key` yo'q |
| T-S-14 | Maxfiy qiymatlar API javoblarida | Hech qachon to'liq |

## 16.8. Qabul sinovlari (UAT) — umumiy yig'ma

Buyurtmachi (Owner) tomonidan o'tkaziladi. Tizim **qabul qilingan** deb hisoblanadi, agar quyidagilar bajarilsa:

| № | Mezon | Tekshiruv |
|---|---|---|
| UAT-01 | Barcha M (Must) talablar bajarilgan | Traceability matritsasi (16.12), har qatorda "o'tdi" |
| UAT-02 | F1 uchun belgilangan testlarning 100% muvaffaqiyatli | CI + qo'lda |
| UAT-03 | 3 ta haqiqiy server (S-00, S-01, S-02) 14 kun uzluksiz kuzatuvda | Ma'lumot bo'shliqlari yo'q; agent resurs mezonlari |
| UAT-04 | Haqiqiy sinov: serverlardan birini (test uchun) ataylab to'xtatish | AL-001 ≤ 2 daqiqada, Telegram'da |
| UAT-05 | Haqiqiy sinov: zaxira cron'ini 27 soat to'xtatish (yoki vaqt siljitish) | AL-068 |
| UAT-06 | Haqiqiy sinov: sinov sertifikat/domen | Muddat ogohlantirishi |
| UAT-07 | 14 kunlik soxta alertlar ulushi | ≤ 5% (qo'lda belgilanadi) |
| UAT-08 | Barcha seed loyihalar uchun tekshiruvlar ishlayapti | 11.2.5 jadvaliga mos |
| UAT-09 | Birinchi restore drill'lar kamida 2 ta loyiha uchun bajarilgan va qayd etilgan | Kartada ko'rinadi |
| UAT-10 | Nazoratchi'ni tiklash mashqi (15.9) RTO ichida | Vaqt qayd etilgan |
| UAT-11 | Hujjatlar: README, runbook'lar (18.2), o'rnatish yo'riqnomasi | Mavjud va tekshirilgan |
| UAT-12 | Xavfsizlik ro'yxati (14.13) | Barcha bandlar o'tdi |

## 16.9. Sinov ma'lumotlari generatori

Generator — alohida kichik dastur (`tools/simulator`, F1 S10 da `simulate.py`: stdlib, token ro'yxati bo'yicha virtual agentlar, p50/p95/p99 va mezon tekshiruvi), konfiguratsiya fayli bo'yicha virtual hostlarni yaratadi.

**Holat (2026-10-28, F1 S10):** xavfsizlik, chaos va unumdorlik testlarining bajarilgan va kutilayotgan qismi `docs/sinov/B13-natijalar.md` da. Avtomatik: `backend/tests/test_security.py` (T-S-04..08, 14; S-09, S-10), `backend/tests/test_chaos.py` (T-C-02, 14, 17); real: T-C-01 va T-P-01 (qisqartirilgan). Haqiqiy serverni talab qiladigan bandlar (ZAP, testssl, T-C-03..13, T-P-02..10) UAT oldidan S-01 da bajariladi.

| Naqsh | Tavsifi |
|---|---|
| `steady` | Doimiy CPU ~10–20%, shovqin bilan |
| `daily` | Kunlik tsikl (kunduzi yuqori, kechasi past) |
| `spike` | Qisqa spayk (2–3 daqiqa, 95%) — alert **bermasligi** kerak (`for` testi) |
| `sustained_high` | 12 daqiqa 93% — alert **berishi** kerak |
| `mem_leak` | RAM sekin o'sadi (+1%/soat) → 90% ga yetadi |
| `disk_growth` | Disk +2 GB/kun → forecast |
| `flapping` | Konteyner har 2 daqiqada tushib-ko'tariladi |
| `crash_loop` | Restart soni tez oshadi |
| `backup_ok` / `backup_fail` / `backup_small` / `backup_stale` | `state.json` va fayllar yaratadi |
| `silent` | Heartbeat to'xtaydi (host o'chdi) |
| `clock_skew` | Agent vaqti siljigan |
| `high_cardinality` | Ko'p noyob label (kardinallik chegarasi testi) |

Generator deterministik (`--seed`) bo'lishi kerak: bir xil seed — bir xil natija (testlar takrorlanishi uchun).

# 16.10. Nuqsonlar va reliz mezonlari

## 16.10.1. Nuqson darajalari

| Daraja | Ta'rif | Misol | Reaksiya |
|---|---|---|---|
| S1 — kritik | Alert yo'qoladi, ma'lumot yo'qoladi, xavfsizlik teshigi, tizim ishlamaydi | Server o'chganda alert chiqmaydi | Reliz bloklanadi; zudlik bilan tuzatish |
| S2 — yuqori | Asosiy funksiya noto'g'ri, lekin aylanib o'tish mumkin | Disk bashorati 2 baravar xato | Reliz oldidan tuzatiladi |
| S3 — o'rta | Qulaylik, kosmetik, kam uchraydigan | Tarjima xatosi | Keyingi relizga ruxsat |
| S4 — past | Takomillashtirish | Rang nuanslari | Ro'yxatda |

## 16.10.2. Reliz chiqish mezonlari (exit criteria)

- S1 va S2 nuqsonlar: 0 ta ochiq.
- F-bosqich uchun belgilangan hamma test o'tgan; CI yashil.
- Yuklama testi (T-P-01, T-P-02, T-P-10) mezonlari bajarilgan.
- Xavfsizlik ro'yxati (14.13) o'tgan.
- Zaxira va tiklash sinovi bajarilgan (T-DEP-05).
- Hujjatlar yangilangan; o'zgarishlar jurnali (CHANGELOG).

## 16.10.3. Regressiya

- Har S1/S2 nuqsonga **regressiya testi** yoziladi (test avval muvaffaqiyatsiz, tuzatishdan keyin o'tadi).
- Har yangi alert qoidasi uchun kamida 2 ta test (rost va yolg'on holat).
- Har o'tgan hodisadan (post-mortem) chiqqan qoida — avtomatik testga aylanadi.

# 16.11. Tugallangan deb hisoblash (Definition of Done)

Har vazifa (user story / talab) tugallangan hisoblanadi, agar:

1. Kod yozilgan va ko'rib chiqilgan (kamida 1 ko'rik; yolg'iz ishlasa — o'z-o'zini ko'rib chiqish ro'yxati bo'yicha, 17.6).
2. Birlik/integratsiya testlari yozilgan va o'tgan; qamrov kamaymagan.
3. Statik tahlil toza (`ruff`, `mypy`, `eslint`, `tsc`, `go vet`).
4. Xavfsizlik jihati ko'rib chiqilgan (kirish tekshiruvi, ruxsat, log).
5. Migratsiya (bo'lsa) qaytariladigan va sinovdan o'tgan.
6. Matnlar `uz` (va `ru`, `en`) tarjimasi qo'shilgan.
7. Hujjat (API sxemasi avtomatik, runbook, TZ da o'zgarish) yangilangan.
8. Tegishli talab ID si PR tavsifida ko'rsatilgan.
9. Staging/dev da qo'lda tekshirilgan.

# 16.12. Izlanuvchanlik matritsasi (traceability) — modul bo'yicha

To'liq "talab → test" jadvali loyiha davomida avtomatik hosil qilinadi (har testda `@requirement("FR-XXX-NNN")` belgisi; skript hisobot yaratadi). Quyida modul darajasidagi xulosa:

| Modul | Talablar | Asosiy testlar |
|---|---|---|
| REG | FR-REG-001..012 | T-I-12, T-E-08, T-U-SEC-10 |
| HST | FR-HST-001..017 | T-U-AGT-01..06, T-P-10, T-E-02, T-E-03 |
| AGT | FR-AGT-001..014 | T-U-AGT-*, T-E-01, T-C-06, T-C-10, T-C-11, T-DEP-06 |
| DKR | FR-DKR-001..010 | T-U-AGT-07, T-SVC-02, T-SVC-09 |
| SVC | FR-SVC-001..006 | T-SVC-03, T-I-13 |
| HTTP | FR-HTTP-001..010 | T-I-03, T-SVC-01, T-P-07 |
| TLS | FR-TLS-001..008 | T-U-TLS-*, T-I-04, T-SVC-04 |
| DNS | FR-DNS-001..009 | T-U-DNS-*, T-I-05, T-E-05, T-SVC-05, T-SVC-06 |
| BKP | FR-BKP-001..020 | T-U-BKP-*, T-BKP-01..10, T-E-04, T-E-09 |
| ALR | FR-ALR-001..020 | T-U-ALR-*, T-ALR-01..13, T-I-02, T-I-14 |
| NTF | FR-NTF-001..009 | T-I-07, T-I-08, T-I-10, T-I-15, T-ALR-09, T-ALR-11 |
| UI | FR-UI-001..019 | T-UI-01..10, T-E-11, T-E-12 |

| Modul | Talablar | Asosiy testlar |
|---|---|---|
| AUTH | FR-AUTH-001..008 | T-S-02..05, T-E-11, T-U-SEC-07 |
| AUD | FR-AUD-001..004 | T-U-SEC-09, T-I-07 |
| INT | FR-INT-001..006 | T-I-06, T-I-13, T-BKP-01 |
| SYS | FR-SYS-001..006 | T-E-10, T-C-01..04, T-DEP-05 |
| NFR-PRF | NFR-PRF-* | T-P-01..10 |
| NFR-REL | NFR-REL-* | T-C-*, T-DEP-02, T-DEP-08 |
| NFR-SEC | NFR-SEC-* | T-S-, *T-U-SEC*- |
| NFR-USA | NFR-USA-* | T-UI-02..04, T-UI-07 |
| NFR-MNT | NFR-MNT-* | CI hisobotlari (qamrov, lint) |

# 17. Loyiha rejasi, resurslar va xavflar

## 17.1. Yondashuv

- **Iterativ (qisqa sikllar).** 2 haftalik sprintlar; har sprint oxirida ishlaydigan natija va qisqa demo (Owner bilan).
- **Vertikal kesimlar.** Avval "yupqa, lekin to'liq" yo'l (agent → core → alert → Telegram), keyin kengaytirish. Birinchi haftalarda Telegram'ga haqiqiy alert kelishi kerak.
- **Eng katta xavfni erta yo'qotish.** Protokol, Timescale agregatlari va `appctl` o'zgartirishlari F0 da sinab ko'riladi.
- **Rollar.** Owner (mahsulot egasi, qabul qiluvchi); Dasturchi(lar) (backend/agent/frontend); Operator (sinov va ekspluatatsiya). Kichik jamoada bir kishi bir necha rolni bajaradi.

## 17.2. Fazalar

| Faza | Maqsad | Asosiy natija | Chiqish mezoni |
|---|---|---|---|
| **F0 — Tayyorgarlik** | Texnik xavflarni kamaytirish, poydevor | Repo + CI; Python prototip agent va minimal ingest; `appctl` o'zgartirishlari (E-01..E-03); Timescale agregatlari sinovi | Prototipda 1 server metrikasi grafikda ko'rinadi; `appctl backup` `state.json` yozadi |
| **F1 — MVP** | Eng muhim ehtiyojni yopish: server o'chsa, disk to'lsa, sayt tushsa, zaxira buzilsa, sertifikat tugasa — Telegram'da xabar | 4–5-bo'limlardagi F1 talablar (18.7); 32 ta F1 alert; asosiy panel | UAT-01..UAT-12 (16.8) |
| **F2 — Kengaytirilgan** | To'liqlik va qulaylik | F2 talablar: 2FA, hisobotlar, kengaytirilgan zaxira/drill, cut-over yordamchisi, qoidalar UI, ru/en, staging va yuklama testlari | F2 testlari; soxta alert ≤ 5% (90 kun) |
| **F3 — Yetuklik** | Qo'shimcha imkoniyatlar | Viewer cheklovi, status sahifa, ko'p lokatsiyali tekshiruv, avto-drill, eskalatsiya, supply-chain | F3 testlari |

## 17.3. Ish paketlari va baho (kishi-kun)

Baho — bir tajribali dasturchi uchun taxminiy, ±30%. Kishi-kun = 1 ta to'liq samarali ish kuni (uchrashuv, kod ko'rigi, nosozlik tuzatish ichida).

### 17.3.1. F0 — Tayyorgarlik (7.5 kun)

| ID | Ish | Kun | Natija |
|---|---|---|---|
| E-01 | `appctl`: `state.json`, `_run.json` (10.3.1–10.3.3) | 1.0 | Zaxira natijasi mashinada o'qiladi |
| E-02 | `appctl`: `list --json`, `status --json` (18.1) | 1.0 | JSON chiqish (bajarildi 2026-10-26) |
| E-03 | `setup.sh`/cron: heartbeat chaqiruvi (10.3.5, 9.14) | 0.5 | Cron signal yuboradi (bajarildi 2026-10-26) |
| P-01 | Repozitoriy, CI skeleti (lint, test), `docker-compose.dev.yml` | 2.0 | Yashil CI |
| P-02 | Protokol prototipi: Python agent (`psutil`) + minimal ingest + Timescale agregat sinovi | 3.0 | Metrika grafikda; agregatlar ishlaydi; R-102 baholandi |

### 17.3.2. F1 — MVP (80 kun)

| ID | Ish paketi | Tarkibi | Kun |
|---|---|---|---|
| B1 | Core skeleti | Django loyiha, sozlamalar, modellar (8-bo'lim asosiy), migratsiyalar, foydalanuvchi/autentifikatsiya, audit asosi, Docker | 5 |
| B2 | Reyestr | Server/Loyiha/Domen CRUD, seed, port tekshiruvi, arxiv | 4 |
| B3 | Ingest | Enroll, metrika qabul qilish, series kesh, `COPY`, `series_latest`, heartbeat, tezlik chegarasi | 4 |
| B4 | Metrika so'rovlari | Query API, agregat marshrutizatsiyasi, caggs, retention | 3 |
| B5a | Agent: tizim | Go: skelet, konfig, scheduler, sender, bufer; CPU/RAM/disk/net kollektorlari | 6 |
| B5b | Agent: Docker | Docker API (proksi orqali), konteyner holati, compose bog'lash | 3 |
| B5c | Agent: systemd/backup/certs | systemd, zaxira skaneri (`state.json`), sertifikat skaneri | 3 |
| B5d | Agent: o'rnatish | `install.sh`, systemd unit, imzo tekshiruvi (o'rnatish), enroll, uninstall | 3 |
| B6 | Prober | HTTP/TCP/TLS/DNS tekshiruvlari, tasdiqlash qoidasi, SSRF filtri | 6 |

| ID | Ish paketi | Tarkibi | Kun |
|---|---|---|---|
| B7 | Alert dvigateli | Qoida modeli, shart turlari, holat mashinasi, incident, bostirish, silence, F1 qoidalar to'plami (32) | 7 |
| B8 | Bildirishnoma | Telegram (xabar, tugmalar, webhook), email zaxira, qayta urinish, shablonlar | 4 |
| B9 | Frontend (F1) | Uskuna, auth, Dashboard, serverlar, server sahifasi, loyihalar, loyiha sahifasi, zaxiralar, domenlar, incidentlar, maintenance (oddiy), sozlamalar (kanallar) | 14 |
| B10 | Zaxira monitoringi | Holat qoidalari, yosh/hajm/offsite (1-daraja), zaxira kartasi | 4 |
| B11 | O'zini kuzatish | `/healthz`, evaluator heartbeat, watchdog, o'z zaxirasi | 3 |
| B12 | Deploy va hujjat | Compose, nginx, CI/CD build+release, bootstrap, runbook'lar (asosiy 15 ta), README | 5 |
| B13 | Sinov va mustahkamlash | Birlik/integratsiya/E2E to'ldirish, yuklama T-P-01, chaos asosiy, xavfsizlik tekshiruvi, UAT tayyorlash | 6 |
| | **Jami F1** | | **80** |

### 17.3.3. F2 — Kengaytirilgan (73 kun)

| ID | Ish paketi | Tarkibi | Kun |
|---|---|---|---|
| C1 | Agent kengaytmalari | O'z-o'zini yangilash (imzo, rollback), diskio/inode/jarayonlar, `apt`/reboot/kernel, mahalliy sertifikat (deploy-hook) | 6 |
| C2 | Docker kengaytmalari | Health, resurs statistikasi, OOM, log hajmi, `system df` | 4 |
| C3 | Tekshiruv kengaytmalari | Kalit so'z, sarlavhalar, yo'naltirish, vaqt komponentlari, kunlik uptime, ko'p resolver/tarqalish rejimi, RDAP, cut-over yordamchisi | 8 |
| C4 | Alert kengaytmalari | Titrash, guruhlash, qoidalar UI, qoida testi, F2 qoidalar (48) | 9 |
| C5 | Bildirishnoma kengaytmalari | Sokin soatlar, kunlik/haftalik hisobot, bot buyruqlari, email shablonlari | 6 |
| C6 | Zaxira kengaytmalari | Yaxlitlik, offsite solishtirish (2-daraja), drill (UI+jurnal+eslatma), siyosat, hajm bashorati asosi | 7 |
| C7 | Frontend (F2) | Kengaytirilgan grafiklar (zoom, belgilar), qoidalar/silence/maintenance UI, hisobotlar, domen sahifasi, qorong'i mavzu, SSE, a11y, ru/en | 15 |
| C8 | Xavfsizlik | 2FA, sessiyalar/token UI, audit UI, ZAP, sarlavhalar, parol siyosati kuchaytirish | 6 |
| C9 | Hisobotlar | Uptime/incident/zaxira hisobotlari, CSV/PDF | 4 |
| C10 | Operatsion | Staging, yuklama va chaos testlari to'liq, runbook'lar to'liq (40+), kuzatuv | 8 |
| | **Jami F2** | | **73** |

### 17.3.4. F3 — Yetuklik (32 kun)

| ID | Ish | Kun |
|---|---|---|
| D1 | Loyiha bo'yicha cheklangan viewer (FR-AUTH-005, NFR-DAT-004) | 3 |
| D2 | Ommaviy holat sahifasi (FR-UI-019) | 3 |
| D3 | Ikkinchi tekshiruvchi tugun (FR-HTTP-008) | 4 |
| D4 | `appctl drill` avtomatik drill (FR-BKP-020) | 4 |
| D5 | Eskalatsiya (FR-ALR-015) | 2 |
| D6 | SSH/apt metrikalari, `nginx -t` tekshiruvi, top-N jarayonlar | 3 |
| D7 | Webhook kanali, deploy belgilari (FR-NTF-007, FR-INT-006) | 3 |
| D8 | Global qidiruv, CSV eksport (FR-UI-015/018) | 2 |
| D9 | Supply-chain: `cosign`, SBOM, Renovate | 3 |
| D10 | Silliqlash va xatolarni tuzatish | 5 |
| | **Jami F3** | **32** |

### 17.3.5. Umumiy baho

| Faza | Kishi-kun | 1 dasturchi (haftasiga 4 samarali kun) | 2 dasturchi (parallel) |
|---|---|---|---|
| F0 | 7.5 | ≈ 2 hafta | ≈ 1 hafta |
| F1 | 80 | ≈ 20 hafta | ≈ 10–11 hafta |
| F2 | 73 | ≈ 18 hafta | ≈ 9–10 hafta |
| F3 | 32 | ≈ 8 hafta | ≈ 4–5 hafta |

| Faza | Kishi-kun | 1 dasturchi (haftasiga 4 samarali kun) | 2 dasturchi (parallel) |
|---|---|---|---|
| **Jami** | **≈ 193** | **≈ 48 hafta (≈ 11 oy)** | **≈ 25 hafta (≈ 6 oy)** |

> **Halol xulosa.** Bu hajm bir kishi uchun jiddiy ish. F1 (MVP) o'zi ham 4–5 oy talab qiladi. Agar maqsad tezroq natija bo'lsa, quyidagi variantlar mavjud:
>
> 1. **Oraliq yechim** (1.9): Uptime Kuma + Healthchecks (o'z-hosting) + `appctl` ga heartbeat — 1–2 kunda ishlaydi va ehtiyojning 60–70% ini yopadi (server/sayt/sertifikat/cron signal).
> 2. **Kichraytirilgan MVP (F1-lite)**: faqat agent (tizim + zaxira + sertifikat) + oddiy panel + Telegram; prober va murakkab alert qoidalarisiz. ≈ 35–40 kun.
> 3. **Mavjud vositalar + o'z qo'shimchasi**: Prometheus/Netdata metrikalar uchun, zaxira/reyestr modulini alohida kichik xizmat qilib yozish. Qaror 17.11-bo'limdagi savollar orqali Owner tomonidan qabul qilinadi.

## 17.4. Muhim bosqichlar (milestones)

1 dasturchi varianti uchun (hafta raqami loyiha boshidan):

| Bosqich | Hafta | Natija | Ko'rsatish (demo) |
|---|---|---|---|
| M0 — Protokol prototipi | 2 | Python agent → ingest → grafik; `appctl state.json` | Bitta server grafigi brauzerda |
| M1 — Birinchi alert | 6 | Go agent (tizim), reyestr, alert dvigateli (asosiy qoidalar), Telegram | Agentni to'xtatib, Telegram'da "Server o'chdi" |
| M2 — Faol tekshiruvlar va zaxira | 11 | Prober, sertifikat/DNS, zaxira kartasi, Docker | Domen/sertifikat/zaxira holati panelda |
| M3 — Panel va ishonchlilik | 16 | To'liq F1 paneli, watchdog, o'z zaxirasi, deploy | S-00/S-02 da agentlar |
| M4 — F1 qabul (UAT) | 20 | S-01 agenti, 14 kunlik UAT | UAT-01..12 |
| M5 — F2 | 38 | Kengaytirilgan funksiyalar | Hisobotlar, 2FA, drill |
| M6 — F3 | 48 | Yetuklik | — |

Eslatma: har bosqichda real serverlarda agent kuzatuvi **mumkin bo'lgan eng erta** boshlanadi (M1 dan S-00, M3 dan S-02, UAT boshida S-01) — shunda real ma'lumot bilan chegaralar sozlanadi.

## 17.5. Resurslar va xarajatlar

### 17.5.1. Odam resursi

| Rol | Ish hajmi | Izoh |
|---|---|---|
| Dasturchi (backend + agent) | F0–F3: ≈ 140 kun | Python/Django + Go |
| Dasturchi (frontend) yoki bir kishi | ≈ 50 kun | React/TS |
| Owner (qabul, qarorlar) | Haftasiga ≈ 2 soat | Demo, savollar |
| Operator (sinov) | UAT davrida ≈ 2 hafta qisman | Haqiqiy serverlar |

(Bir kishi bajarsa, 17.3.5 ning 1-dasturchi ustuni qo'llanadi.)

### 17.5.2. Infratuzilma (taxminiy; narxlar provayderga bog'liq, hujjatda aniq summa yo'q)

| Element | Izoh |
|---|---|
| S-00 (core + offsite) | Mavjud Contabo server (200 GB SSD) qayta o'rnatiladi, yangi xarajat yo'q; S-01 bilan bir provayder (R-110) |
| Watchdog | S-01 da (Owner qarori), qo'shimcha xarajatsiz |
| Offsite zaxira | S-00 da (S-03 ishlatilmaydi); disk hajmi 200 GB ichida nazorat qilinadi |
| Domen | `nazoratchi.<mavjud domen>` — qo'shimcha xarajatsiz |
| Litsenziya | Hamma komponent ochiq manbali (Go, Python/Django, PostgreSQL, TimescaleDB community, Redis 7, nginx, React) |
| Telegram | Bepul (Bot API) |
| CI | GitHub Actions (xususiy repo uchun bepul limit ichida yetarli bo'lishi mumkin) |

### 17.5.3. Ish muhiti

Dev kompyuteri (Docker), GitHub repo, Telegram sinov boti va sinov guruhi, test serveri (vaqtincha VPS yoki lokal VM'lar: Ubuntu 22.04/24.04 amd64+arm64 uchun).

## 17.6. Dasturchi uchun o'z-o'zini ko'rib chiqish ro'yxati

Yolg'iz ishlaganda kod ko'rigi yo'qligi uchun har PR oldidan:

1. Talab ID si bilan bog'langanmi?
2. Test avval muvaffaqiyatsiz bo'lib, keyin o'tdimi (regressiya)?
3. Maxfiy qiymat log/javob/xato matniga tushmaydimi?
4. Yangi so'rov indeks bilan ishlaydimi (`EXPLAIN`)?
5. Nosozlikda (baza/Redis/tarmoq yo'q) nima bo'ladi — jim yutib yubormaydimi?
6. Yangi alert qoidasi uchun "yolg'on rost" holati testlandimi?
7. UI da holat rang bilan birga matn/belgiga egami? Barcha matn tarjima kalitidami?
8. Migratsiya qaytariladimi?
9. 24 soatdan keyin PR ni yangi ko'z bilan qayta o'qish (muhim o'zgarishlar uchun).

## 17.7. Xavf reyestri (loyiha xavflari)

Ehtimol/ta'sir: **Y**uqori / **O**'rta / **P**ast.

| ID | Xavf | Ehtimol | Ta'sir | Oldini olish / Yumshatish | Egasi |
|---|---|---|---|---|---|
| R-101 | **Ish hajmi o'sib ketishi (scope creep)** | Y | Y | MoSCoW; F1 qamrovi muzlatiladi; yangi g'oyalar F3/ro'yxatga | Owner |
| R-102 | **TimescaleDB** tajriba yo'q; ierarxik/join agregatlar kutilganidek ishlamasligi | P | O | **P-02 da hal qilindi (2026-10-26):** ierarxik va JOIN li agregatlar, siqish, siyosatlar 2.17.2 da ishlaydi (docs/P-02-timescale-sinov.md). Zaxira reja (oddiy partitsiya + Celery) kerak emas | Dasturchi |
| R-103 | Go agent: o'rganish egri chizig'i | O | O | F0 da Python prototip; Go agentni kichik (3–5 ming qator) saqlash; yoki Python agent (PyInstaller) zaxirasi | Dasturchi |
| R-104 | Hajm/unumdorlik baholari noto'g'ri (5.1) | O | O | T-P-01..09 erta; baholarni yangilash; kardinallik chegaralari | Dasturchi |
| R-105 | **Soxta alertlar** — ishonch yo'qoladi, odamlar e'tiborsiz qoladi | Y | Y | `for`, tasdiqlash, gisterezis, bostirish; 4 hafta "yumshoq" rejim (faqat kunlik xulosa); oylik ko'rik (12.11) | Operator |
| R-106 | **Bitta odam (bus factor)** | Y | Y | Hujjat (shu TZ + ADR + runbook); sodda texnologiyalar; kodning ochiq tuzilmasi | Owner |
| R-107 | Telegram API vaqtincha ishlamasligi / bloklanishi | P | Y | Email zaxira kanal; ikkinchi kanal (webhook); AL-086 | Dasturchi |
| R-108 | **Agent ishlab turgan serverda yuk/muammo keltiradi** | O | Y | Resurs cheklovlari (`MemoryMax`, `CPUQuota`); bosqichma-bosqich tarqatish (S-00 → S-02 → S-01); 24 soat kuzatuv; tezkor o'chirish (`uninstall`) | Operator |
| R-109 | `appctl` **o'zgartirishi zaxirani buzadi** (eng xavfli) | O | **Y** | O'zgartirishlar kichik va izolyatsiyalangan (state.json yozishdagi xato zaxirani to'xtatmasligi shart: buyruq oxirida `|| true`); avval nusxada sinov; ikki bosqich; o'zgarishdan keyin `appctl backup all` qo'lda tekshiruv; `set -e` ta'siri e'tiborga olinadi | Dasturchi |
| R-110 | Core va kuzatiladigan serverlar bir provayderda — bog'liq nosozlik | O | Y | Qabul qilingan (Owner): core S-00 va S-01 bir provayderda (Contabo); o'zaro kuzatuv (watchdog S-01 da), o'z zaxirasi Telegram'da | Owner |
| R-111 | Vaqt bahosi kam | Y | O | Bufer 20%; har sprintda baholashni yangilash; F1-lite varianti (17.3.5) | Dasturchi |
| R-112 | `docker-socket-proxy` noto'g'ri sozlanishi — agent Docker'ni boshqara oladi | P | Y | Konfiguratsiya testi (T-S-10); `POST=0` majburiy | Dasturchi |
| R-113 | `.uz` domenlar uchun avtomatik WHOIS/RDAP yo'q | Y | P | Qo'lda kiritish + eslatma (FR-DNS-006) | Operator |
| R-114 | S-02 nginx.conf boshqa loyiha tomonidan qayta yoziladi; asadbekashurov.uz yo'qoladi | O | O | Nazoratchi tekshiruvlari darrov topadi (AL-049); aaclub repo egasi bilan kelishuv; qoidani repoga ham qo'shish | Owner |
| R-115 | Maxfiy qiymatlar loglarga tushishi | O | Y | Log filtri + avtomatik skaner testi (T-U-SEC-08) | Dasturchi |
| R-116 | DNS tekshiruvi soxta alert beradi (resolver keshi) | O | P | Tarqalish rejimi, 10 daq. kutish, NS manba haqiqati (11.4) | Dasturchi |
| R-117 | **Restore drill'lar e'tiborsiz qoladi** (jarayon xavfi) | Y | Y | Eslatmalar, panelda ko'rinadigan "hech qachon" belgisi; UAT-09 majburiy | Owner |

| ID | Xavf | Ehtimol | Ta'sir | Oldini olish / Yumshatish | Egasi |
|---|---|---|---|---|---|
| R-118 | Ortiqcha murakkablik: tayyor yechim yetarli edi | O | O | 1.9 va 17.3.5 dagi qarorni F0 oxirida qayta ko'rish ("davom etamizmi?") | Owner |
| R-119 | Redis litsenziyasi / yangi versiya o'zgarishi | P | P | Redis 7.x (BSD) ga qotirish yoki Valkey; Celery/kesh uchun muhim emas | Dasturchi |
| R-120 | Agent yangilanish kanali orqali buzilish (core hujumi) | P | Y | Imzolangan yangilanish (14.5.4) — majburiy talab | Dasturchi |
| R-121 | Telegram bot tokeni sizishi | P | Y | Maxfiy saqlash, rotatsiya yo'riqnomasi (14.12.3) | Owner |
| R-122 | Serverlar soni kutilgandan tez o'sadi | P | P | 15.14 masshtablash bosqichlari | Operator |
| R-123 | Kernel/OS yangilanishi core'ni buzadi (2026-10 saboqi) | O | Y | Avtomatik reboot o'chiq; snapshot; texnik ish oynasi; core'da ham `reboot_required` alerti | Operator |

## 17.8. Bog'liqliklar va taxminlar

| ID | Bog'liqlik/taxmin | Agar bajarilmasa |
|---|---|---|
| DEP-01 | S-00 (core) qayta o'rnatiladi va snapshot olinadi (F0 oxirigacha) | Vaqtincha boshqa serverda (tavsiya etilmaydi, 6.6.1) |
| DEP-02 | `nazoratchi.<domen>` domeni/subdomeni mavjud | Vaqtincha IP orqali (TLS yo'q — tavsiya etilmaydi) |
| DEP-03 | Telegram bot va chat yaratilgan (Owner) | Email bilan boshlanadi |
| DEP-04 | S-00 da offsite papka va SSH kaliti (`rrsync`) sozlanadi | 10.6 offsite qismi kechikadi |
| DEP-05 | `appctl` o'zgartirishlariga Owner roziligi | Papka skaneri (kamroq ishonchli) bilan |
| DEP-06 | Serverlarga root kirish (agent o'rnatish uchun) | — |
| DEP-07 | Owner haftasiga ~2 soat qaror/demo uchun | Kechikish |
| DEP-08 | GitHub repo va CI mavjud | Boshqa CI (GitLab/Gitea) |

## 17.9. Muloqot va hisobot

- Haftalik qisqa hisobot (Telegram/email): bajarilgan, keyingi hafta, to'siqlar, xavflar.
- Har 2 haftada demo (15–20 daqiqa) va talablar bo'yicha qarorlar.
- O'zgarishlar jurnali (`CHANGELOG.md`), talab o'zgarishi TZ hujjat tarixiga yoziladi.
- Har bosqich oxirida qisqa retrospektiva (nima ishladi, nima yo'q).

## 17.10. Hozirgi holatdan o'tish rejasi

| Qadam | Qachon | Nima |
|---|---|---|
| 1 | Hoziroq (1–2 kun) | Oraliq yechim: Uptime Kuma (HTTP/TLS/TCP) + Telegram; `OFFSITE` zaxirani sozlash (ushbu loyihaga bog'liq emas, lekin eng muhim) |
| 2 | F0 | S-00 ni qayta o'rnatish va tayyorlash; `appctl` o'zgartirishlari; protokol prototipi |
| 3 | M1 | S-00 da agent; birinchi Telegram alertlari |
| 4 | M3 | S-02 da agent; Uptime Kuma bilan parallel ishlash |
| 5 | M4 | S-01 da agent; UAT 14 kun |
| 6 | UAT keyin | Uptime Kuma o'chiriladi yoki ikkilamchi tekshiruv sifatida qoladi; 4 haftalik "yumshoq rejim" (soxta alert tuning) |
| 7 | F2/F3 | Qolgan loyihalar (ko'chirilmaganlar) ko'chirilgani sayin reyestrga qo'shiladi |

## 17.11. Owner qarori kerak bo'lgan savollar

| № | Savol | Variantlar | Tavsiya | Ta'siri |
|---|---|---|---|---|
| Q-01 | Nazoratchi'ni yozamizmi yoki tayyor yechimlar bilan boshlaymizmi? | (a) to'liq Nazoratchi; (b) F1-lite; (c) Uptime Kuma + Netdata + o'z zaxira modulingiz | (c) darhol boshlang; (b)/(a) ni F0 natijasi bilan qaror qiling | Muddat 6 oy ↔ 1 hafta |
| Q-02 | Core qayerda? | Alohida VPS (boshqa provayder) / S-02 / S-01 | Alohida VPS | Ishonchlilik |
| Q-03 | Agent tili | Go / Python | Go (F0 da Python prototip) | O'rganish vs ekspluatatsiya soddaligi |
| Q-04 | Telegram: shaxsiy chat yoki guruh; kim ikkinchi kontakt? | | Guruh (2 kishi) | Eskalatsiya |

| № | Savol | Variantlar | Tavsiya | Ta'siri |
|---|---|---|---|---|
| Q-05 | Domen nomi | `nazoratchi.avlo.ai` / boshqa | Mavjud domen ostida subdomen | TLS, DNS |
| Q-06 | Loyiha nomi "Nazoratchi" tasdiqlanadimi? | | Ishchi nom | Brending |
| Q-07 | Offsite server (38.242.143.248) rostan ham sizniki va joy yetarlimi? | | Tekshirish | 10.6 |
| Q-08 | `appctl` ga o'zgartirish kiritishga rozimisiz? | | Ha (ehtiyotkorlik bilan) | R-109 |
| Q-09 | Mijozlarga (masalan, wmg.uz) holat ko'rsatish kerakmi? | | F3 (status sahifa) | Qamrov |
| Q-10 | S-02 (aaclub) serveriga agent o'rnatishga ruxsat egasidan olinganmi? | | Kelishing | Ruxsat |
| Q-11 | Reboot/kernel yangilanishi rejasi (S-01 da 6.8.0-147 kutmoqda) qachon? | | Offsite + snapshotdan keyin | Xavfsizlik |

> **Javoblar (2026-10-26)** — to'liq ro'yxat QARORLAR.md da: Q-01 to'liq F0→F3; Q-02 core S-00 da; Q-03 Go; Q-04 guruh; Q-05 domen nomga qarab; Q-06 "Nazoratchi"; Q-07 offsite S-00 ga (S-03 ishlatilmaydi); Q-08 ha; Q-10 ruxsat kerak emas; Q-11 hozircha reboot yo'q. Q-12 GCP IP yopilgan (`retired_ips`); Q-13 o'z zaxirasi `age` bilan shifrlanib Owner'ning shaxsiy Telegram chatiga; Q-14 watchdog S-01 da (o'zaro kuzatuv).

## 17.12. Topshirish va qabul tartibi

1. Har faza oxirida: ishlaydigan tizim (prod), testlar hisoboti, yangilangan hujjatlar.
2. Owner UAT ro'yxatini (16.8) o'tkazadi; har band uchun imzo (o'tdi/o'tmadi).
3. O'tmagan band — nuqson ro'yxatiga (daraja 16.10.1), tuzatilib qayta tekshiriladi.
4. Faza qabul qilingach, TZ hujjat tarixiga yozuv va keyingi faza boshlanadi.
5. Yakuniy topshirish: manba kod, hujjatlar, runbook'lar, maxfiy qiymatlar ro'yxati (qiymatlarsiz, qayerda saqlanishi), tiklash yo'riqnomasi (15.9) sinalgan.

# 18. Ilovalar

## 18.1. `appctl --json` chiqish sxemalari

Mavjud `appctl` ga `list --json` va `status --json` qo'shiladi (FR-INT-001, FR-INT-002). Qoidalar: faqat stdout'ga JSON (diagnostika stderr'ga); bir qator yoki chiroyli — farqi yo'q; parol/token **hech qachon**; chiqish kodi 0 (muvaffaqiyat); xatoda `{"schema":1,"error":"..."}` va kod 1.

### 18.1.1. `appctl list --json`

```
{
  "schema": 1,
  "host": "vmi3679093",
  "generated_at": "2026-10-22T09:35:00Z",
  "projects": [
    {
      "name": "Nazar",
      "compose_file": "docker-compose.prod.yml",
      "compose_extra": null,
      "state": "running",
      "running": 7,
      "total": 7,
      "health_url": "http://127.0.0.1:8020/healthz/",
      "db": {"service": "db", "user": "nazar", "name": "nazar"},
      "backup_paths": []
    },
    {
      "name": "lutfan_ai",
      "compose_file": "docker-compose.yml",
      "compose_extra": "docker-compose.server.yml",
      "state": "partial",
      "running": 5,
      "total": 6,
      "health_url": null,
      "db": {"service": "postgres", "user": "lutfan", "name": "lutfan"},
      "backup_paths": ["/var/lib/docker/volumes/lutfan_ai_media/_data"]
    }
  ]
}
```

`state`: `running` (hammasi ishlayapti) | `partial` (bir qismi ishlayapti) | `stopped` (hech biri ishlamayapti, shu jumladan konteyner yo'q) | `unknown` (Docker javob bermadi; agent buni "to'xtagan" deb hisoblamasligi shart). Inson rejimidagi `appctl list` 0/N ni "qisman" deb ko'rsatadi; JSON da ogohlantirish mantig'i uchun farqlanadi. `db` — `.appctl` da `DB_SERVICE` bo'lmasa `null`. `health_url` dagi maxfiy qismlar (`token=…`) `***` bilan yopiladi; `HEALTH_CURL_OPTS` (sarlavhalar) chiqarilmaydi. `backup_paths` — `BACKUP_PATHS` bo'yicha massiv.

### 18.1.2. `appctl status --json`

```
{
  "schema": 1,
  "host": "vmi3679093",
  "generated_at": "2026-10-22T09:35:00Z",
  "disk": {"mount": "/", "total_bytes": 85899345920, "used_bytes": 66993537024, "used_percent": 78},
  "memory": {"total_bytes": 8321499136, "used_bytes": 3478923264, "available_bytes": 4842575872},
  "containers": [
    {"name": "nazar-web-1", "status": "Up 2 days", "running": true, "health": "healthy"},
    {"name": "nazar-db-1", "status": "Up 4 days", "running": true, "health": null}
  ],
  "certificates": [
    {"name": "nazar.avlo.ai", "not_after": "2026-12-02T04:11:00Z", "days_left": 41},
    {"name": "avlo.ai", "not_after": "2026-12-05T08:20:00Z", "days_left": 44}
  ],
  "backups": [
    {"project": "BiznesRadar", "last_daily_at": "2026-10-22T03:15:41Z", "daily_count": 7, "weekly_count": 5}
  ]
}
```

Status JSON qoidalari (E-02 da amalga oshirilgan va sinalgan):

- `memory.used_bytes = total_bytes − available_bytes` (`free -b`); `disk` — `df -P -B1 /` (ustunlar oxiridan o'qiladi: fayl tizimi nomida bo'shliq bo'lishi mumkin).
- `containers` — `docker ps -a`: to'xtaganlar ham kiradi (`running: false`). `health`: `healthy` | `unhealthy` | `starting` | `null` (healthcheck yo'q).
- `disk`, `memory`, `containers` olib bo'lmasa `null` ("yo'q" emas, "bilib bo'lmadi": Docker, `df` yoki `free` ishlamasa); `certificates` va `backups` bo'sh massiv bo'lishi mumkin. O'qilmaydigan sertifikat tushib qoladi va stderr'ga ogohlantirish chiqadi.
- `certificates` — `/etc/letsencrypt/live/*/cert.pem` (`APPCTL_LETSENCRYPT` bilan o'zgartiriladi), `not_after` UTC.

- `backups` — `<zaxira>/*/daily` bor loyihalar; `last_daily_at` — eng yangi `*.sql.gz` (`.tmp` siz) `mtime` si; `weekly_count` faqat `*.sql.gz` ni sanaydi (haftalik fayl zaxirasi `*.tar.gz` kirmaydi).
- Xatoda (masalan, buzuq `.appctl`) stdout'ga `{"schema":1,"error":"…"}` chiqadi va kod 1 bo'ladi.

### 18.1.3. Zaxira holati fayllari

`state.json` va `_run.json` sxemalari 10.3.1 va 10.3.2 bo'limlarda.

## 18.2. Runbook'lar

Har runbook: **Belgi** (qanday ko'rinadi), **Tezkor tekshiruv**, **Odatiy sabablar**, **Qadamlar**, **Tekshirish** (hal bo'ldimi), **Keyin** (takrorlanmasligi uchun). Buyruqlar Ubuntu 24.04 va `appctl` ga mos. `<...>` — almashtiriladigan qiymat.

### RB-001. Server o'chdi / aloqa yo'q (AL-001, AL-008)

**Tezkor tekshiruv:** Bu faqat agentmi yoki butun server? Mahalliy kompyuterdan: `ping <ip>`; `ssh <user>@<ip>`; boshqa tarmoq/VPN orqali ham sinab ko'ring (muammo sizning tarmog'ingizda bo'lishi mumkin).

**Odatiy sabablar:** provayder nosozligi; server ichida kernel/boot muammosi; fayl tizimi xatosi; disk to'ldi; DDoS/tarmoq; tasodifiy o'chirish.

**Qadamlar:**

1. Provayder panelida server holati (Contabo: "running"/"stopped"), provayder statusi sahifasi.
2. Panel konsoli (VNC) orqali ekranni ko'rish. Nima ko'rinadi?
   - Login so'rovi → server tirik, tarmoq muammosi: `ip a`, `netplan`, firewall (`ufw status`).
   - `Booting from Hard Disk...` yoki bo'sh ekran → boot muammosi (2-qadamdan keyin: 4).
   - Kernel panic / `emergency mode` → fayl tizimi xatosi (5).
3. Oddiy qayta ishga tushirish (panel "Restart") — **faqat** snapshot olingandan keyin (agar imkon bo'lsa).
4. Yuklanmasa: Rescue System (provayderning "Rescue") rejimiga o'ting: SSH bilan kirib, `lsblk`, `blkid`, `mount /dev/<root> /mnt`, `journalctl -D /mnt/var/log/journal -b -1 -e` bilan oxirgi boot xatosini ko'ring. GRUB: eski ishlaydigan kernelni tanlash (`GRUB_DEFAULT`).
5. Fayl tizimi: `umount`, `fsck -f -y /dev/<root>` (rescue ichida, **mount qilinmagan** holda).
6. Agar hech narsa yordam bermasa: **ma'lumotni boshqa serverga ko'chirish** (rescue'dan `rsync -aHAX` yangi serverga) va loyihalarni qayta ko'tarish (2026-10 tajribasi; `deploy/server/` skriptlari).
7. Offsite zaxira bo'lsa — tiklash (RB-030).

**Tekshirish:** server ping/SSH javob beradi; `appctl list` barcha loyiha `ishlayapti`; Nazoratchi'da AL-001 yopildi.

**Keyin:** sabab post-mortemda; snapshot/zaxira/offsite tekshiruvi; GCP/provayderga xos qoldiq agentlar yo'qligini tekshirish; kernel yangilanishi bo'lgan bo'lsa — RB-015.

### RB-002. Agent javob bermayapti (AL-002, AL-005, AL-006)

**Tezkor tekshiruv:** server tirik (SSH ishlaydi)?

**Qadamlar:**

1. `systemctl status nazoratchi-agent`; `journalctl -u nazoratchi-agent -n 100 --no-pager`.
2. Core'ga yetib boradimi: `curl -sS -m 10 https://<core>/healthz`.
3. Token bekor qilinganmi (401 loglarda)? → panelda yangi enroll tokeni, `install.sh` qayta.
4. Disk to'lganmi (`df -h /var/lib/nazoratchi`)? → RB-004.
5. Soat noto'g'rimi? → RB-016.
6. Oddiy: `systemctl restart nazoratchi-agent`.
7. Yangilanish muammosi: `/usr/local/bin/nazoratchi-agent.old` borligini tekshirib, qaytarish (RB-043).

**Tekshirish:** 1 daqiqada heartbeat; navbat (`agent_queue_files`) kamayadi.

### RB-003. Kutilmagan reboot (AL-003)

1. `last -x reboot | head -n 5`; `last -x shutdown | head`.
2. Oldingi boot oxiri: `journalctl -b -1 -e --no-pager | tail -n 80`.
3. `dmesg -T | grep -i -E "oom|panic|mce|i/o error"`.
4. Provayder hodisalari (xost nosozligi, migratsiya) — panel/email.
5. Kernel o'zgarganmi? `uname -r` (RB-015 bilan solishtiring).

6. Loyihalar tiklandimi: `appctl list`; `docker ps -a | grep -v Up`.
7. Sabab topilmasa — kuzatuvni kuchaytirish (keyingi 7 kun).

### RB-004. Disk to'lmoqda (AL-019..AL-024, AL-028, AL-079)

1. `df -h`; `df -i` (inode).
2. Nima band: `du -xh / --max-depth=2 2>/dev/null | sort -h | tail -n 20`.
3. Docker: `docker system df`; `docker ps -a --size`; katta log: `ls -lhS /var/lib/docker/containers/*/*-json.log | head`.
   - Eski image: `docker image prune` (ehtiyotkorlik bilan; `-a` faqat nima o'chishini bilgan holda).
   - Log limiti yo'q konteynerlar: `docker inspect -f '{{.Name}} {{.HostConfig.LogConfig.Config}}' $(docker ps -q)`; `daemon.json` limiti qo'llanishi uchun konteynerni qayta yaratish (`appctl down X && appctl up X`).
4. Tizim loglari: `journalctl --disk-usage`; `journalctl --vacuum-size=500M`.
5. Eski kernel/paketlar: `apt autoremove --purge` (**faqat yangi kernel muvaffaqiyatli yuklangandan keyin**).
6. Zaxira papkasi: `du -sh /opt/applications/backups/*`; offsite ishlayotgan bo'lsa, eski haftalik nusxalarni qisqartirish.
7. Katta fayllar: `find / -xdev -type f -size +500M -printf '%s %p\n' | sort -n | tail`.
8. Hech narsa yordam bermasa — diskni kengaytirish (provayder) va `growpart`/`resize2fs`.

**Tekshirish:** bandlik < 80%; bashorat > 14 kun.

### RB-005. Fayl tizimi faqat o'qish (AL-025)

**Diqqat:** hech narsa yozishga urinmang; ma'lumot butunligi xavf ostida.

1. `mount | grep -E ' ro[ ,]'`; `dmesg -T | grep -i -E "ext4|xfs|i/o error|remount"`.
2. Provayder panelida snapshot (agar imkon bo'lsa) — avval.
3. Servislarni to'xtatish: `appctl down <loyiha>` (yozish xatolari bo'lmasligi uchun, agar ishlasa).
4. Rescue rejimida `fsck -f -y /dev/<root>` (mount qilinmagan).
5. Muammo takrorlansa — disk/provayder nosozligi: yangi serverga ko'chirish (RB-001/6-qadam).

### RB-006. Sertifikat muddati/yangilanish (AL-057..AL-061)

1. `certbot certificates` (qaysi domen, qachon tugaydi).
2. `certbot renew --dry-run` — xato matnini o'qing.
3. Odatiy sabablar: 80-port yopiq yoki boshqa nginx bloki ushlab turibdi; DNS boshqa serverga qaraydi (RB-007); `server_name` yo'q; `/etc/letsencrypt` ruxsatlari; certbot timer o'chiq (`systemctl list-timers | grep certbot`).
4. Qo'lda: `certbot renew --cert-name <domen>`; so'ng `systemctl reload nginx` (AL-061: yangilangan, lekin reload qilinmagan).
5. S-02 (konteyner variant): `docker run --rm -v /opt/asadbekashurov/letsencrypt:/etc/letsencrypt -v /opt/asadbekashurov/certbot-www:/var/www/certbot certbot/certbot:v2.11.0 renew`; `docker exec asadbekashurov-web-1 nginx -s reload`.
6. Tashqaridan tekshirish: `echo | openssl s_client -connect <domen>:443 -servername <domen> 2>/dev/null | openssl x509 -noout -dates`.

### RB-007. DNS kutilgan IP ga qaramaydi (AL-062..AL-065)

1. Domen qayerda boshqariladi (reyestr: DNS provayder, kim nazorat qiladi).
2. Haqiqat manbai: `dig +short A <domen> @<NS-server>` va ommaviy: `@8.8.8.8`, `@1.1.1.1`, `@9.9.9.9`.
3. Mos kelmasa — provayder panelida A yozuvini kutilgan IP ga o'zgartirish (ahost.uz / Namecheap / Google Cloud DNS / Beget). Mijoz nazoratida bo'lsa (masalan, `*.wmg.uz`) — mijozga aniq xabar: "A yozuvi: `<domen>` → `<IP>` ga o'zgartirilsin".
4. `www.` yozuvi ham bormi tekshiring.
5. O'zgartirishdan so'ng Nazoratchi'da "tarqalish rejimi"ni yoqing. Google DNS keshi: `https://dns.google/cache` orqali tozalash mumkin.
6. Yangi serverda **oldindan** tekshirib qo'yish: `curl -I --resolve <domen>:443:<yangi_ip> https://<domen>/`.
7. Domen **eski** (yopilgan) IP ga qaraydi (AL-063) — ko'chirishdan keyin unutilgan yozuv; xuddi shu qadamlar.

### RB-008. Domen muddati (AL-066, AL-067)

1. Registrar (reyestrda) hisobiga kiring; to'lov usuli va avto-yangilash holatini tekshiring.
2. Domenni yangilang (kamida 1 yil).
3. `.uz` bo'lsa — registrator/provayder (masalan, ahost.uz) orqali; yangi sanani Nazoratchi'da qo'lda kiriting.
4. Avto-yangilash yoqilganini tasdiqlang.

## RB-009. CPU yuqori (AL-009..AL-013, AL-018, AL-040)

1. `top -o %CPU` yoki `htop`; `docker stats --no-stream`; `systemd-cgtop`.
2. `st` (steal) yuqori bo'lsa — provayder resursi yetmayapti: `top` dagi `%st`; vaqt o'tib turaversa provayder bilan bog'laning.
3. Konteyner aybdor bo'lsa: `appctl logs <loyiha> <servis>`; Celery tsikli, cheksiz takrorlanish, qotgan so'rov.
4. Qayta ishga tushirish: `docker restart <konteyner>` (ma'lumot yo'qolmaydi, lekin jarayondagi ish uziladi).
5. Rejalashtirilgan ishlar bilan mos kelishi: 03:15 zaxira (`pg_dump`) — vaqtni siljitishni ko'rib chiqing.
6. Doimiy bo'lsa — resurs chegarasi (`deploy.resources.limits`) yoki server kuchaytirish.

## RB-010. RAM yuqori / OOM (AL-014..AL-017, AL-038, AL-039)

1. `free -h`; `ps aux --sort=-%mem | head -n 10`; `docker stats --no-stream --format 'table {{.Name}}\t{{.MemUsage}}\t{{.MemPerc}}'`.
2. OOM bo'lganmi: `dmesg -T | grep -i -E "out of memory|killed process"`; `docker inspect -f '{{.State.OOMKilled}}' <konteyner>`.
3. Sizish (leak) belgisi: Nazoratchi grafigi — RAM sekin o'sib borsa, jarayonni qayta ishga tushirish vaqtinchalik yordam.
4. Konteynerga xotira limiti qo'ying (bitta konteyner butun serverni o'ldirmasligi uchun).
5. Swap: `swapon --show`; yo'q bo'lsa kichik swap (1–2 GB) xavfsizlik to'ri sifatida.
6. Doimiy — server RAM ini oshirish yoki yukni boshqa serverga ko'chirish.

## RB-011. Disk I/O va yuklama (AL-012, AL-026, AL-027)

1. `iostat -x 1 5` (paket `sysstat`); `iotop -oPa`.
2. Zaxira (`pg_dump`, `tar`) paytimi? `ps aux | grep -E "pg_dump|tar|rsync"`.
3. PostgreSQL: `docker exec <db> psql -U <user> -c "SELECT pid, now()-query_start AS dur, state, left(query,80) FROM pg_stat_activity WHERE state<>'idle' ORDER BY dur DESC LIMIT 10"`.
4. Jarayonni vaqtga siljitish yoki `ionice -c3`/`nice` bilan ishga tushirish.

## RB-012. Nazoratchi'ning o'zi ishlamayapti (watchdog, AL-008)

1. Core serveriga SSH: `docker compose -f /opt/applications/projects/Nazoratchi/docker-compose.yml ps`.
2. Loglar: `docker compose logs --tail 100 web evaluator db`.
3. Disk/RAM: `df -h`, `free -h` (core to'lgan bo'lishi mumkin — RB-004).
4. Qayta ishga tushirish: `appctl up Nazoratchi`; kerak bo'lsa `appctl restart Nazoratchi`.
5. `db` yiqilgan bo'lsa: loglar (`docker compose logs db`), disk, `shared_buffers`/xotira.
6. Server o'lgan bo'lsa — RB-001, so'ng RB-044 (tiklash).
7. **Shu orada** boshqa serverlar kuzatuvsiz: oraliq vosita (Uptime Kuma) va provayder paneli.
8. Tiklangandan keyin agentlar buferdan ma'lumot yuboradi; AL-084/AL-085 yopilishini tekshiring.

## RB-013. Tarmoq xatolari (AL-029)

1. `ip -s link show <iface>`; `ethtool -S <iface> | grep -i -E "err|drop"`.
2. `mtr -rwc 50 <tashqi_manzil>`; MTU: `ip link | grep mtu`.
3. Provayder tarmoq muammosi bo'lishi mumkin — status sahifa/murojaat.

## RB-014. SSH hujumi va xavfsizlik gumoni (AL-031)

1. Urinishlar: `journalctl -u ssh --since "-1h" | grep -E "Failed|Invalid" | awk '{print $(NF-3)}' | sort | uniq -c | sort -rn | head`.
2. `fail2ban-client status sshd` (o'rnatilgan bo'lsa); o'rnatilmagan bo'lsa — `setup.sh --fail2ban`.
3. Parol bilan SSH ochiq bo'lsa — kalitga o'tish (`setup.sh --harden-ssh`, avval kalit bilan kirishni tekshiring).
4. **Muvaffaqiyatli** kirish bormi: `last -a | head`; `grep "Accepted" /var/log/auth.log | tail`.
5. Buzilish gumoni bo'lsa: `authorized_keys`, yangi foydalanuvchilar (`getent passwd`), `crontab -l`, `/etc/cron.d`, noma'lum jarayonlar (`ps auxf`), tashqi ulanishlar (`ss -tnp`). Agar tasdiqlansa — serverni ajratish, barcha maxfiy qiymatlarni almashtirish, agent tokenini bekor qilish (14.12.1).

## RB-015. Kernel yangilanishi va reboot (AL-032..AL-034)

**Reboot oldidan (majburiy):**

1. Snapshot (provayder paneli) — tasdiqlang.
2. Oxirgi zaxira yangi va **offsite'da bor** (Nazoratchi zaxira sahifasi).
3. Texnik ish oynasi yarating (alertlar jim bo'ladi).

4. Eng kamida bitta ishlaydigan eski kernel o'rnatilgan (`dpkg -l | grep linux-image`), GRUB menyusida ko'rinadi.
5. VNC/konsol kirishi tayyor (parol/rescue).

**Bajarish:** `reboot`. 3–5 daqiqa kuting.

**Reboot'dan keyin:** `uname -r`; `docker ps`; `appctl list`; `systemctl --failed`; Nazoratchi'da barcha loyiha yashil.

**Yuklanmasa:** VNC → GRUB menyusi → eski kernel (Advanced options); yoki RB-001. Eski kernelni `autoremove` bilan o'chirishni yangi kernel barqaror ishlagandan keyin (kamida 1 hafta) bajaring.

### RB-016. Vaqt sinxronligi (AL-007)

`timedatectl`; `chronyc tracking` yoki `timedatectl timesync-status`; `systemctl restart systemd-timesyncd` (yoki `chrony`); `timedatectl set-ntp true`.

### RB-020. Konteyner ishlamayapti / crash loop (AL-035..AL-037, AL-042)

1. `docker ps -a --format 'table {{.Names}}\t{{.Status}}'`.
2. `docker logs --tail 100 <konteyner>`; `docker inspect -f '{{.State.ExitCode}} {{.State.OOMKilled}} {{.State.Error}}' <konteyner>`.
3. Loyiha orqali: `appctl ps <loyiha>`; `appctl logs <loyiha> <servis>`.
4. Odatiy sabablar: baza tayyor emas (`depends_on`), noto'g'ri `.env`, port band (`ss -ltnp | grep :<port>`), migratsiya xatosi, xotira (RB-010), disk (RB-004).
5. Tuzatgach: `appctl up <loyiha>`; `appctl status`; sog'liq: `curl -I -H 'Host: <domen>' -H 'X-Forwarded-Proto: https' http://127.0.0.1:<port>/`.
6. Reyestrda yo'q konteyner (AL-042): yangi loyiha reyestrga qo'shilmagan yoki ruxsatsiz konteyner — kim ko'targanini aniqlang.

### RB-021. Konteyner xotira/log muammolari (AL-038, AL-039, AL-041)

1. Log fayli katta: `docker inspect -f '{{.LogPath}}' <konteyner>`; hajm: `ls -lh <yo'l>`.
2. Log limiti yo'q (eski konteyner): `daemon.json` ni tekshiring; konteynerni qayta yaratish: `appctl down <loyiha> && appctl up <loyiha>`.
3. Shoshilinch joy kerak bo'lsa: `truncate -s 0 <LogPath>` (**faqat log**, konteyner ishlayotganda xavfsiz; ma'lumot yo'qoladi).
4. Xotira limiti: compose'da `mem_limit`/`deploy.resources.limits.memory`; ilova talabiga mos qiymat.

### RB-022. systemd servis (AL-043..AL-045)

1. `systemctl status <unit> --no-pager`; `journalctl -u <unit> -n 100 --no-pager`.
2. `EnvironmentFile` ruxsati (servis foydalanuvchisi o'qiy oladimi: `sudo -u <user> cat <fayl>`), `ExecStart` yo'li (`which node`).
3. Port band: `ss -ltnp | grep :<port>`.
4. `systemctl restart <unit>`; takrorlanadigan bo'lsa — `systemctl reset-failed`.
5. Namuna: `avlo-contact` — Node (`/usr/bin/node`), `.env` rejimi `0600`, egasi `avlo`.

### RB-023. nginx konfiguratsiyasi (AL-046)

1. `nginx -t` (xato matni qaysi fayl/qator ekanini aytadi).
2. Odatiy: yo'q sertifikat fayli, yo'q kesh papkasi (`/var/cache/nginx/<nom>` — yaratib `chown www-data`), takroriy `server_name`, `include` yo'li.
3. Tuzatgach: `nginx -t && systemctl reload nginx` (`appctl nginx`).
4. **Reload'dan keyin** 2 soniya kutib tekshiring (reload poygasi tufayli birinchi so'rov 5xx bo'lishi mumkin).

### RB-024. Cron / heartbeat (AL-047, AL-048)

1. Cron ishlayaptimi: `systemctl status cron`; fayl: `cat /etc/cron.d/appctl-backup`.
2. Qo'lda ishga tushirish: `appctl backup all`; chiqish va `tail -n 50 /var/log/appctl-backup.log`.
3. Heartbeat URL ga yetish: `curl -fsS -m 10 https://<core>/hb/<TOKEN>?status=ok` (tokenni hujjatga yozmang).
4. `status=fail` bo'lgan bo'lsa — cron bajargan buyruq xato bergan: RB-031.

### RB-025. Sayt ishlamayapti (AL-049, AL-051, AL-053)

Tashqaridan ichkariga ketma-ket:

1. **DNS:** `dig +short A <domen>` — kutilgan IP mi? Yo'q → RB-007.

2. **TLS:** `curl -vI https://<domen>/ 2>&1 | head -n 25` — sertifikat/ulanish → RB-006.
3. **Server tirikmi:** `ping`/SSH → RB-001.
4. **nginx:** `systemctl status nginx`; `nginx -t` → RB-023; 80/443 ochiqmi: `ufw status`; `ss -ltnp | grep -E ':80|:443'`.
5. **Ilovaga yetadimi (serverda):** `curl -I -H 'Host: <domen>' -H 'X-Forwarded-Proto: https' http://127.0.0.1:<port>/`.
6. **Konteynerlar:** RB-020.
7. Front-proksi (S-02): `/etc/nginx/nginx.conf` dagi `map` qoidalarida domen bormi (`grep -n <domen> /etc/nginx/nginx.conf`).
8. Hammasi ichkarida yaxshi, tashqaridan yo'q → firewall/provayder tarmog'i.

## RB-026. Sayt sekin (AL-050, AL-052)

1. Nazoratchi'da vaqt komponentlari: `dns_ms`, `connect_ms`, `tls_ms`, `ttfb_ms` — qaysi biri katta?
2. TTFB katta bo'lsa — ilova: CPU/RAM (RB-009/010), DB so'rovlari (RB-011/3-qadam), Celery to'planib qolgan navbat.
3. Connect/TLS katta — tarmoq yoki server yuklamasi.
4. Yaqinda deploy bo'lganmi (deploy belgilari, F3)? Qaytarish ko'rib chiqing.

## RB-027. Kutilmagan ochiq port (AL-030)

1. `ss -ltnp` — kim tinglayapti va qaysi manzilda (`0.0.0.0` yoki `127.0.0.1`).
2. Docker portlari: `docker ps --format '{{.Names}} {{.Ports}}'`. Docker nashr qilgan port `ufw` ni chetlab o'tadi.
3. Tuzatish: compose'da `ports: !override ["127.0.0.1:<port>:<port>"]` (yoki `.appctl` dagi `COMPOSE_EXTRA` orqali), `appctl up <loyiha>`.
4. Tashqaridan tasdiqlash: `nmap -Pn -p <port> <ip>` (yopiq bo'lishi kerak).
5. Ruxsatsiz konteyner/jarayon bo'lsa — RB-014 (xavfsizlik gumoni).

## RB-030. Avariya tiklash — ma'lumotni zaxiradan qaytarish

**Diqqat:** `appctl restore` mavjud bazani **o'chiradi**. Faqat tiklash kerak bo'lgan holatda.

1. Maqsadni aniqlang: qaysi loyiha, qaysi vaqtga (RPO).
2. Eng yangi **yaxlit** zaxira: Nazoratchi zaxira kartasi (`gz_ok`, hajm normal) — mahalliy yoki offsite. Offsite'dan olish: `rsync -a <offsite>:/<yo'l>/<loyiha>/daily/<fayl> /opt/applications/backups/<loyiha>/daily/`.
3. Joriy holatni saqlab qo'ying (agar baza hamon o'qilsa): `appctl backup <loyiha>` yoki qo'lda `pg_dump`.
4. `appctl restore <loyiha> /opt/applications/backups/<loyiha>/daily/<fayl>.sql.gz --yes`.
5. Media/fayl zaxirasi kerak bo'lsa: `tar -xzf <fayl>.tar.gz -C /` (yo'llar `BACKUP_PATHS` ga mos).
6. Tekshirish: `appctl ps <loyiha>`, sog'liq URL, ilovada ma'lumot borligi (oxirgi yozuvlar).
7. Ma'lumot tiklangan vaqtdan keyingi yo'qotish haqida foydalanuvchilarga xabar (zarur bo'lsa).
8. Hodisa tahlili va drill natijasini yangilash.

## RB-031. Zaxira kechikdi / muvaffaqiyatsiz (AL-068..AL-070, AL-074)

1. `tail -n 50 /var/log/appctl-backup.log` — xato matni.
2. Cron ishlaganmi: `grep CRON /var/log/syslog | tail`; `cat /etc/cron.d/appctl-backup`.
3. Qo'lda: `appctl backup <loyiha>` — xato aniq ko'rinadi.
4. Odatiy: DB konteyner o'chiq (RB-020); `.appctl` da noto'g'ri `DB_USER`/`DB_NAME`/`DB_SERVICE`; disk to'lgan (RB-004); ruxsat; `pg_dump` versiya nomuvofiqligi.
5. Tuzatgach: qo'lda zaxira → Nazoratchi'da yosh qayta hisoblanadi.

## RB-032. Zaxira shubhali yoki buzuq (AL-071..AL-073)

1. Hajm tarixi: Nazoratchi zaxira kartasida (kunlik hajm grafigi).
2. Fayl: `gzip -t <fayl>`; `zcat <fayl> | head -n 50` (PostgreSQL dump belgilari); `zcat <fayl> | grep -c '^COPY'`.
3. Manba bazada ma'lumot yo'qolganmi? (`SELECT count(*)` asosiy jadvallarda) — "kichik zaxira" haqiqiy ma'lumot yo'qolishining belgisi bo'lishi mumkin.
4. Darhol yangi zaxira oling; ikkalasini solishtiring. Eski yaxshi nusxalar o'chib ketmasligi uchun saqlash (`KEEP_DAILY`) ni vaqtincha oshiring yoki nusxalang.
5. Hajm o'sishi "katta" (AL-072) — odatda yangi ma'lumot yoki log jadvali; sabab topilsa `baseline_reset`.

### RB-033. Offsite zaxira (AL-075..AL-078, AL-080)

1. Sozlanmagan (AL-075): `/opt/applications/appctl.conf` da `OFFSITE=root@<host>:/yo'l`; kalit: `ssh-keygen -t ed25519`, `ssh-copy-id`; sinov: `ssh -o BatchMode=yes <host> 'df -h /yo\'l'`.
2. Eskirgan (AL-076/077): `appctl backup all` oxirida `→ offsite:` qatorini ko'ring; qo'lda: `rsync -a --dry-run -e "ssh -o BatchMode=yes" /opt/applications/backups/ <host>:/yo'l/`.
3. SSH muammo: host kaliti o'zgargan (`ssh-keygen -R`), kalit yo'q, maqsad server o'chiq (RB-001).
4. Maqsadda joy: `ssh <host> df -h`; eski nusxalarni tozalash (maqsadda alohida cron).
5. Mos emas (AL-078): oxirgi fayl nomi/hajmi; rsync oraliq holatida bo'lishi mumkin (30 daq kuting); davom etsa — qo'lda sinxronlash va log.

### RB-034. Restore drill (AL-081..AL-083)

10.8-bo'limdagi qadamlar. Qisqa: vaqtinchalik `postgres:<major>` konteyner → bazani yaratish → `gunzip -c <fayl> | psql ... -v ON_ERROR_STOP=1` → jadval/qator sonini solishtirish → natijani panelga kiritish. **Ishlab turgan bazaga tegmang.**

### RB-040. Alert dvigateli / prober qotdi (AL-055, AL-084, AL-085)

1. `docker compose ps`; `docker compose logs --tail 200 evaluator prober`.
2. Bazaga ulanish: `docker compose exec db pg_isready`; ulanishlar soni: `SELECT count(*) FROM pg_stat_activity`.
3. `advisory lock` ushlanib qolgan bo'lishi mumkin: `SELECT * FROM pg_locks WHERE locktype='advisory'`.
4. Qayta ishga tushirish: `docker compose restart evaluator prober`.
5. Tez-tez qotsa — xotira (`docker stats`), sekin so'rovlar (`EXPLAIN`), qoidalar soni.

### RB-041. Bildirishnoma kanali ishlamayapti (AL-086)

1. Panel → Sozlamalar → Kanallar → "Sinov" tugmasi; xato matni (`last_error`).
2. Telegram: tokenni tekshirish — `curl -sS https://api.telegram.org/bot<TOKEN>/getMe` (natijani/tokenni tashqariga chiqarmang); bot guruhdan chiqarilmaganmi; `chat_id` to'g'rimi.
3. Core'dan tashqariga 443: `curl -sS -m 10 https://api.telegram.org` (firewall/DNS).
4. Webhook: `getWebhookInfo` (`last_error_message`).
5. Email: SMTP ulanishi, parol; spam.
6. Vaqtincha: boshqa kanalni yoqing (email); tuzatgach "sinov".

### RB-042. Nazoratchi bazasi / saqlash muammolari (AL-087..AL-089)

1. Hajm: `SELECT hypertable_name, pg_size_pretty(hypertable_size(format('%I',hypertable_name)::regclass)) FROM timescaledb_information.hypertables;`.
2. Fon vazifalari: `SELECT job_id, application_name, last_run_status, next_start FROM timescaledb_information.job_stats js JOIN timescaledb_information.jobs j USING(job_id);`.
3. Siqish/retention ishlamayotgan bo'lsa — `CALL run_job(<job_id>)`; xato loglari.
4. Hajm o'ssa — retention qisqartirish (sozlama), kardinallik tekshiruvi (qaysi host ko'p `series` yaratyapti), diskni oshirish.
5. Ingest xatolari (AL-088): `docker compose logs web | grep -E "ingest|ERROR"`; agent versiyasi/sxema mosligi.

### RB-043. Agent versiyasi va yangilash (AL-004)

1. Panelda agent versiyasi va "yangilash kerak" belgisi.
2. Yangilash `update` buyrug'i bilan avtomatik; qo'lda: `curl -fsSL https://<core>/install.sh | sudo bash -s -- --upgrade`.
3. Imzo xatosi (`agent_update_rejected`): core'dagi fayl noto'g'ri imzolangan bo'lishi mumkin — **qo'lda o'rnatmang**, CI artefaktini tekshiring (14.5.4).
4. Rollback: `/usr/local/bin/nazoratchi-agent.old` ni qaytarib `systemctl restart nazoratchi-agent`.

### RB-044. Nazoratchi'ni yangi serverda tiklash

15.9-bo'limdagi qadamlar (RTO ≤ 60 daqiqa). Eslatma: `master.key` va `age` maxfiy kaliti parol menejeridan; zaxira fayli Owner'ning Telegram chatidan olinadi.

## 18.3. `state` shart manbalari (12.3.5)

| `source` | Maydonlar | Qiymatlar |
|---|---|---|
| `probe` | `state`, `consecutive_failures`, `days_left`, `total_ms`, `ttfb_ms`, `uptime_30d` | `up/down/unknown`; sonlar |

| source | Maydonlar | Qiymatlar |
|---|---|---|
| backup | status, age_h, size_ratio, gz_ok, offsite_lag_h, offsite_in_sync, drill_age_days, drill_result | ok/fail/stale/...; sonlar; bool |
| cert | days_left, chain_ok, hostname_ok, renew_failures | sonlar; bool |
| dns | status, matches_expected, retired_ip, resolver_disagree_min | match/propagating/mismatch/retired_ip/nxdomain; bool; daqiqa |
| heartbeat | age_s, last_ok | son; bool |
| host | status, last_seen_age_s, tcp22_ok, agent_version_lag, kernel_pending | online/offline/paused; sonlar; bool |
| system | evaluator_age_s, probe_stall_s, db_size_ratio, ingest_error_ratio, channel_failures | sonlar |

## 18.4. Tekshiruv (probe) konfiguratsiya sxemalari

### 18.4.1. http

```
{
  "url": "https://nazar.avlo.ai/healthz/",
  "method": "GET",                      // GET | HEAD
  "expected_status": [200],             // [200] yoki ["200-299"] yoki [200, 301]
  "keyword": {"text": "ok", "mode": "contains"},   // contains | not_contains; null bo'lishi mumkin
  "follow_redirects": true,
  "max_redirects": 5,
  "headers": {"X-Check": "nazoratchi"},     // maxfiy qiymatli sarlavhalar secrets_enc da
  "verify_tls": true,
  "resolve_to": null,                   // "164.68.105.194" — cut-over oldindan tekshiruvi (SNI/Host saqlanadi)
  "latency_warn_ms": 3000,
  "max_body_bytes": 65536,
  "allow_private": false,   // faqat Owner o'rnatadi (SSRF)
  "primary": false          // true: AL-049 (critical), aks holda AL-049H (high)
}
```

### 18.4.2. tcp

```
{ "host": "164.68.105.194", "port": 5432, "expect": "closed" }
```

expect: open (standart) yoki closed (xavfsizlik tekshiruvi). Qo'shimcha: allow_private, primary (http dagidek).

### 18.4.3. tls

```
{ "host": "nazar.avlo.ai", "port": 443, "sni": "nazar.avlo.ai",
  "warn_days": 21, "high_days": 7, "critical_days": 3, "check_chain": true, "check_hostname": true }
```

### 18.4.4. dns

```
{
  "name": "lutfanai.uz",
  "rtype": "A",
  "resolvers": ["8.8.8.8", "1.1.1.1", "9.9.9.9", "ns:auto"],
  "expected": ["164.68.105.194"],
  "retired_ips": ["34.173.220.247"],
  "mismatch_after_min": 10,
  "watch": {"enabled": false, "until": null, "interval_s": 60}
}
```

ns:auto — domenning haqiqiy NS serverlarini aniqlab so'raydi.

### 18.4.5. whois

```
{ "domain": "avlo.ai", "source": "rdap", "warn_days": 30, "critical_days": 7 }
```

.uz uchun "source": "manual" va expires_at qo'lda.

## 18.5. Boshlang'ich reyestr (seed/inventory.yaml) formati va mazmuni

Quyidagi fayl 2026-10-22 holatidagi inventarni (2-bo'lim) Nazoratchi'ga yuklaydi. Qiymatlari aniqlanmagan maydonlar (?) bo'sh qoldirilgan; ularni dasturchi birinchi ishga tushirishdan oldin to'ldiradi.

```
version: 1
retired_ips: [34.173.220.247]   # GCP o'chirilgan (Q-12)

servers:
  - { name: main-2,     provider: Contabo, public_ipv4: 164.68.105.194, role: production, tags: [contabo, docker, appctl, watchdog] }
  - { name: vmd208589,  provider: "",      public_ipv4: 173.249.37.152, role: production, tags: [docker, front-proxy, aaclub] }
  - { name: core-1,     provider: Contabo, public_ipv4: 185.214.135.9,  role: core,       tags: [contabo, core, offsite] }   # S-00, Owner qa
  - { name: backup-1,   provider: "",      public_ipv4: 38.242.143.248, role: unused,     tags: [unused], status: archived }       # S-03 ish

projects:
  - slug: Nazar
    name: Nazar (Beyondlinks)
    kind: compose
    servers: [{ server: main-2, environment: production, ports: [8020, 3020] }]
    domains:
      - { fqdn: nazar.avlo.ai,       primary: true, dns_provider: Namecheap }
      - { fqdn: app.beyondlinks.uz,  primary: true, dns_provider: "" }
    probes:
      - { type: http, url: "https://nazar.avlo.ai/healthz/", expected_status: [200] }
    backup: { expect_daily: true, require_offsite: true, db: postgres16 }

  - slug: BiznesRadar
    name: BiznesRadar
    kind: compose
    servers: [{ server: main-2, environment: production, ports: [8030, 4321] }]
    domains:
      - { fqdn: biznesradar.uz,        primary: true }
      - { fqdn: api.biznesradar.uz }
      - { fqdn: admin.biznesradar.uz }
      - { fqdn: cabinet.biznesradar.uz }
    backup: { expect_daily: true, expect_weekly_files: true, db: postgres17 }

  - slug: beyondlinks_landing
    name: Beyondlinks landing
    kind: static
    servers: [{ server: main-2, environment: production }]
    domains: [{ fqdn: beyondlinks.uz, primary: true }]
    backup: { expect_daily: false }

  - slug: freighthubllc
    name: FreightHub LLC
    kind: static
    servers: [{ server: main-2, environment: production }]
    domains: [{ fqdn: freighthubllc.co, primary: true }, { fqdn: freighthubllc.com }]
    backup: { expect_daily: false }

  - slug: lutfan_ai
    name: Lutfan AI
    kind: compose
    servers: [{ server: main-2, environment: production, ports: [8080, 9000] }]
    domains: [{ fqdn: lutfanai.uz, primary: true }, { fqdn: lutfan.israilov.uz }]
    backup: { expect_daily: true, expect_weekly_files: true, db: postgres15 }

  - slug: wmg
    name: WMG landing sahifalari
    kind: compose
    servers: [{ server: main-2, environment: production, ports: [8011, 8012] }]
    domains:
      - { fqdn: moliyaviyerkinlik.wmg.uz, controlled_by: client, dns_provider: Beget }
      - { fqdn: investitsiya.wmg.uz,      controlled_by: client, dns_provider: Beget }
    backup: { expect_daily: false, expect_weekly_files: true }      # SQLite + media

  - slug: avlo_site
    name: avlo.ai sayt va kontakt formasi
    kind: static
    servers: [{ server: main-2, environment: production, ports: [3017] }]
    domains: [{ fqdn: avlo.ai, primary: true, dns_provider: Namecheap }, { fqdn: www.avlo.ai }]
    services_systemd: [avlo-contact]
    backup: { expect_daily: false }

  - slug: asadbekashurov
    name: asadbekashurov.uz landing
    kind: static
    servers: [{ server: vmd208589, environment: production, ports: [38080, 38443] }]
    domains: [{ fqdn: asadbekashurov.uz, primary: true, dns_provider: ahost.uz }, { fqdn: www.asadbekashurov.uz }]
    backup: { expect_daily: false }

  - slug: aaclub_staging
    name: Aaclub (staging)
    kind: compose
    servers: [{ server: vmd208589, environment: staging, ports: [28080, 28443] }]
    domains: []        # *.staging.<domen> (aniqlanadi)
    backup: { expect_daily: false }    # aniqlanadi

  # Hali ko'chirilmagan (eski serverdan): reyestrda bor, tekshiruvsiz.
  - { slug: Matnchi,   name: Matnchi,   kind: compose, moved: false, domains: [{ fqdn: matnchi.avlo.ai }] }
  - { slug: CallAI,    name: CallAI,    kind: compose, moved: false, domains: [{ fqdn: call.avlo.ai }] }
  - { slug: Transcription, name: Transcription, kind: systemd, moved: false, domains: [{ fqdn: transcription.avlo.ai }] }
  - { slug: avlocorp,  name: avlocorp.com (Quran), kind: other, moved: false, domains: [{ fqdn: avlocorp.com }] }
  - { slug: avloai_uz, name: avloai.uz, kind: static, moved: false, domains: [{ fqdn: avloai.uz }] }

heartbeats:
  - { name: appctl-backup@main-2,         server: main-2,    expected_every_s: 86400, grace_s: 7200 }
  - { name: certbot-renew@main-2,         server: main-2,    expected_every_s: 86400, grace_s: 43200 }
  - { name: asadbekashurov-certbot@vmd208589, server: vmd208589, expected_every_s: 86400, grace_s: 10800 }
  - { name: nazoratchi-self-backup@core-1, server: core-1, expected_every_s: 86400, grace_s: 7200 }
```

## 18.6. Hujjat bilan bog'liq ma'lumotlar

| Mavzu | Qayerda |
|---|---|
| Talablar ro'yxati va sanog'i | 4, 5-bo'limlar va 18.7 |
| Alert qoidalari katalogi | 12.7 |
| Runbook'lar | 18.2 |
| Metrikalar katalogi | 7.4 |
| Ma'lumotlar modeli (DDL) | 8 |
| API | 9 |
| Rejalash va xavflar | 17 |

## 18.7. Talablar sanog'i

Sonlar hujjat jadvallaridan skript bilan sanalgan (ID bo'yicha, takrorlanishsiz) va hujjat nashriga qadar qayta tekshiriladi.

### 18.7.1. Ustuvorlik va bosqich bo'yicha

| | F1 | F2 | F3 | Jami |
|---|---|---|---|---|
| Funksional (FR) | 91 | 63 | 24 | **178** |
| Nofunksional (NFR) | 40 | 22 | 3 | **65** |
| **Jami talablar** | **131** | **85** | **27** | **243** |

| | M | S | C | Jami |
|---|---|---|---|---|
| Funksional (FR) | 95 | 59 | 24 | 178 |
| Nofunksional (NFR) | 47 | 15 | 3 | 65 |
| **Jami** | **142** | **74** | **27** | **243** |

### 18.7.2. Boshqa ro'yxatlar

| Ro'yxat | Soni | Qayerda |
|---|---|---|
| Foydalanuvchi hikoyalari (US) | 43 | 3.3 |
| Foydalanish stsenariylari (UC) | 12 | 3.4 |
| Alert qoidalari (AL) | 89 | 12.7 |
| Runbook'lar (RB) | 34 | 18.2 |
| Loyiha xavflari (R-101…R-123) | 23 | 17.7 |
| Joriy infratuzilma xavflari (R-001…R-009) | 9 | 2.8 |
| Arxitektura qarorlari (ADR) | 14 | 6.3 |
| Metrikalar (nomlangan; faktlarsiz) | ≈ 130 | 7.4 |
| Ma'lumotlar bazasi jadvallari | 32 (+ 3 agregat ko'rinish) | 8 |
| API endpointlari (metod + yo'l juftliklari) | 114 | 9 |
| Ekranlar (SCR) | 20 | 13.3 |
| Owner qarori kerak bo'lgan savollar | 11 | 17.11 |

### 18.7.3. Talablarni bosqichlarga o'tkazish qoidasi

- **F1 ga kiradi:** server/loyiha reyestri, agent (tizim, disk, tarmoq, Docker, systemd, zaxira, sertifikat), heartbeat, HTTP/TLS/DNS tekshiruvlari (asosiy), 35 ta alert (server o'chdi, resurslar, konteyner, sayt, sertifikat, DNS, zaxira, heartbeat, o'zini kuzatish), Telegram + ack, asosiy panel, watchdog, o'z zaxirasi, deploy va asosiy runbook'lar.
- **F2 ga kiradi:** 2FA, kengaytirilgan zaxira (yaxlitlik, offsite 2-daraja, drill), kesib o'tish yordamchisi, qoidalar UI, titrash/guruhlash, hisobotlar, ru/en, qorong'i mavzu, SSE, yangilanish.
- **F3 ga kiradi:** viewer cheklovi, status sahifa, ko'p lokatsiyali tekshiruv, avto-drill, eskalatsiya, qo'shimcha xavfsizlik metrikalari, supply-chain mustahkamlash.